UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH
Escola Superior d'Enginyeries Industrial,
Aeroespacial i Audiovisual de Terrassa

BACHELOR FINAL THESIS

3D Minimum-Time Trajectory Planning for Aircraft Climb, Cruise, and Descent with Accelerated Flight Profiles in a Mission Management System.

Document:

Project Charter

Author:

Pablo Urioste Alarcón

Director/Co-director:

Alex Ferrer Ferre

Degree:

Bachelor in Aerospace Technologies Engineering

Examination session:

Spring 2026

Abstract

This thesis develops and validates a high-performance framework for computing optimal three-dimensional aircraft trajectories under realistic operational constraints, for a Linux-based Mission Management System bound by a real-time threshold. The task is posed as an optimal control problem (OCP): a fixed-wing aircraft must fly from origin to destination while respecting its flight envelope and performance limits, terrain clearance from Digital Elevation Models, no-fly-zone (NFZ) avoidance and ordered waypoint routing, under a minimum-time or a minimum-fuel criterion.

At its core lies the numerical optimization problem. The continuous OCP is transcribed by direct collocation into a large, sparse nonlinear program (NLP) and solved with an interior-point method, in two formulations: a reduced uncoupled formulation (UF) and the full coupled formulation (FC). As the NLP is strongly nonconvex, its convergence, feasibility and runtime are acutely sensitive to the initial guess: a poor seed places the solver in an unfavourable basin of attraction, from which it converges slowly, settles on a suboptimal trajectory, or fails on the very missions operational planning demands.

The central contribution is a physics-informed warm-start pipeline that accelerates and robustifies this NLP. A kinodynamic Hybrid A* planner produces a flyable, environment-aware path in the correct homotopy class; an inverse-dynamics layer lifts it into a dynamically consistent state and control history; this seeds the UF solve, whose optimum in turn conditions the FC solve. Across a benchmark of eight scenarios under both objectives, the warm start keeps the solver feasible where linear and cold initialisations fail or stall, cuts the full-control solve time by roughly forty percent while preserving optimality, and confirms that a well-engineered initial guess, more than the solver configuration alone, is the key enabler of real-time optimal trajectory generation.

Keywords: trajectory optimization; optimal control problem (OCP); nonlinear programming (NLP); direct collocation; interior-point method; warm start; Hybrid A*; inverse dynamics; fixed-wing UAV.

Resum

Aquesta tesi desenvolupa i valida un marc d'alt rendiment per al càlcul de trajectòries òptimes tridimensionals d'aeronaus sota restriccions operatives realistes, per a un Sistema de Gestió de Missió basat en Linux subjecte a un llindar de temps real. La tasca es formula com un problema de control òptim (PCO): una aeronau d'ala fixa ha de volar d'un origen a un destí respectant la seva envolupant de vol i els seus límits de prestacions, la separació del terreny de Models Digitals d'Elevació, l'evitació de zones d'exclusió aèria (NFZ) i el pas ordenat per punts de ruta, sota un criteri de temps mínim o de consum mínim de combustible.

El nucli del treball és el problema d'optimització numèrica. El PCO continu es transcriu mitjançant col·locació directa en un problema de programació no lineal (PNL) gran i dispers, i es resol amb un mètode de punt interior, en dues formulacions: una formulació reduïda i desacoblada (UF) i la formulació acoblada completa (FC). Com que el PNL és fortament no convex, la seva convergència, factibilitat i temps d'execució són molt sensibles a l'estimació inicial: una llavor deficient situa el resolutor en una conca d'atracció desfavorable, des de la qual convergeix lentament, s'estabilitza en una trajectòria subòptima o falla en les missions que la planificació operativa exigeix.

La contribució central és una cadena d'inicialització informada per la física que accelera i robustix aquest PNL. Un planificador cinodinàmic Hybrid A* genera un camí volable i conscient de l'entorn dins la classe d'homotopia correcta; una capa de dinàmica inversa l'eleva a un historial d'estat i control dinàmicament consistent; això inicialitza la resolució UF, l'òptim de la qual condiciona la resolució FC. Sobre un conjunt de vuit escenaris i tots dos objectius, la inicialització informada manté el resolutor factible on les inicialitzacions lineal i freda fallen o s'encallen, redueix el temps de resolució de la formulació completa en aproximadament un quaranta per cent tot preservant l'optimalitat, i confirma que una estimació inicial ben dissenyada, més que la configuració del resolutor per si sola, és el factor clau per generar trajectòries òptimes en temps real.

Paraules clau: optimització de trajectòries; problema de control òptim (OCP); problema no lineal (NLP); col·locació directa; mètode de punt interior; warm start; Hybrid A*; dinàmica inversa; UAV d'ala fixa.

Declaration of Use of Generative Artificial Intelligence Tools

In accordance with the academic integrity policy of the UPC-ESEIAAT, the author declares the following use of generative artificial intelligence (AI) tools in the preparation of this thesis. Two tools were used: **Google Gemini Pro** (web interface, 2024–2026) and **Claude Code by Anthropic** (`claude-sonnet-4-6`, via desktop app and Vscode integration, 2025–2026).

Google Gemini supported literature and concept exploration and occasional phrasing: broad scans of topics such as trajectory optimisation, direct collocation and aircraft dynamics notation, suggested search directions, short summaries of public documentation, and alternative wordings for sentences the author had already drafted. All citations, equations and claims were verified against primary sources by the author, and Gemini output was never treated as authoritative.

Claude Code supported software implementation and targeted technical edits: debugging compiler and runtime errors, refactoring and tracing numerical issues in the OptiFlightC codebase, and assisting with equation environments, table layouts, cross-references and consistency checks across the files. When given a concrete goal, the assistant proposed edits to existing files, but the author retained full responsibility for algorithm choice, physical models, units, notation and the validation of results, and no chapter was accepted without author review.

In summary, generative AI assisted with LaTeX formatting and readability, code debugging and concept organisation, and structural review, while all core simulations, figures and scientific conclusions remain autonomously generated by the author. Under no circumstances has generative AI been treated as a co-author, nor has it shaped the final results and conclusions: every chapter, deduction, decision and outcome was independently conceived, analysed and processed by the author, with AI used strictly as an assistive tool.

Contents

Abstract . . . . . **ii**

Resum . . . . . **iii**

Declaration of Use of Generative Artificial Intelligence Tools . . . . . **iv**

List of Abbreviations and Mathematical Notation . . . . . **xi**

1 Introduction **1**

1.1 Objectives 1

1.2 Project Scope 2

1.2.1 In-Scope: Detailed Developments 2

1.2.2 Out-of-Scope: Project Boundaries 3

1.2.3 Flow Architecture and Work Packages 4

1.3 Requirements 4

1.3.1 Functional and Dynamic Requirements 5

1.3.2 Software and Interface Requirements 5

1.3.3 Numerical and Performance Requirements 6

1.3.4 Safety and Robustness Requirements 6

1.4 Justification 7

1.5 Planning and Scheduling 9

2 State of the art **13**

2.1 Geometric and Graph-Based Planning 13

2.2 Numerical Trajectory Optimization 15

2.3 Acceleration and Seeding Techniques 17

2.3.1 Numerical Structure and Gradient Information 18

2.4 Synthesis and Research Gap . . . 19

3 Methodology . . . 20

3.1 The Challenge of Trajectory Optimization with IPOPT . . . 20

3.2 Three-Tier Incremental Development Strategy . . . 21

3.2.1 Tier 1 — Analytical and Semi-Analytical Exploration . . . 21

3.2.2 Tier 2 — Benchmark Optimal-Control Problem with Minimal Constraints . . . 22

3.2.3 Tier 3 — Fully Developed Problem with External Constraints and Warm Start . . . 23

3.3 Validation Procedure . . . 23

3.4 Warm-Start Performance Assessment . . . 24

4 Alternative Solutions and Decision . . . 27

4.1 Overview of Candidate Approaches . . . 27

4.1.1 Geometric Path Planner (Warm-Start Producer) . . . 27

4.1.2 Trajectory Optimization Transcription . . . 28

4.1.3 Aircraft Dynamic Model Used Inside the Optimizer . . . 29

4.2 Decision Summary and Trade-offs . . . 29

5 Theoretical background . . . 32

5.1 Problem Statement . . . 32

5.1.1 The Continuous-Time Optimal Control Problem (OCP) . . . 32

5.1.2 Numerical Transcription to Nonlinear Programming (NLP) . . . 34

5.1.3 Constraints and the Lagrangian Formulation . . . 36

5.1.4 Karush-Kuhn-Tucker (KKT) Optimality Conditions . . . 36

5.2 Uncoupled Trajectory Dynamics . . . 37

5.2.1 Reference Frames and Coordinate Systems . . . 37

5.2.2 State and Control Vectors . . . 38

5.2.3 Equations of Motion . . . 39

5.2.4 Aerodynamic Model . . . 41

5.2.5 Summary of the Governing Equations . . . 42

5.3 Full-Control Formulation . . . 42

5.3.1 State and Control Vectors . . . 43

5.3.2 The Full-Control Coupled Resolution . . . 44

5.3.3 External Wind and Aerodynamic Angles . . . 45

5.3.4 Aerodynamic Coefficient Model . . . 46

5.3.5 Rotation of Forces and Moments to Body Axes . . . 49
5.3.6 Translational Dynamics . . . 49
5.3.7 Rotational Dynamics . . . 50
5.3.8 Euler Angle Kinematics . . . 50
5.3.9 Position Kinematics . . . 51
5.3.10 Mass and Energy Evolution . . . 51
5.3.11 Summary of the Governing Equations . . . 52
5.3.12 Comparison of the Two Formulations . . . 52

6 Numerical Implementation . . . 54

6.1 Numerical Transcription of the OCP . . . 55
6.1.1 Decision-Vector Layout . . . 55
6.1.2 Trapezoidal Direct Collocation . . . 55
6.1.3 Residual Normalization . . . 56

6.2 Constraint Architecture . . . 56
6.2.1 Domain and Environment Constraints . . . 58
6.2.2 Collocation and Dynamic Consistency Constraints . . . 60
6.2.3 Boundary Equality Constraints . . . 61
6.2.4 Performance Constraints . . . 62
6.2.5 Summary of the Constraint Architecture . . . 63

6.3 Cost Function Definitions . . . 64
6.3.1 Objective Function: The Time–Fuel Trade-off . . . 64
6.3.2 Cost regularization . . . 66

6.4 Validation Results . . . 69
6.4.1 Test Scenarios for Numerical Validation . . . 69
6.4.2 Scenario Definitions . . . 69
6.4.3 Validation Metrics . . . 70
6.4.4 Trajectory Observation and Analysis . . . 71
6.4.5 Criterion Comparison: Minimum-Time vs Minimum-Fuel . . . 74
6.4.6 Formulation Comparison: UF vs FC . . . 77
6.4.7 Constraint Satisfaction . . . 81

6.5 Impact of Cost Regularization on the Optimum . . . 88
6.5.1 Headline Metrics . . . 89
6.5.2 Qualitative Impact on the Trajectory . . . 90

6.6 Warm-Start Acceleration . . . 96
6.6.1 Role of the Initial Guess in Interior-Point Optimization . . . 96

6.6.2 Hybrid A* Path Planner . . . 97
6.6.3 Inverse-Dynamics Trajectory Reconstruction . . . 99
6.6.4 UF → FC State Projection . . . 100
6.6.5 Numerical Pipeline Overview . . . 100
6.6.6 Warm-Start Performance Results . . . 102

7 Budget and Economic Viability . . . 118
7.1 Costing Methodology . . . 118
7.2 Personnel Costs . . . 118
7.3 Software, Equipment and Indirect Costs . . . 119
7.4 Total Budget . . . 120

8 Environmental and Social Impact . . . 121
8.1 Environmental Impact . . . 121
8.2 Social Impact . . . 122
8.3 Balance and Outlook . . . 122

9 Conclusions . . . 124
9.1 General Conclusions . . . 124
9.2 Fulfillment of Requirements . . . 125
9.3 Limitations . . . 127
9.4 Future Work and Next Steps . . . 127

References . . . 129

Appendices . . . 136

Appendix A Derivation of the Equations of Motion . . . 136
A.1 General Body-Axis Dynamics . . . 136
A.2 Transition to the Wind-Axis Frame . . . 137
A.2.1 Reference frames and rotation angles . . . 137
A.2.2 From body-axis forces to drag, sideforce, and lift . . . 138
A.2.3 Thrust from body axes to wind axes . . . 138
A.2.4 Weight and other forces . . . 139

A.2.5 Summary . . . 139

A.3 Final Trajectory Equations (UF Formulation) . . . 139
A.3.1 Dynamic Equations . . . 140
A.3.2 Kinematic Equations . . . 140
A.3.3 Mass and Energy Relation . . . 140

A.4 Problem Statement and Degrees of Freedom . . . 140
A.4.1 Comparison of the Two Formulations . . . 140

Appendix B Coordinate Frame Transformations . . . 142

B.1 Motivation . . . 142

B.2 Reference ellipsoid: WGS-84 . . . 143

B.3 Geodetic to ECEF (*toECEF*) . . . 143

B.4 Geodetic to NED (*geodeticToNed*) . . . 144

B.5 Altitude curvature correction . . . 144

B.6 NED to geodetic (*nedToGeodetic*) . . . 145

B.7 Algorithm schema . . . 146

B.8 Summary table . . . 146

Appendix C Aircraft Specifications — MSN012 (Flyox I) . . . 148

C.1 Airframe and performance envelope . . . 149

C.2 Aerodynamic and propulsion data . . . 150

Appendix D Regularisation penalties: full per-scenario validation . . . 152

D.1 Weight tuning protocol . . . 152

D.2 Reading guide . . . 153

D.3 Scenario SC-00 . . . 154

D.4 Scenario SC-01 . . . 156

D.5 Scenario SC-02 . . . 160

D.6 Scenario SC-03 . . . 162

D.7 Cross-scenario tables . . . 165

Appendix E Per-run constraint detail . . . 170

E.1 Decision-variable and hard-constraint bounds 170
E.2 Worst-case margin comparison across criteria and formulations 172
E.3 SC-00 — straight-line level flight 177
E.3.1 Uncoupled formulation (UF) 177
E.3.2 Full-control formulation (FC) 181
E.4 SC-01 — straight-line steady climb 185
E.4.1 Uncoupled formulation (UF) 185
E.4.2 Full-control formulation (FC) 189
E.5 SC-02 — 180° horizontal turn 193
E.5.1 Uncoupled formulation (UF) 193
E.5.2 Full-control formulation (FC) 197
E.6 SC-03 — combined climb-and-turn 201
E.6.1 Uncoupled formulation (UF) 201
E.6.2 Full-control formulation (FC) 205

Appendix F Min-time vs min-fuel comparison: per-scenario tables 209

F.1 Scenario SC-00 — straight-and-level baseline 210
F.2 Scenario SC-01 — straight climb 211
F.3 Scenario SC-02 — 180° heading reversal at level flight 212
F.4 Scenario SC-03 — climb plus heading reversal 213

List of Abbreviations and Mathematical Notation

Abbreviations

Abbreviation	Full term or meaning
3-DOF	Three degrees of freedom
6-DOF	Six degrees of freedom
AoA	Angle of attack
BADA	Base of Aircraft Data (EUROCONTROL)
Bolza problem	Optimal control with Mayer (terminal) plus Lagrange (running) cost
CAGR	Compound annual growth rate
CFD	Computational fluid dynamics
CIMNE	International Centre for Numerical Methods in Engineering
DEM	Digital elevation model
DSM	Dynamic safety margin
ECEF	Earth-centred Earth-fixed (Cartesian frame)
ECTS	European Credit Transfer and Accumulation System
ERG	Explicit reference governor
FC	Full-Control Formulation (body-axis dynamics with control surfaces)
FMS	Flight management system
GNSS	Global navigation satellite system
IPM	Interior point method
Ipopt	Interior Point Optimizer (open-source NLP solver)
ISA	International Standard Atmosphere
KKT	Karush–Kuhn–Tucker (optimality conditions)
MMS	Mission management system
NED	North–East–Down (local geodetic frame)
NFZ	No-fly zone

Abbreviation	Full term or meaning
NLP	Nonlinear programming
NMPC	Nonlinear model predictive control
OCP	Optimal control problem
ODE	Ordinary differential equation
PRM	Probabilistic roadmap
RRT	Rapidly-exploring random tree
SFC	*Context-dependent:* (1) specific fuel consumption in fuel-burn models; (2) safe flight corridor in path planning
SiL	Software-in-the-loop
SLAM	Simultaneous Localization and Mapping
SoC	System-on-chip (state of charge when used for batteries)
SOTA	State of the art
TAS	True airspeed
TFG	Bachelor's thesis (*Treball de Final de Grau*)
UAM	Urban air mobility
UAV	Unmanned aerial vehicle
UF	Uncoupled Formulation (wind-axis point mass, C_L as control)
UPC	Universitat Politècnica de Catalunya
WGS84	World Geodetic System 1984
WP	Work package

Mathematical notation

Symbol	Description
Optimal control problem and NLP transcription	
$\mathbf{x}$	State vector of the generic continuous-time OCP ($\mathbb{R}^n$)
$\mathbf{u}$	Control vector of the generic continuous-time OCP ($\mathbb{R}^m$)
J	Scalar cost functional (Bolza form)
Φ	Mayer (terminal) cost
$\mathcal{L}$	Lagrange (running) integrand; also used for rolling moment in *wind* axes (see **Aerodynamics** below)
t_0, t_f	Initial and final time; t_f may be a free optimisation parameter
$\mathbf{f}$	Right-hand side of state ODEs, $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}, t)$
Ψ	Terminal constraint vector
$\mathbf{h}$, $\mathbf{h}_{\min}$, $\mathbf{h}_{\max}$	Path constraint functions and bounds

Symbol	Description
$\mathbf{z}$	Stacked NLP decision vector after direct-collocation transcription: node samples of the state $\mathbf{x}$ and control $\mathbf{u}$ together with the free final time t_f
$F,\ g,\ h$	Discrete objective and constraint functions in NLP notation
$\lambda,\ \nu$	Lagrange multipliers (equalities / inequalities)
N	Number of discretisation intervals in direct transcription
Frames, positions, and kinematic angles	
$\mathcal{F}_{\text{NED}},\ \mathcal{F}_W$	Local North–East–Down frame; wind-axis frame
$\varphi,\ \lambda,\ h$	Geodetic latitude, longitude, ellipsoidal height
$x,\ y,\ z$	NED position components (UF); altitude $h = -z$
$x_e,\ y_e,\ z_e$	NED position (FC notation)
$u,\ v,\ w$	Body-axis velocity components (FC)
$u_r,\ v_r,\ w_r$	Wind-relative body velocities
$V_{wb,1},\ V_{wb,2},\ V_{wb,3}$	Components of wind in body axes
$\mathbf{R}_{bn},\ \mathbf{R}_{nb}$	Direction-cosine matrices (body $\leftrightarrow$ NED)
$\phi,\ \theta,\ \psi$	Euler roll, pitch, yaw (FC); ψ replaces heading ξ of UF
$\gamma,\ \xi,\ \mu$	Flight-path angle, heading angle, bank angle (UF)
V	True airspeed (UF)
V_a	Aerodynamic speed (FC), $V_a = \sqrt{u_r^2 + v_r^2 + w_r^2}$
$\alpha,\ \beta$	Angle of attack and sideslip (algebraic in FC)
Uncoupled formulation: states, controls, and drag polar	
$\mathbf{x}$	$[x,\ y,\ z,\ V,\ \mu,\ \gamma,\ \xi,\ m,\ C]^T$ (UF state vector)
$\mathbf{u}$	$[T,\ C_L,\ p]^T$ (UF controls: thrust, lift coefficient, roll rate command)
T	thrust [N]
C_L	Lift coefficient (direct control input in UF)
p	Roll rate command (bank kinematics) [rad s^{-1}]
m	Mass [kg]; $m_0,\ m_{\text{dry}}$ initial and dry masses
C	Battery charge (electric aircraft)
q	Dynamic pressure $q = \frac{1}{2}\rho V^2$ (UF)
S	Wing reference area
ρ	Air density (ISA)
η_p	Propulsive efficiency
P	Shaft power [W]
c_{sfc}	Specific fuel consumption [g kW^{-1} h^{-1}]

Symbol	Description
C_{D_0}, k, e, $\mathcal{A}$	Parasite drag, induced-drag factor, Oswald factor, aspect ratio; $k = 1/(\pi e \mathcal{A})$
Full-control formulation: states, controls, and non-dimensional rates	
$\mathbf{x}$	$[x_e, y_e, z_e, m, u, v, w, \phi, \theta, \psi, p, q, r, C]^T$ (FC state vector)
$\mathbf{u}$	$[T, \delta_e, \delta_a, \delta_r]^T$ (FC controls: thrust and surface deflections)
δ_e, δ_a, δ_r	Elevator, aileron, rudder deflections [rad]
p, q, r	Body-axis angular rates [rad s^{-1}] (not UF roll-rate control)
q_S	Dynamic pressure × area, $q_S = \frac{1}{2}\rho(z_e)\, V_a^2\, S_w$
S_w	Wing reference area in FC notation
b	Wing span
$\bar{c}$	Mean aerodynamic chord
$\hat{p}$, $\hat{q}$, $\hat{r}$, $\hat{u}$, $\hat{\dot{\alpha}}$	Non-dimensional rates: $\hat{q} = \bar{c}q/(2V_a)$, etc.; $\hat{u} = u_r/V_a$
$I_{xx}, I_{yy}, I_{zz}, I_{xz}$	Moments and product of inertia (including roll–yaw coupling)
d_{Tz}	Thrust arm for pitch moment from thrust offset
Aerodynamics: coefficients, forces, moments, and stability derivatives	
C_L, C_D, C_Y	Lift, drag, side force coefficients
C_l, C_m, C_n	Rolling, pitching, yawing moment coefficients (bold C_l vs. lift C_L)
L, D, Y	Lift, drag, side force [N]
$\mathcal{L}_W$, $\mathcal{M}_W$, $\mathcal{N}_W$	Rolling, pitching, yawing moments in wind axes [N m]
L_b, M_b, N_b	Body-axis moment components
$F_{x,a}$, $F_{y,a}$, $F_{z,a}$	Body-axis aerodynamic force components
$C_{L_0}, C_{L_\alpha}, C_{L_{\delta_e}}, C_{L_q}$	Lift: base, α, δ_e, and q effects.
$C_{D_0}, C_{D_{k1}}, C_{D_{k2}}, C_{D_q}, C_{D_{CT}}$	Drag: parasitic, induced, q, and thrust (C_T).
$C_{Y_0}, C_{Y_\beta}, C_{Y_{\delta_a}}, C_{Y_{\delta_r}}, C_{Y_p}, C_{Y_r}$	Side force: β, δ_a, δ_r, p, and r.
$C_{l_0}, C_{l_\beta}, C_{l_{\delta_a}}, C_{l_{\delta_r}}, C_{l_p}, C_{l_r}$	Rolling moment: β, δ_a, δ_r, p, and r.
$C_{m_0}, C_{m_\alpha}, C_{m_{\delta_e}}, C_{m_q}$	Pitching-moment derivatives: trim offset, static stability with angle of attack, elevator control effectiveness, and pitch damping associated with the non-dimensional rate $\hat{q}$. Higher-order unsteady and speed-stability terms (commonly written as $C_{m_{\dot{\alpha}}}$ and C_{m_u}) are acknowledged in the literature but are set to zero in the numerical implementation used in Section 6.4.
$C_{n_0}, C_{n_\beta}, C_{n_{\delta_a}}, C_{n_{\delta_r}}, C_{n_p}, C_{n_r}$	Yawing moment: β, δ_a, δ_r, p, and r.

List of Figures

1.1 Bachelor thesis and software development Gantt chart by work package 12

2.1 Dijkstra shortest-path search on a 2D grid with static NFZ constraints 14
2.2 Sampling-based RRT variants: comparative path-planning results 14

5.1 Continuous-time OCP: dynamics, cost, boundary and path constraints 33
5.2 Direct single shooting: discrete controls and forward-integrated states 35
5.3 ECEF frame, geodetic coordinates, and construction of local NED 38
5.4 FC algorithm: sequential right-hand-side evaluation 45

6.1 Trajectory overview — all criteria and formulations 72
6.2 UF altitude profiles per scenario and criterion 73
6.3 FC altitude profiles per scenario and criterion 74
6.4 Cost base term, min-time vs min-fuel 75
6.5 Min-time vs min-fuel comparison, UF 76
6.6 Min-time vs min-fuel comparison, FC 77
6.7 UF vs FC comparison, Min-time 78
6.8 UF vs FC comparison, Min-fuel 79
6.9 UF vs FC altitude overlay, SC-00 80
6.10 UF vs FC altitude overlay, SC-01 81
6.11 Hard-path margins, min-time vs min-fuel 83
6.12 Speed-envelope margins, min-time vs min-fuel 84
6.13 Attitude margins, min-time vs min-fuel 85
6.14 FC control margins, min-time vs min-fuel 86
6.15 Hard-path margins, UF vs FC 87
6.16 Control-surface margins, UF vs FC 88
6.17 UF regularization dashboard for SC-03 91
6.18 FC regularization dashboard for SC-03 92
6.19 FC peak control deflections across scenarios 93
6.20 FC cost sensitivity to regularization weights 95
6.21 Numerical pipeline overview 101
6.22 Benchmark scenarios 5–8 ground tracks 104

6.23 UF initializations, scenario 8 . . . 107
6.24 UF initializations, scenario 5 . . . 108
6.25 UF guess distance from the solution . . . 109
6.26 UF initializations, scenario 5 (min-fuel) . . . 111
6.27 UF initializations, scenario 3 (min-fuel) . . . 112
6.28 UF guess distance from the solution, min_fuel . . . 112
6.29 FC solve time, full pipeline versus direct seed . . . 114
6.30 FC guess distance from the solution . . . 115
6.31 FC solve time, full pipeline versus direct seed, min_fuel . . . 116
6.32 FC guess distance from the solution, min_fuel . . . 116

A.1 Body ($\mathcal{F}_B$), aerodynamic ($\mathcal{F}_A$) and wind ($\mathcal{F}_W$) frames of the aircraft, with the angle of attack α_B, the sideslip angle β, the flight-path angle γ, and the aerodynamic forces L, D and the weight mg. . . . 137

B.1 Forward and inverse geodetic, ECEF, and local NED conversion flow . . . 146

C.1 Flyox I platform used as MSN012 . . . 149

D.1 Penalty-ablation dashboards, SC-00 . . . 154
D.2 Penalty-ablation dashboards, SC-01 . . . 157
D.3 Penalty-ablation dashboards, SC-02 . . . 160
D.4 Penalty-ablation dashboards, SC-03 . . . 163

E.1 UF worst-case constraint margins, all scenarios . . . 175
E.2 FC worst-case constraint margins, all scenarios . . . 176

List of Tables

1.1 Work packages and milestone structure 4
1.3 Tasks and estimated hours by work package 9

2.1 Path-planning methods: algorithms and trade-offs 14
2.2 Trajectory derivatives: naming and physical interpretation 17

3.1 Three-tier methodology: scope and role of each formulation 21

4.1 Design decisions 30

5.1 Uncoupled formulation: state variables 39
5.2 Uncoupled formulation: controls and terminal time 39
5.3 Uncoupled formulation: parabolic polar parameters 41
5.4 Uncoupled formulation: nine governing ODEs by physical category 42
5.5 Full-Control formulation: state variables 44
5.6 Full-Control formulation: controls and terminal time 44
5.7 Aerodynamic stability and control derivatives appearing in Equations (5.24a)–(5.24f). Signs in the last column refer to a statically stable, conventionally controlled aircraft. 48
5.8 Full-Control formulation: fourteen governing differential equations 52
5.9 Uncoupled vs. Full-Control: structural comparison 53

6.1 NLP size of the two formulations 55
6.2 NLP reference scaling factors 57
6.3 Performance constraints: UF vs. FC 63
6.5 Cost weight α regimes 65
6.6 Validation benchmark scenarios 70
6.7 SC-03 cross-mode headline metrics 89
6.8 Benchmark scenario suite 103

6.9 UF initialisation comparison (min_time): warm start (Hybrid A* and inverse dynamics), linear interpolation (straight seed with analytic trim) and cold start (boundary state held at every node). Total time and IPOPT iterations are absolute; the gain columns give the warm-start reduction relative to each seed (baseline = that seed), i.e. (seed – warm)/seed × 100, so a positive value means the warm start is faster / needs fewer iterations than that seed. Runs that did not converge or saturated the computational-time cap (overtime) are shown as – and are excluded from the means. 106

6.10 Initial-guess distance from the converged solution (UF, min_time): per-scenario (x, y, z, V) deviation [%] of each seed from the optimum (warm = Hybrid A*+inverse-dynamics seed, no warm start = linear-interpolation seed, cold = hold-initial-state seed). * marks runs that did not converge, whose guess is measured against the warm-start converged UF solution. 107

6.11 UF initialisation comparison (min_fuel): warm start (Hybrid A* and inverse dynamics), linear interpolation (straight seed with analytic trim) and cold start (boundary state held at every node). Total time and IPOPT iterations are absolute; the gain columns give the warm-start reduction relative to each seed (baseline = that seed), i.e. (seed – warm)/seed × 100, so a positive value means the warm start is faster / needs fewer iterations than that seed. Runs that did not converge or saturated the computational-time cap (overtime) are shown as – and are excluded from the means. 110

6.12 Initial-guess distance from the converged solution (UF, min_fuel): per-scenario (x, y, z, V) deviation [%] of each seed from the optimum (warm = Hybrid A*+inverse-dynamics seed, no warm start = linear-interpolation seed, cold = hold-initial-state seed). * marks runs that did not converge, whose guess is measured against the warm-start converged UF solution. 110

6.13 Computational and iteration overview (FC, min_time): WS→UF→FC (baseline) vs. WS→FC (skip the UF solve). Iteration counts are the FC-solve iterations only on both sides, so the comparison is like-for-like. Runs that did not converge or saturated the computational-time cap (overtime) are shown as – and are excluded from the means. 113

6.14 Computational and iteration overview (FC, min_fuel): WS→UF→FC (baseline) vs. WS→FC (skip the UF solve). Iteration counts are the FC-solve iterations only on both sides, so the comparison is like-for-like. Runs that did not converge or saturated the computational-time cap (overtime) are shown as – and are excluded from the means. 115

7.1 Personnel cost breakdown 119

7.2 Software and license cost 119

7.3 Equipment amortization 119

7.4 Total project budget 120

9.1 Requirement fulfillment status 126

A.1 Body-to-wind-axis variable substitutions 139

B.1 Summary of geodetic, ECEF, and NED conversion routes 147

C.1 MSN012 geometry and mass properties 150
C.2 MSN012 performance limits 150
C.3 MSN012 aerodynamic derivatives 151
C.4 MSN012 propulsion specifications 151

D.1 UF penalty ablation, SC-00 155
D.2 FC penalty ablation, SC-00 156
D.3 UF penalty ablation, SC-01 158
D.4 FC penalty ablation, SC-01 159
D.5 UF penalty ablation, SC-02 161
D.6 FC penalty ablation, SC-02 162
D.7 UF penalty ablation, SC-03 164
D.8 FC penalty ablation, SC-03 165
D.9 UF control total variation (min-time), reg. vs ablation 166
D.10 UF control total variation (min-fuel), reg. vs ablation 166
D.11 FC control total variation (min-time), reg. vs ablation 166
D.12 FC control total variation (min-fuel), reg. vs ablation 167
D.13 UF defect breakdown (min-time), reg. vs ablation 167
D.14 UF defect breakdown (min-fuel), reg. vs ablation 168
D.15 FC defect breakdown (min-time), reg. vs ablation 168
D.16 FC defect breakdown (min-fuel), reg. vs ablation 169

E.1 UF variable boundaries 171
E.2 UF hard-constraint boundaries 171
E.3 FC variable boundaries 172
E.4 FC hard-constraint boundaries 172
E.5 Binding margins: min-time vs min-fuel 173
E.6 FC attitude-control margins: min-time vs min-fuel 173
E.7 Binding margins: UF vs FC 174
E.8 Group A var. boundaries (UF, SC-1, min-time) 177
E.9 Group B hard constraints (UF, SC-1, min-time) 178
E.10 Group C dynamic residuals (UF, SC-1, min-time) 178
E.11 Group A var. boundaries (UF, SC-1, min-fuel) 179
E.12 Group B hard constraints (UF, SC-1, min-fuel) 179
E.13 Group C dynamic residuals (UF, SC-1, min-fuel) 180
E.14 Group A var. boundaries (FC, SC-1, min-time) 181

E.15 Group B hard constraints (FC, SC-1, min-time) . . . 182
E.16 Group C dynamic residuals (FC, SC-1, min-time) . . . 182
E.17 Group A var. boundaries (FC, SC-1, min-fuel) . . . 183
E.18 Group B hard constraints (FC, SC-1, min-fuel) . . . 184
E.19 Group C dynamic residuals (FC, SC-1, min-fuel) . . . 184
E.20 Group A var. boundaries (UF, SC-2, min-time) . . . 185
E.21 Group B hard constraints (UF, SC-2, min-time) . . . 186
E.22 Group C dynamic residuals (UF, SC-2, min-time) . . . 186
E.23 Group A var. boundaries (UF, SC-2, min-fuel) . . . 187
E.24 Group B hard constraints (UF, SC-2, min-fuel) . . . 187
E.25 Group C dynamic residuals (UF, SC-2, min-fuel) . . . 188
E.26 Group A var. boundaries (FC, SC-2, min-time) . . . 189
E.27 Group B hard constraints (FC, SC-2, min-time) . . . 190
E.28 Group C dynamic residuals (FC, SC-2, min-time) . . . 190
E.29 Group A var. boundaries (FC, SC-2, min-fuel) . . . 191
E.30 Group B hard constraints (FC, SC-2, min-fuel) . . . 192
E.31 Group C dynamic residuals (FC, SC-2, min-fuel) . . . 192
E.32 Group A var. boundaries (UF, SC-3, min-time) . . . 193
E.33 Group B hard constraints (UF, SC-3, min-time) . . . 194
E.34 Group C dynamic residuals (UF, SC-3, min-time) . . . 194
E.35 Group A var. boundaries (UF, SC-3, min-fuel) . . . 195
E.36 Group B hard constraints (UF, SC-3, min-fuel) . . . 195
E.37 Group C dynamic residuals (UF, SC-3, min-fuel) . . . 196
E.38 Group A var. boundaries (FC, SC-3, min-time) . . . 197
E.39 Group B hard constraints (FC, SC-3, min-time) . . . 198
E.40 Group C dynamic residuals (FC, SC-3, min-time) . . . 198
E.41 Group A var. boundaries (FC, SC-3, min-fuel) . . . 199
E.42 Group B hard constraints (FC, SC-3, min-fuel) . . . 200
E.43 Group C dynamic residuals (FC, SC-3, min-fuel) . . . 200
E.44 Group A var. boundaries (UF, SC-4, min-time) . . . 201
E.45 Group B hard constraints (UF, SC-4, min-time) . . . 202
E.46 Group C dynamic residuals (UF, SC-4, min-time) . . . 202
E.47 Group A var. boundaries (UF, SC-4, min-fuel) . . . 203
E.48 Group B hard constraints (UF, SC-4, min-fuel) . . . 203
E.49 Group C dynamic residuals (UF, SC-4, min-fuel) . . . 204
E.50 Group A var. boundaries (FC, SC-4, min-time) . . . 205
E.51 Group B hard constraints (FC, SC-4, min-time) . . . 206
E.52 Group C dynamic residuals (FC, SC-4, min-time) . . . 206

E.53 Group A var. boundaries (FC, SC-4, min-fuel) 207
E.54 Group B hard constraints (FC, SC-4, min-fuel) 208
E.55 Group C dynamic residuals (FC, SC-4, min-fuel) 208

F.1 Min-time vs min-fuel (UF, SC-1) 210
F.2 Min-time vs min-fuel (FC, SC-1) 210
F.3 Min-time vs min-fuel (UF, SC-2) 211
F.4 Min-time vs min-fuel (FC, SC-2) 211
F.5 Min-time vs min-fuel (UF, SC-3) 212
F.6 Min-time vs min-fuel (FC, SC-3) 212
F.7 Min-time vs min-fuel (UF, SC-4) 213
F.8 Min-time vs min-fuel (FC, SC-4) 213

Chapter 1

Introduction

A fundamental pillar of aircraft trajectory planning is the preliminary analysis of feasible flight paths. Beyond simple pathfinding, modern aviation requires robust algorithms capable of optimizing routes from an initial point A to a destination B. In the context of Unmanned Aerial Vehicles (UAVs), this becomes even more critical due to the need for precise guidance through complex sequences of waypoints. Furthermore, should the onboard systems detect any environmental interference or operational constraints—such as obstacles, forbidden zones, or potential collision threats—the algorithm must be capable of dynamically computing an alternative feasible trajectory in real-time.

By leveraging differential equations and advanced optimization mathematics, this study develops a responsive framework within the field of flight mechanics. The proposed approach implements optimal routes that strictly account for control factors and the aircraft's specific performance parameters. Modern trajectory planning has evolved beyond manual drafting; it is now an intelligent process that integrates precise performance limitations and aerodynamic constraints, which ultimately dictate the trajectory and velocity of various flight maneuvers.

1.1 Objectives

The primary goal of this thesis is the design and implementation of a high-performance 3D trajectory solver integrated into a Linux-based Mission Management System (MMS). This software is intended to compute optimal flight paths for a generic aircraft, moving beyond traditional constant-velocity models by incorporating accelerated flight profiles during climb, cruise, and descent. A key functional requirement is the seamless integration of environmental and operational constraints, specifically utilizing Digital Elevation Models (DEM) [1] for terrain clearance and strict logic for No-Fly Zone (NFZ) [2] avoidance, ensuring that the resulting trajectories are not only time-optimal but also safety-compliant.

Central to this objective is the pursuit of computational efficiency to meet a strict real-time operational threshold of under 4 seconds. Consequently, the thesis focuses on the mathematical simplification of the Optimal Control Problem (OCP) to facilitate rapid convergence. A significant portion of the research is dedicated to the development of a hierarchical optimization strategy that utilizes semi-analytical solutions and geometric path planning. These simplified models serve as a robust "warm start" for the high-fidelity solver, providing a high-quality initial guess that significantly reduces the search space and avoids the pitfalls of random initialization.

Furthermore, this work aims to exploit an Inverse Dynamics [3] problem formulation as a core acceleration technique. By mapping desired spatial trajectories directly to the required control inputs, the system bypasses the heavy computational burden associated with traditional forward integration during the seeding phase. The final objective is to validate this dual-layer approach, ensuring that the transition from simplified analytical guesses to full-scale numerical optimization maintains physical consistency with the aircraft's flight envelope and performance parameters .

1.2 Project Scope

1.2.1 In-Scope: Detailed Developments

The following areas constitute the core technical contributions of this research and are developed in full:

- Development of a fully coupled state-control architecture integrating the aircraft equations of motion and dynamic transitions within a Linux-based Software environment.
- Implementation of high-fidelity performance models accounting for variable mass from fuel consumption and the influence of wind vectors on ground-speed relationships.
- Design of a modular numerical optimization framework that utilizes both direct collocation and multiple shooting transcription methods for simultaneous state and control optimization.
- Implementation of a bidirectional coordinate transformation layer between internal analytical frames and the North-East-Down (NED) geodetic system to ensure full compatibility with standard autopilot waypoint protocols.
- Formulation of a geospatial constraint engine that translates digital elevation models and no-fly zone polygons into mathematical inequality constraints for terrain clearance and geofencing.
- Research into acceleration algorithms and semi-analytical techniques to optimize the computational pipeline for real-time compliance within a sub-4-second threshold.
- Integration of heuristic seeding methods, specifically using A* pathfinding and Dubins curves,

to provide the non-linear solver with a robust warm start and improved convergence.

- Migration of the core optimization logic to C++ to achieve a production-grade deliverable capable of handling high-frequency trajectory updates.
- Execution of a systematic analysis across several representative usecases to validate the model's reliability in diverse mission profiles and varying environmental conditions.
- Comparison of performance metrics and time optimization results between different solvers, evaluating the computational gains of warm start versus cold start strategies.
- Development of analytical comparisons to verify the numerical solver's output against theoretical aircraft performance limits and established flight envelopes.
- Generation of a technical report assessing the environmental impact benefits, specifically focusing on fuel-burn reduction and emission optimization through efficient trajectory design.
- Assessment of the technology's scalability and market potential, exploring the possibilities for massive deployment within the U-Space and urban air mobility sectors.

1.2.2 Out-of-Scope: Project Boundaries

To ensure the technical depth remains focused on the optimization framework within the allocated ECTS timeframe, the following areas are excluded:

- The SiL (Software-in-the-Loop) analysis and model validation are performed by Singular Aircraft; therefore, the execution of these tests and their primary results are outside the scope of this project. While the author will incorporate external feedback into the system design, the formal validation process will not be personally conducted.
- The framework assumes pre-processed and clean data inputs for state estimation and mapping, omitting the development of SLAM, LiDAR processing, or computer vision algorithms.
- Aerodynamic modeling is restricted to established performance databases and simplified coefficient models, excluding the execution of new CFD simulations or wind tunnel testing.
- The project focuses strictly on the backend solver architecture and Mission Management System logic, excluding the design and implementation of high-fidelity graphical user interfaces.
- While the solver incorporates Digital Elevation Models and No-Fly Zone polygons as operational constraints, the project does not include a comprehensive assessment of the accuracy or quality of these geospatial datasets.

1.2.3 Flow Architecture and Work Packages

The thesis is organized into six Work Packages (WPs), representing a workflow where each phase builds the necessary infrastructure to reach the next milestone. Table 1.1 maps each package to its technical focus and milestone deliverable.

Table 1.1: Structure of Work Packages and Project Milestones.

WP	Technical Description	Milestone Achievement
WP1	Development of uncoupled kinematic models and point-force formulations. Focuses on non-transient optimization and familiarization with NLP solvers.	Capability to generate simplified 3D trajectories using uncoupled point-mass dynamics in static test cases.
WP2	Implementation of fully coupled 6-DOF dynamics utilizing a full trapezoidal integrator. Integration of No-Fly Zones (NFZ) and environmental constraints into the solver.	Convergence of high-fidelity trajectories that respect simultaneous translational and rotational constraints.
WP3	Verification of all numerical constraints and optimization of the source code for performance, focusing on computational speed and stability.	Validation of the solver's numerical consistency and achievement of improved execution latency.
WP4	Development of warm-start logic utilizing Dubins curves and A* pathfinding to generate high-quality initial seeds for the optimization layer.	Significant reduction in solver iterations and prevention of local minima traps during initialization.
WP5	Development of inverse dynamics to mesh warm-start trajectories with the solver, enabling the synthesis of full state and control vectors from geometric seeds.	Successful reconstruction of feasible state-space profiles from kinematic path primitives.
WP6	Synthesis of experimental results, formal validation of the mission management framework through and external party, and completion of the thesis manuscript.	Formal verification of the system's operational envelope and final submission of the research.

1.3 Requirements

The following technical requirements define the operational envelope, data interfaces, and functional constraints of the proposed Mission Management System (MMS).

1.3.1 Functional and Dynamic Requirements

F1 The MMS shall solve simultaneous translational and rotational equations of motion using a fully coupled 6-DOF formulation to optimize control inputs and vehicle dynamics as a single system.

F2 The optimization solver shall be decoupled from the specific vehicle plant, accepting external datasets for aircraft performance envelopes, mass properties, and aerodynamic coefficients.

F3 The solver shall enforce state and control constraints, including maximum actuator deflections (δ) and rate limits ($\dot{\delta}$), as defined by the vehicle input data.

F4 The equations of motion shall incorporate a dynamic wind-vector field to ensure trajectory accuracy and energy-state reliability under non-zero ambient flow conditions.

F5 The solver shall enforce environmental constraints—including Digital Elevation Models (DEM), No-Fly Zones (NFZ), and Geofencing—as formal boundaries within the optimization problem, in addition to vehicle-specific state and control variable limitations.

F6 The framework shall implement multiple *warm start* techniques. The system must be capable of dynamically selecting and activating specific initialization strategies based on the operational use case to ensure high-speed convergence.

F7 The optimization solver shall be designed to function independently of the transcription method. It must support both *Direct Collocation* and *Multiple Shooting* formulations to allow for varied levels of numerical robustness and problem sparsity.

F8 The architecture shall support multiple numerical integration schemes for state propagation, including *Forward Euler*, *Runge-Kutta 4 (RK4)*, and *Trapezoidal* rules, allowing for configurable precision-to-performance ratios.

F9 For initial mission activation or recovery from infeasible states, the system shall generate an analytical trajectory seed using kinematic path primitives (e.g., Dubins curves or straight-line segments) to initialize the solver.

1.3.2 Software and Interface Requirements

S1 The system shall utilize a standardized JSON-based input architecture for parsing mission parameters, environmental constraints, and geodetic waypoints.

S2 The system shall export 3D trajectories in geodetic coordinates compliant with the WGS84 reference

frame for integration with industrial GNSS and autopilots.

S3 The core solver and mission management logic shall be developed to execute natively within a Linux-based SiL environment. The architecture must ensure seamless data exchange between the optimization framework and the simulated aircraft plant to allow for industrial validation of the 6-DOF control laws before deployment.

1.3.3 Numerical and Performance Requirements

N1 The solver shall converge on a feasible trajectory solution within a maximum latency of 4.0 seconds per tactical look-ahead window.

N2 The optimization engine shall achieve a solution with a fixed convergence tolerance of 10^{-4}, where results exceeding this residual error are discarded as non-feasible.

N3 The solver shall utilize analytical Jacobians for constraints and cost functions, maintaining a precision error relative to finite-difference checks of less than 10^{-6}.[1]

N4 The C++ production core shall utilize static or pre-allocated memory structures for all optimization-related matrices, with no dynamic heap allocation permitted during the solver loop.

1.3.4 Safety and Robustness Requirements

R1 The system shall perform automated range-checking on all JSON-parsed inputs, rejecting data that exceeds defined structural or geodetic boundaries.

R2 In the event of solver non-convergence or a timeout exceeding 4.0 seconds, the MMS shall output the last valid trajectory segment or the analytical backup seed.

R3 The output trajectory shall maintain C^2 continuity to ensure the path is physically achievable by the aircraft's actuators without inducing structural jitter.

R4 The system shall log the solver's internal performance metrics, including iteration count and residual error, for every optimization cycle.

[1]The results reported in this thesis were obtained with the solver version that still built the Jacobians by forward finite differences. Exact derivatives were introduced afterwards (see the Conclusions chapter).

1.4 Justification

Aerospace engineering is currently undergoing a paradigm shift driven by the development of **Mission Management Systems (MMS)**, which transition unmanned platforms from simple navigational aids to autonomous decision-making structures. As Unmanned Aerial Vehicles (UAVs) integrate into commercial logistics and Urban Air Mobility (UAM) networks, the aviation industry is experiencing an unprecedented industrial boom, with global market forecasts indicating a Compound Annual Growth Rate (CAGR) exceeding 25% [4], [5]. Consequently, the industry is moving away from basic heuristic waypoint navigation toward complex real-time trajectory optimization algorithms capable of functioning in crowded, non-segregated airspace.

Despite these advancements, a significant technological rift persists between idealized geometric planning and the rigorous physical realities of flight dynamics. Current industrial standards often favor "kinematic decoupling," adhering to simplified 3-DOF (Three Degrees of Freedom) point-mass models such as the EUROCONTROL Base of Aircraft Data (BADA) [6]. Historically, this abstraction has been a necessary concession to the limited computational throughput of flight-certified hardware and the requirements of safety-critical software certification. Because solving a fully coupled 6-DOF optimization in real-time is a non-convex, intensive task, most commercial systems utilize high-speed search algorithms (such as A* [7] or RRT* [8]) that essentially defer the burden of aerodynamic complexity to the lower-level autopilot.

This reliance on simplified models is increasingly inadequate for high-performance or heavy-payload platforms, such as the amphibious *Flyox I* [9]. By ignoring the high-frequency coupling between control surface deflections and aerodynamic stability, solvers frequently output geometric paths under the risky assumption that the autopilot possesses the control authority to compensate for dynamic residuals. This can lead to *tracking divergence*[2], occurring when a trajectory is mathematically plotted but physically unachievable due to actuator saturation, shifting centers of gravity, or severe cross-coupled moments.

The origins of this thesis are fundamentally rooted in a professional collaboration with the *International Center for Numerical Methods in Engineering (CIMNE)* through a series of curricular and extracurricular internships. The primary objective of this engagement was the development of a robust Mission Management System for *Singular Aircraft* [10]. The development of the core algorithms and codebase was driven by industrial R&D requirements, necessitating an early start prior to the official academic term in March. This chronological overlap was essential to ensure the depth and quality of the technical work, positioning the thesis as the final developmental phase of a project already integrated into a professional aerospace pipeline.

Ultimately, this thesis serves as a technical booster for the existing MMS architecture by developing a

[2]**Tracking divergence:** A state in which the error between the commanded trajectory and the actual dynamic state increases monotonically over time, preventing asymptotic stability.

fully coupled state-control optimization framework [11]. By formulating the trajectory generation as a *Nonlinear Optimal Control Problem (OCP)* and targeting high-performance embedded Systems-on-Chip (SoC), the proposed system embeds the aircraft's true attitude and actuator constraints directly into the planning phase. While the current work focuses on solving these OCPs, the framework is architected for eventual deployment as a *Nonlinear Model Predictive Control (NMPC)* scheme [12], intended for instantaneous planning and continuous updates via onboard sensors. To ensure real-time feasibility for such future closed-loop applications, the research implements specific acceleration techniques, most notably *Warm Start* features and *Inverse Dynamics*. These features allow the solver to handle the complexity of the 6-DOF model while remaining computationally agile, ensuring that autonomous operations are strictly bounded by the true aerodynamic and structural limits of the airframe.

1.5 Planning and Scheduling

Given that developing a 3D trajectory optimizer is a highly complex and non-trivial undertaking, the project delivered to Singular officially commenced in September 2025. This early start was essential to establish the core algorithmic framework required for the subsequent analysis.

The following schedule integrates the code development process conducted during the autumn semester, highlighting the direct relationship between these foundational tasks and the final objectives of this Bachelor's Thesis (TFG). The initial phase focused on defining the optimality problems and building the robust software architecture necessary to support the advanced computations presented in the latter stages of the study.

Table 1.3: Detail of tasks classified in each work package of the thesis.

ID	Type	Detailed Description and Technical Objectives	Hours
WP1: Uncoupled Formulation			
T-001	Theory	**State of the Art:** Research on transcription methods (Direct Collocation) and parabolic trajectory discretization.	8
C-001	Coding	**Aircraft Class Composition:** OOP architecture for aerodynamic and mechanical parameters.	3
C-002	Coding	**Analytical Non-Opt:** Implementation of constant speed use-cases without optimization.	12
C-003	Coding	**Analytical Opt:** min Time/Fuel optimization for uncoupled systems.	12
C-004	Coding	**3D Analytical Extension:** Integration of coordinate turns into the analytical model.	4
T-002	Theory	**Optimization Formulation:** Definition of cost functions and study of NLP solvers.	2
C-005	Coding	**Numerical Opt (Scalar):** Solver for $V = const$ uncoupled formulations.	4
C-006	Coding	**Numerical Opt (Vector):** Solver for transient states ($V = variable$).	12
C-007	Coding	**Horizontal Simplification:** Acceleration/deceleration logic validation.	5
C-008	Coding	**3D Transient Uncoupled:** Complex 3D problem for min Time/Fuel.	6
T-003	Theory	**NFZ Approach:** Mathematical formulation of geofencing constraints.	4

Continues on next page...

ID	Type	Detailed Description and Technical Objectives	Hours
C-010	Coding	**NFZ Integration:** Inequality constraints implementation in the solver.	8
WP2: Coupled Formulation			
T-004	Theory	**Coupled Dynamics:** Theory of 6-DOF stability and aircraft coupling.	12
C-011	Coding	**Analytical Non-Opt (Coupled):** Non-linear solving with `fsolve`.	6
C-012	Coding	**Wind Integration:** Equation modification for aerodynamic speed vs ground speed.	8
C-013	Coding	**Mass Loss Integration:** Power-based fuel consumption iterative loop.	4
C-014	Coding	**Analytical Opt (Coupled):** Static optimization with full performance restrictions.	12
C-015	Coding	**Transient Opt (Coupled):** Full vector optimization for coupled dynamics.	20
C-016	Coding	**NFZ in Coupled:** Validation of geofencing in the 6-DOF environment.	14
WP3: Performance Study & Optimization Core			
C-022	Coding	**Performance Study:** Assessment of solver robustness and accuracy.	10
C-023	Coding	**C++ Migration:** Porting Matlab core to C++ for computational speedup.	40
C-024	Coding	**Adjoint & Gradient Formulation:** Analytical sensitivity analysis to improve convergence.	35
WP4: Warm Start			
C-017	Coding	**Dubins Trajectories:** Geometrically feasible analytical paths.	40
C-018	Coding	**Warm Start (A*):** Heuristic trajectory proposal as initial guess for NLP.	35
WP5: Inverse Dynamics			
C-019	Coding	**Inverse Dynamics:** Control vector computation from trajectory discretization.	20
C-020	Coding	**Orbit Here:** Target entry point trajectory computation.	10
C-021	Coding	**Discretization Output:** Resolution criteria for autopilot performance.	6
WP6: TFG Writing & Documentation			

Continues on next page...

ID	Type	Detailed Description and Technical Objectives	Hours
TFG-01	Mgmt	**Project Charter & Planning:** Scope definition and ECTS scheduling.	10
TFG-02	Writing	**Technical Body I:** Documentation of dynamics and uncoupled/coupled models.	15
TFG-03	Writing	**Technical Body II:** Adjoint methods, C++ architecture, and sensitivity theory.	15
TFG-04	Writing	**Results & Analysis:** Discussion on Warm Starts and performance metrics.	14
TFG-05	Writing	**Final Review:** Formatting, bibliography, and abstract preparation.	10
		TOTAL ESTIMATED HOURS (12 ECTS)	415*

*The total estimated hours include a preliminary stage of the project which took place between September 2025 and February 2026. During this period, several activities and code development tasks were completed (approximately until the completion of Work Package 2 and the beginning of Work Package 3). This early start was necessary due to the extensive work required to develop and optimize the algorithm and codebase. Through an internship at CIMNE, these tasks were carried out prior to the official Bachelor's thesis period. The remaining tasks have been scheduled to satisfy the 360 hours corresponding to the 12 ECTS credits.

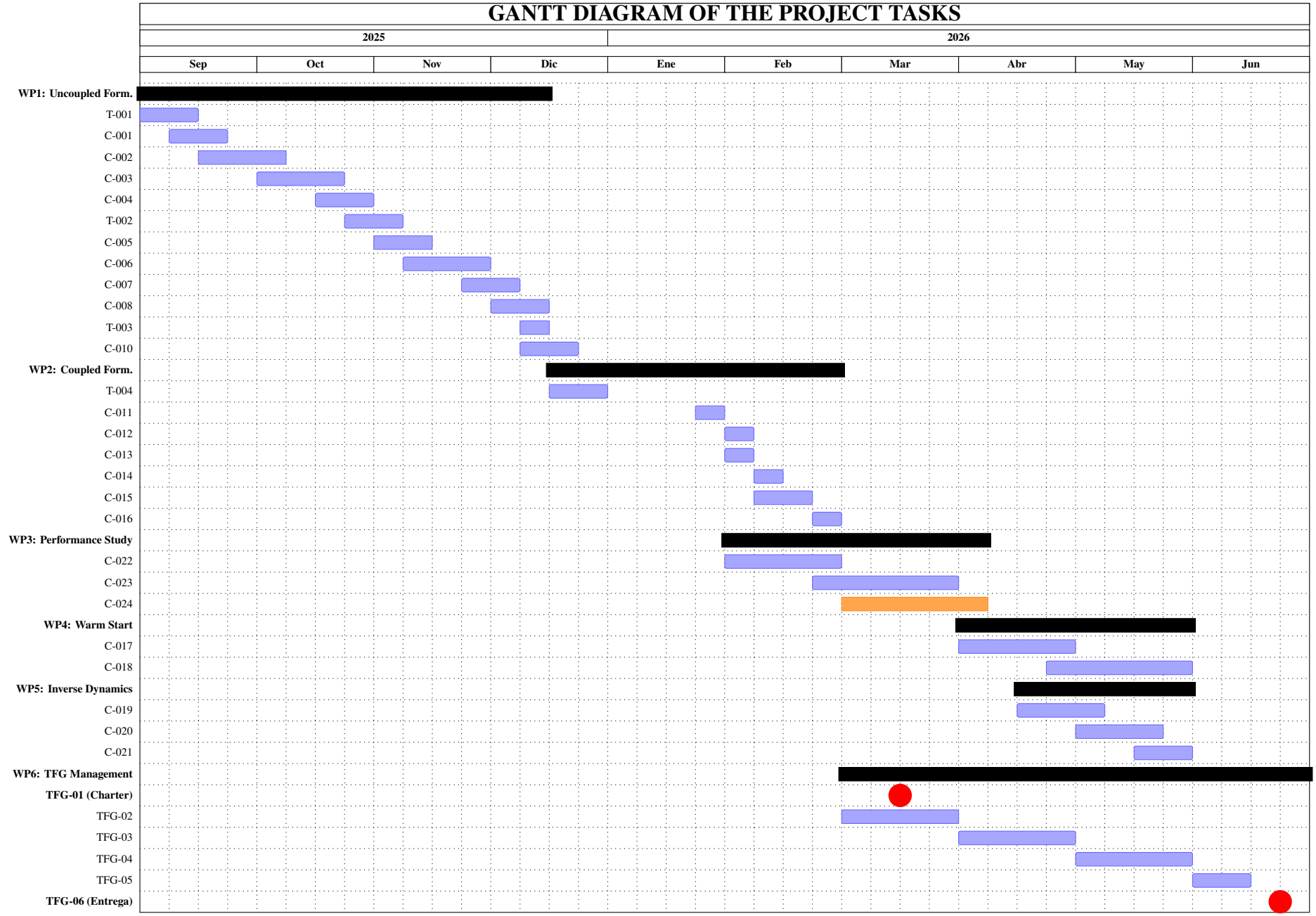


Figure 1.1: Project schedule for the bachelor's thesis and code development, categorized by Work Packages (WPs). (*Own Work*)

Chapter 2

State of the art

The transition of aviation from human-piloted systems to fully autonomous Unmanned Aerial Vehicles (UAVs) has highlighted a fundamental tension between computational geometry and aeronautical reality. In traditional flight management systems, the challenge of navigating an aircraft through three-dimensional space was largely solved through the lens of obstacle avoidance and topological connectivity [13]. However, as the operational envelope of UAVs expands to include aggressive maneuvers, dense urban environments, and complex multi-agent coordination, the limitations of purely geometric and graph-based planning have become critical points of failure. The core of the issue lies in the fact that most ubiquitous path planning protocols rely on simplified kinematic models that treat the aircraft as a point mass or a unicycle-like vehicle, effectively ignoring the transient dynamics, actuator saturation limits, and aerodynamic forces that dictate the actual flight path [14].

2.1 Geometric and Graph-Based Planning

Currently, the vast majority of commercial and proprietary flight management systems (FMS) utilize a hierarchical approach where a high-level planner generates a baseline topological route. These planners typically operate on a discretized representation of the airspace, such as a 3D raster or a visibility graph, to find a collision-free path from an origin to a destination.

The Dijkstra algorithm remains a foundational technique in this domain, providing a single-source shortest-path solution on weighted graphs. By exploring the state space in order of increasing cost, Dijkstra's algorithm guarantees a globally optimal solution in static environments. Nevertheless, the exhaustive nature of the search leads to significant computational overhead, especially in large-scale maps where memory consumption and processing time scale poorly with resolution [13].

To mitigate these computational challenges, the A* algorithm is frequently employed. By integrating a

heuristic function that estimates the remaining cost to the goal, A* can reduce the search space by several orders of magnitude while maintaining optimality guarantees [7]. Despite its efficiency, A* inherits the fundamental limitation of grid-based methods: the resulting path is inherently piecewise-linear and restricted to the connectivity of the graph. In a 3D environment, this often results in a "jagged" trajectory with sharp corners that require infinite acceleration to follow exactly—a physical impossibility for any aircraft. Consequently, while the path is geometrically feasible, it is not "flyable" in its raw state, necessitating a post-processing or smoothing phase that may compromise the original collision-free guarantees [15].

In high-dimensional configuration spaces where grid-based discretization becomes intractable, sampling-based algorithms such as Rapidly-exploring Random Trees (RRT) [8] and Probabilistic Roadmaps (PRM) [16] have gained prominence. RRT algorithms are particularly effective for aircraft because they can integrate differential constraints directly into the tree expansion process, allowing for the discovery of paths that respect minimum turning radii or climb rates. Variants such as RRT* and Informed RRT* focus on refining these paths toward an optimal solution, yet they still struggle with "noise" in the trajectory and the need for expensive post-optimization to eliminate sharp turns and random oscillations.

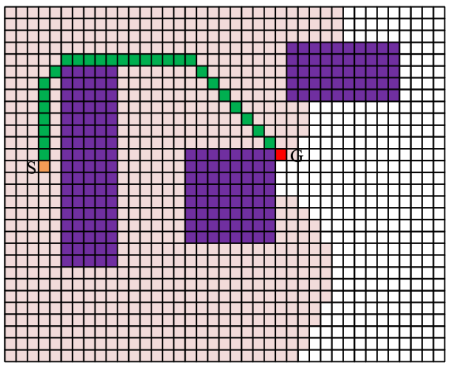


Figure 2.1: Dijkstra algorithm example for NFZ avoidance using 2D grid discretization [13].

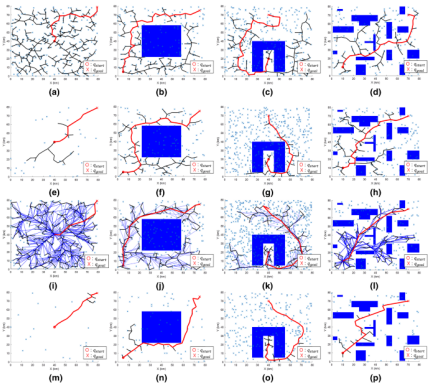


Figure 2.2: Comparative results of various sampling-based RRT algorithms [17].

The differences and trade-offs of these methods are shown in Table 2.1.

Table 2.1: Comparative Analysis of Path Planning Methodologies

Algorithm	Search Mechanism	Optimality	Constraint Handling	Primary Limitation
Dijkstra	Breadth-first / Weighted	Global	Static Obstacles	Computational Complexity
A*	Heuristic-guided	Global (Admissible)	Static Obstacles	Jagged Piecewise-Linear
RRT	Random Sampling	Probabilistic	Nonholonomic	Suboptimal / Randomness
RRT*	Rewiring Random Tree	Asymptotic	Nonholonomic	High Iteration Count
PRM	Probabilistic Roadmap	Probabilistic	Multi-query	Static Map Dependence

2.2 Numerical Trajectory Optimization

While geometric and probabilistic methods provide feasible routes, they often fail to capture the complex kinodynamic constraints of the aircraft, particularly during transient phases of flight where speed and heading change rapidly. To address this limitation, modern Mission Management Systems are increasingly adopting Numerical Trajectory Optimization. This approach formulates the path planning task as an Optimal Control Problem (OCP), which is subsequently transcribed into a finite-dimensional Nonlinear Programming (NLP) problem using direct methods, such as direct collocation or multiple shooting [18].

By utilizing robust Non-Linear Programming (NLP) solvers, it is possible to optimize continuous trajectories while strictly adhering to vehicle dynamics, fuel capacity, and complex spatial constraints, such as dynamic no-fly zones. Among state-of-the-art numerical solvers, Interior Point Methods (IPM) have proven highly effective for large-scale, heavily constrained aerospace applications [19]. Specifically, the open-source solver Ipopt (Interior Point Optimizer) is widely recognized in the literature for its computational efficiency and reliability [20]. To interface with such solvers, modern frameworks often utilize CasADi, an algorithmic differentiation (AD) tool that automates the transcription of Optimal Control Problems (OCPs) into NLP form by computing exact Jacobians and Hessians [21].

However, while CasADi is indispensable for rapid prototyping, it introduces several limitations for high-performance, real-time aerospace applications. Its reliance on symbolic expression trees [1] and an internal virtual machine can lead to significant memory overhead and increased latency compared to hand-optimized, compiled code. Furthermore, in safety-critical environments, the complexity of CasADi's memory management can be a hurdle for deterministic execution. To overcome this, integrating Ipopt directly within a compiled C++ architecture allows the Mission Management System (MMS) to manage memory efficiently and achieve the low-latency performance required for real-time or near real-time mission recalculations [23], offering a significant operational advantage over the semi-interpreted or graph-based environments traditionally used in preliminary academic research.

The implementation of the trajectory optimizer is modular, separating the numerical engine from the vehicle-specific physics. The NLP formulation is generally composed of the following core components:

- **Numerical Solver (IPOPT):** At the heart of the system is the *Interior Point Optimizer*. It utilizes a primal-dual interior-point algorithm to solve large-scale non-linear problems. When exact (analytical or automatically differentiated) Jacobians and Hessians are available, it navigates the

[1] A Symbolic Expression Tree is a hierarchical data structure where internal nodes represent mathematical operators (e.g., +, ×, sin) and leaf nodes represent variables or constants. While this allows for Exact Algorithmic Differentiation (AD), navigating these pointer-based structures during function evaluation introduces significant computational overhead compared to the direct execution of compiled, hard-coded arithmetic [22].

high-dimensional state space to find a solution that satisfies all constraints while minimizing the objective functional.

- **System Dynamics:** These represent the fundamental "rules of flight" expressed as differential equations (ODEs). In a direct collocation framework, these equations are discretized into equality constraints, ensuring that the transition between state x_k and x_{k+1} is physically consistent with the aircraft's motion model.

- **Environmental Constraints:** These define the "safe" configuration space. They include No-Fly Zones (NFZs), terrain elevation models, and dynamic obstacles. To ensure robustness, these are often modeled as inequality constraints that keep the vehicle within a safe distance from prohibited volumes.

- **Aircraft Performance and Physics:** Beyond basic motion, the solver must respect the structural and aerodynamic limits of the airframe. This includes constraints on maximum thrust, stall speeds, load factor limits (g-load), and maximum bank angles to prevent loss of control or structural failure.

- **Cost Function (Objective):** This defines the "quality" of the trajectory. While many solvers aim for minimum time or minimum fuel, modern flight management systems often prioritize smoothness to protect avionics and actuators. Optimization criteria depend on the specific mission priorities; however, common physical objectives are detailed in Table 2.2.

In practice, a trajectory is rarely optimized for a single criterion. Instead, a composite cost function is constructed as a weighted sum of various objectives, penalties, and soft constraints. For example, a typical objective functional J might take the following form:

$$J = w_t \int_{t_0}^{t_f} 1 \, dt + w_s \int_{t_0}^{t_f} \|\sigma^{(4)}(t)\|^2 dt$$

Where:

- w_s and w_t are weight coefficients for **Snap** (smoothness) and **Time** (efficiency if the objective of the simulation is to obtain a *min-time* solution).

Adopting this approach, though, significantly escalates the problem's difficulty, introducing a higher degree of complexity and a vast increase in the degrees of freedom. Unlike geometric path-finding, the simultaneous integration of dynamic equations and state variables is computationally intensive. Given that Non-Linear Programming (NLP) solvers are hypersensitive to the initial guess, implementing robust seeding and acceleration techniques is essential to meeting the strict convergence windows required for real-time applications.

Table 2.2: Physical Significance of Trajectory Derivatives in Aerial Planning

Derivative	Name	Physical Significance
$\sigma^{(0)}$	Position	Obstacle clearance and waypoint tracking
$\sigma^{(1)}$	Velocity	Kinetic energy and total travel time
$\sigma^{(2)}$	Acceleration	Required thrust magnitude and lift forces
$\sigma^{(3)}$	Jerk	Rate of change of acceleration (stress on airframe)
$\sigma^{(4)}$	Snap	Angular rates and attitude smoothness

2.3 Acceleration and Seeding Techniques

While numerical solvers like `Ipopt` are mathematically robust, they operate on the principle of local descent, making them highly sensitive to the provided initial state. In non-convex environments—specifically those cluttered with No-Fly Zones (NFZs) or terrain obstacles—a "cold start" (such as a simple linear interpolation between waypoints) often causes the solver to converge to a poor local minimum or fail entirely due to a "locally infeasible" solution [24], [25]. To bridge the gap between global discrete planning and local continuous optimization, a hierarchical three-stage pipeline is typically employed:

1. **Global Skeleton Generation:** A graph-based algorithm (e.g., A^*, Dijkstra, or Theta*) is used to compute an obstacle-free spatial skeleton. This ensures that the high-level routing is topologically correct and avoids all static constraints.

2. **Kinematic Refinement:** To prevent the optimizer from wasting iterations smoothing the "jagged" piecewise-linear segments of the skeleton, the path is refined using analytical curves such as *Dubins curves* [26], *Reeds-Shepp paths* [27] or *Clothoids* [28]. By enforcing the minimum turning radius (R_{min}) of the aircraft, this stage transforms the skeleton into a G^1 or G^2 continuous path that respects the vehicle's basic maneuverability.

3. **Inverse Dynamics Mapping:** To bridge the gap between a 3-DOF kinematic guess and a full 6-DOF dynamic solution, *Inverse Dynamics* are employed to estimate initial control surface deflections and thrust requirements. By mapping desired accelerations and curvatures back to the aircraft's actuator space, the solver avoids "tracking divergence" caused by initial attitude (ϕ, θ, ψ) inconsistencies.

4. **NLP Warm-Starting:** The refined kinematic path is sampled to provide the continuous solver with a "warm start." This involves initializing the primal variables—position $\mathbf{p}(t)$, velocity $\mathbf{v}(t)$, and control inputs $\mathbf{u}(t)$—near their expected optimal values. Furthermore, advanced implementations estimate the dual variables (Lagrange multipliers) associated with the system's

constraints, facilitating a rapid satisfaction of the Karush-Kuhn-Tucker (KKT)[2] optimality conditions.

This synergy between global geometric seeding and local NLP warm-starting drastically reduces the iterations required for convergence. Within a C++ `Ipopt` framework, this approach targets the generation of locally optimal, flyable profiles within the short re-planning windows required for autonomous missions.

2.3.1 Numerical Structure and Gradient Information

Beyond providing a high-quality initial guess, the convergence rate of a Nonlinear Program (NLP) is fundamentally governed by the accuracy and sparsity of the problem's derivatives. Relying on the solver to estimate gradients via finite differences is computationally expensive and numerically unstable, particularly in high-speed flight regimes where small perturbations in control inputs $\mathbf{u}(t)$ can lead to divergent state trajectories. Therefore, a critical acceleration technique involves providing the analytical Jacobian of the constraints and the Gradient of the cost function.

In the context of **Direct Collocation**, the state and control trajectories are discretized into a large-scale but sparse NLP. By providing the exact sparsity pattern of the Jacobian $\nabla\mathbf{h}(\mathbf{x})$, solvers such as `Ipopt` can leverage sparse linear algebra to evaluate the KKT conditions efficiently [29]. This sparsity is a direct result of the local nature of collocation polynomials, where a change in a single collocation point only affects neighboring constraints [30].

Conversely, in **Single Shooting** formulations—where only control inputs are decision variables—the resulting numerical problem is smaller but dense [30]. To maintain scalability in these cases, the **Adjoint Method** is frequently employed. By solving the adjoint equations backward in time, the sensitivity of the cost function with respect to all parameters can be computed with a computational cost independent of the number of decision variables. This offers a significant advantage for high-dimensional problems where the mission duration or the number of obstacles would otherwise lead to the "curse of dimensionality."[31], [32]

Furthermore, to handle the "saddle-point" nature of constrained optimization, techniques such as the Uzawa method [33] or augmented Lagrangian iterations are utilized to decouple the primal and dual updates. These methods ensure that the dual variables (Lagrange multipliers) evolve alongside the primal trajectory, facilitating the satisfaction of *Complementary Slackness*—whereby multipliers for inactive constraints are driven to zero [29]. When combined with adjoint-based gradient computation

[2]The KKT conditions represent the mathematical requirements for a point to be considered an optimal solution in nonlinear programming. Beyond ensuring that all physical limits are respected (*Primal Feasibility*), they require the gradient of the objective function to be perfectly balanced by the gradients of the constraints (*Stationarity*), while ensuring that only active constraints influence the solution (*Complementary Slackness*). Formally: (i) **Stationarity**: $\nabla f(x) + \sum \mu_i \nabla g_i(x) + \sum \lambda_j \nabla h_j(x) = 0$; (ii) **Primal Feasibility**: $g_i(x) \leq 0, h_j(x) = 0$; (iii) **Dual Feasibility**: $\mu_i \geq 0$; and (iv) **Complementary Slackness**: $\mu_i g_i(x) = 0$. [29]

and a warm-start initialization, these numerical strategies ensure that the flight management system remains responsive and capable of real-time trajectory re-planning.

2.4 Synthesis and Research Gap

The progression from discrete graph-based search to continuous numerical optimization reflects the increasing demand for UAV trajectories that are not merely "collision-free," but dynamically feasible and energy-optimal. As established in this review, while global planners like A* and RRT* provide topological robustness, they lack the high-fidelity awareness of aircraft limits such as load factor ($\sigma^{(2)}$) and attitude smoothness ($\sigma^{(4)}$). Conversely, while Nonlinear Programming (NLP) solvers like `Ipopt` offer the precision required for these "accelerated flight profiles," their heavy dependence on high-quality initial guesses and analytical gradients remains a significant barrier to real-time, deterministic execution in safety-critical environments.

The current literature suggests a clear hierarchy for path planning, yet a critical gap remains in the seamless integration of these layers for high-performance maneuvers. Specifically, the transition from a "jagged" geometric skeleton to a "warm-started" NLP often introduces numerical artifacts or excessive latencies that can render a system unresponsive during dynamic re-planning.

This thesis addresses this gap by developing a high-performance, unified C++ framework that integrates a complete aircraft dynamic model within the `Ipopt` ecosystem. By moving away from semi-interpreted environments and leveraging direct memory management, the proposed system facilitates the use of advanced warm-starting techniques—specifically the seeding of primal-dual variables derived from kinematically refined skeletons. By exploiting the numerical structure of the problem (sparsity, residual scaling and, in its latest version, exact derivatives), this research shows how the tension between computational geometry and aeronautical reality can be reduced. The result is a mission management architecture that generates complex, flyable trajectories reliably, and an assessment of how close it comes to the strict temporal windows required for autonomous operations.

Chapter 3

Methodology

3.1 The Challenge of Trajectory Optimization with IPOPT

Building an aircraft-trajectory optimizer around a general-purpose nonlinear-programming solver such as IPOPT is not a one-shot procedure [19], [20]. Direct collocation transforms the original continuous-time Optimal Control Problem (OCP) into a large sparse Nonlinear Program (NLP) whose decision vector simultaneously encodes states and controls at every collocation node, whose equality constraints encode the discretized dynamics, and whose inequality constraints encode the entire flight envelope together with the spatial mission constraints [34], [35]. For a realistic aircraft mission the resulting NLP contains thousands of decision variables and constraints and is strongly non-convex[1] [36], [37], with a curvature that is dominated by the aerodynamic model and by the trigonometry of the body-to-wind rotations. IPOPT is a primal-dual interior-point algorithm and is only locally convergent [38], so whether it reaches the global solution, converges to a poor local minimum or simply fails depends almost entirely on how the problem is presented to it—which is precisely the reason why an informed warm-start strategy becomes a first-class design concern [39], [40].

Three properties of the formulation control this outcome and they interact strongly with each other. The first is scaling: the decision variables span many orders of magnitude, from mass measured in tons to control-surface deflections measured in fractions of a radian and from positions measured in kilometers to angles measured in radians, and without a careful non-dimensionalization of both variables and residuals the interior-point barrier sub-problem becomes ill-conditioned and the quasi-Newton Hessian approximation degrades after a handful of iterations [37], [41]. The second is the quality of the initial guess. An interior-point method spends its early iterations driving the trial point into the interior of the feasible set, and a guess that is dynamically inconsistent or that already violates several path constraints

[1] An optimization problem is non-convex if its objective function or its feasible region does not satisfy the conditions of convexity; in a non-convex NLP a local optimum is not guaranteed to be the global one.

will force the solver into a long detour through infeasibility and very often into a non-physical local minimum, with symptoms such as negative-thrust segments or large bank reversals [24]. The third is sparsity and the availability of exact derivatives. A dense Jacobian obtained by finite differences makes each IPOPT iteration prohibitively expensive on a thousand-variable problem, and the theoretical complexity of the algorithm is only recovered when the sparsity pattern is exposed explicitly and the derivatives are provided exactly, for example through the algorithmic differentiation that CasADi performs on its symbolic expression graph [21], [42].

Because of these three sensitivities the realistic six-degree-of-freedom trajectory-optimization problem cannot be handed to IPOPT as a single monolithic NLP and expected to converge. The methodology followed in this thesis therefore grows the formulation in complexity one block at a time and fully understands and validates each stage before the next one is enabled. That staged construction is the subject of Section 3.2; the corresponding validation protocol and the warm-start analysis are described in Sections 3.3 and 3.4.

3.2 Three-Tier Incremental Development Strategy

The implementation pipeline is organized as a hierarchy of three increasingly faithful formulations. Each tier produces solutions that are directly usable as sanity checks, as warm starts, or as final mission-ready trajectories for the next tier, so that any departure from the expected behavior can be isolated to the block that was most recently introduced. This block-by-block construction is the standard practice recommended for large-scale direct-collocation solvers [34]. Table 3.1 summarizes the role of each tier; the rest of this section discusses them in turn.

Table 3.1: Three incremental tiers of the development methodology. Each row reflects what is solved, why it is built, and how its output feeds the next tier.

Tier	Formulation	Role in the methodology	Output used as
1	Analytical / semi-analytical	Qualitative understanding, preliminary sanity checks	Reference for tier 2
2	Benchmark NLP (UF, minimal constraints)	Solver tuning, weight calibration, debugging	Warm start for tier 3
3	Full constrained NLP (UF $\rightarrow$ FC) with warm start	Mission-ready trajectories under real environment	Final deliverable

3.2.1 Tier 1 — Analytical and Semi-Analytical Exploration

The first tier collects closed-form and semi-analytical solutions of restricted regimes of the point-mass dynamics, namely constant-airspeed cruise, constant-mass climb, equilibrium turn at fixed bank and the Riccati-type mass evolution that arises under a quadratic-drag and quadratic-thrust polar [43], [44].

These derivations rely on strong simplifying assumptions — no wind, frozen geometry, fixed speed — and they are not part of the final deliverable of this thesis. They are explicitly out of scope as a stand-alone result: the indirect adjoint route was discarded early because the dynamics overhead of deriving and maintaining the costate equations of the Full-Control model would have been prohibitive, and direct collocation has become the de-facto standard for aerospace trajectory optimisation in any case [34], [45]. A complete exposition of the adjoint formalism, with its application to optimal-control problems of the type considered here, is given by Bradley [32].

Their role was instead methodological. The closed-form expressions produced quasi-stationary reference trajectories against which the early optimizer runs of Tier 2 could be sanity-checked, which made it possible to recognize immediately when IPOPT had converged to a non-physical local minimum simply by comparing the predicted cruise-optimal airspeed or steady-climb angle with the one returned by the solver. The same derivations also clarified the optimality structure of the OCP through the adjoint and Hamiltonian formalism, even though the final solver is direct rather than indirect [45], and they established the ranking of forces, with drag dominating over induced drag at cruise and gravity dominating the energy balance in climb, that later informed the residual-normalization step of Tier 2. In other words, Tier 1 acted as a scaffold rather than as a result, and its derivations remain in the working notes of the project.

3.2.2 Tier 2 — Benchmark Optimal-Control Problem with Minimal Constraints

The second tier is the first genuinely numerical formulation. It applies direct trapezoidal collocation to the Uncoupled point-mass dynamics introduced in Section 5.2 [30], [35], reduces the cost function to a pure minimum-time term and keeps only the indispensable bounds active, namely positive mass, positive airspeed, bounded lift coefficient and bounded flight-path angle. The mission boundary conditions are kept deliberately simple — straight-and-level transit or single climb leg between two waypoints in empty airspace — so that the optimal answer is intuitive and can be compared term by term against the Tier 1 references.

The purpose of this tier is not to produce mission-ready trajectories but to regulate the optimizer itself. It is on these benchmark cases that the residual normalization, the decision-vector scaling, the Jacobian sparsity pattern and the IPOPT options — barrier parameter strategy, ℓ_1 penalty for infeasibility and L-BFGS Hessian approximation — were tuned [19], [20], [41]. The cost-weight α that balances time and fuel (Section 6.3) was first calibrated here in its two pure endpoints, $\alpha = 0$ and $\alpha = 1$, precisely on cases where the qualitative shape of each Pareto extreme is known in advance and where a deviation from the expected control profile is therefore an unambiguous symptom of mis-scaling rather than of a poor solver setting [46].

The validation suite that ships with the thesis is built directly on top of this benchmark formulation: the same four reference scenarios used in Chapter 6 are the natural continuation of the Tier 2 benchmarks,

with the constraint set being gradually expanded in Tier 3. The numerical results reported in Section 6.4 thus inherit by construction the solver settings that were tuned at this stage.

3.2.3 Tier 3 — Fully Developed Problem with External Constraints and Warm Start

The third tier is the final formulation that is intended to be deployed and it adds three blocks of complexity on top of the Tier 2 baseline. The first block is the external environment: the path constraints corresponding to the operational environment are switched on, namely the Digital-Elevation-Model terrain clearance [1], the no-fly-zone exclusion [2], the geofence containment and the intermediate-waypoint hard constraints, all of which are detailed in Chapter 6. The second block is the performance envelope and the regularization: the aircraft performance bounds ($V_{\max}$, V_{stall}, $C_{L,\max}$, $P_{\max}$ and $h_{\min}$) are enforced, the formulation is promoted from Uncoupled to Full-Control (Section 5.3) and the smoothness penalties on control rates and actuator deflections (Section 6.3.2) are enabled [47]. The third block is the warm start: a Hybrid A* path planner [48] followed by an inverse-dynamics reconstruction [3] provides a kinematically feasible initial guess, and the converged UF solution is projected onto the FC state space to seed the rigid-body NLP. The full pipeline is described in Section 6.6 and its empirical performance is studied separately in Section 3.4.

Activating these three blocks one at a time, rather than all together, is what makes the methodology tractable. Each block is regression-tested against the Tier 2 baseline before the next one is enabled, so that any loss of convergence or any drift in the optimum can be attributed to a specific source rather than to an opaque interaction between several newly introduced features at once [49].

3.3 Validation Procedure

The validation of the third-tier solver is concentrated in Chapter 6 and is built on the same benchmark scenarios that were used to tune Tier 2. The purpose of this section is to describe what is verified and along which axes; the full numerical results, the plots and the tables are deferred to the corresponding section of the numerical implementation chapter and to the appendices. The four-axis cross-checking protocol described below is the qualitative-then-quantitative discipline that is universally recommended for direct-collocation solutions [34].

Every comparison reported in the validation chapter is run on the same four reference scenarios, SC-00 to SC-03, whose geometric and environmental parameters are listed in Section 6.4.1. The scenarios were designed as a 2×2 factorial in empty airspace, contrasting straight against turning missions and level against climbing ones, precisely so that the effects of the cost function, of the dynamic model and of the constraint set can be isolated rather than mixed. Each scenario is solved twice per formulation, once under the minimum-time criterion and once under the minimum-fuel criterion, which corresponds to the two extremes $\alpha = 0$ and $\alpha = 1$ of the cost-weight introduced in Section 6.3 [46].

A core element of the methodology is that the same problem is solved with two dynamic models of increasing fidelity. The Uncoupled point-mass formulation (UF, Section 5.2) carries three translational degrees of freedom and treats the lift coefficient as a control input, while the Full-Control rigid-body formulation (FC, Section 5.3) [50] carries the full six degrees of freedom and exposes the elevator, aileron and rudder commands explicitly. Both formulations are introduced in full in Chapter 5; here it is enough to anticipate that running them on the same scenario suite serves two complementary purposes. It cross-validates the simpler model against the higher-fidelity one, since any qualitative disagreement between UF and FC on a benchmark case is in itself a finding that has to be explained, and it quantifies the cost paid in iteration count, wall clock and final objective when the model is promoted from UF to FC. The cross-formulation comparison is reported in Section 6.4.

For each combination of scenario, formulation and criterion the verification proceeds along four redundant axes. The qualitative trajectory inspection (Section 6.4.4) overlays the top-view ground tracks and the altitude profiles of the four combinations of formulation and criterion and checks the expected qualitative signatures, such as the minimum-fuel optimum trading altitude for airspeed on climbing legs or the minimum-time solution saturating the propulsive envelope [43]. The constraint-satisfaction verdict (Section 6.4.7) computes a normalized worst-case margin for every decision variable, every hard path constraint and every collocation defect, with the full per-run numerical margins tabulated in Annex E. The Pareto trade-off between criteria (Section 6.4.5) compares the minimum-time and minimum-fuel runs of the same formulation head-to-head on flight time, fuel burn, control effort and constraint activity, producing a formulation-specific Pareto endpoint pair. The impact of the regularization penalties (Section 6.5) finally repeats every run with the smoothness penalties of Section 6.3.2 enabled and disabled, which isolates the cost paid for physically smooth controls and quantifies the corresponding change in the optimum.

The four axes are intentionally redundant. A failure or an unexpected behavior visible in any one of them is cross-checked against the others before being attributed to a specific source, be it the solver settings, the model fidelity, the constraint conditioning or the penalty calibration. This redundancy is what guards the methodology against the silent failure modes catalogued for direct-collocation transcriptions in recent literature [49].

3.4 Warm-Start Performance Assessment

Once the third-tier solver converges reliably on the validation suite, the next concern of the methodology is computational cost. Interior-point methods are notoriously sensitive to the quality of the initial guess [39], and the difference between a kinematically feasible seed and a naive linear interpolation between waypoints can be of several orders of magnitude in iteration count and in wall-clock time [24]. The warm-start pipeline introduced in Section 6.6 — Hybrid A* path planning [48], followed by inverse-dynamics reconstruction [3] and by the UF $\rightarrow$ FC state projection — is therefore evaluated

against a cold start in a dedicated experiment. The cold reference is built by feeding IPOPT a straight-line interpolation between the initial and final waypoints, with constant nominal airspeed and zero attitude; it is the worst credible guess that is still admissible by the bounds, and it is the kind of guess that a non-specialized user of the solver would supply.

Before any iteration-count comparison is performed the warm-start guess is assessed on its own merits, independently of IPOPT. The trapezoidal collocation defect of the guess — a direct measure of how close the seed already is to a feasible trajectory [35] — is computed and compared against the same defect of the cold-start guess. This step verifies that the warm-start pipeline is producing a useful seed in the first place: a guess whose defect is comparable to the cold one would not justify the additional planning and inverse-dynamics machinery, no matter how much IPOPT seemed to benefit from it downstream.

For every combination of scenario and formulation the comparison then proceeds along four quantitative metrics. The first is the initial-guess quality just described, expressed as a root-mean-square collocation defect of the seed before any IPOPT iteration. The second is the number of primal-dual iterations required to reach the same convergence tolerance under the warm and the cold guess [19]. The third is the wall-clock breakdown of the total solve time, decomposed by pipeline layer (Hybrid A*, inverse dynamics, UF NLP, UF $\rightarrow$ FC projection and FC NLP) so that the cost of the warm-start machinery itself is not hidden in the comparison. The fourth is the sensitivity to the number of collocation nodes, evaluated on a single representative scenario in order to expose the trade-off between discretization accuracy and solve time as the problem size grows [51]. A complementary strategy comparison runs the same scenarios with several warm-start variants (a straight-line cold start, the Hybrid A* path alone without inverse-dynamics reconstruction, and the full warm-start pipeline) so that the marginal contribution of each block can be quantified rather than merely conjectured.

A point that deserves to be stated explicitly is that the workflow chosen for the production pipeline, namely Hybrid A* $\rightarrow$ inverse dynamics $\rightarrow$ UF $\rightarrow$ FC, is not necessarily the fastest one in every individual scenario. The experimental campaign reported in Chapter 6 shows that on some easy geometries a leaner warm start (for example, the Hybrid A* path used directly without inverse-dynamics reconstruction) can shave a few IPOPT iterations off the FC solve. The reason that pipeline is nevertheless retained as the default is one of robustness: across the full complexity span swept by the experiments, including obstacle-rich and NFZ-rich missions, it is the only configuration that converges in every single case [40]. In the operational context of this thesis a slightly slower but uniformly convergent solver is preferred to a faster solver that occasionally fails, because a failure in production would have to be handled by the mission planner and would erase any average speed advantage on its first occurrence.

The cases selected for the warm-start study are deliberately picked to span the full complexity envelope of the solver. The lower end of the sweep consists of the SC-00 and SC-01 baseline scenarios already used in the validation suite, where even the cold start converges and the comparison is purely about

speed. The upper end adds NFZ-rich and obstacle-dense missions in which the cold start either does not converge, converges to a poor local minimum, or takes prohibitively long to do so; in those regimes the warm-start pipeline is what makes the problem tractable at all. This deliberate span of complexity is what allows the warm-start analysis to report not a single context-free speed-up factor but the full operating range over which the pipeline remains the recommended option. The corresponding empirical results — convergence curves, timing breakdowns, sensitivity plots and strategy-comparison tables — are reported in the warm-start results section of Chapter 6.

Chapter 4

Alternative Solutions and Decision

4.1 Overview of Candidate Approaches

This chapter discusses the main technical alternatives that were considered for each of the three core decisions of the project — (i) the geometric path planner used as warm start, (ii) the trajectory optimization transcription, and (iii) the aircraft dynamic model that the optimizer sees — and justifies the option finally chosen. The detailed mathematical development of the selected solutions is given in Chapters 5 and 6; here the focus is the decision rationale rather than the formulation itself. Throughout the chapter the reader should keep in mind that the final user of the solver is *Singular Aircraft* [10], whose engineering and delivery requirements (mature codebase, deterministic solve times on embedded hardware, certifiable behavior at delivery) have acted as a strong filter on the design choices and, in particular, have constrained the time available for the exploration of experimental transcription methods within the scope of this thesis.

4.1.1 Geometric Path Planner (Warm-Start Producer)

Three families of methods were screened for the role of geometric warm-start producer. The first family is the grid-based shortest-path approach, with Dijkstra, A* and Hybrid A* as its main representatives [7], [48]. These methods are deterministic, optimal on the discretized graph, naturally handle obstacles and no-fly zones, and —most importantly for the present application— Hybrid A* additionally embeds the minimum-turn-radius kinematic constraint, so that the resulting path is already flyable as a first guess for the NLP [48]. The second family is sampling-based planning, principally RRT/RRT* [8] and probabilistic roadmaps [16]. These methods scale better in high dimensions but produce non-deterministic paths whose quality depends on the sampling budget, which is less suited to an offline mission-planning loop that must be reproducible run-to-run. The third family groups the lightweight heuristic seeders — direct potential fields and the like — which are the cheapest at runtime but

cannot guarantee obstacle avoidance and produce un-smooth seeds that destabilize the interior-point solver downstream. Hybrid A* was selected for its determinism, its native obstacle handling and the turn-radius kinematic embedding that makes the warm start dynamically consistent before the NLP is ever called [52].

4.1.2 Trajectory Optimization Transcription

Three transcription families were considered for the optimal-control problem itself. The first is the family of indirect methods, in which the Pontryagin maximum principle is used to derive the costate equations and the problem is solved as a two-point boundary-value problem, typically via shooting [32]. These methods provide rigorous first-order optimality conditions and can be very accurate, but they require deriving the costate equations in closed form for every change in the cost function or the dynamic model and are notoriously sensitive to the initial guess of the costates [45]. The second family is direct shooting and multiple shooting, where the controls are discretized and the dynamics are integrated forward in time inside the optimizer [53]. They are easy to set up but the resulting Jacobian is dense in time because every state at the end of the horizon depends on every preceding control, which hurts large-scale convergence and exposes the solver to a known set of transcription-induced failure modes [49]. The third family is direct collocation — trapezoidal, Hermite-Simpson, pseudo-spectral — which decouples state and control via collocation nodes and produces a large but very sparse NLP that is well suited to modern interior-point solvers such as IPOPT and that admits warm starts at the primal-variable level [34], [35].

Direct collocation was chosen as the transcription method for this project, in the simplest trapezoidal variant. The choice is motivated by three reasons that are widely documented in the optimal-control literature and that are particularly relevant for the mission-planning workloads delivered to Singular Aircraft. First, direct collocation produces an NLP whose constraint Jacobian is block-banded and extremely sparse, so that the cost of one IPOPT iteration scales essentially linearly in the number of nodes rather than quadratically as in dense direct shooting [21], [34]. This sparsity is what makes trapezoidal collocation competitive on the thousand-variable problem sizes considered here. Second, direct collocation is robust against discontinuous or only mildly feasible initial guesses, because every node carries its own slack and any infeasibility is absorbed locally instead of accumulating along a shooting integration [30], [51]. This property is what makes the upstream Hybrid A* and inverse-dynamics seeds usable as warm starts in the first place. Third, direct collocation is by now the state-of-the-art transcription for six-degree-of-freedom aerospace trajectory-optimization problems of the size considered in this thesis: recent work on rocket landing, hypersonic re-entry and unmanned-aircraft mission planning all uses direct collocation in commercial or open-source NLP solvers [45], [49], and the standard reference implementations (CasADi, GPOPS-II, PSOPT) have made the transcription itself essentially routine to set up [19], [21].

A deliberate scope decision is associated with this choice. The mathematical formulation and the

procedural outline of the indirect adjoint method are not reproduced in this thesis; a complete exposition is given by Bradley [32], and a survey of its use in aerospace optimal control is provided by [31], [45]. The direct-single-shooting transcription is summarized at the theoretical level in Chapter 5 [30]; however, *the present thesis has not yet incorporated a working implementation using either of those two formulations*. The only formulation that has been numerically realized end-to-end is direct trapezoidal collocation. The reasons for this are partly technical and partly contractual. On the technical side, single shooting would require a tightly coupled forward integrator with sensitivity propagation that does not reuse the CasADi expression graph already built for the collocation pipeline, and an adjoint formulation would require deriving and verifying the costate equations of the FC model by hand, which is a multi-week effort on its own. On the contractual side, the development of the solver has been bounded by the time and delivery constraints imposed by Singular Aircraft as the final user of the solver [9]: integration into their existing mission-management pipeline, the corresponding test campaign and the deliverable deadlines have all had higher priority than the exploration of additional transcription methods. Single shooting and adjoint methods are therefore identified explicitly as future work in Chapter 9.

4.1.3 Aircraft Dynamic Model Used Inside the Optimizer

Two dynamic-model fidelities were considered for the inner loop of the optimizer. The Uncoupled point-mass model (UF) is a three-degrees-of-freedom formulation written in wind axes, in which the lift coefficient enters as a direct control input and attitude is not propagated explicitly [43]. It has a compact state vector and is fast to optimize, but it ignores the attitude dynamics and the actuator limits that the autopilot will ultimately have to enforce. The Full-Control rigid-body model (FC) is a six-degrees-of-freedom formulation written in body axes, with explicit elevator, aileron and rudder commands and with actuator-rate limits enforced as path constraints [50]. It is physically realistic and produces trajectories that an autopilot can execute directly, but the state vector and the Jacobian grow accordingly and the NLP becomes harder to converge from scratch. Rather than choose one of the two models, the project uses both in a hierarchy: the UF formulation solves first, on a smaller and better-conditioned NLP, and its converged solution is then used as a structured warm start for the FC formulation, which inherits a near-feasible iterate and only has to refine the attitude states and the actuator commands. This two-stage workflow exploits the strengths of both fidelities and is what makes the FC solve tractable in the wall-clock budgets imposed by the production pipeline [39].

4.2 Decision Summary and Trade-offs

Table 4.1 consolidates the final design decision for each block of the solver together with the dominant trade-off behind it. The chosen pipeline is therefore Hybrid A* path → inverse-dynamics seed → UF trapezoidal-collocation NLP solved by IPOPT → FC trapezoidal-collocation NLP warm-started from the UF solution [19], [35], [48]. A few remarks justify why the table reads the way it does, beyond the

specific decision of each row.

Table 4.1: Summary of the alternatives considered and the chosen option for each block of the solver, together with the dominant driver of the decision. The chosen options are mutually consistent and form the pipeline described in Chapter 6.

Block	Chosen option	Main driver of the decision
Path planner / warm start	Hybrid A* + inverse dynamics	Deterministic, obstacle-aware, kinematically feasible seed; robust across the full complexity envelope of the mission set.
Optimization method	Direct trapezoidal collocation + IPOPT	Sparse NLP, warm-start friendly, robust to discontinuous guesses; state-of-the-art transcription for 6-DOF aerospace problems.
Dynamic model	UF → FC hierarchy	Fast and well-conditioned convergence on UF; physical fidelity on FC; FC inherits a near-feasible iterate from UF.
Cost function	Bolza form (time + fuel + regularization)	Smooth Pareto sweep between minimum-time and minimum-fuel; regularization absorbs collocation noise without biasing the optimum.

The first remark concerns the choice of direct collocation as the transcription method. As argued in Section 4.1.2, the deciding factors were the sparsity of the resulting NLP, which scales the per-iteration cost linearly with the number of nodes, and the robustness of the method against the only mildly feasible warm-start guesses produced by the upstream Hybrid A* and inverse-dynamics layers [34], [51]. Direct collocation is also the de facto state of the art for six-degree-of-freedom trajectory-optimization problems of comparable size in adjacent aerospace domains [49], which gives the selected transcription a substantial body of reference implementations to draw on. The thesis explicitly does not deploy direct single shooting or the indirect adjoint method, even though both are formalized in the appendices [32]; the development time was prioritized on the production-grade direct collocation pipeline because of the delivery constraints imposed by Singular Aircraft as the final user of the solver [10].

The second remark concerns the warm-start pipeline. The empirical campaign reported in Chapter 6 demonstrates that the chosen workflow — Hybrid A* → inverse dynamics → UF → FC — is not always the absolute fastest configuration on any single scenario; on the simplest geometries a stripped-down warm start (for example, the Hybrid A* path used directly without inverse-dynamics reconstruction) can save a handful of IPOPT iterations. The reason that the full pipeline is nevertheless adopted as the production default is one of robustness: it is the only configuration in the experimental set that converges in every scenario considered, including the obstacle-rich and NFZ-rich missions where the cold start and the leaner warm-start variants either diverge, converge to a poor local minimum, or hit

the iteration limit [24], [40]. For an operational solver that must be handed over to an industrial end user, a slightly slower but uniformly convergent pipeline is preferred over a faster but occasionally failing one, because in production a single failure cannot be compensated by an average speed-up.

Aspects that were deliberately left out of the present scope — real-time re-planning, MPC closure [12], stochastic wind, multi-aircraft coordination, and the indirect/adjoint and direct-single-shooting transcriptions discussed in the appendices — are identified as future work in Chapter 9.

Chapter 5

Theoretical background

5.1 Problem Statement

The development of a Trajectory Optimizer for a Mission Management System (MMS) requires a rigorous mathematical definition to transition from high-level mission requirements to a flyable numerical solution. This section formalizes the optimization task by first defining the continuous-time Optimal Control Problem (OCP) governing the vehicle's motion, and subsequently detailing its transcription into a discrete, constrained Nonlinear Programming (NLP) problem solvable by computational algorithms.

5.1.1 The Continuous-Time Optimal Control Problem (OCP)

Before introducing numerical discretization, the trajectory generation must be formulated in the continuous time domain, $t \in [t_0, t_f]$. The system's degrees of freedom are governed by two primary, time-dependent vector functions:

- **State Vector ($\mathbf{x}(t) \in \mathbb{R}^n$):** Describes the dynamic condition of the vehicle at any given instant. This typically encapsulates the 3D position, velocity vectors, attitude (e.g., Euler angles or quaternions), and angular rates.

- **Control Vector ($\mathbf{u}(t) \in \mathbb{R}^m$):** Represents the independent input variables commanded by the MMS to manipulate the state trajectory (e.g., thrust magnitude, control surface deflections).

The core objective of the continuous OCP is to determine the optimal control history $\mathbf{u}(t)$—and potentially the final time t_f—that minimizes a predefined scalar performance metric J, while strictly adhering to the physical laws of motion and operational boundaries. Mathematically, the standard continuous OCP is formulated as a Bolza problem:

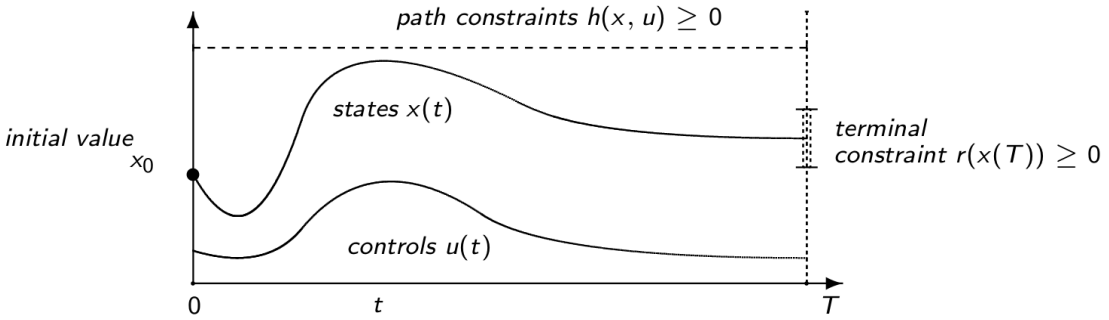


Figure 5.1: Visual representation of the continuous-time Optimal Control Problem (OCP). The optimizer seeks continuous control inputs $u(t)$ that drive the states $x(t)$ from a fixed initial value x_0 to a state satisfying the terminal constraints at time T, while continuously strictly adhering to the defined path constraints. [53]

$$
\begin{aligned}
\min_{\mathbf{u}(t),t_f} \quad & J = \underbrace{\Phi(\mathbf{x}(t_f), t_f)}_{\text{Mayer Term}} + \int_{t_0}^{t_f} \underbrace{L(\mathbf{x}(t), \mathbf{u}(t), t)}_{\text{Lagrange Term}} \, dt \\
\text{subject to} \quad & \dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t), t), \quad \forall t \in [t_0, t_f] \\
& \mathbf{x}(t_0) = \mathbf{x}_0 \\
& \Psi(\mathbf{x}(t_f), t_f) = \mathbf{0} \\
& \mathbf{h}_{\min} \leq \mathbf{h}(\mathbf{x}(t), \mathbf{u}(t), t) \leq \mathbf{h}_{\max}, \quad \forall t \in [t_0, t_f]
\end{aligned}
\tag{5.1}
$$

In this formulation, the components are defined as follows:

- **Cost Function (J):** The metric being minimized.This scalar functional evaluates the overall quality of the generated flight path and is typically expressed in the standard Bolza form, which separates the objective into boundary and continuous components:
  - **Mayer Term (Φ):** Represents the terminal cost or boundary penalty, evaluated exclusively at the final time t_f. This term is designed to optimize discrete conditions at the conclusion of the mission. Typical applications include minimizing the total time of flight (t_f) for time-critical operations, or minimizing the terminal distance/error relative to a specific target landing zone.
  - **Lagrange Term (L):** Represents the running cost, which is integrated continuously over the entire flight horizon from t_0 to t_f. This term penalizes path-dependent behaviors and accumulated costs throughout the mission. It is frequently utilized to minimize total fuel or energy consumption, as well as to penalize aggressive maneuvering (e.g., minimizing acceleration or jerk) to ensure structural safety and passenger comfort.

Then, by assigning weights to different cost function terms, the MMS balances competing mission priorities—like speed versus energy efficiency—to calculate the optimal flight path.

- **System Dynamics ($\dot{\mathbf{x}} = \mathbf{f}$):** This is the governing Ordinary Differential Equation (ODE). It represents the highly non-linear kinematics and kinetics of the aircraft. The optimizer must find a control $\mathbf{u}(t)$ whose resulting integrated state $\mathbf{x}(t)$ perfectly satisfies this ODE at every instant.

- **Boundary Conditions ($\mathbf{x}_0, \Psi$):** Equality constraints anchoring the trajectory. The initial condition ties the solution to the vehicle's current telemetry, while the terminal condition Ψ dictates the mission's required end state.

- **Path Constraints (h):** Continuous inequality constraints that enforce safety and structural limits (e.g., maximum load factor, actuator saturation) and spatial boundaries (e.g., terrain avoidance, No-Fly Zones) throughout the entire flight.

5.1.2 Numerical Transcription to Nonlinear Programming (NLP)

Because numerical solvers cannot compute infinite-dimensional continuous functions, the continuous OCP expressed in Equation 5.1 must be discretized. Direct methods "transcribe" the infinite-dimensional control problem into a finite-dimensional NLP by partitioning the continuous time domain $[t_0, t_f]$ into N intervals on a discrete grid.

Depending on how the states $\mathbf{x}(t)$ and controls $\mathbf{u}(t)$ are handled during this discretization, direct methods are broadly categorized into three distinct approaches:

1. **Direct Single Shooting (Sequential Approach):** Only the control variables $\mathbf{u}(t)$ are discretized into a finite set of parameters (e.g., piecewise constant or linear). The states $\mathbf{x}(t)$ are treated as dependent variables, obtained by explicitly integrating the system dynamics ODE ($\dot{\mathbf{x}} = \mathbf{f}$) forward in time using a numerical simulator (like Runge-Kutta, Euler or Trapezoidal).

    - *Advantages:* Results in a very small NLP, as the optimizer only searches the control parameter space. It is relatively simple to implement if a reliable ODE solver is already available.

    - *Disadvantages:* The forward integration makes it highly sensitive to initial guesses, especially in unstable systems (where there is a lot of coupling between the states in different time steps). Furthermore, evaluating continuous path constraints is difficult and computationally expensive.

2. **Direct Multiple Shooting (Hybrid Approach):** The time horizon is split into multiple shorter intervals. Both states and controls are discretized at the boundaries of these intervals (nodes). The ODE is integrated independently within each short interval, and the NLP solver enforces

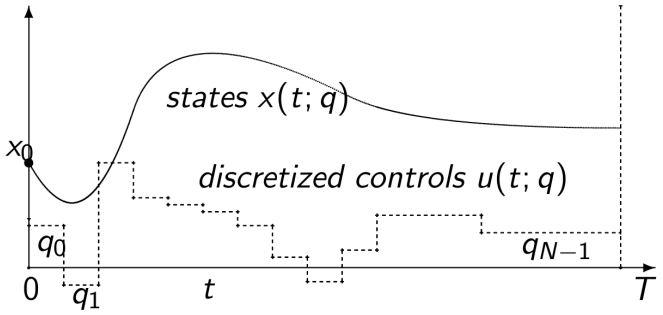


Figure 5.2: Illustration of the Direct Single Shooting transcription method. The infinite-dimensional control trajectory is discretized into a finite set of piecewise constant parameters $q = (q_0, q_1, \ldots, q_{N-1})$. The state trajectory $x(t;q)$ is treated as a dependent variable, obtained by integrating the system dynamics forward in time based on the discrete control inputs.[53]

continuity between intervals using algebraic "defect" equality constraints. This limits the propagation of integration errors compared to single shooting.

3. **Direct Collocation (Simultaneous Approach):** Both the control and state histories are fully discretized over the time grid, creating a large, simultaneous parameter space. Instead of using explicit numerical forward integration, the system dynamics ODEs are approximated using polynomial interpolation (e.g., Hermite-Simpson). The governing ODEs are then enforced as algebraic equality constraints (defects) strictly at specific collocation points.

    - *Advantages:* Highly robust to poor initial guesses and handles unstable dynamics well, since errors do not compound over time. It is exceptionally efficient at handling path constraints. The resulting NLP is very large but highly sparse, which modern solvers exploit for rapid computation.

    - *Disadvantages:* Requires large-scale sparse NLP solvers and is mathematically more complex to formulate than shooting methods.

By applying one of these transcription methods (often Direct Collocation in modern aerospace applications), the continuous problem is reformulated into a discrete parameter optimization task. The optimizer must now find the optimal finite-dimensional **Decision Vector**, $\mathbf{z}$. In a simultaneous approach, this vector typically encompasses all discrete states, controls, and potentially the final time:

$$\mathbf{z} = \left[\mathbf{x}_0, \mathbf{u}_0, \mathbf{x}_1, \mathbf{u}_1, \ldots, \mathbf{x}_N, \mathbf{u}_N, t_f\right]^T \in \mathbb{R}^{(n+m)(N+1)+1} \tag{5.2}$$

5.1.3 Constraints and the Lagrangian Formulation

The optimization search space for the decision vector $\mathbf{z}$ is fundamentally restricted by the physical and operational boundaries of the mission. Once discretized, these boundaries are mathematically encoded into two distinct constraint vectors:

- **Equality Constraints ($g(\mathbf{z}) \in \mathbb{R}^p$):** Typically represent the discretized vehicle dynamics (defect constraints) and the necessary initial/terminal boundary conditions.
- **Inequality Constraints ($h(\mathbf{z}) \in \mathbb{R}^q$):** Safeguard the vehicle by enforcing structural limits (e.g., maximum aerodynamic loads) and operational boundaries (e.g., collision avoidance and designated No-Fly Zones).

To solve this constrained optimization problem, Interior-Point solvers, such as `Ipopt`, transform it into a sequence of unconstrained sub-problems utilizing the **Lagrangian** function, $\mathcal{L}$. This function effectively internalizes the constraints by augmenting the original discrete objective function $F(\mathbf{z})$ with weighted penalty terms. These weights are the Lagrange multipliers (λ for equality constraints and ν for inequality constraints), which act as mathematical restoring forces that pull the solution back into the feasible region whenever a bound is violated.

The Lagrangian is defined as:

$$\mathcal{L}(\mathbf{z}, \lambda, \nu) = F(\mathbf{z}) + \sum_{i=1}^{p} \lambda_i g_i(\mathbf{z}) + \sum_{j=1}^{q} \nu_j h_j(\mathbf{z}) \tag{5.3}$$

The solver's primary objective is to identify a stationary point where the gradients of the Lagrangian vanish ($\nabla\mathcal{L} = \mathbf{0}$). Reaching this stationary point signifies that the system has achieved a local mathematical balance, formally described by the Karush-Kuhn-Tucker (KKT) optimality conditions.

5.1.4 Karush-Kuhn-Tucker (KKT) Optimality Conditions

For a solution to be considered a local optimum in a nonlinear programming problem, it must satisfy the KKT conditions. These conditions represent the first-order necessary criteria for optimality and consist of the following principles:

- **Primal Feasibility ($g(\mathbf{z}) = \mathbf{0}$, $h(\mathbf{z}) \leq \mathbf{0}$):** Ensures that the resulting trajectory is physically possible. It dictates that the solution must strictly respect all discretized environmental boundaries, structural limits, and system dynamics.
- **Dual Feasibility ($\nu_j \geq 0, \forall j$):** Requires that the Lagrange multipliers associated with the inequality constraints be non-negative. This confirms that the gradient of the cost function is properly counterbalanced by the gradients of the active constraints, preventing the optimizer

from moving the solution outside the feasible region to artificially lower the cost.

- **Complementary Slackness ($\nu_j h_j(\mathbf{z}) = 0, \forall j$):** Acts as a logical bridge between the constraints and the cost. It mathematically enforces that an inequality constraint (e.g., a minimum altitude limit) only influences the optimization cost ($\nu_j > 0$) when the vehicle is actively operating exactly on that boundary ($h_j(\mathbf{z}) = 0$). If the constraint is inactive (the vehicle is safely above minimum altitude), its corresponding multiplier must be zero ($\nu_j = 0$).

5.2 Uncoupled Trajectory Dynamics

The Uncoupled Formulation (UF) models the aircraft as a point mass whose path is governed directly by the resultant aerodynamic forces, without resolving the internal attitude dynamics [50]. The lift coefficient C_L enters as a direct control input, bypassing the need to compute the angle of attack α from aerodynamic databases or to propagate any rotational equations of motion. The physical justification for this simplification is a separation of timescales: the short-period and roll-subsidence modes of a typical fixed-wing aircraft decay in tenths of a second, whereas trajectory-level changes evolve over tens of seconds to minutes [44]. The optimizer specifies the quasi-steady lift level that an inner attitude control loop is assumed to track instantaneously; consequently, while this simplification remains to be validated against high-fidelity models, it possesses an inherent inner coherence as shown in the comparison between models done in Section 6.4.

The result is a compact formulation that is tractable for real-time or near-real-time mission planning, while retaining the physical accuracy needed to predict fuel consumption and satisfy hard operational constraints.

5.2.1 Reference Frames and Coordinate Systems

Two primary reference frames are used throughout the formulation [50], [54].

1. **Local NED frame ($\mathcal{F}_{\text{NED}}$).** A North-East-Down inertial frame fixed to the Earth's surface. The aircraft position is expressed as $(x,\ y,\ z)_{\text{NED}}$, where z is positive downward. Altitude follows as $h = -z$.

2. **Wind-axis frame ($\mathcal{F}_W$).** Centered at the center of mass, with the x_w axis aligned with the velocity vector $\vec{V}$. In this frame, aerodynamic forces decompose cleanly: drag D acts along $-x_w$ and lift L acts along $-z_w$ (upward in steady level flight).

The orientation of the velocity vector in the NED frame is described by the flight-path angle γ (positive nose-up) and the heading angle ξ (clockwise from North). The bank angle μ rotates the lift vector within the plane perpendicular to $\vec{V}$, enabling lateral trajectory curvature.

Mission-level inputs (waypoints, airspace boundaries) are often specified in geodetic latitude, longitude, and altitude; the optimizer states (x, y, z) are Cartesian offsets in NED relative to a reference point on the WGS-84 ellipsoid. Figure 5.3 sketches the Earth-centered Earth-fixed (ECEF) frame and its relation to geodetic coordinates; Annex B documents the complete geodetic–ECEF–NED transformation chain, including practical curvature corrections for long horizontal legs.

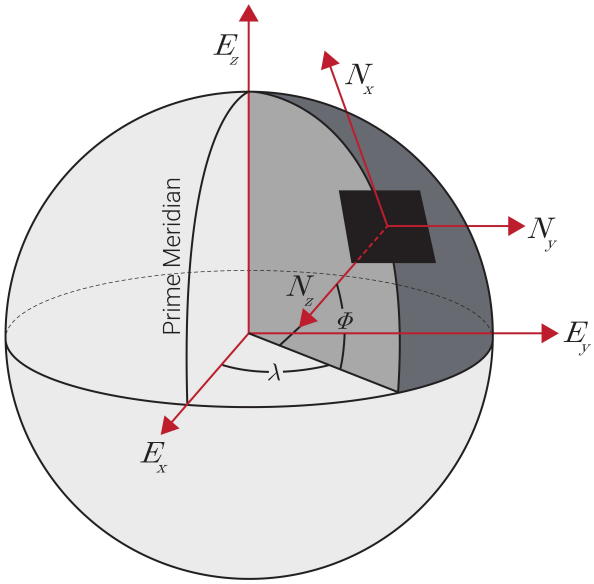


Figure 5.3: Earth-centered Earth-fixed (ECEF) Cartesian frame (X, Y, Z) and geodetic coordinates: latitude φ, longitude λ, and ellipsoidal height h along the ellipsoid normal. The local NED frame used in the trajectory model is obtained by projecting the ECEF displacement from a reference point into North, East, and Down axes fixed to that point (see Annex B).

5.2.2 State and Control Vectors

The UF formulation is defined by a nine-dimensional state vector and a three-dimensional control vector:

$$\mathbf{x}(t) = \left[x,\ y,\ z,\ V,\ \mu,\ \gamma,\ \xi,\ m,\ C\right]^T \in \mathbb{R}^9 \tag{5.4}$$

$$\mathbf{u}(t) = \left[T,\ C_L,\ p\right]^T \in \mathbb{R}^3 \tag{5.5}$$

together with the free terminal time t_f as an additional scalar optimization parameter. Here $\mathbf{x}(t)$ is the instantaneous state of the aircraft and $\mathbf{u}(t)$ its physical control inputs (thrust, lift coefficient and roll-rate command in this formulation). These continuous histories are not optimized as functions: under the direct-collocation transcription of Chapter 6, the values of $\mathbf{x}$ and $\mathbf{u}$ at the N discretization

nodes are stacked, together with t_f, into the single NLP *decision vector* $\mathbf{z}$ (Equation (6.1)) that the optimizer actually solves for. Tables 5.1 and 5.2 describe each variable, its physical meaning, and its operating domain.

Table 5.1: State variables of the Uncoupled formulation.

Symbol	Description	Units	Domain
x	Position along North axis (NED)	m	$\mathbb{R}$
y	Position along East axis (NED)	m	$\mathbb{R}$
z	Position along Down axis (NED), $h = -z$	m	$\mathbb{R}$
V	True airspeed (TAS)	$\mathrm{m\,s^{-1}}$	$[V_{\text{stall}},\, V_{\max}]$
μ	Bank angle	rad	$[-\mu_{\max},\, \mu_{\max}]$
γ	Flight-path angle	rad	$[-\gamma_{\max},\, \gamma_{\max}]$
ξ	Heading angle	rad	$[0,\, 2\pi)$
m	Aircraft mass	kg	$[m_{\text{dry}},\, m_0]$
C	Battery state of charge (electric a/c)	–	$[0,\, 1]$

Table 5.2: Control variables and optimization parameter of the Uncoupled formulation.

Symbol	Description	Units	Domain
T	Engine thrust	N	$[0,\, T_{\max}(h)]$
C_L	Lift coefficient (direct control input)	–	$[C_{L,\min},\, C_{L,\max}]$
p	Roll rate (bank angle rate)	$\mathrm{rad\,s^{-1}}$	$[p_{\min},\, p_{\max}]$
t_f	Final time (free parameter)	s	$(0,\, +\infty)$

5.2.3 Equations of Motion

The equations of motion are derived by applying Newton's second law in the wind-axis frame and projecting the kinematic relations into the NED frame [44], [50]. The full system comprises nine equations, organized below by physical character.

Trajectory Kinematics

The rate of change of position in the NED frame follows from the definition of the velocity vector:

$$\begin{aligned} \dot{x} &= V \cos\gamma \cos\xi \\ \dot{y} &= V \cos\gamma \sin\xi \\ \dot{z} &= -V \sin\gamma \end{aligned} \tag{5.6}$$

These three relations are purely geometric; they hold regardless of the aerodynamic model.

Translational Dynamics

Projecting the forces along the three wind-axis directions and applying Newton's second law, the scalar equations for speed, flight-path angle, and heading angle are:

$$\dot{V} = \frac{T - D - mg\sin\gamma}{m} \tag{5.7}$$

$$\dot{\gamma} = \frac{L\cos\mu - mg\cos\gamma}{mV} \tag{5.8}$$

$$\dot{\xi} = \frac{L\sin\mu}{mV\cos\gamma} \tag{5.9}$$

Equation (5.7) governs the longitudinal energy balance: thrust accelerates the aircraft along the path while drag and the weight component along $\vec{V}$ (equal to $mg\sin\gamma$) oppose it. Equation (5.8) controls the vertical curvature of the path: the lift component $L\cos\mu$, directed normal to $\vec{V}$ within the vertical plane, changes the flight-path angle against the restoring effect of gravity. Equation (5.9) controls the horizontal curvature: the lateral component $L\sin\mu$ turns the heading. These three equations reflect the well-known result that in the absence of sideforce, a banked turn is the only way to change heading in coordinated flight. The full mathematical development of these force equations can be found in Annex A.3

Bank Angle Kinematics

The bank angle evolves at the commanded roll rate:

$$\dot{\mu} = p \tag{5.10}$$

Within the uncoupled model, p is treated as a direct control input rather than as a state with its own moment equation. This is consistent with the decoupling assumption: the roll time constant is much shorter than the trajectory timescale, so the optimizer can specify p as if μ tracks any desired value without dynamic lag.

Mass and Energy Dynamics

For a fuel-burning aircraft, the mass decreases at a rate proportional to the propulsive power delivered by the engine. Expressing the specific fuel consumption in $\mathrm{g\,kW^{-1}\,h^{-1}}$:

$$\dot{m} = -\frac{c_{\text{sfc}} \cdot P}{3.6 \times 10^6}, \qquad P = \frac{T\,V}{\eta_p} \tag{5.11}$$

where P is the shaft power in watts and η_p is the propulsive efficiency. For an electrically powered aircraft, $\dot{m} = 0$ and the battery state of charge depletes as:

$$\dot{C} = -\frac{P}{U_{\text{batt}} \cdot 3.6} \tag{5.12}$$

where U_{batt} is the battery voltage in volts and C is in ampere-hours.

5.2.4 Aerodynamic Model

The lift L and drag D forces are computed from the standard aerodynamic definitions using dynamic pressure $q = \frac{1}{2}\rho V^2$:

$$L = q\,S\,C_L, \qquad D = q\,S\,C_D \tag{5.13}$$

The drag coefficient follows the parabolic (quadratic) polar:

$$C_D = C_{D_0} + k\,C_L^2 \tag{5.14}$$

where C_{D_0} is the zero-lift parasite drag coefficient and k is the induced drag factor, related to the wing aspect ratio $\mathcal{A}$ and Oswald span efficiency e by $k = 1/(\pi\,e\,\mathcal{A})$ [43]. Both coefficients are taken from the aircraft aerodynamic database and held constant (no high-lift device corrections at this fidelity level); Table 5.3 lists their typical ranges. The air density ρ is computed from an ISA model as a function of altitude $h = -z$ [55].

Table 5.3: Aerodynamic polar parameters used in the Uncoupled formulation.

Parameter	Physical meaning	Typical range (fixed-wing UAV)
C_{D_0}	Zero-lift (parasite) drag coefficient	0.01 – 0.05
k	Induced drag factor	0.03 – 0.12
S	Wing reference area	aircraft-specific [m^2]
e	Oswald efficiency factor	0.70 – 0.95

The parabolic polar is exact for an elliptic lift distribution and remains a good approximation for most

moderate-aspect-ratio wings at subsonic speed [43]. Its analytical simplicity is particularly valuable in the context of trajectory optimization, as it yields closed-form expressions for the lift-to-drag ratio and the fuel-optimal airspeed.

5.2.5 Summary of the Governing Equations

Table 5.4 collects the nine governing differential equations of the UF formulation — one per state variable — together with the two algebraic aerodynamic closures.

Table 5.4: The nine governing differential equations of the Uncoupled formulation, one per state variable, grouped by physical category, followed by the two algebraic aerodynamic closures. The mass and battery-charge equations are mutually exclusive: $\dot{m}$ is active for fuel-burning aircraft and $\dot{C}$ for electric ones, with the inactive equation held at zero.

Category	**#**	**Equation**
Trajectory kinematics	1	$\dot{x} = V\cos\gamma\cos\xi$
	2	$\dot{y} = V\cos\gamma\sin\xi$
	3	$\dot{z} = -V\sin\gamma$
Force (translational) equations	4	$\dot{V} = (T - D - mg\sin\gamma)/m$
	5	$\dot{\gamma} = (L\cos\mu - mg\cos\gamma)/(mV)$
	6	$\dot{\xi} = L\sin\mu\,/(mV\cos\gamma)$
Bank-angle kinematics	7	$\dot{\mu} = p$
Mass evolution	8	$\dot{m} = -\mathrm{c}_{\mathrm{sfc}}\cdot P/(3.6\times 10^6)$
Energy (battery) evolution	9	$\dot{C} = -P/(U_{\mathrm{batt}}\cdot 3.6)$
Total (ODE states)	**9**	
Algebraic closures	+2	$L = \frac{1}{2}\rho V^2 S\,C_L$ $C_D = C_{D_0} + k\,C_L^2$

5.3 Full-Control Formulation

The Full-Control Formulation (FC) extends the UF model by replacing the direct C_L input with physical control-surface deflections and by explicitly propagating both the translational and rotational dynamics in the *body-axis frame* [44], [50]. Unlike the UF model, which operates in the wind-axis frame and treats C_L as a quasi-steady control input, the FC formulation integrates Newton's and Euler's equations directly in body coordinates, computing the aerodynamic angles α and β algebraically from the instantaneous body velocities at every integration step. This added fidelity comes at the cost of a substantially larger state space and a tightly coupled system of equations, but it produces trajectories in which every control input corresponds to a physical actuator command that an autopilot can execute

directly.

The FC formulation is best suited to situations where attitude dynamics matter — steeper climbs, coordinated turns at high bank angles, or tight maneuver sequences — and where a trajectory intended for direct upload to an autopilot must include explicit surface commands.

5.3.1 State and Control Vectors

The body-axis state vector comprises the NED position, the aircraft mass, the three body-axis velocity components, the three Euler orientation angles, the three body-axis angular rates, and the battery state of charge:

$$\mathbf{x}(t) = \left[x_e,\ y_e,\ z_e,\ m,\ u,\ v,\ w,\ \phi,\ \theta,\ \psi,\ p,\ q,\ r,\ C\right]^T \in \mathbb{R}^{14} \tag{5.15}$$

Here (u, v, w) are the projections of the velocity vector onto the body x_b, y_b, z_b axes respectively, and (ϕ, θ, ψ) are the classical roll–pitch–yaw Euler angles that orient the body frame relative to the NED frame. The yaw angle ψ plays the role of the heading ξ of the UF formulation, but is now a proper ODE state driven by the rotational equations rather than a wind-axis angle. The battery state C is omitted for fuel-burning aircraft (where $\dot{m} \neq 0$) and activated for electrically powered ones.

The control vector consists of the engine thrust and the three primary control-surface deflections:

$$\mathbf{u}(t) = \left[T,\ \delta_e,\ \delta_a,\ \delta_r\right]^T \in \mathbb{R}^4 \tag{5.16}$$

with t_f retained as a free scalar optimization parameter. As in the UF formulation, $\mathbf{x}(t)$ is the instantaneous aircraft state and $\mathbf{u}(t)$ its physical control inputs (here the thrust and the three actuator deflections); the direct-collocation transcription again stacks their node samples, together with t_f, into the single NLP decision vector $\mathbf{z}$ (Equation (6.1)). The state grows from nine to fourteen components, and the controls from three to four, but the optimization variable solved for remains the stacked $\mathbf{z}$.

Tables 5.5 and 5.6 list each variable with its physical meaning and operating domain.

Table 5.5: State variables of the Full-Control formulation. All translational and rotational quantities are expressed in the body-axis frame; position is in the NED frame.

Symbol	Description	Units	Notes
x_e, y_e, z_e	NED position (North, East, Down)	m	Integration in NED
m	Aircraft mass	kg	Shared with UF
u	Body-axis forward velocity	$\mathrm{m\,s^{-1}}$	Replaces V of UF
v	Body-axis lateral velocity	$\mathrm{m\,s^{-1}}$	New in FC
w	Body-axis normal velocity	$\mathrm{m\,s^{-1}}$	New in FC
ϕ	Euler roll angle	rad	Replaces μ of UF
θ	Euler pitch angle	rad	Replaces γ of UF
ψ	Euler yaw/heading angle	rad	Replaces ξ of UF
p	Body-axis roll rate	$\mathrm{rad\,s^{-1}}$	New in FC
q	Body-axis pitch rate	$\mathrm{rad\,s^{-1}}$	New in FC
r	Body-axis yaw rate	$\mathrm{rad\,s^{-1}}$	New in FC
C	Battery charge (electric a/c)	Ah	Active only if $\mathrm{c_{sfc}} = 0$

Table 5.6: Control variables and optimization parameter of the Full-Control formulation.

Symbol	Description	Units	Domain
T	Engine thrust	N	$[0,\ T_{\max}(h)]$
δ_e	Elevator deflection	rad	$[\delta_{e,\min},\ \delta_{e,\max}]$
δ_a	Aileron deflection	rad	$[\delta_{a,\min},\ \delta_{a,\max}]$
δ_r	Rudder deflection	rad	$[\delta_{r,\min},\ \delta_{r,\max}]$
t_f	Final time (free param.)	s	$(0,\ +\infty)$

5.3.2 The Full-Control Coupled Resolution

The central challenge of the FC formulation — the *full-control coupled dilemma* — is that the aerodynamic forces depend on the instantaneous incidence angles α and β, which are themselves algebraic functions of the body velocities (u, v, w), which in turn are the output of the translational dynamics driven by those same aerodynamic forces. In the UF formulation this circularity is broken by treating C_L as a free control input; in the FC formulation it is resolved step by step at each collocation node, evaluating the right-hand side of every differential equation in a fixed sequential order. The procedure is summarized in Algorithm 5.4 and detailed in the subsections that follow.

Algorithm 5.4: FC Right-Hand Side Evaluation at Node k

Input: state $\mathbf{x}_k = [x_e, y_e, z_e, m, u, v, w, \phi, \theta, \psi, p, q, r, C]_k$, control $\mathbf{u}_k = [T, \delta_e, \delta_a, \delta_r]_k$, wind $\mathbf{V}_w^{\text{NED}}$

1. **Build rotation matrix.** Compute $\mathbf{R}_{bn}(\phi, \theta, \psi)$ (NED $\rightarrow$ body) and its trigonometric scalars $\{c_\phi, s_\phi, c_\theta, s_\theta, c_\psi, s_\psi\}$.
2. **Rotate wind to body frame.** Project $\mathbf{V}_w^{\text{NED}}$ into body coordinates; compute wind-relative velocities $u_r = u - V_{wb,1}$, $v_r = v - V_{wb,2}$, $w_r = w - V_{wb,3}$ and aerodynamic speed $V_a = \sqrt{u_r^2 + v_r^2 + w_r^2}$.
3. **Compute aerodynamic angles.** $\alpha = \text{atan2}(w_r, u_r)$, $\beta = \arcsin(v_r/V_a)$.
4. **Evaluate aerodynamic coefficients.** Compute $C_L, C_D, C_Y, C_l, C_m, C_n$ from the linear stability-derivative model (Section 5.3.4).
5. **Compute wind-axis forces and moments.** L, D, Y from dynamic pressure; $\mathcal{L}_W, \mathcal{M}_W, \mathcal{N}_W$ from span and chord.
6. **Rotate to body axes.** Apply the (wind $\rightarrow$ body) rotation $\mathbf{R}(\alpha, \beta)$ to obtain body-axis force components $F_{x,a}, F_{y,a}, F_{z,a}$ and moment components L_b, M_b, N_b (including the thrust pitch moment $T\, d_{T_z}$).
7. **Translational dynamics.** Evaluate $(\dot{u}, \dot{v}, \dot{w})$ from Newton's law in body axes, including gravitational projections and Coriolis cross-products.
8. **Rotational dynamics.** Evaluate $(\dot{p}, \dot{q}, \dot{r})$ from the coupled Euler moment equations with inertial cross-coupling I_{xz}.
9. **Euler angle kinematics.** Evaluate $(\dot{\phi}, \dot{\theta}, \dot{\psi})$ from the kinematic differential equations.
10. **Position kinematics.** Evaluate $(\dot{x}_e, \dot{y}_e, \dot{z}_e)$ by projecting (u, v, w) through $\mathbf{R}_{nb}$ into the NED frame.
11. **Energy/mass evolution.** Evaluate $\dot{m}$ (fuel) or $\dot{C}$ (electric).

Output: $\dot{\mathbf{x}}_k \in \mathbb{R}^{14}$

Figure 5.4: Sequential evaluation of the FC right-hand side at a single collocation node. Steps 2–6 break the aerodynamic coupling: α and β are obtained algebraically from the current body velocities before being fed into the coefficient model and the body-axis force rotation.

5.3.3 External Wind and Aerodynamic Angles

The aircraft's aerodynamic response depends not on the inertial body velocities (u, v, w) but on the velocity of the body *relative to the ambient air* [44]. Given a wind vector $\mathbf{V}_w^{\text{NED}} = (W_N, W_E, W_D)^T$ expressed in the NED frame, its body-axis projection is obtained through the direction-cosine matrix:

$$\begin{pmatrix} V_{wb,1} \\ V_{wb,2} \\ V_{wb,3} \end{pmatrix} = \mathbf{R}_{bn}\, \mathbf{V}_w^{\mathrm{NED}} \tag{5.17}$$

where $\mathbf{R}_{bn}$ is the 3-2-1 Euler rotation matrix:

$$\mathbf{R}_{bn} = \begin{pmatrix} c_\theta c_\psi & c_\theta s_\psi & -s_\theta \\ s_\phi s_\theta c_\psi - c_\phi s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & s_\phi c_\theta \\ c_\phi s_\theta c_\psi + s_\phi s_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi & c_\phi c_\theta \end{pmatrix} \tag{5.18}$$

using the shorthand $c_\phi \equiv \cos\phi$, $s_\phi \equiv \sin\phi$, and likewise for θ and ψ. The wind-relative body velocities are then:

$$u_r = u - V_{wb,1}, \qquad v_r = v - V_{wb,2}, \qquad w_r = w - V_{wb,3} \tag{5.19}$$

from which the aerodynamic speed and incidence angles follow:

$$V_a = \sqrt{u_r^2 + v_r^2 + w_r^2} \tag{5.20}$$

$$\alpha = \operatorname{atan2}(w_r,\, u_r), \qquad \beta = \arcsin\!\left(\frac{v_r}{V_a}\right) \tag{5.21}$$

Equations (5.21) are purely *algebraic*: α and β are not ODE states — they have no associated differential equations — but are recomputed at every evaluation of the right-hand side from the current values of $(u, v, w, \phi, \theta, \psi)$. This is a key structural difference from some alternative formulations in which α and β are treated as additional dynamic states with their own kinematic differential equations (cf. Section 5.3.12).

5.3.4 Aerodynamic Coefficient Model

The aerodynamic forces and moments are expressed through six non-dimensional coefficients using the standard linear stability-and-control derivative model [44], [50]. Static terms, primary control-surface terms, damping derivatives with respect to non-dimensional angular rates, and a thrust-related drag increment are retained in the formulation below.

Introduce the non-dimensional angular rates (based on the aerodynamic speed V_a, mean aerodynamic

chord $\bar{c}$, and wing span b):

$$\hat{q} = \frac{\bar{c}}{2V_a}\, q, \qquad \hat{p} = \frac{b}{2V_a}\, p, \qquad \hat{r} = \frac{b}{2V_a}\, r \tag{5.22}$$

Dynamic pressure is

$$q_S = \tfrac{1}{2}\, \rho(z_e)\, V_a^2\, S_w \tag{5.23}$$

with $\rho(z_e)$ the ISA air density at altitude $h = -z_e$ [55]. The coefficients are assembled in implementation order: C_L first, then C_D (which may depend on C_L, $\hat{q}$, and thrust T), then the remaining four.

$$C_L = C_{L_0} + C_{L_\alpha}\, \alpha + C_{L_{\delta_e}}\, \delta_e + C_{L_q}\, \hat{q} \tag{5.24a}$$

$$C_D = C_{D_0} + C_{D_{k1}}\, C_L + C_{D_{k2}}\, C_L^2 + C_{D_q}\, \hat{q} + C_{D_{CT}}\, \frac{T}{q_S} \tag{5.24b}$$

$$C_m = C_{m_0} + C_{m_\alpha}\, \alpha + C_{m_{\delta_e}}\, \delta_e + C_{m_q}\, \hat{q} \tag{5.24c}$$

$$C_Y = C_{Y_0} + C_{Y_\beta}\, \beta + C_{Y_{\delta_a}}\, \delta_a + C_{Y_{\delta_r}}\, \delta_r + C_{Y_p}\, \hat{p} + C_{Y_r}\, \hat{r} \tag{5.24d}$$

$$C_l = C_{l_0} + C_{l_\beta}\, \beta + C_{l_{\delta_a}}\, \delta_a + C_{l_{\delta_r}}\, \delta_r + C_{l_p}\, \hat{p} + C_{l_r}\, \hat{r} \tag{5.24e}$$

$$C_n = C_{n_0} + C_{n_\beta}\, \beta + C_{n_{\delta_a}}\, \delta_a + C_{n_{\delta_r}}\, \delta_r + C_{n_p}\, \hat{p} + C_{n_r}\, \hat{r} \tag{5.24f}$$

Implementation remark. The presence of the non-dimensional angular rates $\hat{p}$, $\hat{q}$, and $\hat{r}$ in Equations (5.24a)–(5.24f) reflects the general linear stability-derivative structure retained at the theoretical level. However, in the numerical examples and case studies reported in Section 6.4, all damping and unsteady derivatives associated with these rates are set to zero in the aerodynamic database following the quasi-steady stability-derivative practice catalogued in the USAF DATCOM [56]. In particular, the unsteady and speed-stability contributions that would be associated with $C_{m_{\dot{\alpha}}}$ and C_{m_u} are neglected in the implementation, so that the pitching moment reduces in practice to the quasi-steady combination of C_{m_0}, C_{m_α}, $C_{m_{\delta_e}}$, and C_{m_q} around the chosen trim condition.

The drag polar reduces to the UF parabolic form $C_D = C_{D_0} + k\, C_L^2$ when $C_{D_{k1}} = 0$, $C_{D_{k2}} = k$, and $C_{D_q} = C_{D_{CT}} = 0$. The wind-axis forces and aero moments are then:

$$L = q_S\, C_L, \quad D = q_S\, C_D, \quad Y = q_S\, C_Y \tag{5.25}$$

$$\mathcal{L}_W = q_S\, b\, C_l, \qquad \mathcal{M}_W = q_S\, \bar{c}\, C_m, \qquad \mathcal{N}_W = q_S\, b\, C_n \tag{5.26}$$

with b the wing span and $\bar{c}$ the mean aerodynamic chord. Table 5.7 lists the principal derivatives with their expected signs for a statically stable aircraft.

Table 5.7: Aerodynamic stability and control derivatives appearing in Equations (5.24a)–(5.24f). Signs in the last column refer to a statically stable, conventionally controlled aircraft.

	Coefficient	Brief meaning	Sign
Lift	C_{L_0}	Lift bias / trim intercept	varies
	C_{L_α}	Lift curve slope ($\partial C_L/\partial\alpha$)	+
	$C_{L_{\delta e}}$	Elevator → lift	varies
	C_{L_q}	Pitch-rate lift derivative	varies
Drag	C_{D_0}	Zero-lift (parasite) drag	+
	$C_{D_{k_1}}$	Linear induced-drag term in C_L	+
	$C_{D_{k_2}}$	Quadratic induced-drag term in C_L^2	+
	C_{D_q}	Pitch-rate drag increment	+
	$C_{D_{c_T}}$	Thrust-related drag increment	+
Side force	C_{Y_0}	Sideforce bias / trim intercept	≈ 0
	C_{Y_β}	Sideslip stability ($\partial C_Y/\partial\beta$)	−
	$C_{Y_{\delta a}}$	Aileron → sideforce	varies
	$C_{Y_{\delta r}}$	Rudder → sideforce	+
	C_{Y_p}	Roll-rate sideforce derivative	varies
	C_{Y_r}	Yaw-rate sideforce derivative	varies
Roll moment	C_{l_0}	Roll-moment bias / trim intercept	≈ 0
	C_{l_β}	Dihedral effect ($\partial C_l/\partial\beta$)	−
	$C_{l_{\delta a}}$	Aileron effectiveness in roll	+
	$C_{l_{\delta r}}$	Rudder → roll coupling	varies
	C_{l_p}	Roll damping ($\partial C_l/\partial\hat{p}$)	−
	C_{l_r}	Yaw-rate → roll coupling	varies
Pitch mom.	C_{m_0}	Pitch-moment bias / trim intercept	varies
	C_{m_α}	Static longitudinal stability	−
	$C_{m_{\delta e}}$	Elevator effectiveness in pitch	−
	C_{m_q}	Pitch damping ($\partial C_m/\partial\hat{q}$)	−
Yaw moment	C_{n_0}	Yaw-moment bias / trim intercept	≈ 0
	C_{n_β}	Weathercock stability ($\partial C_n/\partial\beta$)	+
	$C_{n_{\delta a}}$	Aileron adverse yaw	varies
	$C_{n_{\delta r}}$	Rudder effectiveness in yaw	+
	C_{n_p}	Roll-rate → yaw coupling	varies
	C_{n_r}	Yaw damping ($\partial C_n/\partial\hat{r}$)	−

5.3.5 Rotation of Forces and Moments to Body Axes

The aerodynamic forces (L, D, Y) and moments $(\mathcal{L}_W, \mathcal{M}_W, \mathcal{N}_W)$ of Equations (5.25)–(5.26) are expressed in the wind-axis frame. Before entering the body-axis equations of motion they must be projected into body coordinates through the wind→body rotation $\mathbf{R}(\alpha, \beta)$, parameterized by the same incidence angles computed in Section 5.3.3.

Using the shorthand $c_\alpha \equiv \cos\alpha$, $s_\alpha \equiv \sin\alpha$, $c_\beta \equiv \cos\beta$, $s_\beta \equiv \sin\beta$:

$$\begin{aligned} F_{x,a} &= -D\, c_\alpha c_\beta + L\, s_\alpha - Y\, c_\alpha s_\beta \\ F_{y,a} &= Y\, c_\beta - D\, s_\beta \\ F_{z,a} &= -D\, s_\alpha c_\beta - L\, c_\alpha - Y\, s_\alpha s_\beta \end{aligned} \tag{5.27}$$

$$\begin{aligned} L_b &= c_\alpha c_\beta\, \mathcal{L}_W - c_\alpha s_\beta\, \mathcal{M}_W - s_\alpha\, \mathcal{N}_W \\ M_b &= s_\beta\, \mathcal{L}_W + c_\beta\, \mathcal{M}_W + T\, d_{Tz} \\ N_b &= s_\alpha c_\beta\, \mathcal{L}_W - s_\alpha s_\beta\, \mathcal{M}_W + c_\alpha\, \mathcal{N}_W \end{aligned} \tag{5.28}$$

The term $T\, d_{Tz}$ in M_b accounts for the pitching moment generated when the thrust line passes at a vertical offset d_{Tz} from the center of gravity. It is the only way thrust enters the rotational equations; in all other channels the engine contributes solely to the translational dynamics.

5.3.6 Translational Dynamics

With the body-axis forces known, Newton's second law in the rotating body frame gives the accelerations of each velocity component [50]. Three additional terms appear compared to the inertial-frame equations: the gravitational acceleration projected onto the body axes, and the Coriolis cross-product $\boldsymbol{\omega} \times \mathbf{v}$ arising from the non-inertial frame:

$$\begin{aligned} \dot{u} &= \frac{T + F_{x,a}}{m} - g\sin\theta - (qw - rv) \\ \dot{v} &= \frac{F_{y,a}}{m} + g\cos\theta\sin\phi - (ru - pw) \\ \dot{w} &= \frac{F_{z,a}}{m} + g\cos\theta\cos\phi - (pv - qu) \end{aligned} \tag{5.29}$$

The gravity projections $(-g s_\theta,\ g c_\theta s_\phi,\ g c_\theta c_\phi)$ are the components of the gravitational acceleration vector $\mathbf{g}^{\mathrm{NED}} = (0, 0, g)^T$ resolved in the body frame through $\mathbf{R}_{bn}$. The Coriolis terms $(qw - rv)$, $(ru - pw)$, $(pv - qu)$ are the components of $\boldsymbol{\omega} \times \mathbf{v}$ and account for the apparent acceleration due to the rotation of the body frame.

5.3.7 Rotational Dynamics

The angular accelerations $(\dot{p}, \dot{q}, \dot{r})$ follow from Euler's equations for a rigid body [44], [50]. The implementation uses the general form that retains the I_{xz} product of inertia (non-zero for most real aircraft due to wing dihedral and structural asymmetries). Defining the augmented moment components:

$$\begin{aligned} \tilde{L} &= L_b + \left(I_{yy} - I_{zz}\right) q\, r + I_{xz}\, p\, q \\ \tilde{M} &= M_b - \left(I_{xx} - I_{zz}\right) p\, r - I_{xz}\left(p^2 - r^2\right) \\ \tilde{N} &= N_b + \left(I_{xx} - I_{yy}\right) p\, q - I_{xz}\, q\, r \end{aligned} \tag{5.30}$$

and the inertia determinant:

$$\Delta = I_{xx}\, I_{zz} - I_{xz}^2 \tag{5.31}$$

the angular accelerations are obtained by inverting the 2×2 coupled system in $\dot{p}$ and $\dot{r}$:

$$\dot{p} = \frac{I_{zz}\tilde{L} + I_{xz}\tilde{N}}{\Delta} \tag{5.32a}$$

$$\dot{q} = \frac{\tilde{M}}{I_{yy}} \tag{5.32b}$$

$$\dot{r} = \frac{I_{xz}\tilde{L} + I_{xx}\tilde{N}}{\Delta} \tag{5.32c}$$

Note that if $I_{xz} = 0$ (perfectly symmetric aircraft), Equations (5.32a) and (5.32c) reduce to the familiar form $\dot{p} = \tilde{L}/I_{xx}$ and $\dot{r} = \tilde{N}/I_{zz}$. The general form is used here because although the aircraft models in the database have non-negligible I_{xz}, and omitting it would introduce a systematic bias in roll–yaw coupling, the aim of the thesis is to develop a generic optimizer for any kind of fixed-wing aircraft.

5.3.8 Euler Angle Kinematics

The Euler angles (ϕ, θ, ψ) are not force quantities; their evolution is governed by the purely kinematic relation between the body-axis angular rates (p, q, r) and the Euler angle rates $(\dot{\phi}, \dot{\theta}, \dot{\psi})$ [50]:

$$\begin{aligned} \dot{\phi} &= p + \tan\theta\,(q \sin\phi + r \cos\phi) \\ \dot{\theta} &= q \cos\phi - r \sin\phi \\ \dot{\psi} &= \frac{q \sin\phi + r \cos\phi}{\cos\theta} \end{aligned} \tag{5.33}$$

These equations are the standard 3-2-1 Euler kinematic equations. They contain the well-known gimbal-lock singularity at $\theta = \pm 90°$; in practice, a warning is issued when $|c_\theta| < 10^{-4}$ and the trajectory optimizer enforces a pitch angle path constraint that keeps the solution away from this singularity.

5.3.9 Position Kinematics

The NED position rates follow from projecting the body-axis velocity vector (u, v, w) into the NED frame through the transpose of $\mathbf{R}_{bn}$, i.e. through the body→NED rotation $\mathbf{R}_{nb} = \mathbf{R}_{bn}^T$:

$$\begin{aligned}
\dot{x}_e &= c_\theta c_\psi\, u + \left(s_\phi s_\theta c_\psi - c_\phi s_\psi\right) v + \left(c_\phi s_\theta c_\psi + s_\phi s_\psi\right) w \\
\dot{y}_e &= c_\theta s_\psi\, u + \left(s_\phi s_\theta s_\psi + c_\phi c_\psi\right) v + \left(c_\phi s_\theta s_\psi - s_\phi c_\psi\right) w \\
\dot{z}_e &= -s_\theta\, u + s_\phi c_\theta\, v + c_\phi c_\theta\, w
\end{aligned} \tag{5.34}$$

These three equations are the body-axis counterpart of the simple wind-axis kinematic relations $\dot{x} = V\cos\gamma\cos\xi$ of the UF model. They are more expensive to evaluate but are exact for any combination of (ϕ, θ, ψ) without requiring the auxiliary wind-axis angles γ and ξ.

5.3.10 Mass and Energy Evolution

The mass equation is identical in structure to that of the UF formulation. The propulsive power is now computed using the wind-relative axial velocity u_r rather than the inertial airspeed, because u_r is the velocity component that the engine thrust actually does work against:

$$P = \frac{T\, u_r}{\eta_p} \tag{5.35}$$

For a fuel-burning aircraft considering that the specific fuel consumption is given in g/KWh:

$$\dot{m} = -\frac{\mathrm{c_{sfc}} \cdot P}{3.6 \times 10^6} \tag{5.36}$$

For an electrically powered aircraft ($\mathrm{c_{sfc}} = 0$, $\dot{m} = 0$), the battery state of charge evolves as:

$$\dot{C} = -\frac{P}{U_{\mathrm{batt}} \cdot 3.6} \tag{5.37}$$

where $U_{\mathrm{batt}} = U_{b,\min} + (U_{b,\max} - U_{b,\min})\,(C/C_{\mathrm{nom}})$ is the state-of-charge-dependent terminal voltage and C is in ampere-hours. Note that the use of u_r rather than $V_a = \sqrt{u_r^2 + v_r^2 + w_r^2}$ is consistent with

the assumption that the propeller thrust acts primarily along the body x_b axis.

5.3.11 Summary of the Governing Equations

Table 5.8 organizes the fourteen differential equations of the FC formulation by physical category.

Table 5.8: Fourteen governing differential equations of the FC formulation, one per state variable. The aerodynamic angles α and β are algebraic quantities; the six aerodynamic coefficient relations (5.24a)–(5.24f) are also algebraic once q_S and the non-dimensional rates (5.22) are evaluated. Including these algebraic relations raises the effective equation count to 22, but only 14 integration states are advanced per time step. As in the UF formulation, the mass and battery-charge equations are mutually exclusive: $\dot{m}$ is active for fuel-burning aircraft and $\dot{C}$ for electric ones, with the inactive equation held at zero.

Category	**#**	**Variables evolved**
Position kinematics (NED)	3	$\dot{x}_e,\ \dot{y}_e,\ \dot{z}_e$
Translational dynamics	3	$\dot{u},\ \dot{v},\ \dot{w}$
Rotational dynamics	3	$\dot{p},\ \dot{q},\ \dot{r}$
Euler angle kinematics	3	$\dot{\phi},\ \dot{\theta},\ \dot{\psi}$
Mass evolution	1	$\dot{m}$
Energy (battery) evolution	1	$\dot{C}$
Total (ODE states)	**14**	
Algebraic closures	+8	$\alpha,\ \beta,\ C_L,\ C_D,\ C_Y,\ C_l,\ C_m,\ C_n$

5.3.12 Comparison of the Two Formulations

The UF and FC formulations represent different balances within the fidelity–complexity trade-off. Table 5.9 summarizes their key structural differences, while Annex A.4.1 provides a complementary derivation-based comparison. This annex clearly demonstrates that the UF formulation is a direct simplification of the complete problem.

Table 5.9: Comparison of the Uncoupled and Full-Control formulations. Both solve a Bolza OCP; the columns highlight where the two formulations differ structurally.

	Uncoupled (UF)	**Full-Control (FC)**
Reference frame	Wind axis	Body axis
ODE states	9	14
State variables	$x, y, z, V, \mu, \gamma, \xi, m, C$	$x_e, y_e, z_e, m, u, v, w,$ $\phi, \theta, \psi, p, q, r, C$
Control variables	3+1 (t_f, T, C_L, auxiliary p)	4+1 (t_f, T, δ_e, δ_a, δ_r)
Aero angles α, β	Not computed (absorbed into C_L)	Algebraic, from (u_r, v_r, w_r)
Inertia coupling	None (decoupled channels)	Full I_{xz} cross-coupling
Wind model	Implicit (speed V is TAS)	Explicit NED→body rotation
Path constraints*	6	8

* Actuator bounds on individual surfaces are not counted; the listed numbers refer to the operational path constraints (stall, load factor, thrust, sideslip, angular rates, and terrain clearance). The FC formulation adds the pitch singularity avoidance constraint and an explicit sideslip limit absent from the UF model.

The UF model offers a compact, fast-to-solve representation well suited to mission-level planning, warm starting, and analytical approximations (see Annex A.3). The FC model resolves the full coupling between control surfaces, attitude, and path that is needed when trajectories are intended for direct autopilot upload or when attitude constraints are binding. In the numerical implementation of Chapter 6 (pipeline overview in Figure 6.21) the UF solution is used as the warm start for the FC optimization: the body velocities (u, v, w) are initialized from the UF airspeed and heading, and the Euler angles from the UF flight-path and heading angles, through the analytical state projection introduced in Section 6.6.4 and validated in Section 6.4.

Chapter 6

Numerical Implementation of Optimal Trajectory Planning

The theoretical background developed in Chapter 5 established the continuous-time Optimal Control Problem (OCP) and the two flight-mechanics models governing the optimization: the nine-state Uncoupled Formulation (UF) and the fourteen-state Full-Control Formulation (FC). While these models provide a mathematically rigorous description of the vehicle dynamics, they are inherently infinite-dimensional. Because they describe the evolution of continuous functions $\mathbf{x}(t)$ and $\mathbf{u}(t)$ over a closed interval $[t_0, t_f]$, they exist in a functional space that standard numerical optimizers cannot navigate directly.

To bridge this gap, the continuous OCP must be transformed into a finite-dimensional Nonlinear Programming (NLP) problem. This process, known as transcription, involves discretizing the state and control trajectories at specific nodes. However, this transition is not merely a computational formality; it introduces a critical trade-off [49], [51] between:

- **Numerical Fidelity:** Ensuring the discretized solution accurately honors the underlying physics of the UF and FC models.
- **Computational Tractability:** Managing the "curse of dimensionality," particularly for the 14-state FC model, to ensure the solver converges within a reasonable timeframe.

In this chapter, it is detailed the selection of the *Direct Collocation* method employed to solve the trajectory optimization problem. The chapter starts with the discretization schemes used to handle the differential constraints, followed by the formulation of the algebraic path constraints and objective functions. Finally, it is outlined the algorithmic structure of the NLP solver and the initialization strategies required to ensure robust convergence for both flight-mechanics formulations.

6.1 Numerical Transcription of the Continuous OCP

The continuous OCP defined in equation (5.1) is transcribed into a finite-dimensional NLP through a *Direct Collocation* method [21], [30], [35], [37], [53]. The time horizon $[0, t_f]$ is partitioned into $N-1$ equal sub-intervals of duration $\Delta t = t_f/(N-1)$; the continuous state and control histories are then represented by their values at the N resulting collocation nodes $\{t_k\}_{k=0}^{N-1}$. This choice converts the search for two infinite-dimensional functions into the search for a finite, structured decision vector and replaces the continuous-time dynamic constraint $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}, t)$ by an algebraic *defect* equation evaluated on each segment.

6.1.1 Decision-Vector Layout

For both formulations the decision vector $\mathbf{z}$ stacks the final time, the state samples, and the control samples in a state-then-control block layout:

$$\mathbf{z} \;=\; \left[\, t_f,\; \mathbf{x}_0^\top,\; \ldots,\; \mathbf{x}_{N-1}^\top,\; \mathbf{u}_0^\top,\; \ldots,\; \mathbf{u}_{N-1}^\top \right]^\top \;\in\; \mathbb{R}^{n_z}, \tag{6.1}$$

where the dimension of the state and control blocks differs between the two formulations. Table 6.1 summarizes the resulting NLP size as a function of the node count N. The state-then-control ordering produces a block-banded sparsity pattern in the constraint Jacobian: the collocation rows couple only adjacent nodes, while every nodewise constraint (terrain clearance, performance bounds, box bounds) is block-diagonal in the node index. The free final time t_f enters as an additional scalar variable that allows both minimum-time and minimum-fuel objectives to be expressed within the same NLP template.

Table 6.1: Dimension of the decision vector (6.1) for the UF and FC formulations as a function of the node count N.

Formulation	**States N_X**	**Controls N_U**	**Decision dim. n_z**	**Equality block**
UF (point-mass)	9	3	$1 + 12N$	$9(N-1)$
FC (rigid body)	14	4	$1 + 18N$	$14(N-1)$

6.1.2 Trapezoidal Direct Collocation

On each sub-interval $[t_k, t_{k+1}]$ the state trajectory is approximated by a piecewise linear function and the time derivative $\dot{\mathbf{x}}$ by the trapezoidal rule. Requiring the resulting approximation to satisfy the dynamics $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ at both endpoints of every segment produces the *collocation defect*

$$\boldsymbol{\Delta}_k \;=\; \mathbf{x}_{k+1} - \mathbf{x}_k \;-\; \frac{\Delta t}{2}\left[\mathbf{f}(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{f}(\mathbf{x}_{k+1}, \mathbf{u}_{k+1})\right] \;=\; \mathbf{0}, \qquad k = 0, \ldots, N-2, \tag{6.2}$$

which appears in the NLP as a block of $N_X(N-1)$ equality constraints. Each component of $\mathbf{\Delta}_k$ corresponds to a single state variable: position, velocity, attitude, mass, or energy. The defect block (6.2) is structurally identical for the two formulations; what differs is the dynamics function $\mathbf{f}$, which is the nine-state point-mass field for UF and the fourteen-state rigid-body field for FC. The trapezoidal scheme is a $O(\Delta t^2)$ method whose accuracy is governed by the node density along the trajectory [30], [45]; the choice of N is therefore a free parameter of the implementation and is studied in Section 6.6.

6.1.3 Residual Normalization

The constraint and objective rows of the Nonlinear Programming (NLP) problem combine quantities of heterogeneous physical dimensions, ranging from lengths in meters to force in Newtons and dimensionless angular ratios. Without appropriate preprocessing, the condition number of the constraint Jacobian becomes artificially large. This numerical ill-conditioning causes the descent directions computed by the interior-point algorithm to be poorly scaled, often leading to stalled convergence or termination at a spurious local minimum [34], [57]

To ensure numerical robustness, every residual produced by the implementation is divided by a representative reference scale before being inserted into the NLP. We define a normalized residual vector $\tilde{\mathbf{r}}_i$ as:

$$\tilde{\mathbf{r}}_i = \mathrm{diag}(\boldsymbol{s}_{\mathrm{ref}})^{-1}\mathbf{r}_i \tag{6.3}$$

where $\boldsymbol{s}_{\mathrm{ref}}$ is a vector of scaling factors chosen such that every component of $\tilde{\mathbf{r}}_i$ is $O(1)$ irrespective of the underlying units. This implicit preconditioning is independent of the optimizer's internal scaling heuristics [19], [20] and is essential for the robust convergence observed in the case studies presented later.

The specific reference scales used for each state and control variable are derived from the aircraft's physical limits and mission characteristics, as summarized in Table 6.2.

6.2 Constraint Architecture of the Discretized NLP

The discretized optimal control problem solved in this chapter is the augmented Bolza form (extension from Equation (5.1)), in which the Mayer/Lagrange cost is supplemented by a container of *soft-penalty* terms that act on mission-level preferences (intermediate-point tracking, terminal shaping, slack relaxations):

$$J = \Phi\big(\mathbf{x}(t_f), t_f\big) + \int_0^{t_f} \mathcal{L}\big(\mathbf{x}, \mathbf{u}, t\big), dt + J_{\mathrm{pen}} \tag{6.4}$$

Table 6.2: Reference scaling factors s_{ref} for NLP variables.

Category	Variable(s)	Reference Scale
Translational	Position (x, y, z) Velocity (u, v, w)	L_{ref} (Total mission distance) V_{ref} (Cruise speed / V_{NE})
Attitude	Bank angle (ϕ) Pitch angle (θ) Yaw angle (ψ)	$\phi_{\max}$ (Active limit) $\theta_{\max}$ 2π
Angular Rates	Roll/Yaw rate (p, r) Pitch rate (q)	$2V_{\text{ref}}/b$ (Span-based rate) $2V_{\text{ref}}/\bar{c}$ (Chord-based rate)
Actuation	Control surfaces $(\delta_e, \delta_a, \delta_r)$ Thrust (T) Mass (m)	$\delta_{\max}$ (Max deflection) $T_{\max}$ m_{initial}
Temporal	Final time (t_f)	$L_{\text{ref}}/V_{\text{ref}}$

The grouped form of $\mathcal{L}$ and the explicit form of J_{pen}, together with the discretized regularization weights actually used by the NLP solver, are introduced in Section 6.3.2; the heavy algebra is developed in Section 6.3.2.

At this point, however, the cost functional alone is insufficient to characterize a physically meaningful solution: an unconstrained NLP would freely exploit every degree of freedom in the decision space and produce trajectories that violate gravity, penetrate terrain, or demand thrust levels far beyond any realized powerplant. It is the constraint layer that transforms the discretized cost minimization into a genuine flight-planning problem.

The constraint architecture is partitioned into four mathematical layers, ordered from the broadest spatial scale to the finest aerodynamic detail:

- **Domain and environment constraints** enforce geometric and map-based feasibility. They prevent the trajectory from intersecting the terrain, entering restricted airspace volumes, or leaving the authorized operational area, and they ensure that prescribed intermediate waypoints are visited in the correct order. These constraints depend only on the position state and are therefore structurally identical in UF and FC, differing only in the numerical values of certain safety margins.

- **Collocation and dynamic consistency constraints** constitute the equality backbone of the NLP. They force the sequence of decision variables to satisfy the equations of motion $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ on every time segment, coupling adjacent nodes and ensuring that the optimized trajectory is dynamically realisable.

- **Boundary equality constraints** anchor the beginning and end of the trajectory to the mission-

prescribed states. They encode the take-off conditions, the landing attitude and position, and optionally any intermediate state fixings required by the mission profile.

- **Performance constraints** confine the aircraft to its aerodynamic and propulsive operating envelope. Because UF and FC represent aircraft motion at fundamentally different fidelity levels, these constraints differ materially between the two formulations: UF constraints are expressed in terms of the scalar airspeed V, while FC constraints operate on the full wind-corrected aerodynamic velocity vector. A detailed comparison of the two sets is presented in Section 6.2.4.

These four layers map one-to-one onto the structural ingredients of the continuous OCP defined in Equation (5.1). The collocation defects (6.2) are the discretized counterpart of the governing ODE $\dot{\mathbf{x}} = \mathbf{f}$; the boundary equality constraints correspond to $\mathbf{x}(t_0) = \mathbf{x}_0$ and $\Psi(\mathbf{x}(t_f), t_f) = \mathbf{0}$; and the domain and performance constraints together populate the path constraint vector $\mathbf{h}$. Once written in finite-dimensional form they are incorporated into the NLP Lagrangian $\mathcal{L}(\mathbf{z}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = F(\mathbf{z}) + \boldsymbol{\lambda}^\top \mathbf{g}(\mathbf{z}) + \boldsymbol{\nu}^\top \mathbf{h}(\mathbf{z})$ through dual variables $\boldsymbol{\nu} \geq \mathbf{0}$, and the KKT stationarity conditions $\nabla_{\mathbf{z}} \mathcal{L} = \mathbf{0}$ [29], [36] become the numerical target pursued by the interior-point algorithm [19], [38].

Throughout this section the residual-normalization principle of Section 6.1.3 (Equation (6.3)) is assumed to be applied to every constraint row before insertion into the NLP.

6.2.1 Domain and Environment Constraints

Domain and environment constraints define the spatial region within which a trajectory is admissible. They are assembled from geospatial data — digital elevation models (DEM), restricted-airspace polygons, operational geofences — and from mission waypoint specifications. All constraints in this category act *nodewise*: they are evaluated independently at each of the N collocation nodes $i = 0, 1, \ldots, N-1$, and no coupling between adjacent nodes is introduced. This structure preserves the block-diagonal sparsity pattern of the constraint Jacobian.

1) Terrain Clearance via Digital Elevation Model

The ground surface beneath the planned flight path is represented by a *Digital Elevation Model* (DEM) [1], a gridded raster dataset that associates a terrain height $h_{\mathrm{DEM}}(x, y)$ with every horizontal coordinate pair in the mission area. At query points that fall between grid nodes the height is recovered by bilinear interpolation, providing a continuous and differentiable surface whose gradients are compatible with the analytic Jacobian assembly.

The terrain clearance constraint requires the geometric altitude of the aircraft z_i to exceed the interpolated DEM height by at least a prescribed safety buffer ϵ_{DEM}:

$$z_i \;\geq\; h_{\mathrm{DEM}}(x_i,\, y_i) \;+\; \epsilon_{\mathrm{DEM}}, \qquad \epsilon_{\mathrm{DEM}} = 5\ \mathrm{m}, \qquad \forall\, i = 0, \ldots, N-1. \tag{6.5}$$

The 5 m safety buffer in (6.5) is selected to absorb both the finite resolution of the DEM (500 m voxel size in the use case considered here) and the short-period altitude excursions that may occur between collocation nodes where the dynamics are not explicitly enforced. Under the North-East-Down (NED) sign convention — in which altitude increases with decreasing z — the constraint is implemented in the standard inequality form

$$-z_i \; - \; h_{\mathrm{DEM}}(x_i,\, y_i) \;\leq\; 0, \tag{6.6}$$

with one-sided bounds $(-\infty,\, 0]$.

2) No-Fly Zone Exclusion

No-fly zones (NFZs) represent volumes of airspace in which the operation of unmanned aerial vehicles is forbidden or restricted [2] — examples include airport terminal control areas, military exercise zones, and urban overflights. In the present implementation NFZs are modeled as *extruded polygon prisms*: a horizontal polygon footprint is defined by a set of vertices, and the zone is then extruded vertically between a lower altitude $z_j^{\min}$ and an upper altitude $z_j^{\max}$ to form a closed 3D exclusion volume.

For each zone $j = 1, \ldots, N_{\mathrm{NFZ}}$ and each outward-facing polygon face ℓ, an individual signed-distance inequality is generated:

$$d_{ij\ell}(x_i,\, y_i) \;\geq\; \epsilon_{\mathrm{NFZ}}, \qquad z_i \in [z_j^{\min},\, z_j^{\max}], \quad \forall\, i, \tag{6.7}$$

where $d_{ij\ell}$ denotes the perpendicular distance from node i to face ℓ of zone j, measured positively outward. The altitude condition activates the constraint only when the node lies within the vertical extent of the zone, ensuring that the formulation is not penalized for flying over or under an NFZ that does not reach the current flight level.

The safety margin ϵ_{NFZ} is set in both formulations to

$$\epsilon_{\mathrm{NFZ}}^{\mathrm{UF}} = 25 \text{ m}, \qquad \epsilon_{\mathrm{NFZ}}^{\mathrm{FC}} = 25 \text{ m}. \tag{6.8}$$

3) Geofence Containment

While NFZs exclude the trajectory from specific volumes, the *geofence* performs the complementary role of confining the trajectory to an authorized operational area. Structurally, the geofence is identical to an NFZ with reversed sign: rather than requiring the aircraft to remain *outside* a polygon, the aircraft must remain *inside* the geofence boundary at all times.

The feasibility condition is expressed as

$$\mathbf{r}_i \in \Omega_{\text{GEO}}, \qquad i = 0, \ldots, N-1, \tag{6.9}$$

where Ω_{GEO} denotes the interior of the geofence polygon prism. The constraint reuses the polygon-distance infrastructure of (6.7) with reversed sign on each face. A common inward safety margin $\epsilon_{\text{GEO}} = 5$ m is applied in both UF and FC, ensuring that the trajectory does not graze the boundary.

4) Intermediate Waypoints

Many missions require the trajectory to pass through or over a series of prescribed intermediate points — inspection targets, relay stations, or airspace entry/exit gates — before reaching the terminal state. These conditions are encoded as *waypoint constraints*, which impose simultaneous requirements on position, attitude, and speed at specific points along the trajectory.

To avoid introducing non-smooth conditional logic into the NLP, each waypoint condition is enforced through an *auxiliary arc-length slack variable* $s_{wp,k} \geq 0$ that is appended to the optimization vector. The three coupled conditions per waypoint k are:

1. **Position matching.** The Euclidean distance between the trajectory position and the prescribed waypoint location must not exceed a small tolerance:

$$\left\| \mathbf{r}(s_{wp,k}) - \mathbf{r}^{\text{ref}}_{wp,k} \right\|_2 \leq \epsilon_{wp}.$$

   The arc-length parameter $s_{wp,k}$ is determined implicitly by the optimizer as the closest point on the trajectory to the waypoint.

2. **Attitude matching.** Heading, bank angle, and pitch are required to match the prescribed waypoint attitude, with residuals normalized by the respective attitude reference scale.

3. **Speed matching.** The local airspeed at the waypoint must equal the prescribed speed, with the residual normalized by V_{ref} in UF or by the corresponding body-velocity scale in FC.

The slack-variable formulation ensures that the waypoint constraint degrades gracefully under infeasibility: if the prescribed point cannot be reached exactly, the optimizer converges to a minimum-violation solution rather than declaring the problem infeasible outright.

6.2.2 Collocation and Dynamic Consistency Constraints

The domain constraints of the previous subsection confine the trajectory spatially, but they say nothing about *how* the aircraft moves between nodes. It is the collocation constraints that enforce the physical

laws of motion along the entire trajectory.

The trapezoidal direct-collocation scheme introduced in (6.2) [30], [34] divides the time horizon $[0, t_f]$ into $N-1$ equal segments of duration $\Delta t = t_f/(N-1)$. On each segment $[t_k, t_{k+1}]$, the state trajectory is approximated by a piecewise linear function and the time derivative is approximated by the trapezoidal rule. Requiring this approximation to satisfy the equations of motion $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ at both endpoints of every segment gives rise to the defect vector

$$\boldsymbol{\Delta}_k \;=\; \mathbf{x}_{k+1} - \mathbf{x}_k \;-\; \frac{\Delta t}{2}\Big[\mathbf{f}(\mathbf{x}_k, \mathbf{u}_k) \;+\; \mathbf{f}(\mathbf{x}_{k+1}, \mathbf{u}_{k+1})\Big] \;=\; \mathbf{0}, \qquad k = 0, \ldots, N-2. \tag{6.10}$$

Each component of the defect $\boldsymbol{\Delta}_k$ corresponds to one state variable: positions, velocities, angles, mass, and battery charge. Setting all components to zero — as required by the equality constraint with zero bounds — is the mechanism through which Newton's laws, aerodynamic lift and drag, and propulsive thrust are all embedded into the optimization. Without this constraint block, the decision variables at adjacent nodes would be completely decoupled.

The defect equation (6.10) is the same in structure for both formulations, but the dynamics function $\mathbf{f}$ differs: it is the nine-state point-mass field in UF and the fourteen-state rigid-body field in FC. Each residual component is normalized as described in Section 6.1.3 before insertion into the NLP.

6.2.3 Boundary Equality Constraints

The collocation constraints enforce internal consistency, but they do not by themselves specify where the trajectory starts or ends. Boundary equality constraints anchor the first and last nodes to the mission-prescribed initial and terminal conditions.

Initial State

At node $k = 0$, all state components are fixed to the take-off conditions specified in the mission file:

$$\mathbf{x}_0 \;=\; \mathbf{x}^{\text{init}}. \tag{6.11}$$

This is implemented as a block of equality constraints with zero-tolerance bounds [0, 0] on each normalized residual, effectively removing the initial state from the optimization and replacing it with a fixed parameter.

Final State

At the terminal node $k = N-1$, the landing position and attitude are enforced as hard equalities:

$$\mathbf{r}_{N-1} = \mathbf{r}^{\text{end}}, \qquad \boldsymbol{\eta}_{N-1} = \boldsymbol{\eta}^{\text{end}}, \tag{6.12}$$

where $\boldsymbol{\eta}$ represents the attitude sub-vector: (ϕ, θ, ψ) in FC or (μ, γ, ξ) in UF. Terminal speed and mass may optionally be fixed to prescribed values as well, depending on the mission configuration.

6.2.4 Performance Constraints

Performance constraints confine the aircraft to its aerodynamic and propulsive operating envelope. Unlike the spatial constraints of Section 6.2.1, which depend only on position, performance constraints couple speed, altitude, mass, and — in the rigid-body formulation — attitude and angular rates. Their mathematical form therefore differs between the two formulations.

In UF all performance limits are expressed in terms of the scalar airspeed V. In FC the governing quantity is the *wind-corrected aerodynamic velocity*

$$\mathbf{V}_{\text{aer},i} = \begin{pmatrix} u_i \\ v_i \\ w_i \end{pmatrix} - R_{EB}(\phi_i, \theta_i, \psi_i)\, \mathbf{V}_{\text{wind},i}, \qquad V_{\text{aer},i} = \|\mathbf{V}_{\text{aer},i}\|, \tag{6.13}$$

where $R_{EB} \in SO(3)$ is the Earth-to-body rotation matrix and $U_{r,i} = \mathbf{V}_{\text{aer},i} \cdot \hat{\mathbf{e}}_x$ is the forward wind-relative component on the propeller axis. Using $\mathbf{V}_{\text{aer}}$ instead of the raw body velocity ensures that lift, drag, and thrust limits are evaluated in the correct aerodynamic reference frame regardless of the ambient wind.

Table 6.3 provides a side-by-side comparison of all performance constraints. A "—" entry indicates that the constraint is not applicable to that formulation.

Table 6.3: Performance constraints enforced at each collocation node ($i = 0, \ldots, N-1$) for the UF ($N_X = 9$) and FC ($N_X = 14$) formulations.

Constraint	UF — Point-Mass	FC — Rigid Body
Anti-Stall Margin	$\frac{V_i}{V_{\text{stall}}(h_i, m_i)} \geq 1$	$\frac{V_{\text{aer},i}}{V_{\text{stall}}(h_i, m_i)} \geq 1$
	Scalar airspeed bounded by mass-corrected stall speed.	Wind-relative speed used to prevent aerodynamic stall.
Power Availability	$\frac{T_i\, V_i}{\eta_p\, P_{\max}(h_i)} \leq 1$	$\frac{T_i\, U_{r,i}}{\eta_p\, P_{\max}(h_i)} \leq 1$
	Propulsive power limited by altitude-derated shaft power.	Power computed using forward body-velocity U_r.
Structural Limit	$V_i \leq V_{NE}$	$\frac{\|\mathbf{V}_{b,i}\|_2}{V_{NE}} \leq 1$
	Box bound on scalar airspeed.	Norm of the full body-frame velocity vector.
C_L **Saturation**	*Implicit (Direct Control)*	$C_L(\alpha_i, \delta_{e,i}) \in [C_{L,\min},\, C_{L,\max}]$
	Enforced via box bounds on the C_L control variable.	Nonlinear mapping of AoA and elevator deflection.
Min. Altitude	$\frac{-z_i}{h_{\min}} \geq 1$	$\frac{-z_i}{h_{\min}} \geq 1$
	Absolute floor to avoid ground-effect or impact.	Absolute floor to avoid ground-effect or impact.
Control Surfaces	*N/A (Kinematic)*	$\|\delta_e\|, \|\delta_a\|, \|\delta_r\| \leq \delta_{\max}$
	Roll rate is the primary control.	Physical limits on aerodynamic actuators.
Angular Rates	$\|p_{\text{rate},i}\| \leq p_{\max}$	$\|p_i\|, \|q_i\|, \|r_i\| \leq (\cdot)_{\max}$
	Bound on single-axis roll kinematics.	Individual bounds on all three body-frame rates.

6.2.5 Summary of the Constraint Architecture

The complete constraint architecture of the discretized NLP reflects a deliberate balance between physical fidelity and computational tractability. All residuals are normalized to $O(1)$ before being handed to the optimizer, and the constraint Jacobian inherits the block-sparse structure imposed by the nodewise and segmentwise formulations.

The two formulations share the domain constraints (terrain, NFZ, geofence, waypoints), the collocation

defect equations, and the boundary equalities. They diverge in the performance layer, where the rigid-body FC model requires wind-corrected speed definitions, an explicit C_L-saturation constraint, a minimum-altitude floor, and individual actuator bounds — none of which are necessary in the point-mass UF model.

Together, these four constraint layers populate the equality block $\mathbf{g}(\mathbf{z}) = \mathbf{0}$ (collocation defects, boundary equalities) and the inequality block $\mathbf{h}(\mathbf{z}) \leq \mathbf{0}$ (domain, box and performance constraints) of the discretized OCP. The two blocks, together with the cost functional defined in Section 6.4 for UF and Section 6.4 for FC, constitute the full mathematical specification of the NLP solved throughout the remainder of this chapter.

6.3 Cost Function Definitions

While a generalized NLP cost function was introduced in Section 6.2, both the UF and FC formulations require distinct modifications to tailor the objective to their specific kinematic and dynamic constraints. Despite these individual variations, both formulations preserve a core Bolza-form structure, augmented by the numerical penalties detailed in Section 6.3.2.

This shared Bolza-form objective consists of a Mayer term that penalizes the final time and a Lagrange term that integrates propulsive energy along the trajectory. The resulting template is flexibly adapted to represent two limiting operational regimes—the minimum-time and minimum-fuel objectives—by zeroing the complementary weighting factors. Accordingly, this section formalizes the mathematical framework for the base cost function (Section 6.3.1) and details the smoothness regularization used to stabilize the problem at the NLP level (Section 6.3.2).

6.3.1 Objective Function: The Time–Fuel Trade-off

The continuous-time Bolza objective function is transcribed into a discrete Non-Linear Programming (NLP) framework using the trapezoidal collocation grid. This mathematical architecture splits the optimization criteria into two distinct physical components:

- **The Mayer Term:** This term evaluates only the final snapshot of the trajectory. It isolates the total final time t_f, scaled by the weight w_{t_f}. Minimizing this term forces the solver to treat the flight path like a drag race.
- **The Lagrange Term:** This term accumulates across the entire flight duration. It integrates instantaneous propulsive power—calculated as the product of thrust and velocity $(T \cdot V)$—approximated via a discrete trapezoidal sum. Minimizing this integral directly optimizes fuel conservation.

To balance these competing operational demands, a scalar weighting parameter $\alpha \in [0, 1]$ is introduced

as a control dial. The mathematical actions and resulting aircraft behaviors under this parameterization are summarized in Table 6.5.

Table 6.5: Operational regimes driven by the scalar weight α.

Setting	Objective	Mathematical Action	Flight Behavior
$\alpha = 0$	Min-Time	Minimizes t_f ($w_{t_f} > 0$)	Max thrust ($T_{\max}$), shortest path
$\alpha = 1$	Min-Fuel	Minimizes power sum ($T \cdot V$)	Max efficiency (L/D), saves fuel
$0 < \alpha < 1$	Pareto	Blended compromise	Sweet spot: speed vs. throttle

It is critical to emphasize that both the UF and FC formulations share an identical constraint set (e.g., identical obstacle boundaries and aerodynamic limits). Consequently, if the two formulations produce differing converged trajectories under the exact same value of α, the variance is not caused by differing physical laws. Instead, it is driven solely by the specific subset of decision variables that each formulation exposes to the NLP solver, with the UF model occasionally exploiting coordinate shortcuts hidden from its attitude-restricted counterpart.

Throughout this section the trapezoidal mid-node average and the forward increment are denoted

$$\bar{X}_k \;=\; \tfrac{1}{2}\big(X_k + X_{k+1}\big), \qquad \Delta(\,\cdot\,)_k \;=\; (\,\cdot\,)_{k+1} - (\,\cdot\,)_k.$$

Both formulations are augmented by a smoothness-regularization term that penalizes node-to-node oscillations of the controls and of selected state components. This term is kept abstract as $J_{\text{pen}}^{\text{UF}}$ and $J_{\text{pen}}^{\text{FC}}$ in Sections 6.3.1–6.3.1 and is developed in vectorized form in Section 6.3.2. Its numerical motivation, the per-aircraft weight-selection protocol and the role of the gates β_c and β_s are covered in Section 6.3.2 and Annex D.1.

Uncoupled formulation (UF)

Of the point-mass state and control defined in Section 5.2.2 of Chapter 5, only one quantity shapes the running cost: the velocity that feeds the Lagrange term. Here the scalar airspeed V is carried explicitly in the state (Equation (5.4)), so the propulsive power is evaluated directly as $T\,V$, with no reconstruction required.

Assembling the Mayer, Lagrange and regularization contributions, the UF cost reads

$$J_{\text{UF}} \;=\; w_{t_f}\, t_f \;+\; \sum_{k=0}^{N-2} \bar{T}_k\, \bar{V}_k\, \Delta t \;+\; J_{\text{pen}}^{\text{UF}}, \tag{6.14}$$

where the smoothness term $J_{\text{pen}}^{\text{UF}}$ is expanded in Equation (6.16) below.

Full-control formulation (FC)

The FC reuses the same Bolza template; the single substantive change is, once again, the velocity entering the propulsive power. Unlike the UF state, the rigid-body state of Section 5.3 of Chapter 5 exposes no scalar airspeed, so the Lagrange term is instead built from the body-frame velocity norm $\|\mathbf{V}_b\|_2 = \sqrt{u^2 + v^2 + w^2}$, the same quantity that already drives the mass-flow equation (Equation (5.35)). In the no-wind validation suite (Section 6.4.1) this norm coincides numerically with the airspeed magnitude, so the two costs remain commensurable on a per-mission basis; algebraically, however, the input is distinct and reflects the fidelity at which each formulation samples the trajectory.

Carrying this substitution into the template gives

$$J_{\mathrm{FC}} \;=\; w_{t_f}\, t_f \;+\; \sum_{k=0}^{N-2} \bar{T}_k\, \overline{\|\mathbf{V}_b\|}_k\, \Delta t \;+\; J_{\mathrm{pen}}^{\mathrm{FC}}, \tag{6.15}$$

where the overline denotes the same trapezoidal mid-node average as in the UF, and $J_{\mathrm{pen}}^{\mathrm{FC}}$ is expanded in Equation (6.17) below.

6.3.2 Cost regularization: smoothness penalties

The Bolza cost J in (6.4), introduced in Section 6.2, already exposes a family of soft-penalty terms J_{pen}. In the discretized NLP these terms are joined by a second family of cost-level regularization weights that act on the first differences of the controls and on a small subset of the state. Their algebraic form is collected in Section 6.3.2 (Equations (6.16) and (6.17)); the present subsection is devoted to the rationale. The regularization is not part of the underlying optimal control problem: it is a numerical device [46], [47] that compensates for two artefacts of direct collocation with a finite-difference Jacobian.

Two empirical phenomena motivate J_{pen}. The first is finite-difference Jacobian noise: the pipeline version used for the results of this thesis computes the constraint Jacobian by forward differences (a later version replaces them with exact derivatives), whose truncation error scales as $\sqrt{\varepsilon_{\mathrm{mach}}} \approx 10^{-8}$ in the best case and in practice closer to 10^{-6} once the compound nonlinearities $\rho(h)$, $C_D(C_L)$ and $T \cdot V$ propagate [42], [58]. At this noise floor the KKT residual oscillates and IPOPT cannot certify convergence; the quadratic increment penalties smooth the search direction enough for the line search to progress. The second is high-frequency control chatter: with no rate limit in the continuous OCP, the optimizer may switch controls between adjacent bounds at adjacent nodes, yielding physically uninstantiable trajectories—pulse-like δ_e, saw-tooth thrust—that satisfy collocation yet cannot be flown. The regularization acts as an implicit, cheaper substitute for explicit $\dot{u}$-bounds, and replaces no path inequality or mission-level constraint.

Weight selection procedure

The weights are tuned per aircraft following the standard practice in multi-objective NLP weighting [46]. The procedure, described in detail in Annex D.1, is heuristic: starting from $w_{(\cdot)} = 0$, each weight is raised in geometric steps until two conditions hold simultaneously,

1. the total variation $\mathrm{TV}(u) = \sum_k |\Delta u_k|$ of the control history drops below an acceptable threshold (typically an order of magnitude below the unregularized value); and

2. the regularization contribution $P_c + P_s$ at the converged solution remains below $\sim 15\,\%$ of J_{total}, which ensures the solver converges to a similar local minimum. If the penalty term were too high relative to the base cost, it would dominate the objective function, introducing significant errors when minimizing the primary cost functional (e.g., minimum time or minimum fuel consumption).

The nominal split of the objective into time, fuel, regularization ($P_c + P_s$), and baseline terms at the converged iterate is the auditing quantity used to check condition (2).

Regularization penalty terms

The placeholders $J_{\text{pen}}^{\text{UF}}$ and $J_{\text{pen}}^{\text{FC}}$ that close Equations (6.14) and (6.15) share a common architecture. Each is the sum of two families:

- a **control-increment family** that penalizes node-to-node jumps of the control vector and is activated by a global gate $\beta_c \in [0, 1]$;

- a **state regularization family** that penalizes node-to-node jumps (and, for γ, the absolute value) of a small subset of the state and is activated by a second global gate $\beta_s \in [0, 1]$.

The two gates are convenience switches at the NLP level: the production configuration is $\beta_c = \beta_s = 1$, while the ablation $\beta_c = \beta_s = 0$ recovers a pure Bolza objective and is used in the with-/without-regularization comparison of Section 6.5. Mixed settings make either family inspectable in isolation.

UF formulation This formulation penalizes three control variables: thrust, lift coefficient (C_L), and roll rate (p). For the state variables, penalties are applied strictly to the derivatives of the flight-path angle and the bank angle. In this specific scenario, penalizing the control variable p is functionally equivalent to penalizing the derivative of the bank angle state. Although both approaches were evaluated, weighting the penalty of the finite difference terms was chosen as it is more straightforward to tune.

$$J_{\text{pen}}^{\text{UF}} = \beta_c \sum_{k=0}^{N-2} \left\| \boldsymbol{\xi}_{c,k}^{\text{UF}} \right\|_{\mathbf{W}_c^{\text{UF}}}^2 + \beta_s \sum_{k=0}^{N-2} \left\| \boldsymbol{\xi}_{s,k}^{\text{UF}} \right\|_{\mathbf{W}_s^{\text{UF}}}^2, \tag{6.16}$$

with the per-family regularization vectors and their diagonal weight matrices given by

$$\boldsymbol{\xi}_{c,k}^{\mathrm{UF}} = \begin{bmatrix} \Delta T_k & \Delta C_{L,k} & \Delta p_k \end{bmatrix}^\top,$$

$$\mathbf{W}_c^{\mathrm{UF}} = \mathrm{diag}\left(w_{\Delta T},\, w_{\Delta C_L},\, w_{\Delta p}\right),$$

$$\boldsymbol{\xi}_{s,k}^{\mathrm{UF}} = \begin{bmatrix} \Delta \gamma_k & \Delta \mu_k \end{bmatrix}^\top,$$

$$\mathbf{W}_s^{\mathrm{UF}} = \mathrm{diag}\left(\, w_{\Delta \gamma},\, w_{\Delta \mu}\right).$$

FC formulation In contrast, the full-control (FC) formulation penalizes a broader set of control variables, specifically the aerodynamic surface deflections (δ). Additionally, a penalty on the rudder deflection (δ_r) has been incorporated to prevent the optimization solver from converging on uncoordinated flight conditions. This uncoordinated tendency arises from the solver's increased degrees of freedom and must be restricted to ensure a physically natural trajectory and flight dynamics. Regarding the state variables, penalties were applied to ensure the smoothness of the equivalent aerodynamic and body Euler angles.

$$J_{\mathrm{pen}}^{\mathrm{FC}} \;=\; \beta_c \sum_{k=0}^{N-2} \left\| \boldsymbol{\xi}_{c,k}^{\mathrm{FC}} \right\|_{\mathbf{W}_c^{\mathrm{FC}}}^2 \;+\; \beta_s \sum_{k=0}^{N-2} \left\| \boldsymbol{\xi}_{s,k}^{\mathrm{FC}} \right\|_{\mathbf{W}_s^{\mathrm{FC}}}^2, \tag{6.17}$$

where the control-increment family targets the four FC actuators and the state family the Euler angles tracked by the rigid-body attitude dynamics,

$$\boldsymbol{\xi}_{c,k}^{\mathrm{FC}} = \begin{bmatrix} \Delta T_k & \Delta \delta_{e,k} & \Delta \delta_{a,k} & \Delta \delta_{r,k} \end{bmatrix}^\top,$$

$$\mathbf{W}_c^{\mathrm{FC}} = \mathrm{diag}\left(w_{\Delta T},\, w_{\Delta \delta_e},\, w_{\Delta \delta_a},\, w_{\Delta \delta_r}\right),$$

$$\boldsymbol{\xi}_{s,k}^{\mathrm{FC}} = \begin{bmatrix} \Delta \phi_k & \Delta \theta_k \end{bmatrix}^\top,$$

$$\mathbf{W}_s^{\mathrm{FC}} = \mathrm{diag}\left(w_{\Delta \phi},\, w_{\Delta \theta}\right).$$

The tuning rationale of the weights $\{w_{(\,\cdot\,)}\}$ are detailed in Annex D as well as the per-aircraft tuned values used in this work. With the base cost and its regularization introduced in this compact form, the UF and FC objectives remain directly comparable across the minimum-time and minimum-fuel limits.

6.4 Validation Results

Verifying the initial optimization results against a suite of fundamental use cases is essential to evaluate the capabilities and numerical robustness of the developed framework. Accordingly, this section presents the numerical validation of the Uncoupled Formulation (UF) and the Full-Control Formulation (FC) across the four reference scenarios established in Section 6.4.1.

Each scenario is evaluated along two independent axes: the optimization *criterion* (minimum-time, $\alpha = 0$, versus minimum-fuel, $\alpha = 1$) and the *formulation* (UF versus FC). The analysis first examines the converged trajectories to compare the optimization criteria with the formulation held fixed. It then contrasts the two formulations under identical criteria, evaluates a detailed constraint satisfaction suite, and concludes with a brief assessment of the practical impacts of the penalty framework on the optimization process.

6.4.1 Test Scenarios for Numerical Validation

The numerical implementation introduced in the previous section is tested using a structured set of scenarios defined in Table 6.6. This suite uses a 2×2 factorial design in empty airspace to isolate specific aircraft movements:

- **Lateral:** Straight flight versus a 180° heading reversal.
- **Longitudinal:** Level flight versus a 200 m climb between the entry and exit gates.

Aircraft `MSN012`, whose performance specifications and parameters are detailed in Annex C, is used throughout this study. To isolate the longitudinal and lateral dynamics along with their intrinsic coupling, environmental and mission-level constraints—including the digital elevation model (DEM), no-fly zones, geofencing, wind, and intermediate waypoints—were deactivated. This ensures that the baseline validation focuses strictly on the underlying physics of the trajectory rather than geospatial feasibility layers. However, more complex and detailed scenarios are introduced in Table 6.8 to thoroughly assess the performance and robustness of the optimization framework.

6.4.2 Scenario Definitions

The validation suite consists of four distinct flight profiles designed to stress-test the solver's ability to handle vertical transitions, lateral maneuvers, and combined 3D trajectory optimization. These benchmarks provide the basis for the performance metrics analyzed in Chapter 6. Each scenario is evaluated under both the UF and FC formulations across the optimization spectrum, ranging from minimum-time ($\alpha = 0$) to minimum-fuel ($\alpha = 1$) objectives.

Table 6.6: Geometric and environmental configurations for validation benchmarks, specifying mission layout, vertical displacement, and trajectory complexity.

Label	Mission Geometry	Δ Alt. [m]	Dist. [km]	Wind	Primary Objective
SC-00	Straight-line, level flight	0	3.35	None	Baseline horizontal transit
SC-01	Straight-line, steady climb	+200	3.36	None	Vertical energy efficiency
SC-02	180° horizontal turn	0	4.09	None	Lateral maneuverability
SC-03	Combined climb-and-turn	+200	4.10	None	Full 3D trajectory coupling

6.4.3 Validation Metrics

To ensure the integrity of the numerical results, a converged solution is subjected to an *external* validation process. A trajectory is deemed valid only if the following metrics remain within their prescribed admissible ranges:

- **Maximum Normalized Defect:** The infinity norm of the state residuals, $\|\boldsymbol{\Delta}\|_\infty \leq 10^{-4}$, is computed *a posteriori* using the converged decision vector. This ensures the collocation scheme is consistent with the underlying vehicle dynamics and that the discretization density is sufficient to suppress integration errors.

- **Constraint Feasibility Margins:** Every inequality constraint is mapped using the negative convention, requiring the condition $h_j(\mathbf{z}) \leq 0$ for compliance. For constraints expressed as dimensionless ratios—such as the aerodynamic stall limit ($1 - V/V_{\text{stall}} \leq 0$) or the power availability threshold—the constraint residual $h_j(\mathbf{z})$ directly quantifies the proximity of the trajectory to the edge of the flight envelope. A non-positive residual is strictly mandatory across all discrete nodes for the optimized flight path to be considered physically realizable.

- **Cross-Formulation Consistency:** This metric evaluates the deviation between the UF and FC trajectories. The constrained solution must remain within an expected spatial tolerance of the baseline, proportional to the reference length and the number of nodes. Consistency across these formulations validates that the optimizer has reached a global or high-quality local optimum rather than an algorithmic artifact.

- **Minimum-Time versus Minimum-Fuel Performance:** A quantitative comparison across different optimization criteria, highlighting the fundamental variations in aircraft performance when prioritizing competing operational metrics (transit speed versus fuel conservation).

- **Influence of Regularization Penalties:** An analysis of how smoothing penalties regularize the NLP objective function. This comparison, which is expanded in Appendix D, demonstrates the distinct numerical effects of applying regularization terms to either the control inputs or the state variables to suppress unphysical chattering.

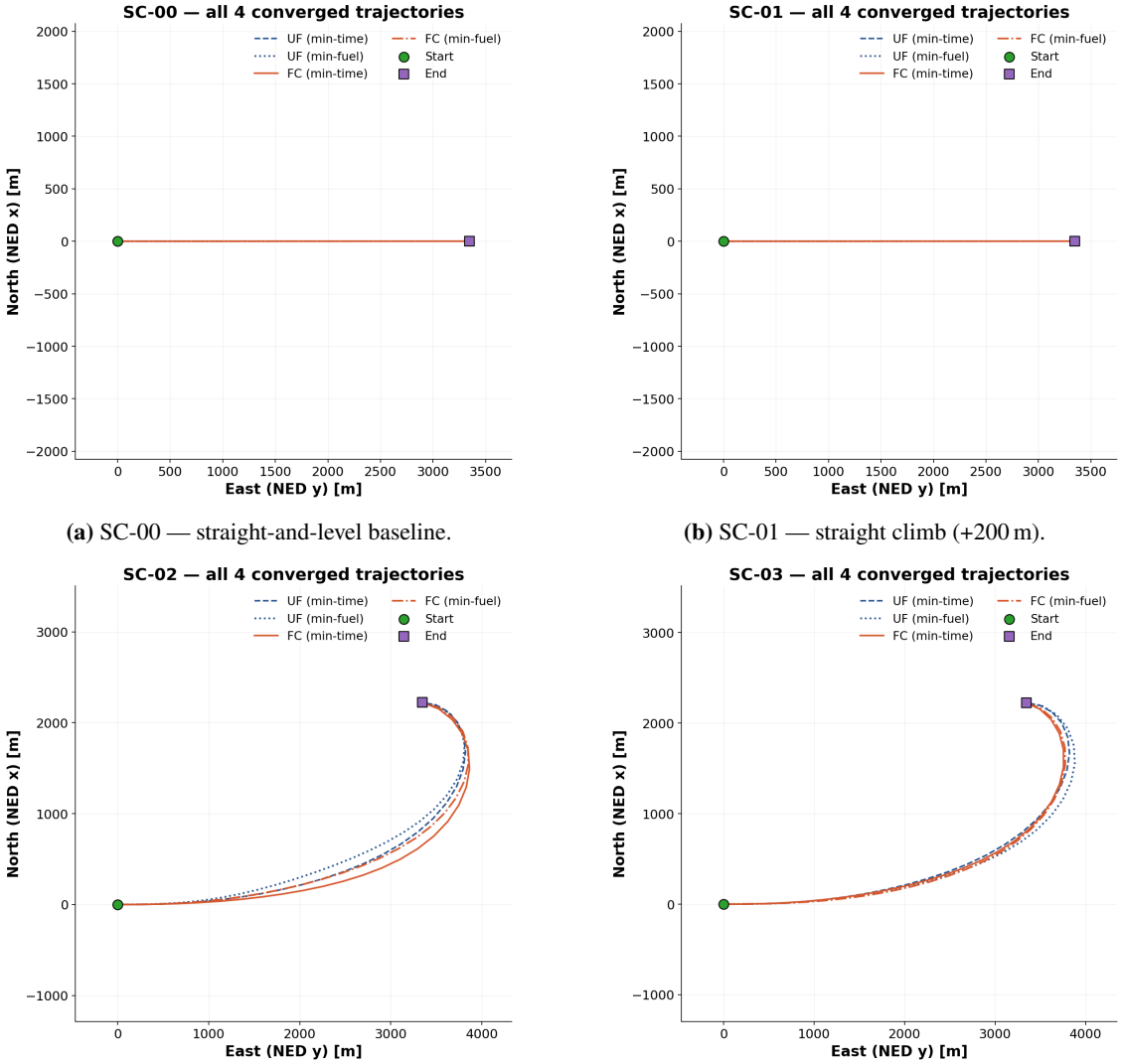
These five key metrics—dynamic consistency (defects), constraint feasibility (margins), objective parameterization (time versus fuel), regularization effects (penalties), and formulation sensitivity—provide a comprehensive framework for the numerical validation presented in the following results.

6.4.4 Trajectory Observation and Analysis

The four solutions are initially inspected qualitatively, following the "visualize first, then quantify" methodology outlined by Betts [34]. This sequence is standard practice for direct collocation results, as the mathematical trajectories possess clear physical representations. Figure 6.1 overlays the four converged ground tracks for each scenario, comparing the UF and FC formulations under both the minimum-time and minimum-fuel criteria.

Within each scenario, the four ground tracks are nearly coincident. This represents the first structural validation check: the entry and exit gates, combined with the prescribed mission geometry, tightly constrain the horizontal corridor, preventing either the optimization criterion or the mathematical formulation from significantly altering the planimetric path. The heading-reversal cases exhibit only a small ground-track spread (below 5.5% in path length), where the optimizer trades a marginally wider turning radius for a more favorable conservation of kinetic energy. The most pronounced discrepancies between the two formulations occur in scenario SC-02.

Throughout this section, these differences are analyzed and primarily attributed to the divergent behavior of the penalty terms within the respective cost functions of the UF and FC models. Despite possessing a similar relative weight in the overall cost function ($\approx 15\%$), their practical effects on the resulting trajectory differ significantly. It is important to highlight this distinction; if strictly congruent trajectories across both formulations were desired, the penalty structures and weight distributions would need to be meticulously co-tuned to achieve such close alignment.



(a) SC-00 — straight-and-level baseline.

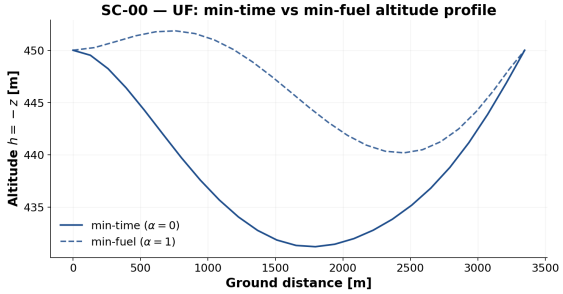
(b) SC-01 — straight climb (+200 m).

(c) SC-02 — level 180° turn.

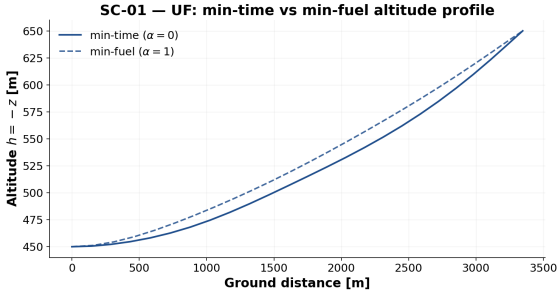
(d) SC-03 — climb-and-turn (+200 m).

Figure 6.1: Top-view overlay of all four converged trajectories per scenario (UF and FC, min-time and min-fuel). All thumbnails are rendered at the same physical height so that the relative scale of each scenario's operational corridor is preserved across the grid. The geometric similarity across criteria confirms that the optimization criterion modifies the flight profile, not the ground track.

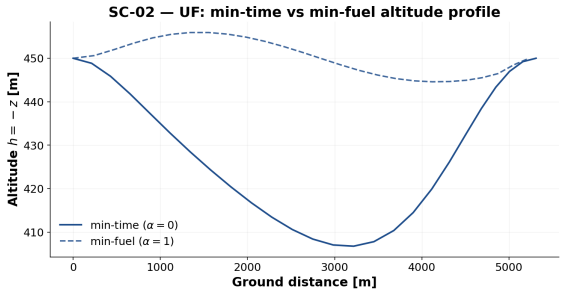
Since the horizontal path is essentially fixed, the information lives in the vertical and speed schedule. Figures 6.2 and 6.3 show the altitude profiles of the UF and FC solutions, overlaying the two criteria on each scenario. Two readings emerge that the rest of the section makes quantitative. Along the criterion axis, the objective reshapes the vertical envelope, most visibly on the level missions where the optimizer has room to choose its altitude freely and least on the climbs where the terminal altitude constraint ties it down. Along the formulation axis, resolving the rigid-body dynamics refines the same profiles without moving the corridor. These two observations are exactly the two comparisons that follow.



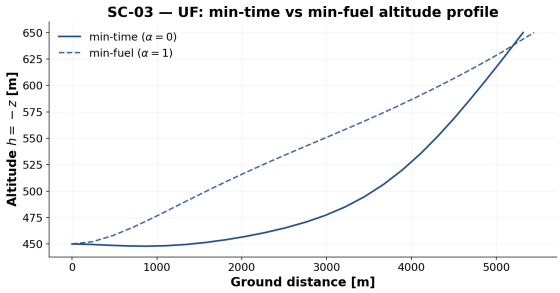
(a) SC-00 — straight-and-level baseline.



(b) SC-01 — straight climb (+200 m).

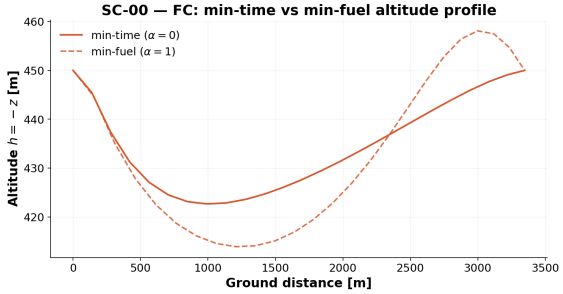


(c) SC-02 — level 180° turn.

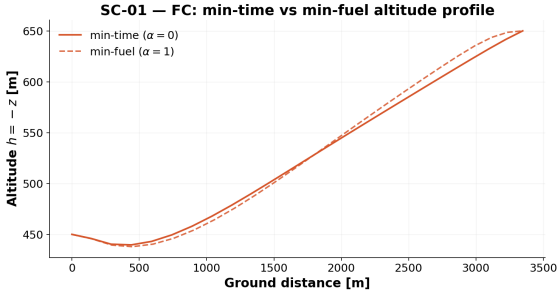


(d) SC-03 — climb-and-turn.

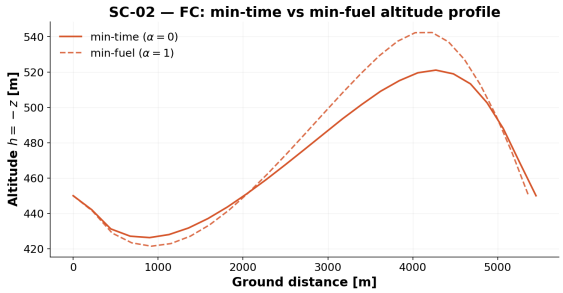
Figure 6.2: Altitude profiles of the converged UF solution on every reference scenario, overlaying the min-time ($\alpha = 0$) and min-fuel ($\alpha = 1$) criteria. SC-00 and SC-02 are level missions; SC-01 and SC-03 add the suite's +200 m terminal climb. SC-02 and SC-03 impose the 180° heading reversal.



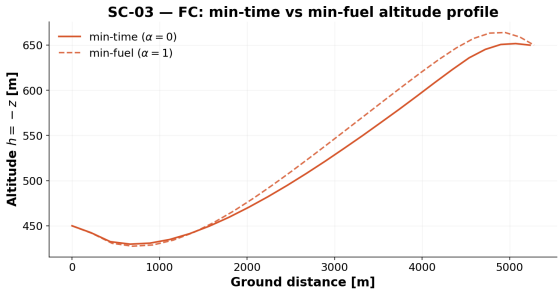
(a) SC-00 — straight-and-level baseline.



(b) SC-01 — straight climb (+200 m).



(c) SC-02 — level 180° turn.



(d) SC-03 — climb-and-turn.

Figure 6.3: Altitude profiles of the converged FC solution on every reference scenario, overlaying the min-time ($\alpha = 0$) and min-fuel ($\alpha = 1$) criteria.

6.4.5 Criterion Comparison: Minimum-Time vs Minimum-Fuel

With the formulation held fixed, the weighting parameter α in the cost functional (6.14) interpolates between a pure minimum-time objective ($\alpha = 0$) and a pure minimum-fuel one ($\alpha = 1$). Figure 6.4 isolates the two quantities the objective function trades off against each other—flight time and energy—for each formulation. The trade-off aligns with expectations: shifting toward a minimum-fuel objective lengthens the flight time but reduces total energy expenditure.

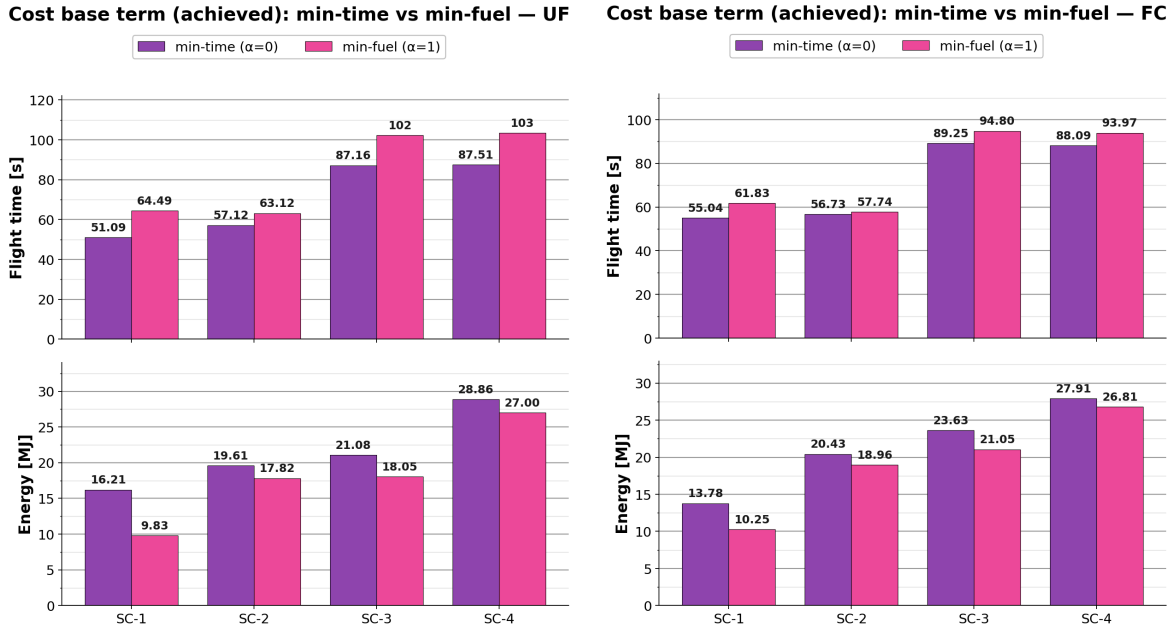
The magnitude of this trade-off, however, heavily depends on the specific maneuver. The level-flight missions yield the largest savings because the aircraft can decelerate and reduce its bank angle; conversely, the climb missions yield much lower savings because the required potential energy is irreducible and cannot be recovered by flying at lower speeds.

Furthermore, a comparison of the cost function results across both formulations conclusively demonstrates that both models reflect similar trends across the optimization targets. Nonetheless, minor discrepancies arise, particularly in scenarios SC-1 and SC-3. These correspond to the level-flight configurations where the optimizer possesses greater freedom to maneuver. The author has identified two primary reasons for these discrepancies:

- **Inherent formulation divergence:** The FC formulation incorporates significantly more variables

and constraints than the UF model, such as explicit limitations and penalties on control variables. While the uncoupled (UF) formulation primarily relies on velocity to modulate flight path time, the coupled FC system must solve for the exact trim conditions at each node to satisfy simultaneous lateral and longitudinal equilibrium. Consequently, velocity in the FC model is merely one additional decision variable within a highly complex trimming problem.

- **Misalignment of penalty weights:** Section 6.3.2 details the effects of these penalties on the system variables, highlighting a fundamental divergence in how penalties are integrated into the cost function of each formulation. The penalty weights were obtained through an iterative, manual tuning process rather than being mapped to objective, physical values. Because each formulation penalizes solutions differently, it is highly challenging to establish a unified penalty structure and weight distribution that yields uniform results across disparate formulations and scenarios.



(a) UF. **(b)** FC.

Figure 6.4: Achieved cost base term with the formulation held constant and the criterion swept: flight time (top) and energy (bottom), minimum-time versus minimum-fuel, across the four scenarios. The trade is widest on the level missions and muted on the climbs.

The same trade read across every performance, kinematic and propulsive metric is condensed, per formulation, in Figures 6.5 and 6.6; the per-scenario tables are collected in Annex F. The common tendency is that the fuel objective consistently prefers slower mean airspeeds and gentler maneuvering, and that the most economical way to manage energy is to couple the altitude and the speed schedules—banking potential against kinetic energy so the engine avoids the abrupt thrust transients an uncoupled profile would force. This is the vertical divergence already visible between the two criteria in Figures 6.2

and 6.3. A last, less intuitive point closes the comparison: the minimum-time path is not the shortest one. A time-optimal solution often flies a slightly longer, higher route, because trading altitude for speed lets it sustain a faster transit and reach the gate sooner despite the extra distance.

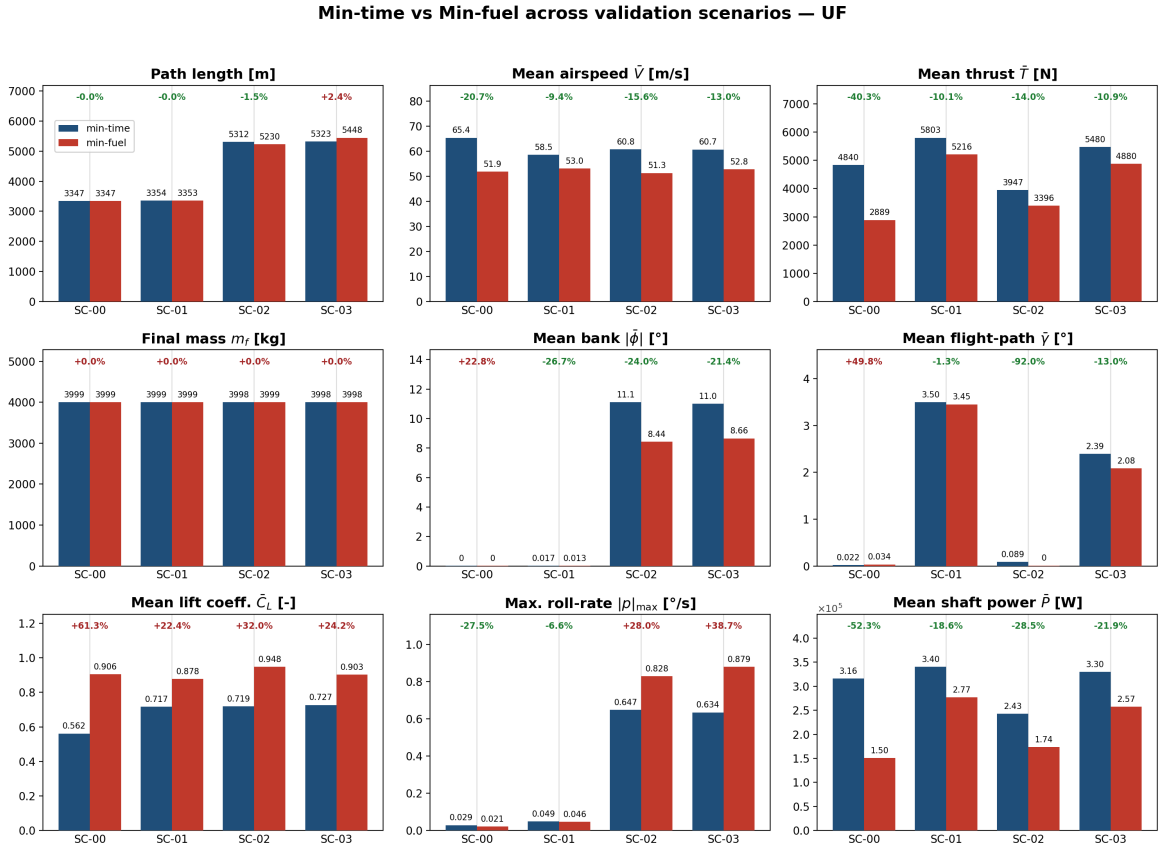


Figure 6.5: Global comparison of the minimum-time and minimum-fuel solutions across the four reference scenarios for the UF. Each panel reports a key performance, kinematic, propulsive or computational metric; bars in blue and red correspond to $\alpha = 0$ and $\alpha = 1$ respectively.

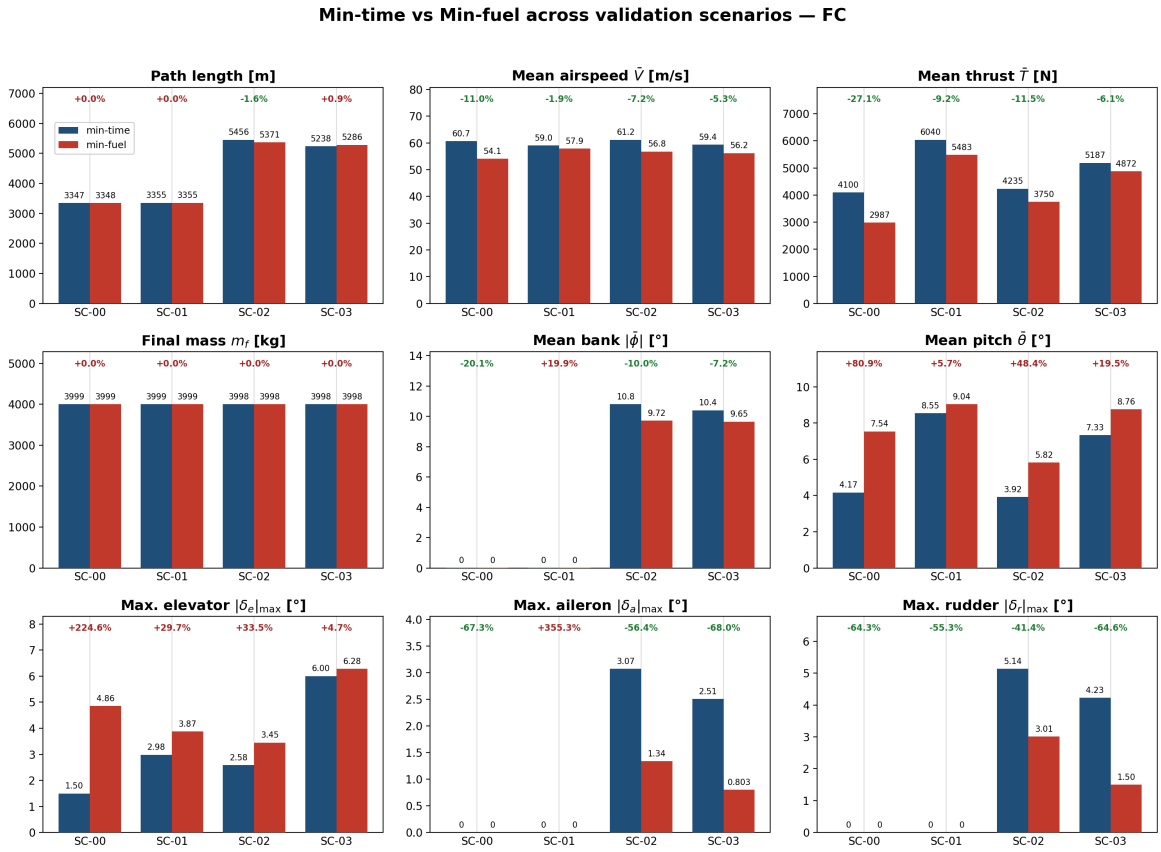


Figure 6.6: Global comparison of the minimum-time and minimum-fuel solutions across the four reference scenarios for the Full-Control Formulation (FC).

6.4.6 Formulation Comparison: UF vs FC

Holding the criterion fixed isolates the effect of model fidelity. It is worth recalling that the two formulations are not expected to coincide: the UF is a three-degree-of-freedom point-mass model in the wind frame and abstracts the aircraft's orientation away, whereas the FC carries the full rigid-body attitude state and the actuator dynamics that go with it. They search genuinely different solution spaces. Figures 6.7 and 6.8 compare their cost scenario by scenario, one panel per criterion.

Under the minimum-time criterion, the two formulations yield nearly indistinguishable results regarding flight time, energy expenditure, and overall path length. Because a time-optimal solution forces the aircraft to operate along the upper boundary of its performance envelope—bound tightly by the mission's boundary conditions and the thrust ceiling—the additional fidelity of the FC model has little operational room to alter the trajectory.

The primary distinction under this criterion is computational: the FC pipeline is significantly slower due to the numerical stiffness of solving the underlying trim equations. This added computational burden is

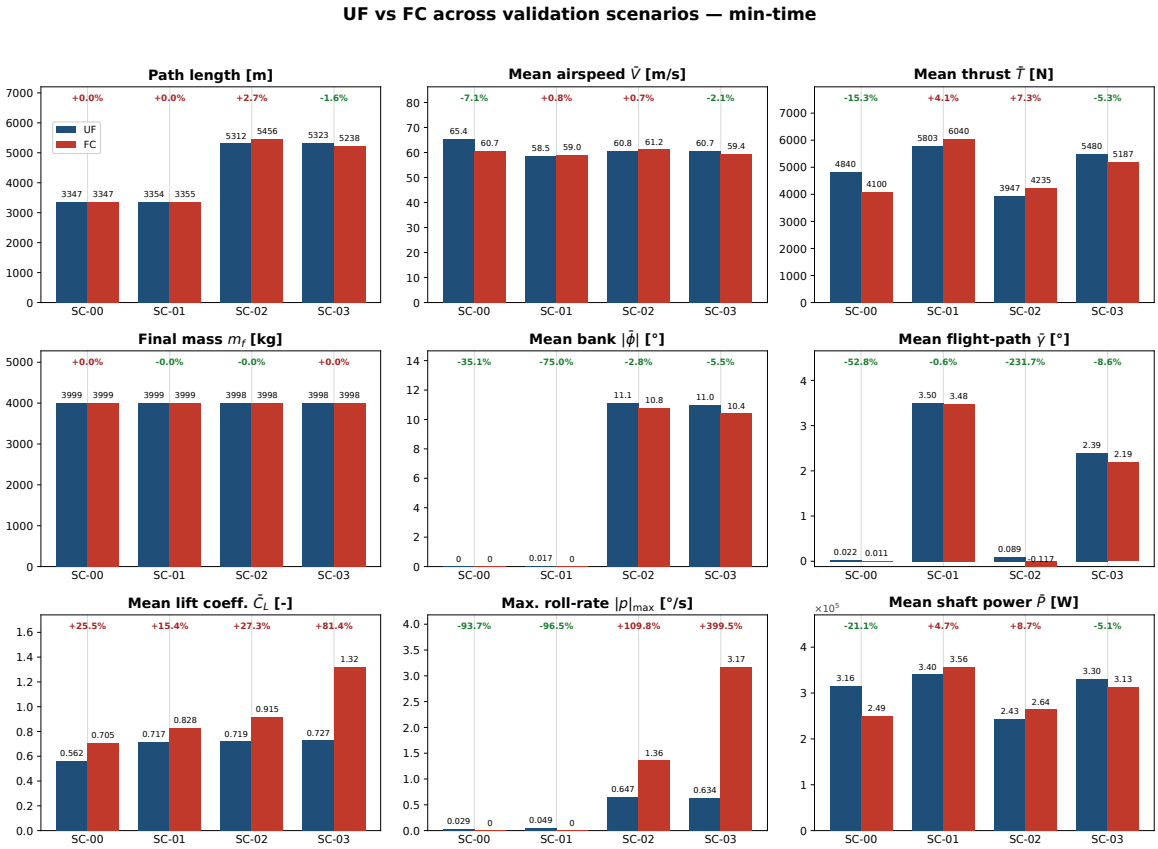


Figure 6.7: Global comparison of the UF and FC formulations solutions across the four reference scenarios for the Min-time optimization criteria.

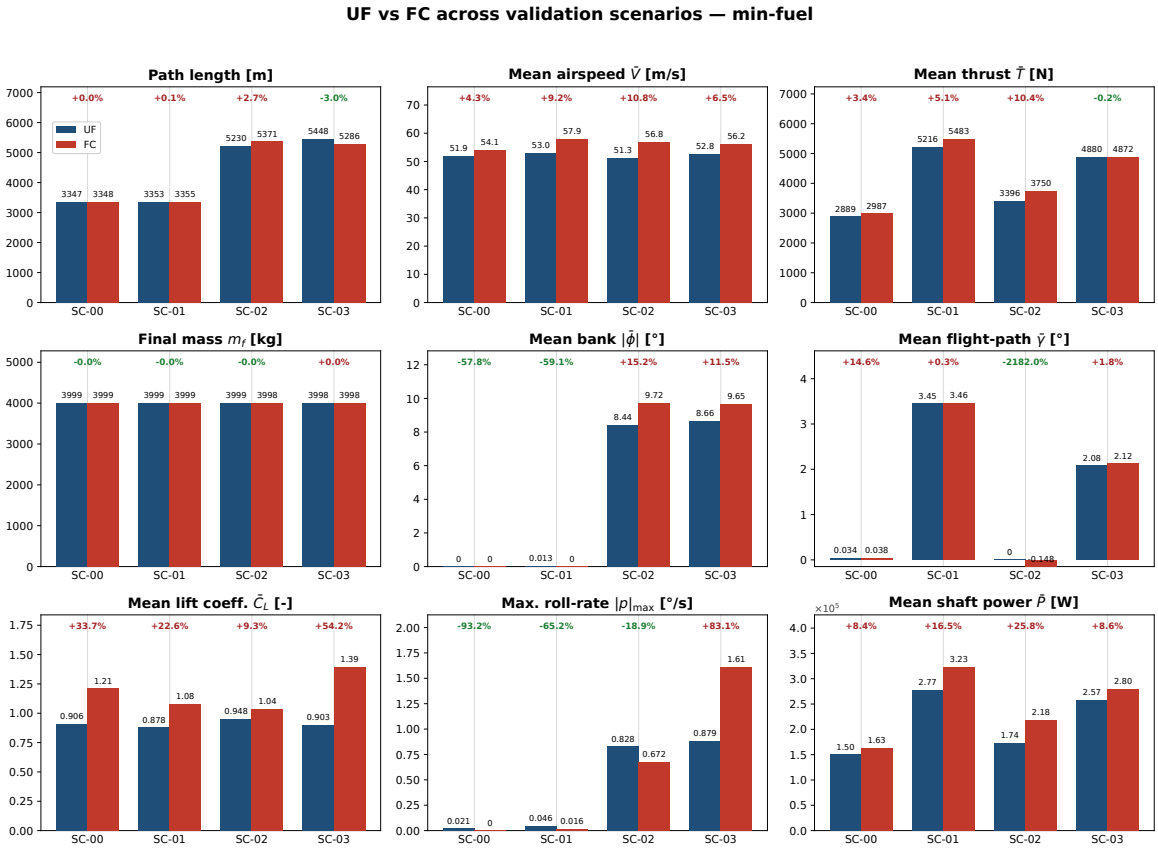


Figure 6.8: Global comparison of the UF and FC formulations solutions across the four reference scenarios for the Min-fuel optimization criteria.

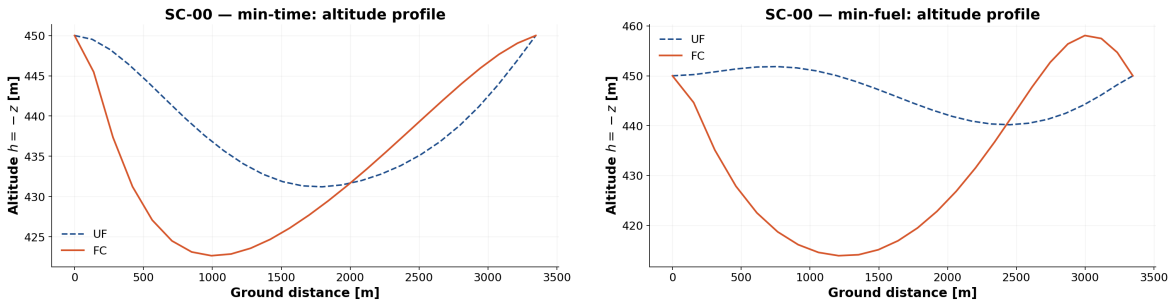
precisely what motivates the sequential UF → FC warm-start strategy introduced in Section 6.6.6.

Conversely, the formulations diverge noticeably under the minimum-fuel criterion due to distinct physical mechanisms. To minimize fuel consumption, the optimizer naturally drives the aircraft toward lower flight speeds; however, these lower velocity regimes expose the core disagreement between the two models. The UF model balances aerodynamic forces solely by manipulating airspeed, allowing it to shed velocity freely and report overly optimistic fuel savings.

In contrast, the FC model must trim the actual physical airframe. In this coupled formulation, flying at lower speeds demands a higher angle of attack and corresponding elevator deflections—control states that are explicitly penalized within the cost function. Consequently, the FC formulation converges onto a more restrictive, physically realistic operational envelope.

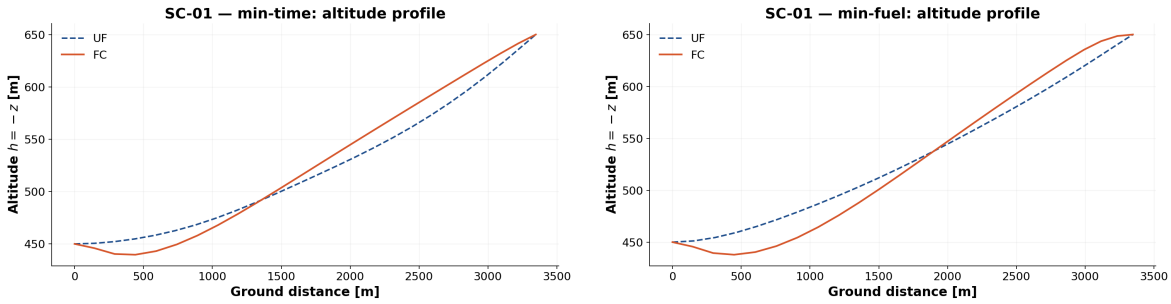
Figures 6.9 and 6.10 illustrate this behavior directly. While the UF and FC altitude profiles show a further similairty under the minimum-time objective, they decouple significantly under the minimum-fuel objective—a trend most prominent during the coupled climb-and-turn maneuver.

A similar discrepancy applies to aircraft banking: the UF model reports an idealized, wind-frame bank angle (μ), whereas the FC model tracks the true Euler roll angle (ϕ). Because these variables cannot be compared directly one-to-one, the rigid-body trim constraints prevent the FC roll angle from relaxing as deeply as the idealized UF abstraction when optimizing for fuel.



(a) Min-time. **(b)** Min-fuel.

Figure 6.9: UF and FC altitude traces overlaid on SC-00 (straight-and-level baseline) under both criteria.



(a) Min-time. **(b)** Min-fuel.

Figure 6.10: UF and FC altitude traces overlaid on SC-01 (straight climb) under both criteria.

6.4.7 Constraint Satisfaction

Having compared what the solutions cost, it remains to confirm they are physically valid. To this end every converged solution is subjected to a three-tier verification pipeline, condensed here into the worst-case residual within each category:

- **Tier A: Decision-Variable Box Bounds** — search-space limits imposed on the individual components of the state and control vectors. The specific non-dimensional bounds are listed in Tables E.1 and E.3.
- **Tier B: Hard Path Constraints** — physical and operational envelopes (e.g. structural limits or safety margins) enforced during the optimization. The limits for UF and FC are detailed in Tables E.2 and E.4.
- **Tier C: Collocation Defect Residuals** — dynamic-fidelity residuals quantifying how exactly the trapezoidal collocation scheme satisfies the underlying differential equations. By construction these equality residuals are satisfied to within the solver tolerance and are reported as absolute magnitudes rather than margins.

For Tiers A and B the pipeline reports a *worst-case margin*. To provide a uniform metric for both upper and lower limits, the margin is computed as

$$M = \begin{cases} \text{value} - \bar{x} & \text{(upper bounds)} \\ \underline{x} - \text{value} & \text{(lower bounds)} \end{cases} \tag{6.18}$$

Under this convention a **negative margin (OK)** indicates that the constraint is satisfied with a remaining slack of $|M|$, whereas a **positive margin (ACTIVE)** indicates that it is exactly met or slightly violated within the allowed numerical tolerance. Tier C residuals are under the tolerance values, since an interior-point solver satisfies its equality constraints whenever it converges. The exhaustive per-run margins, together with the consolidated comparison tables across criteria and formulations, are gathered

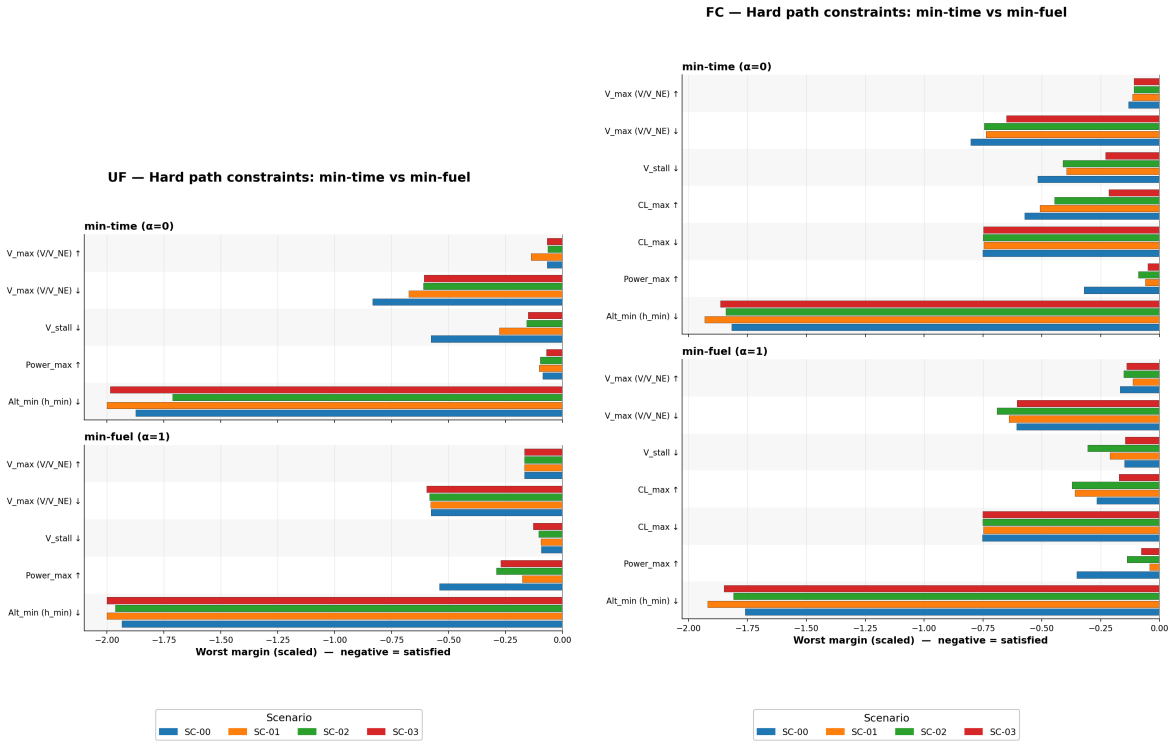
in Annex E.

A note on how these margins are scaled before they reach the plots. The constraints are first non-dimensionalized internally, following the criteria of Section 6.1.3 and Table 6.2, so that quantities with very different physical units (speeds, angles, powers) share a common numerical footing inside the solver. On top of that, for the per-formulation overview bars collected in Annex E (Figures E.1 and E.2) the margins are additionally renormalized—as a pure post-processing step—against the margin distribution itself, which places every bar on a common visual scale. Under this max–min renormalization the bars are confined to $[-1, 0]$ and should be read against three reference points: -1 is a wide absolute margin (constraint far from binding), 0 is a near-violation (the solver operating at the boundary), and an intermediate value such as ≈ -0.5 is *not* a tight margin but the signature of a family of variables whose absolute margins are uniformly small, which the renormalization simply collapses around the midpoint of its range. The per-category comparison plots of the following subsections keep the same sign convention—negative denotes a satisfied constraint—but plot the scaled worst margin directly, without the $[-1, 0]$ collapse, so the relative tightness across categories stays visible.

The global verdict is unambiguous: every margin is on the satisfied side for both formulations, across all scenarios and both criteria, so no box bound, envelope limit or actuator constraint is violated anywhere in the suite. The complete per-formulation overview bars, which gather every constraint of each model on a single renormalized chart, are collected in Annex E (Figures E.1 and E.2). The analysis here turns instead to the two comparison axes, which slice the same data to expose how the criterion and the formulation each load the envelope.

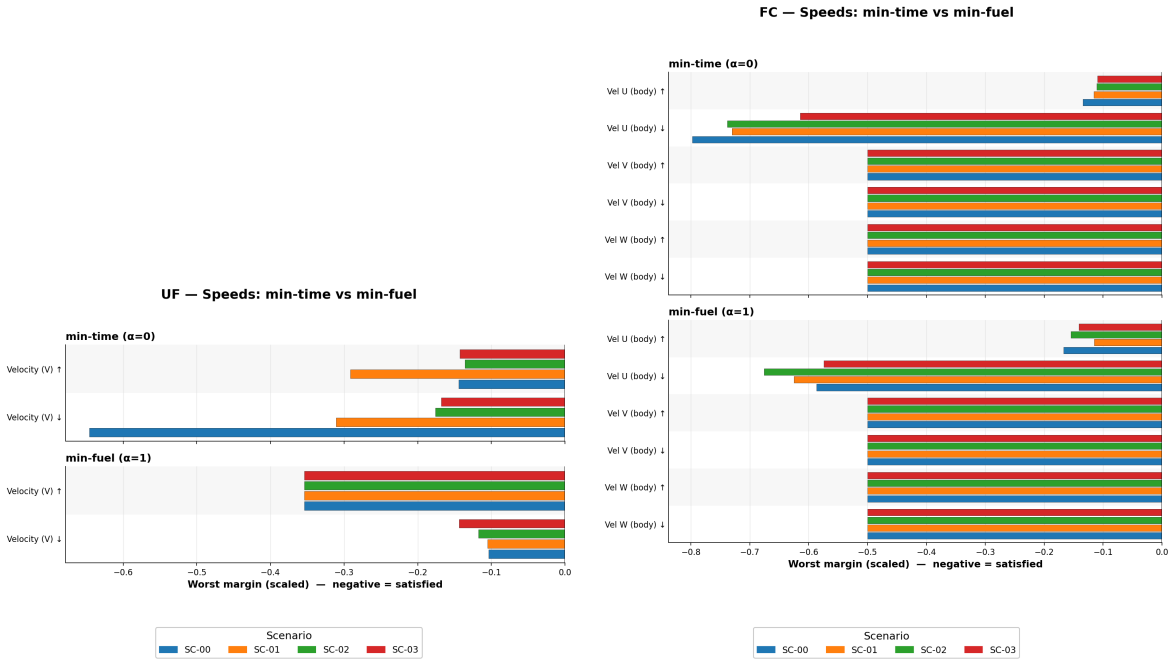
Criterion Comparison: Minimum-Time vs Minimum-Fuel

With the formulation fixed, the two criteria load opposite edges of the same envelope, and they do so consistently. Figure 6.11 reads this off the hard-path constraints, and Figure 6.12 isolates the speed envelope where the effect is cleanest. Under minimum time the aircraft holds as much airspeed as the structure allows, so the velocity upper bound and the power ceiling are the tight constraints while the stall margin stays wide. Under minimum fuel the picture inverts: the solver descends toward the stall to minimize drag work, so the lower speed and lift-coefficient limits tighten while the power backs well off. The two objectives thus approach the same physical envelope from opposite sides—the constraint-level expression of the cost trade-off seen earlier. The numbers behind this flip are tabulated in Annex E (Table E.5).



(a) UF. **(b)** FC.

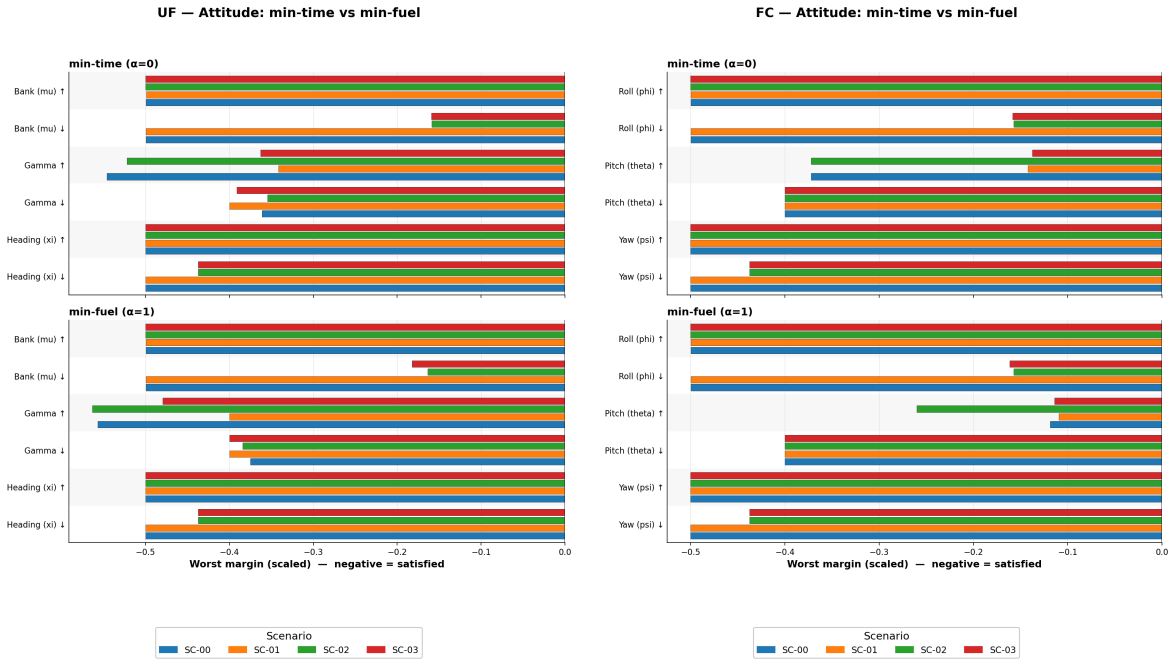
Figure 6.11: Hard-path constraint margins with the formulation held constant and the criterion swept (min-time top panel, min-fuel bottom panel of each chart). Minimum-time loads the power ceiling; minimum-fuel loads the stall and lift-coefficient limits.



(a) UF. **(b)** FC.

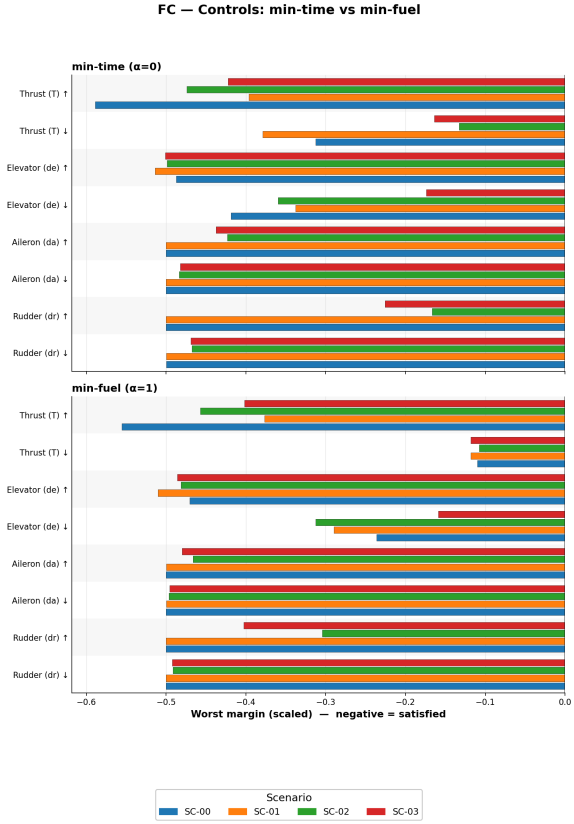
Figure 6.12: Speed-envelope margins, criterion swept. The active bound flips from the upper limit under min-time to the lower (stall) limit under min-fuel.

The attitude and the actuator channels tell the matching story. Figure 6.13 shows that turns are flown at a gentler bank under the fuel objective, since a slower turn needs less of one, and Figure 6.14 shows the rigid-body control surfaces and body rates: they are essentially idle on the straight missions and load up only in the turning ones, yet remain far from saturation under either criterion. The decision-variable position and state box bounds stay slack throughout; their per-run Group A margins are listed in Annex E.

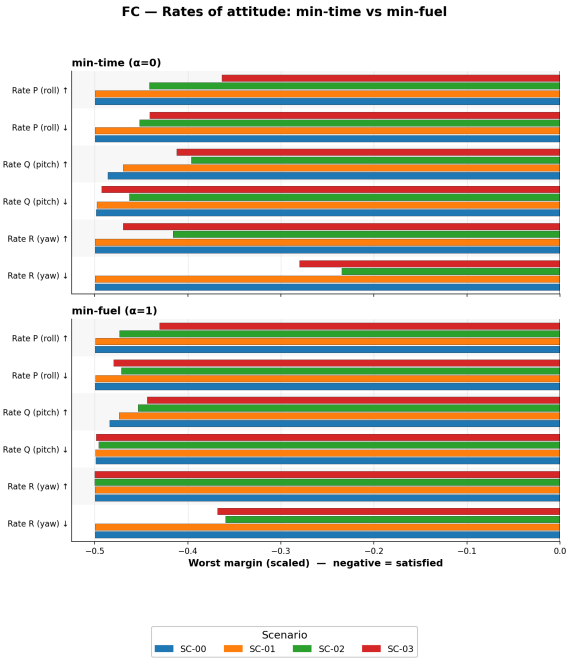


(a) UF. **(b)** FC.

Figure 6.13: Attitude margins, criterion swept. The fuel objective flies the turns at a gentler bank than the time objective.



(a) Control surfaces ($\delta_e, \delta_a, \delta_r$).

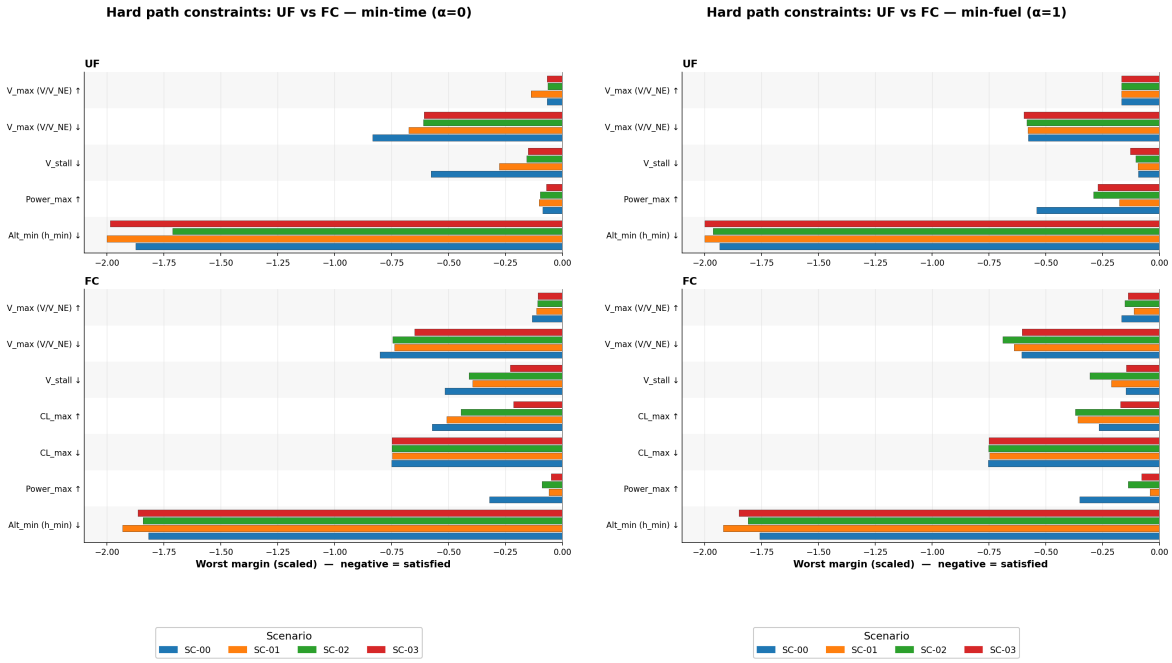


(b) Body rates (p, q, r).

Figure 6.14: Full-control actuator margins, criterion swept. Control surfaces and body rates stay well clear of saturation under both objectives, activating only in the turning scenarios.

Formulation Comparison: UF vs FC

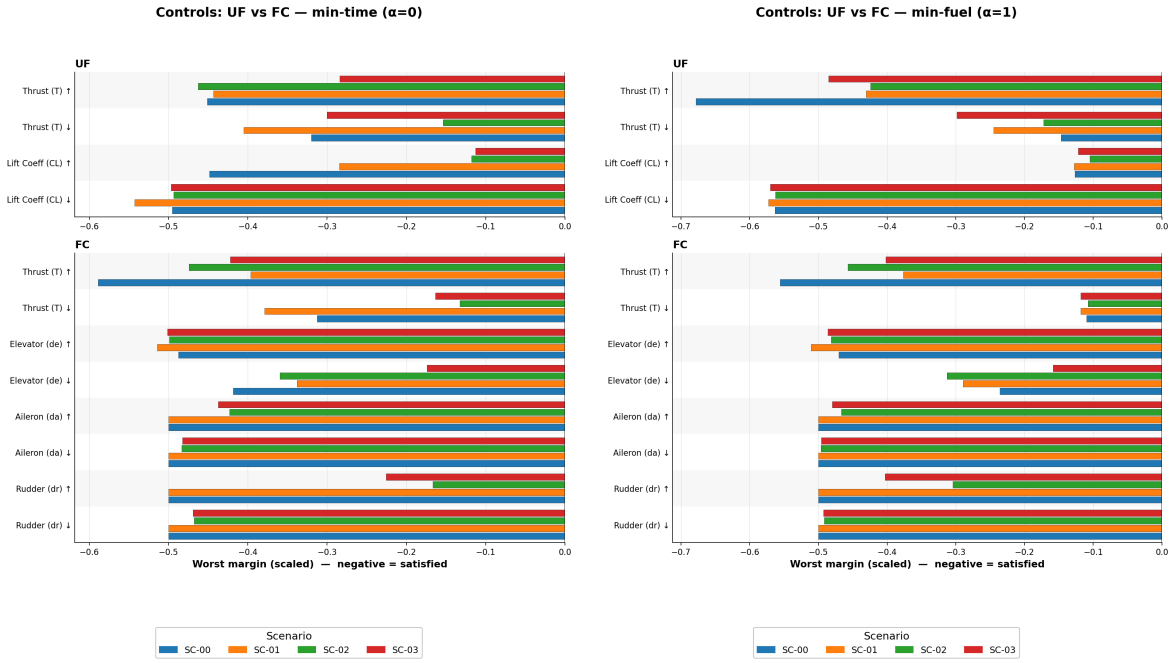
With the criterion fixed instead, the question is whether the higher-fidelity model respects the same physics as the point-mass one. Figure 6.15 stacks the two formulations on the shared hard-path envelope: they sit at comparable distances from the same limits, so the FC does not buy its extra realism by crowding some boundary the UF leaves open. It simply distributes the same margins over a richer set of variables, resolving for instance the lift-coefficient envelope explicitly where the UF folds it into its reduced state. The accompanying numbers are in Annex E (Table E.7).



(a) min-time ($\alpha = 0$). **(b)** min-fuel ($\alpha = 1$).

Figure 6.15: Hard-path margins with the criterion held constant and the formulation swept (UF top panel, FC bottom panel of each chart). Both formulations respect the same envelope with comparable slack.

The only genuinely new content of the FC lives in the variables the UF does not carry—its control surfaces and body rates. Figure 6.16 confirms these remain comfortably within their limits across the suite, so the rigid-body solver reaches the same objectives without aggressive or unrealisable actuator commands. This is in large part the work of the control-rate penalties of Section 6.3.2, whose effect is examined next.



(a) min-time ($\alpha = 0$). **(b)** min-fuel ($\alpha = 1$).

Figure 6.16: Control-surface margins, formulation swept. The UF panel is sparse by construction; the FC actuators stay far from saturation under both criteria.

6.5 Impact of Cost Regularization on the Optimum

To verify that the regularization introduced in Section 6.3.2 acts mainly on smoothness rather than on the location of the constrained optimum, every benchmark scenario was solved twice: once with the nominal gates $\beta_c = \beta_s = 1$ and once with the ablation $\beta_c = \beta_s = 0$. All other numerical settings, including the warm-start strategy and the interior-point tolerances, were held fixed. The discussion below uses scenario SC-03 under the minimum-time criterion as a highly explicit representative case—its climb-and-turn geometry loads the longitudinal and lateral axes at once, so the effect of the penalties is most visible there; the dashboards, tables and per-state defect breakdowns for the remaining scenarios are reported scenario by scenario in Annex D, with the SC-03 entries collected in Section D.6.

6.5.1 Headline Metrics

Table 6.7: SC-03 (climb-and-turn), min-time ($\alpha = 0$): comparison of global quantities between the regularized production run ($\beta_c = \beta_s = 1$) and the unregularized ablation ($\beta_c = \beta_s = 0$). Δ is the unregularized value relative to the regularized one. The bank-angle row is the clearest smoothness discriminator: the unregularized solver reaches a faster, lower-energy optimum but only by banking roughly twice as hard.

Form.	Metric	With reg.	No reg.	Δ [%]
UF	Flight time t_f [s]	68.21	62.43	-8.5
UF	Energy [J]	2.70×10^7	2.46×10^7	-9.0
UF	Path length [m]	4438	4133	-6.9
UF	Max bank [deg]	12.77	29.72	+132.8
UF	TV(p)	0.019	1.03	+5395
FC	Flight time t_f [s]	69.73	62.91	-9.8
FC	Energy [J]	2.78×10^7	2.50×10^7	-9.8
FC	Path length [m]	4473	4087	-8.6
FC	Max bank [deg]	11.99	28.99	+141.7
FC	TV(δ_a)	0.0059	5.65	+95300

The regularization and control smoothing, however, come at a modest and bounded expense in optimality. This behavior is reflected in the key metrics of Table 6.7:

- **Global Cost.** For the SC-03 minimum-time scenario both formulations remain feasible, and both converge to a slightly more conservative optimum once the penalties are active: the unregularized runs reach a faster, lower-energy solution by roughly nine percent in time and energy alike. The price of that marginal gain is a far more aggressive trajectory, whose clearest symptom is the bank angle—the unregularized solver banks close to twice as hard. The regularization therefore buys a markedly smoother, more realistic maneuver for a few percent of optimality, rather than relocating the constrained optimum.

- **Control Total Variation.** Table 6.7 shows the effect directly on SC-03: TV(p) rises by two orders of magnitude for UF and TV(δ_a) by nearly three for FC once the increment penalties are removed. The cross-scenario aggregate in Section D.7 reports the same pattern on the remaining benchmarks.

- **Collocation Defect.** The per-state defect norms reported in the same annex section remain below the prescribed external tolerance for both formulations. The unregularized runs simply exhibit more aggressive profile oscillations without violating the defect bounds.

The comprehensive, per-scenario data for all four benchmarks is reported in Annex D.

6.5.2 Qualitative Impact on the Trajectory

Figures 6.17 and 6.18 illustrate, for scenario SC-03, the comprehensive trajectory dashboards obtained with the Uncoupled (UF) and Full-Control (FC) formulations, respectively. Rather than isolating the actuator behaviors in separate diagrams, these unified dashboards integrate the control history tracks directly alongside the system states. This enables a direct visualization of how active penalization simultaneously shapes the global trajectory, smooths out control inputs, and conditions highly sensitive attitude states such as the pitch angle (θ) and the bank or roll angle (μ for UF, ϕ for FC).

The plots highlight the close agreement between both formulations when regularization is active, as well as the successful suppression of high-frequency oscillations across all channels. While both models track the same global mission geometry, they differ fundamentally in the transient rigidity of their attitude dynamics, as discussed in Section 6.4. On this coupled climb-and-turn mission, the unregularized solver reaches a marginally faster or more fuel-efficient optimum only by forcing aggressive, highly oscillatory variations in pitch and banking roughly twice as hard as its regularized counterpart.

This introduces a classical Pareto trade-off between pure mathematical optimality and physical feasibility: activating the regularization penalties yields longer, significantly smoother control histories at the expense of the primary cost functional. By restricting the solver from exploiting high-frequency chatter to squeeze out marginal numerical gains, the regularized runs trade a few percent of baseline performance (e.g., flight time or fuel mass) for a far gentler, structurally sound flight profile. A detailed quantitative comparison of these regularized and unregularized runs—including cost summaries, total variation metrics, and defect breakdowns—is provided in the SC-03 block of Annex D.6.

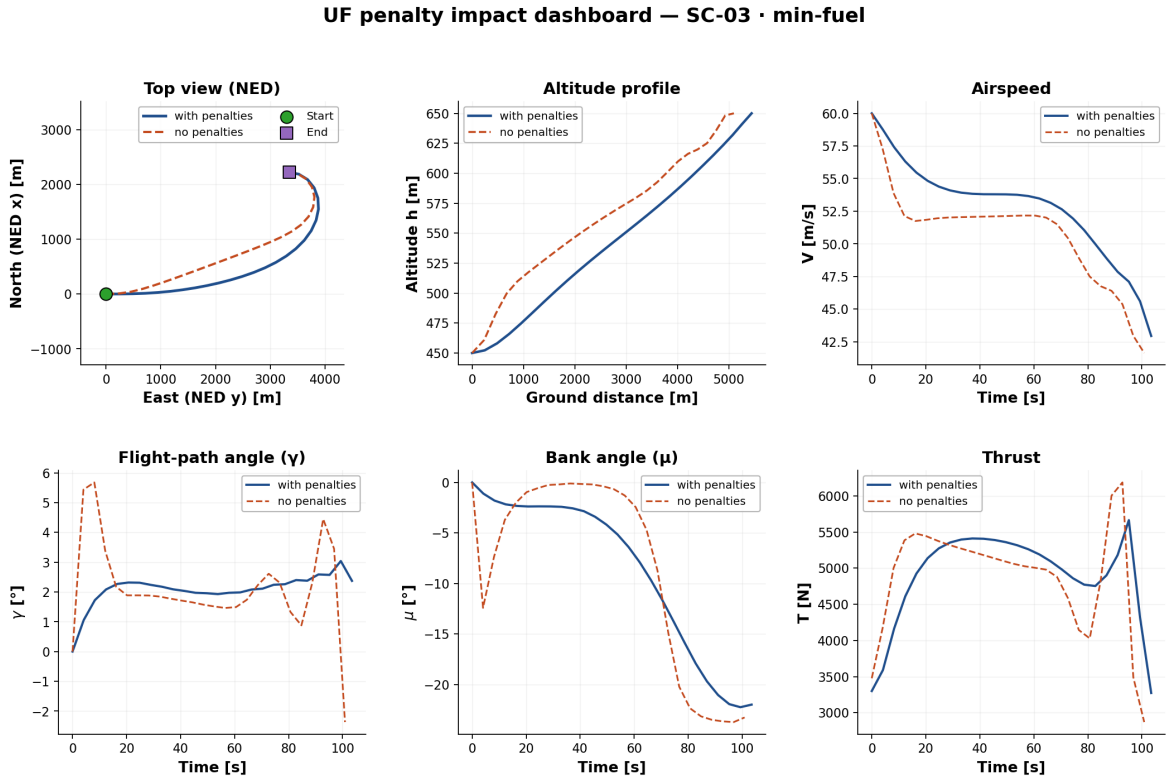


Figure 6.17: Trajectory for Scenario SC-03 (climb-and-turn) using the UF: comparison of the solution with active penalty regularization ($\beta_c = \beta_s = 1$) versus the unregularized ablation ($\beta_c = \beta_s = 0$). The panels display the altitude profile, the wind-frame bank angle (μ), the flight-path angle, the airspeed, and the underlying penalized control variables.

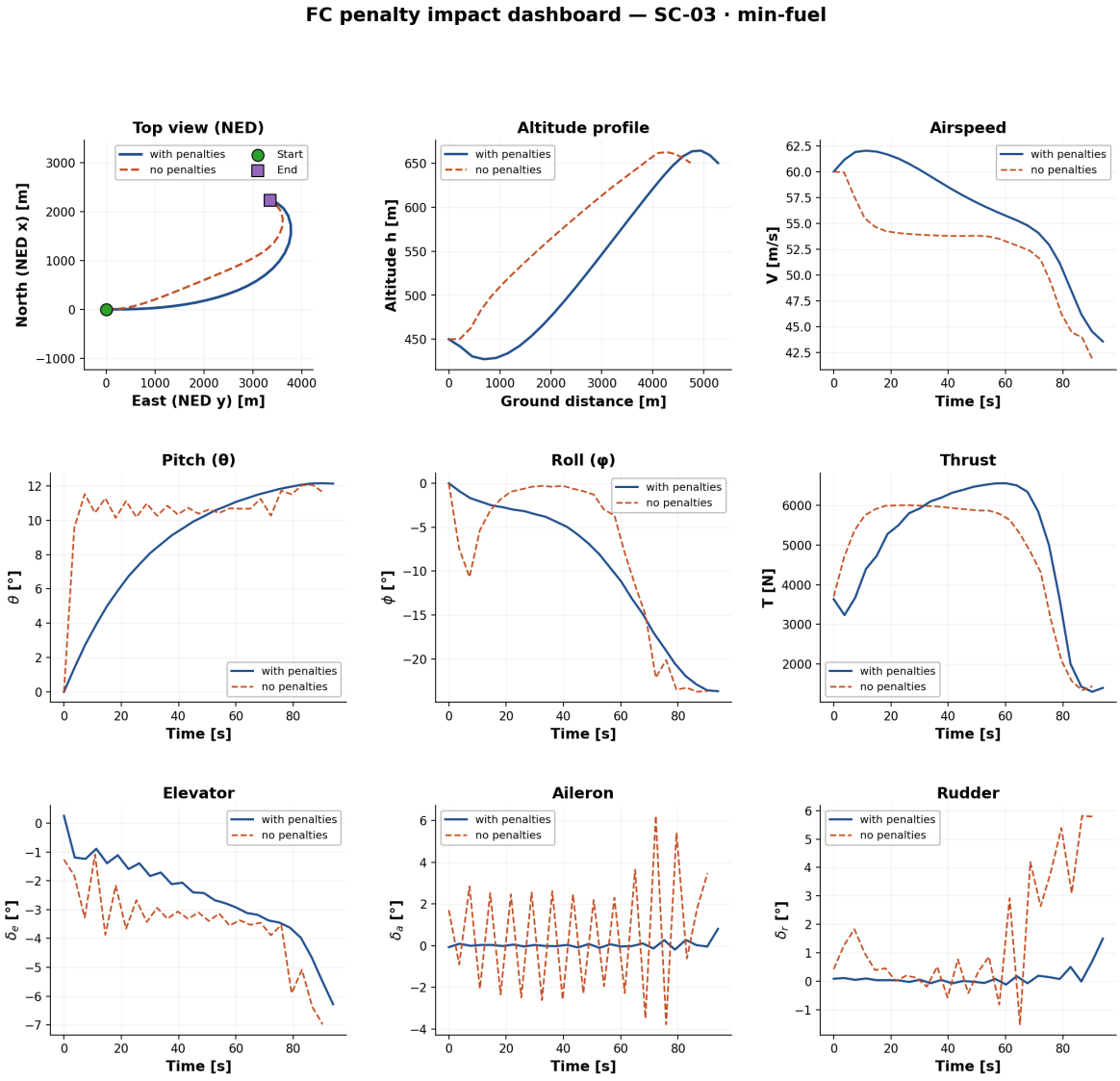
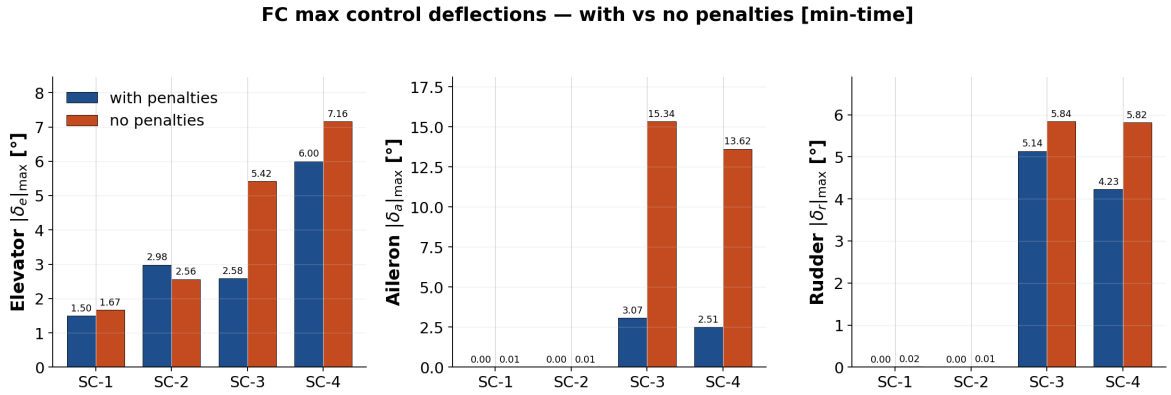
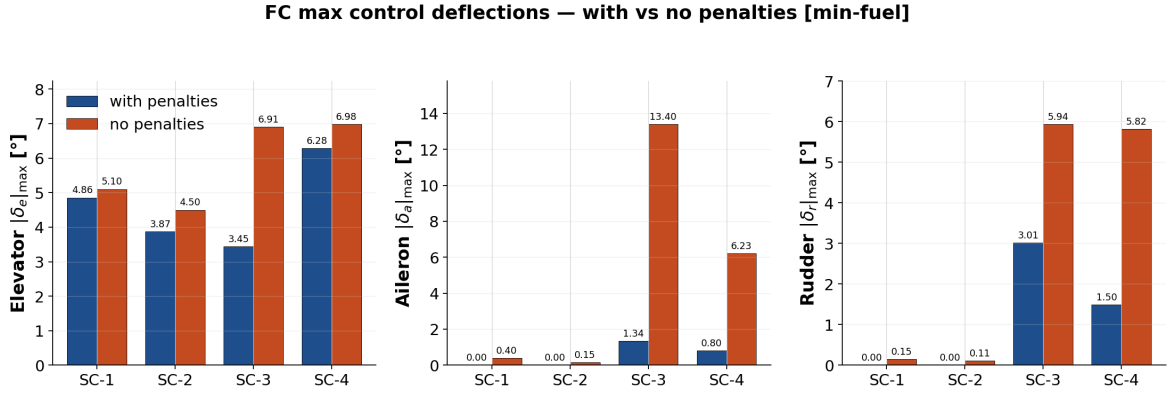


Figure 6.18: Trajectory for Scenario SC-03 (climb-and-turn) using the FC, under the same conventions as Figure 6.17. The rigid-body panels display the stabilized pitch angle (θ), the true roll angle (ϕ), and the integrated control-surface deflections where active penalties successfully eliminate high-frequency actuator chatter.

Beyond this single representative mission, the suppression of actuator chatter is systematic across the entire benchmark suite. Figure 6.19 provides a cross-scenario comparative analysis, tracking the peak FC control-surface deflections reached with and without increment penalties. The pattern is unambiguous: removing the regularization inflates the maximum deflection magnitudes several-fold across all control surfaces. This trend is most dramatic on the lateral axes (aileron and rudder) of the turning missions, where the unregularized solver swings the actuators by close to an order of magnitude more than the smoothed run. The framework successfully minimizes both the absolute peak and the mean deflection transitions, ensuring steady control tracking.



(a) Minimum-time ($\alpha = 0$).



(b) Minimum-fuel ($\alpha = 1$).

Figure 6.19: Peak FC control-surface deflections (elevator δ_e, aileron δ_a, rudder δ_r) across the four reference scenarios, comparing the regularized production run ($\beta_c = \beta_s = 1$, blue) against the unregularized ablation ($\beta_c = \beta_s = 0$, red). Removing the penalties inflates the peak deflections several-fold, confirming that the increment penalties keep the actuators within smooth, physically realistic bounds.

Crucially, the comparison between Figure 6.19a and Figure 6.19b reveals that the regularization effects are noticeably more significant in the minimum-fuel trajectories ($\alpha = 1$). Because minimum-fuel formulations naturally favor energy-efficient, smoother aerodynamic profiles, the regularization framework operates in harmony with the underlying physics, vastly reducing unnecessary control effort.

Conversely, minimum-time trajectories ($\alpha = 0$) are inherently more exigent and aggressive by nature. To satisfy severe temporal constraints, the aircraft must execute rapid, highly demanding maneuvers that dictate a higher baseline minimum deflection. Consequently, while the penalty framework still successfully strips away non-physical high-frequency chatter in minimum-time runs, the remaining smoothed profiles must maintain larger peak magnitudes to comply with the brutal requirements of the time-optimal trajectory.

Of the regularization weights defined in Equations (6.16) and (6.17), the controls contributing most to

the total penalty mass at the converged solution are $w_{\Delta\delta_e}$ and $w_{\Delta T}$ for the FC formulation, and $w_{\Delta C_L}$ for the UF formulation. As observed in both the dashboard panels and the per-axis deflection bars of Figure 6.19, these specific axes correspond directly to the channels where the unregularized discrete optimum exhibits its largest oscillation amplitudes, aligning with the structural diagnosis detailed in Section 6.3.2.

Mitigating these high-frequency control and state oscillations is paramount for practical aerospace engineering applications. In an unregularized numerical formulation, the mathematical solver frequently introduces rapid, non-physical control chattering (bang-bang behavior) to exploit idealized mathematical sensitivities. By explicitly penalizing these sharp variations within the cost functional, the regularization framework ensures smooth control deflection profiles and prevents erratic, high-frequency excitation of the pitch (θ) and roll (ϕ) attitude states. Consequently, this yielding of a unified penalty structure eliminates unrealistic actuator fatigue and thermal stress, resulting in physically realisable flight paths that conform tightly to true aircraft operational limitations.

Lastly, it must be stated that as the relative weight of the penalization terms within the total cost increases compared to the baseline tracking term, a distinct discrepancy is introduced regarding the true underlying minimization objective. Higher values of β_c and β_s impose such a severe restriction on the permissible state dynamics and control variations that the distinct optimization landscapes begin to merge. Consequently, as illustrated in Figure 6.20, heavily regularized runs tend to converge toward virtually identical trajectory profiles, regardless of whether a minimum-time ($\alpha = 0$) or a minimum-fuel ($\alpha = 1$) performance criterion is selected. In this asymptotic regime, the primary driver of the trajectory generation shifts from mission performance optimization to absolute control smoothing, masking the unique physical behaviors inherent to each distinct flight-mechanics objective.

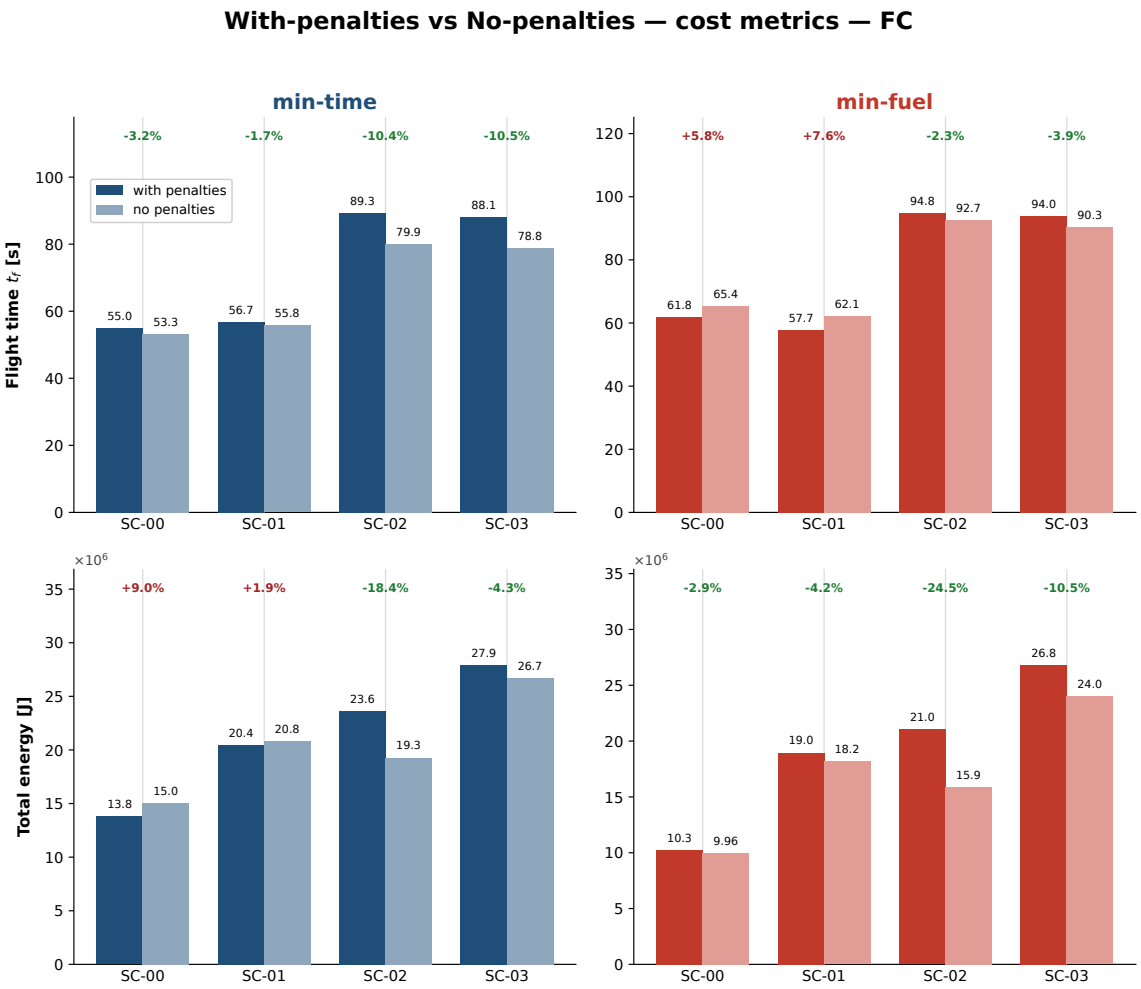


Figure 6.20: Trajectory convergence behavior and cost functional sensitivity under escalating regularization weights (β_c, β_s) for the FC formulation, illustrating the merging of minimum-time and minimum-fuel profiles.

6.6 Computational Acceleration via Warm-Start Strategy

The numerical implementation developed in Sections 6.1–6.4 is mathematically complete: any feasible initial guess together with a sufficiently small barrier parameter would in principle drive the interior-point algorithm to a KKT point of the discretized NLP. In practice, however, the convergence rate of an interior-point method is highly sensitive to the quality of the initial iterate $\mathbf{z}_0$, and the cost of each linear sub-problem grows with the dimension and the conditioning of the constraint Jacobian. This section presents the *warm-start* chain employed to construct a near-feasible initial guess prior to any NLP call, together with the additional algorithmic choices that reduce the per-iteration cost of the optimizer. The section closes with the quantitative analysis of how these choices affect convergence and wall-clock time on the four reference scenarios of Section 6.4.1.

6.6.1 Role of the Initial Guess in Interior-Point Optimization

Building upon the theoretical treatment of equality and inequality constraints, as well as the first-order Karush–Kuhn–Tucker (KKT) optimality conditions detailed in Section 5.1.3, the practical numerical framework translates the continuous OCP into a discretized NLP. Specifically, interior-point methods transform the constrained NLP into a sequence of unconstrained barrier sub-problems parameterized by a decreasing barrier parameter $\mu > 0$. For a given μ, the algorithm introduces a vector of positive slack variables $\mathbf{s} > 0$ to convert the physical inequalities into equalities, subsequently seeking a stationary point of the log-barrier augmented Lagrangian:

$$\mathcal{L}_\mu(\mathbf{x}, \mathbf{s}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \;=\; F(\mathbf{x}) \;-\; \mu \sum_{j=1}^{N_I} \ln(s_j) \;+\; \boldsymbol{\lambda}^\top \mathbf{g}(\mathbf{x}) \;+\; \boldsymbol{\nu}^\top \big(\mathbf{h}(\mathbf{x}) - \mathbf{s}\big), \tag{6.19}$$

which balances four core mathematical components: the primary cost function $F(\mathbf{x})$ to be minimized; a logarithmic barrier term weighted by μ that penalizes proximity to the boundary constraints; a vector of problem equality constraints $\mathbf{g}(\mathbf{x})$ (such as collocation dynamics) scaled by the Lagrange multipliers $\boldsymbol{\lambda}$; and the slack-implicit equalities scaled by $\boldsymbol{\nu}$. As an iterate approaches an inequality boundary, the corresponding slack s_j approaches zero, driving the logarithmic penalty to infinity and forcing the variables to remain strictly within the interior of the feasible region. As $\mu \to 0$, the influence of this barrier vanishes, and the solution sequence of (6.19) satisfies first-order optimality, converging to a KKT point of the original NLP [19], [39], [40].

The convergence rate of this sequence depends critically on how close the initial primal-dual point is to the feasible manifold. If the initial guess violates many constraints simultaneously, the optimizer must enter an expensive recovery phase to establish feasibility before it can meaningfully minimize the objective. For trajectory NLPs in particular—where the dominant equality block is the $(N-1) \times N_X$ collocation defect—a random or linear initial guess typically produces defect norms several orders of

magnitude above the solver tolerance, triggering heavy computational overhead.

A warm start addresses this by constructing an initial trajectory where both the defect norms and key inequality violations are already minimized, placing the initial iterate in the immediate neighborhood of the feasible manifold. The strategy adopted here builds this initial guess through two sequential layers: a geometric path-planning stage in $\mathbb{R}^2$, followed by an inverse-dynamics trajectory-reconstruction stage that lifts the geometric path into a physically consistent state-control sequence.

6.6.2 Hybrid A* Path Planner

The first layer of the warm-start chain is a Hybrid A* search [48], whose objective is to deliver, as cheaply as possible, a kinematically feasible two-dimensional center-line in the horizontal plane that connects the mission start and goal poses while avoiding all declared no-fly zones. The output is a sequence of poses (x_k, y_k, ψ_k) in the $\mathbb{R}^2$ configuration space [59], with the heading ψ retained as a third search coordinate; altitude, speed, attitude and control-surface commands are deliberately left to the inverse-dynamics layer and to the NLP. Restricting the search to a planar graph keeps it cheap enough to be invoked at the very top of the warm-start chain while replacing the piecewise-linear, infinite-curvature corners returned by classical grid A^* [7] by a path that already respects the minimum-turn-radius constraint of the aircraft [15], [48], [60].

The search domain is a 2-D occupancy grid built from the bounding box of the start, goal and NFZ vertices, expanded by $1.5\,R_{\min}$ and discretized with 25 m cells. The minimum turn radius itself is fixed by the active bank-angle limit,

$$R_{\min} = \frac{V_{\text{cruise}}^2}{g \tan \phi_{\text{active}}}, \tag{6.20}$$

and is enforced at every node expansion through the unicycle kinematic model

$$\dot{x} = V\cos\psi, \qquad \dot{y} = V\sin\psi, \qquad \dot{\psi} = V/R, \tag{6.21}$$

integrated in closed form over a fixed arc length Δs for $R \in \{-R_{\min}, \infty, +R_{\min}\}$ (left turn, straight, right turn). The resulting successor of a node $n_k = (x_k, y_k, \psi_k)$ is

$$\begin{aligned} x_{k+1} &= x_k + R\big[\sin(\psi_k + \Delta s/R) - \sin\psi_k\big], \\ y_{k+1} &= y_k - R\big[\cos(\psi_k + \Delta s/R) - \cos\psi_k\big], \\ \psi_{k+1} &= \psi_k + \Delta s/R, \end{aligned} \tag{6.22}$$

so that the descendants of n_k live in the continuous plane and only their containing cell is used for closed-set bookkeeping and collision queries. (This is the main concept that separates appart a regular A* algorithm from the Hybrid approach) Successor nodes that fall within an obstructed cell are immediately discarded, enforcing strict, collision-free feasibility at every expansion step. Valid nodes populated in the open set are then prioritized according to a standard best-first total cost evaluation

function:

$$f(n) \;=\; g(n) \;+\; h(n). \tag{6.23}$$

To ensure an optimal search sequence, this cost function cleanly decouples the actual flight history from the predictive forward-looking estimates:

- **The Path-History Cost, $g(n)$:** This term represents the exact arc length already traversed along the integrated motion primitives. To actively discourage erratic maneuvering and prevent high-frequency control chattering during the path-finding phase, a small additive penalty is injected into $g(n)$ whenever the commanded turning curvature changes sign (i.e., when transitioning between left and right turns). The search successfully terminates as soon as an expanded node lands within a predefined tolerance region of the target goal pose.

- **The Heuristic Cost-to-Go, $h(n)$:** This term provides an admissible lower-bound estimate of the remaining distance from node n to the goal.

To simultaneously account for both geometric proximity and aircraft kinematic limits, the heuristic $h(n)$ is computed as the maximum of two distinct admissible baselines:

$$h(n) \;=\; \max\left(h_{\text{eucl}}(n),\; h_{\text{dubins}}(n)\right) \tag{6.24}$$

where:

1. **The holonomic Euclidean distance (h_{eucl})** calculates a direct, straight-line path to the goal while completely ignoring obstacles and curvature limits.

2. **The non-holonomic Dubins lower bound (h_{dubins})** is defined as the length of the shortest continuous curve of bounded minimum turn radius $R_{\min}$ joining the current pose (x, y, ψ) to the final target pose $(x_{\text{g}}, y_{\text{g}}, \psi_{\text{g}})$ [61].

As proven by Dubins [61], this kinematic minimizer is strictly restricted to one of six closed-form analytical primitives composed of circular arcs (L/R) and straight segments (S): {LSL, LSR, RSL, RSR, LRL, RLR}. Because these paths are evaluated analytically, their computational overhead is virtually negligible compared to a primitive expansion step. Crucially, this term injects the minimum-turn-radius constraints of fixed-wing aircraft directly into the priority ordering of the framework [26]. Reeds–Shepp paths [27] are omitted from the formulation since aircraft cannot execute backward maneuvers.

In practice, these two heuristic components complement each other dynamically across the mission profile. The Euclidean branch dominates during the wide, free-space cruise portions of long trajectories,

where geometric distance is the primary cost driver. Conversely, the Dubins branch takes precedence near the goal region, where precise heading alignment becomes highly constraining. Because taking the maximum of two admissible and consistent heuristics preserves both admissibility and consistency, the search algorithm is guaranteed to expand nodes in an optimal sequence while aggressively pruning unfeasible dead-ends early.

6.6.3 Inverse-Dynamics Trajectory Reconstruction

The Hybrid A* output provides a kinematically consistent geometric path, but it lacks critical dynamic information regarding speed, aerodynamic forces, angular states, or mass. The inverse-dynamics layer [3] bridges this gap by mapping the kinematic waypoints into a full state–control trajectory that satisfies the aircraft equations of motion.

Under a coordinated-turn assumption (zero sideslip and wings level with respect to the local lift vector), the bank angle ϕ_k and the body-frame aerodynamic angular rates (p_k, q_k, r_k) are determined analytically from the flight-path characteristics:

$$\phi_k = \arctan\left(\frac{V_0\,\dot{\chi}_k}{g}\right), \qquad q_k = \dot{\gamma}_k, \qquad r_k = \dot{\chi}_k \cos\gamma_k, \qquad p_k = \dot{\phi}_k - r_k \tan\gamma_k, \tag{6.25}$$

where χ_k is the heading, γ_k is the flight-path angle, and $V_0 = \|\mathbf{V}_{\text{gnd},0}\|$ represents the constant ground speed. These aerodynamic rates directly dictate the rotational velocity of the airframe and drive the resulting aerodynamic moments. To ensure physical viability prior to optimization, the computed rates are clamped to the structural limits $(p_{\max}, q_{\max}, r_{\max})$ matching the rigid-body box constraints in Section 6.2.4.

With the kinematic and rotational states fully defined at each node, a Newton-based 6-DoF trim solver is invoked sequentially for $k = 0, \ldots, N-1$ [50]. The solver explicitly manipulates specific degrees of freedom (DoFs) to enforce a rigid-body force and moment balance. Specifically, it solves for the control inputs—thrust T_k and elevator deflection $\delta_{e,k}$—alongside the body-frame velocity components (u_k, v_k, w_k) to satisfy the trim equation:

$$\mathbf{f}_{\text{trim}}(T_k,\ \delta_{e,k},\ u_k,\ v_k,\ w_k;\ \phi_k, \gamma_k, \chi_k, p_k, q_k, r_k,\ m_k) \ = \ \mathbf{0}. \tag{6.26}$$

Within this formulation, the velocity DoFs (u_k, v_k, w_k) establish the local aerodynamic angles (angle of attack α_k and sideslip $\beta_k = 0$), while T_k and $\delta_{e,k}$ act as the primary balancing mechanisms to counteract the aerodynamic forces and moments induced by the angular rates (p_k, q_k, r_k).

Concurrently, the aircraft mass is updated by integrating the propulsive fuel-burn model forward in time:

$$m_{k+1} \ = \ m_k \ - \ \frac{T_k}{\eta_p\, V_0\, c_{\text{sfc}}}\,\Delta t, \tag{6.27}$$

where c_{sfc} is the specific fuel consumption coefficient. While trimming each node in isolation does not yield global optimality, it dramatically reduces the collocation defect compared to a standard linear initial guess, providing a highly effective warm start for the NLP.

6.6.4 UF → FC State Projection

After the UF NLP converges, its solution must be mapped into the fourteen-dimensional FC state space before the rigid-body NLP can be initialized. A direct linear interpolation of the nine UF states would leave the body angular rates, the cross-track velocities and the control-surface deflections undefined, producing a large initial defect in the FC collocation constraints (6.10). The projection used here performs this mapping analytically at every FC node.

For each node i the UF state $(V_i, \mu_i, \gamma_i, \xi_i)$ is used to reconstruct the aerodynamic angle of attack from the linearized lift relation:

$$\alpha_i \;=\; \theta_i - \gamma_i, \qquad \theta_i \approx \gamma_i + \frac{C_{L_0} + C_{L_\alpha}\alpha_i - C_{L,\mathrm{eq}}}{C_{L_\alpha}}, \qquad C_{L,\mathrm{eq}} = \frac{2m_i g}{\rho V_i^2 S_w}, \tag{6.28}$$

where $C_{L,\mathrm{eq}}$ is the equilibrium lift coefficient required to balance weight at the current speed. The Euler angles are then assigned as $(\phi_i, \theta_i, \psi_i) = (\mu_i, \theta_i, \xi_i)$; the body velocities (u_i, v_i, w_i) follow from the standard aerodynamic-to-body rotation; the angular rates are derived from the coordinated-turn relations (6.25); and the elevator deflection is computed from the longitudinal trim condition. This analytical projection places the FC initial guess in the near-feasible region of the rigid-body state space, so that the FC NLP requires far fewer iterations than a cold start would.

6.6.5 Numerical Pipeline Overview

The numerical solution of the discretized OCP follows a sequential pipeline that exploits the increasing fidelity of the two flight models as a built-in initial-guess refinement. Figure 6.21 summarizes the four numerical layers involved.

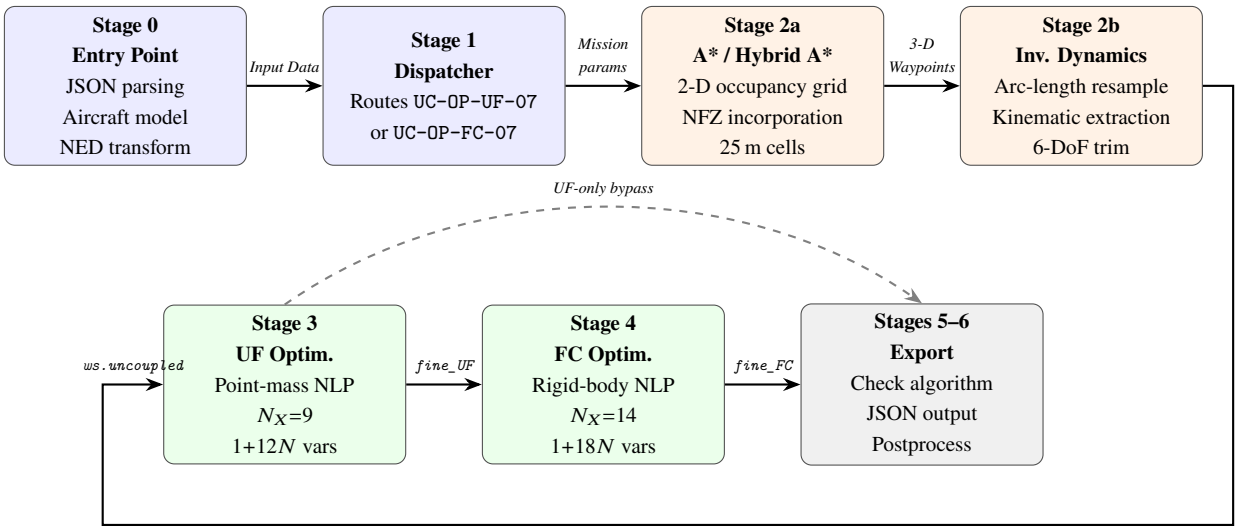


Figure 6.21: Schematic of the numerical pipeline used to transcribe and solve the continuous OCP. The geometric layer and inverse-dynamics stage produce a near-feasible initial guess that warm-starts the UF NLP; the converged UF solution, after a state-space projection, in turn warm-starts the FC NLP. The dashed arrow indicates the lower-fidelity UF-only output used in scenarios that do not require rigid-body validation. The mathematical content of each block is detailed in the corresponding section of this chapter.

Ideally, the optimization pipeline would directly transition from the warm-start trajectory generators—specifically the Hybrid A* geometric planner and the inverse-dynamics module—to the final target optimizer (either the UF or FC solver) to initialize the decision vector z_0. However, as detail the results from Table 6.13, directly seeding the complex full-control (FC) solver from the inverse-dynamics layer proved to be computationally inefficient. To mitigate this performance penalty, the final architectural layout enforces a sequential coupling where the FC optimizer is invariably preceded and conditioned by an intermediate unconstrained-force (UF) solution. This architectural trade-off underpins the final solver pipeline illustrated in Figure 6.21, balancing overall computational tractability with trajectory fidelity.

Consequently, this final pipeline is structured into four distinct layers of increasing mathematical fidelity. First, the geometric layer searches for a kinematically feasible centerline in $\mathbb{R}^2$ subject to a minimum turn radius constraint. Second, the inverse-dynamics stage solves a sequence of nodewise, 6-DoF trim problems to construct a dynamically consistent initialization guess. Third, the UF NLP optimizes the discretized point-mass optimal control problem with all path and performance constraints active. Finally, the FC NLP refines the converged UF solution to satisfy the additional rigid-body constraints. The first two layers are formalized as the warm-start strategy in Section 6.6, while the mathematical formulations for the UF and FC optimization layers are detailed in Sections 5.2–5.3.

6.6.6 Warm-Start Performance Results

The performance framework of this thesis evaluates an extended suite of eight benchmark scenarios. This suite comprises the four baseline validation cases previously detailed in Section 6.4 alongside four advanced missions featuring highly complex tracking geometries. To systematically analyze the performance of the optimization algorithms, these scenarios are categorized into four distinct operational families based on their geometric complexity and environmental constraints:

- `empty`: Obstacle-free baselines designed to test core flight mechanics within the baseline validation suite.
- `nfz_only`: Missions requiring active 2-D avoidance of designated no-fly zones.
- `waypoint_only`: Missions requiring the sequential intersection of intermediate path targets.
- `nfz_waypoint`: Fully constrained environments combining both spatial avoidance and waypoint routing features.

These environmental constraints introduce localized non-convexities, which impose significant computational overhead on the numerical solver. Each scenario is evaluated under both minimum-time ($\alpha = 0$) and minimum-fuel ($\alpha = 1$) optimization objectives, across two independent performance dimensions. Unlike the compact account, this section develops *both* criteria side by side: the minimum-time analysis carries the primary narrative, and the minimum-fuel analysis is reported in the same place so the two can be read against each other rather than across an annex.

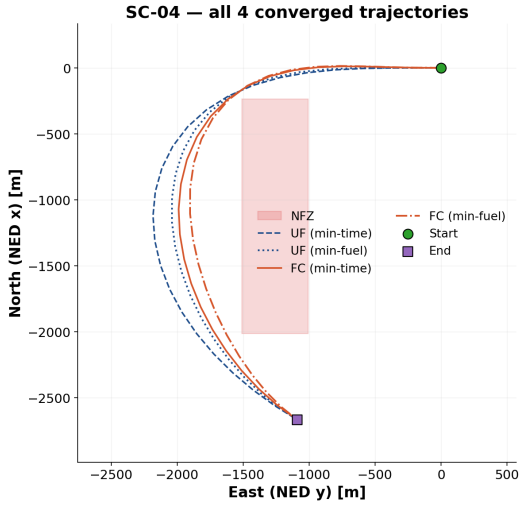
The first benchmark dimension isolates the direct numerical benefit of the warm-start mechanism within the reduced Uncoupled Formulation (UF). The high-fidelity seed generated by the Hybrid A* algorithm and refined via inverse dynamics is tested against two classical initializations: a linear interpolation baseline spanning the boundary states (augmented with a basic trim guess), and a static cold start where the initial boundary conditions are held constant across all collocation nodes.

The second benchmark dimension evaluates the systemic utility of the multi-stage execution pipeline. It contrasts the proposed cascading approach (WS $\rightarrow$ UF $\rightarrow$ FC)—where the intermediate UF solution serves as a warm-start trajectory for the full rigid-body problem—against a direct monolithic approach (WS $\rightarrow$ FC) that completely bypasses the simplified UF optimization phase. Table 6.8 details the structural parameters, transit distances, and collocation node distributions characterizing each benchmark configuration.

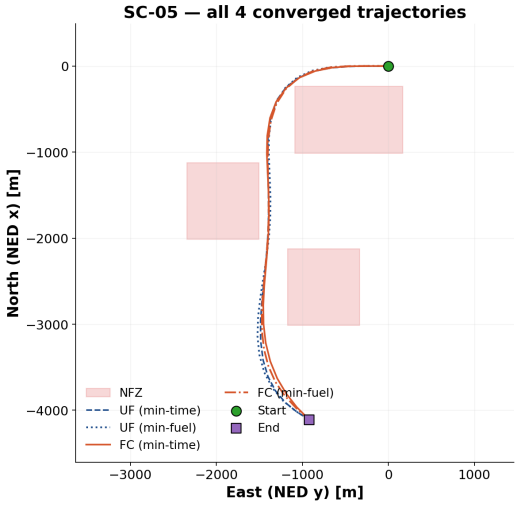
Table 6.8: The eight benchmark scenarios stratified by operational mission family, straight-line transit distance, allocated collocation nodes, and specific geometric features.

Scenario	Distance [km]	Nodes	Geometric / Operational Description
Family I: Unconstrained Environments (`empty`)			
Scenario 1	3.35	25	Straight-and-level baseline with zero altitude change or heading variations. Establishes core UF–FC numerical consistency on a trivial trajectory.
Scenario 2	3.35	25	Pure longitudinal climb baseline featuring a +200 m altitude allocation without heading changes. Isolates and exercises thrust and pitch dynamics.
Scenario 3	4.09	26	Pure lateral heading-reversal baseline performing a sharp 180° turn at a constant altitude. Stresses bank angles and roll-rate bounds.
Scenario 4	4.10	26	Fully coupled climb-and-turn maneuver, combining the longitudinal stresses of Scenario 2 and the lateral demands of Scenario 3 into a single 3-D path.
Family II: Spatial Avoidance Environments (`nfz_only`)			
Scenario 5	2.88	25	Basic no-fly-zone avoidance layout. The trajectory must actively route around defensive polygons; the linear baseline guess intersects the obstacle and fails.
Scenario 6	4.22	26	Advanced no-fly-zone avoidance featuring a dense, highly restrictive obstacle topology. This case completely prevents convergence when using a linear initial guess.
Family III: Target Interception Environments (`waypoint_only`)			
Scenario 7	6.10	31	Sequential waypoint routing. The vehicle is forced to intercept multiple intermediate spatial targets in a fixed order within an open flyway.
Family IV: Fully Constrained Environments (`nfz_waypoint`)			
Scenario 8	6.70	27	Simultaneous no-fly-zone avoidance and waypoint routing. Represents the most restrictive mission profile, tightly coupling all geometric constraints.

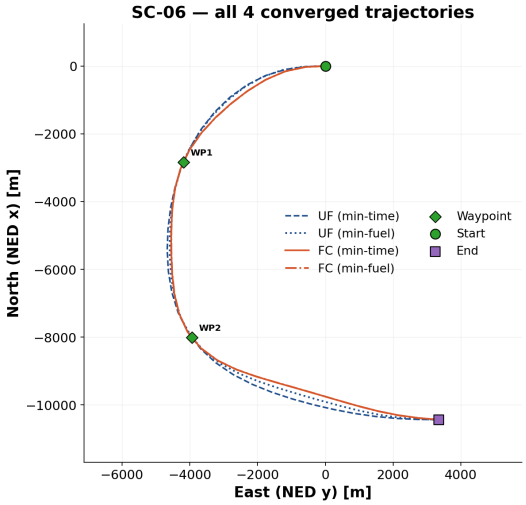
The four complex missions (Scenarios 5–8) serve as the primary metrics for evaluating the robustness of the warm-start architecture. Their planimetric ground tracks are illustrated in Figure 6.22, mapping the four converged solution variants (UF and FC models under both tracking criteria) against the active no-fly-zone boundaries and designated waypoint intersections.



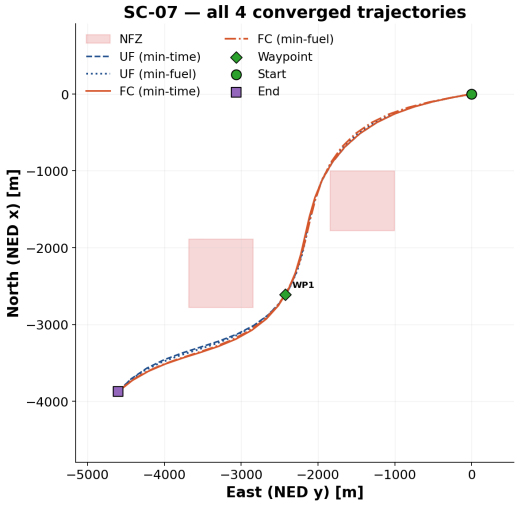
(a) Scenario 5 (`nfz_only`): single NFZ avoidance.



(b) Scenario 6 (`nfz_only`): multi-NFZ corridor.



(c) Scenario 7 (`waypoint_only`): two-waypoint routing.



(d) Scenario 8 (`nfz_waypoint`): NFZ + waypoint.

Figure 6.22: Top-view (East × North) overlay of the four converged trajectories (UF and FC, min-time and min-fuel) for the geometry-rich benchmark missions, with NFZ polygons and waypoints. The tight clustering of the four curves confirms that the criterion and formulation shape the flight profile, not the ground track.

Before analyzing the results and tables, the reader should keep the following context in mind:

1. **Performance Metrics:** Throughout this subsection, a *positive* percentage denotes an improvement. Specifically, a positive *speedup* indicates that the corresponding pipeline is faster, whereas a positive iteration figure means it requires fewer IPOPT iterations.

2. **Repetitions and Timing:** Each scenario was run twice to ensure analytical accuracy. The number of iterations was verified to be identical between repetitions, and the reported computational

time is established as the *mean* of both runs.

3. **Iteration Counts:** For the UF comparison the reported figure is the IPOPT iteration count of the single UF solve. For the FC comparison it is the FC-solve iteration count *only*, on both the WS→UF→FC and WS→FC sides, so the two pipelines are compared on the same final solve rather than on a UF+FC total.

4. **Non-Convergence and Timeouts:** A cell marked with "−" indicates a run that either failed to converge or saturated the computational-time cap (overtime). Such runs are retained as rows but are *excluded from all mean calculations* so that the aggregate figures only compare scenarios where both configurations produced a usable solution.

5. **Feasibility Conditions:** Feasibility is determined solely by the *final optimization stage* of each pipeline (the UF solve for a UF comparison, and the FC solve for an FC comparison). Consequently, a UF failure inside an FC run serves only as a diagnostic on the seed and does not, by itself, render the final FC result infeasible.

Warm Start in the UF Pipeline: Effect of the Initial Guess

The first analysis evaluates the effect of initial-guess quality within the reduced UF formulation. The same UF problem is solved three times, changing only the trajectory supplied to the solver as a starting point. The cold start holds the boundary state at every node and provides no path at all. The linear start draws a straight line from the origin to the goal and trims the airspeed and thrust along it, so it is inexpensive to construct but blind to the environment. The warm start is the Hybrid A* path refined by inverse dynamics, a guess that already routes around the no-fly zones and threads the waypoints. Because the solver is a local method, it does not search for the global optimum [1]; it converges into whatever basin its guess lands in. The quality of the guess therefore governs the cost of the solve, and in some cases whether a solution is obtained at all. This is true regardless of the tracking criterion, which is why the two objectives are read together below.

Minimum-time objective. Table 6.9 lays the three initializations side by side under the minimum-time criterion, giving the solve time and the iteration count of each, together with the increment of the linear and cold seeds relative to the warm start.

The clearest signal is feasibility. The warm start converges on all eight scenarios. The linear seed manages only the five missions with open geometry and becomes infeasible the moment a no-fly zone or a waypoint blocks the straight line. The cold start is weaker still. It clears the easy missions but, on the hardest geometries, it either drifts far from any sensible trajectory or does not converge at all. So

[1] IPOPT is a local interior-point method that cannot guarantee global optimality for non-convex problems. Without an external multi-start framework, the standalone solver remains strictly bound to the local basin of attraction defined by the initial guess.

Table 6.9: UF initialisation comparison (min_time): warm start (Hybrid A* and inverse dynamics), linear interpolation (straight seed with analytic trim) and cold start (boundary state held at every node). Total time and IPOPT iterations are absolute; the gain columns give the warm-start reduction relative to each seed (baseline = that seed), i.e. (seed − warm)/seed × 100, so a positive value means the warm start is faster / needs fewer iterations than that seed. Runs that did not converge or saturated the computational-time cap (overtime) are shown as – and are excluded from the means.

	Total time [ms]			IPOPT iterations			Warm gain: time [%]		Warm gain: iters [%]	
#	Warm	Linear	Cold	Warm	Linear	Cold	Linear	Cold	Linear	Cold
1	650	564	807	17	15	22	-15.2%	+19.5%	-13.3%	+22.7%
2	811	880	801	20	23	21	+7.8%	-1.2%	+13.0%	+4.8%
3	1705	891	2383	39	22	63	-91.4%	+28.5%	-77.3%	+38.1%
4	1478	1538	11716	33	35	200	+3.9%	+87.4%	+5.7%	+83.5%
5	1994	–	–	39	–	–	–	–	–	–
6	3853	–	–	33	–	–	–	–	–	–
7	2950	–	–	39	–	–	–	–	–	–
8	4133	5616	–	61	67	–	+26.4%	–	+9.0%	–
Mean	**2197**	**1898**	**3927**	**35**	**32**	**77**	**-13.7%**	**+33.6%**	**-12.6%**	**+37.3%**

before any timing is considered, the warm start is already the only seed that solves every mission an operational planner would actually face.

Where the seeds do converge, the effort each one demands follows the same order. On the open `empty` missions the linear interpolation already sits near the answer, so it converges in a handful of iterations and the warm start, carrying a small planning overhead, gives no speed advantage there. The cold start shows the opposite behavior. With nothing to exploit it wanders, and its iteration count climbs into the hundreds on the climbing and turning cases while the warm start stays in the low tens. The lesson is consistent across the table. An informed guess converges cheaply, and an uninformed one pays for every node it has to discover for itself.

Cost is only half the story; the converged trajectories must also agree. Table 6.10 reports the flight time each initialization reaches and its deviation from the warm-started solution. On the open missions the seeds land on the same trajectory, but where the cold start does converge on a constrained geometry it settles on a far longer path, deviating by tens of percent from the warm optimum.

The geometry behind these numbers is easiest to see directly. Figures 6.23 and 6.24 overlay, for two constrained missions, the raw Hybrid A* path, the inverse-dynamics guess, and the UF trajectory each seed converges to, with the no-fly zones and waypoints drawn in.

Table 6.10: Initial-guess distance from the converged solution (UF, min_time): per-scenario (x, y, z, V) deviation [%] of each seed from the optimum (warm = Hybrid A*+inverse-dynamics seed, no warm start = linear-interpolation seed, cold = hold-initial-state seed). * marks runs that did not converge, whose guess is measured against the warm-start converged UF solution.

#	Warm start	No warm start	Cold start
1	2.6	2.0	85.4
2	6.3	3.8	85.5
3	18.0	34.6	90.2
4	18.1	33.6	89.9
5	13.9	54.2*	89.0*
6	6.1	46.7*	89.8*
7	8.4	68.5*	95.5*
8	7.5	16.6	91.4*
Mean	**10.1**	**32.5**	**89.6**

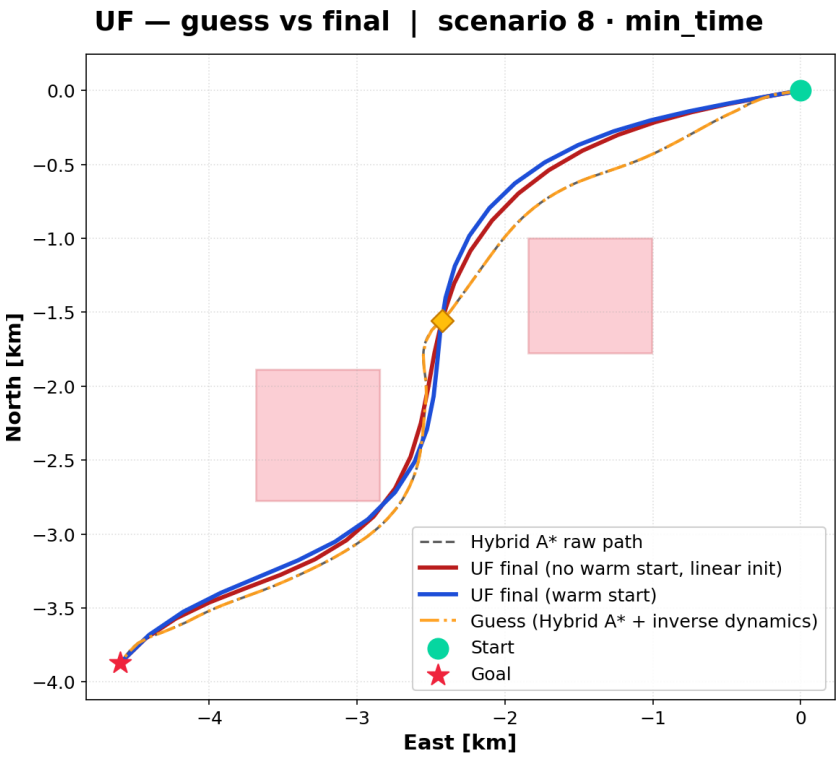


Figure 6.23: UF initialization comparison for scenario 8 (`nfz_waypoint`, `min_time`), top-down East × North. The warm-started UF follows the geometry-aware guess smoothly around the NFZ and through the waypoints. The simpler seeds start from a straight segment and must bend it into a feasible path as the solve proceeds.

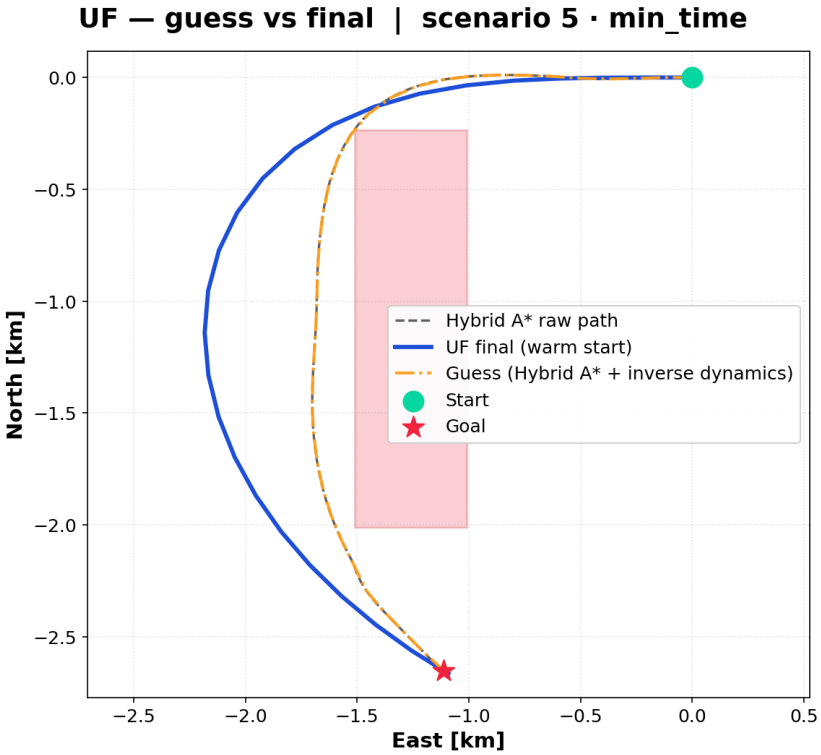


Figure 6.24: UF initialization comparison for scenario 5 (`nfz_only`, `min_time`), with the same curve set as Figure 6.23. The warm start is already routed around the zone and converges without trouble. The straight linear seed cuts through the forbidden region, and the solver cannot recover a feasible trajectory from it.

The two overlays tell the same story in pictures. In scenario 8 the warm start bends gently around the obstacle and the simpler seeds have to be dragged into shape. In scenario 5 the straight seed runs right through the no-fly zone, and from there the solver never finds its way out. Table 6.9 confirms that this is not particular to two missions. Across the whole suite the warm start keeps the UF solve time low and steady, while the cold start grows by an order of magnitude exactly where the geometry turns difficult.

All of these outcomes trace back to a single quantity: the distance between each guess and the solution it converges to. Figure 6.25 measures that distance directly, as the deviation of each seed from the converged UF trajectory.

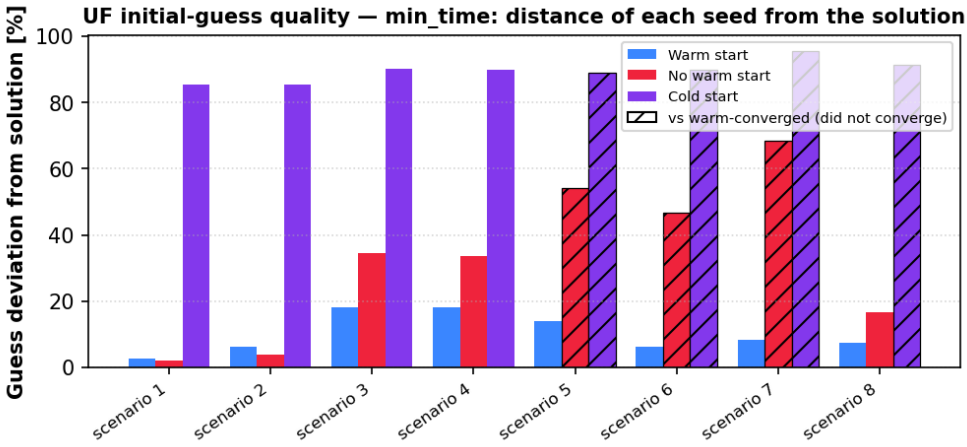


Figure 6.25: Distance of each UF seed from the converged `min_time` solution, per scenario. The warm start begins close to the optimum throughout. The linear seed starts moderately far on open geometry and very far once an obstacle appears. The cold start begins almost entirely apart from the answer.

The ordering is unambiguous and it explains everything above. On average the warm start begins about ten percent from its solution, the linear seed about a third of the way off, and the cold start almost ninety percent apart. The averages hide the part that matters most. The linear seed stays tolerable on the open missions but jumps past half the distance the moment a no-fly zone forces a detour, which is precisely where it stops converging. A guess that starts close is already inside the right basin, so the solver only has to polish it. A guess that starts far is either confined to a poor region or, when an obstacle lies between it and the answer, trapped in an infeasible one. Guess distance, feasibility, and iteration count are three views of the same fact.

Minimum-fuel objective. Switching the cost from time to fuel does not change the role of the guess; it only changes the trajectory it converges to. Table 6.11 repeats the three-way comparison for the minimum-fuel criterion, and Table 6.12 with Figure 6.28 report the guess distances on this objective.

Two further examples, seen from the top-view perspective, are shown in Figures 6.26 and 6.27.

Table 6.11: UF initialisation comparison (min_fuel): warm start (Hybrid A* and inverse dynamics), linear interpolation (straight seed with analytic trim) and cold start (boundary state held at every node). Total time and IPOPT iterations are absolute; the gain columns give the warm-start reduction relative to each seed (baseline = that seed), i.e. (seed − warm)/seed × 100, so a positive value means the warm start is faster / needs fewer iterations than that seed. Runs that did not converge or saturated the computational-time cap (overtime) are shown as – and are excluded from the means.

	Total time [ms]			IPOPT iterations			Warm gain: time [%]		Warm gain: iters [%]	
#	Warm	Linear	Cold	Warm	Linear	Cold	Linear	Cold	Linear	Cold
1	974	759	1915	18	14	33	-28.3%	+49.1%	-28.6%	+45.5%
2	730	1454	1011	21	26	19	+49.8%	+27.8%	+19.2%	-10.5%
3	2204	1222	2015	42	23	52	-80.4%	-9.4%	-82.6%	+19.2%
4	911	1318	16635	18	27	262	+30.9%	+94.5%	+33.3%	+93.1%
5	2065	–	38817	47	–	899	–	+94.7%	–	+94.8%
6	2378	–	20151	32	–	317	–	+88.2%	–	+89.9%
7	2471	–	–	36	–	–	–	–	–	–
8	3449	4242	15016	53	57	251	+18.7%	+77.0%	+7.0%	+78.9%
Mean	**1898**	**1799**	**13651**	**33**	**29**	**262**	**-1.9%**	**+60.3%**	**-10.3%**	**+58.7%**

Table 6.12: Initial-guess distance from the converged solution (UF, min_fuel): per-scenario (x, y, z, V) deviation [%] of each seed from the optimum (warm = Hybrid A*+inverse-dynamics seed, no warm start = linear-interpolation seed, cold = hold-initial-state seed). * marks runs that did not converge, whose guess is measured against the warm-start converged UF solution.

#	**Warm start**	**No warm start**	**Cold start**
1	7.2	3.9	86.7
2	7.5	4.3	85.8
3	16.6	32.9	90.4
4	19.7	34.2	90.0
5	13.8	52.8*	90.9
6	6.6	46.4*	92.3
7	7.6	67.2*	95.7*
8	6.0	16.4	94.0
Mean	**10.6**	**32.3**	**90.7**

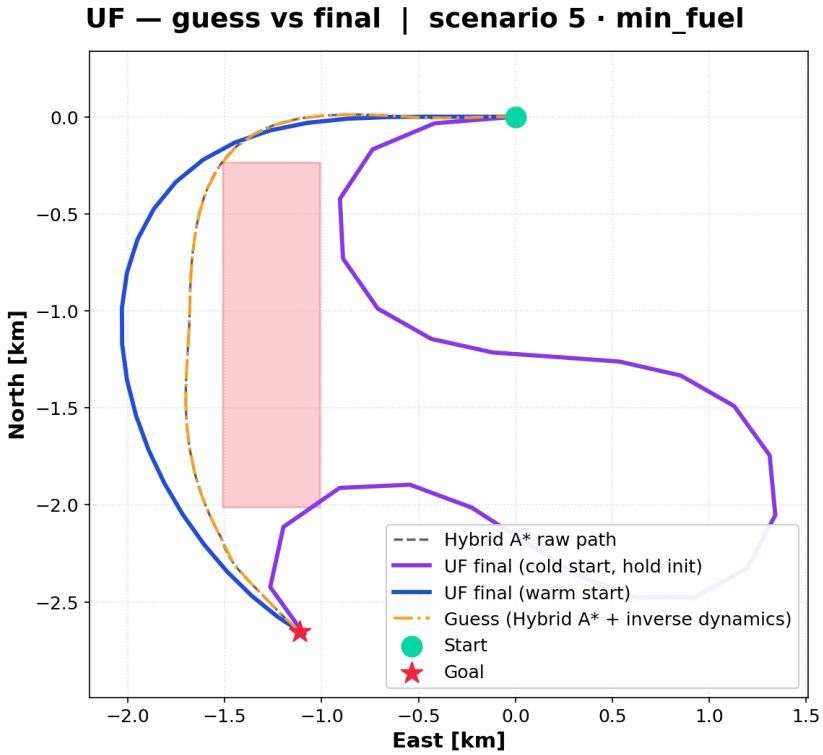


Figure 6.26: UF initialization comparison for scenario 5 (`nfz_only`, `min_fuel`). As in the minimum-time case, the warm start is already routed around the no-fly zone and converges without trouble, whereas the straight linear seed cuts through the forbidden region and the solver cannot recover a feasible trajectory from it.

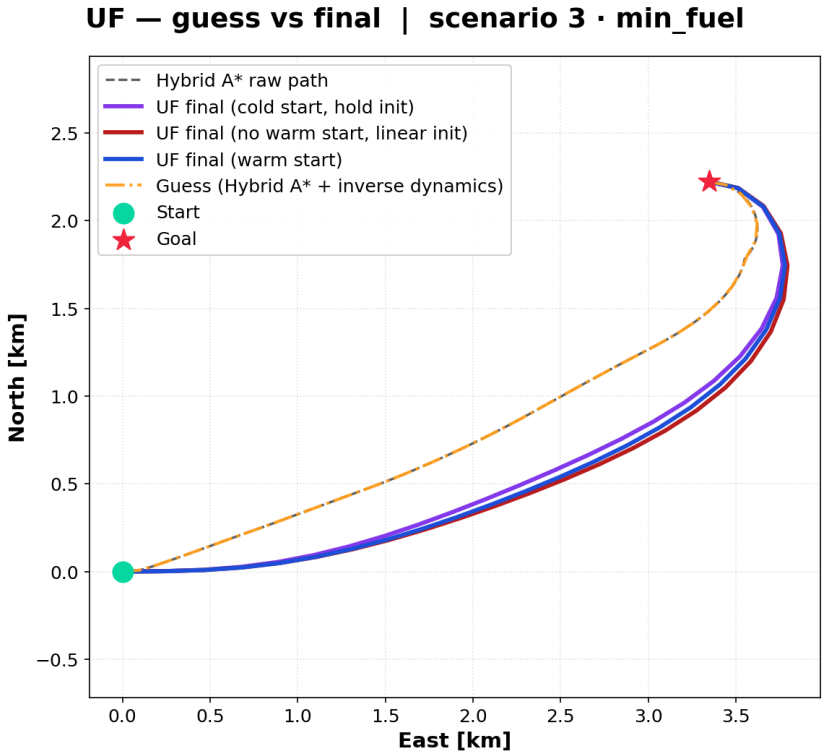


Figure 6.27: UF initialization comparison for scenario 3 (sharp 180° heading reversal, `min_fuel`). The warm start follows the curved reversal from the outset, while the linear seed must be bent from a straight segment into the U-turn and the cold start begins far from the maneuver.

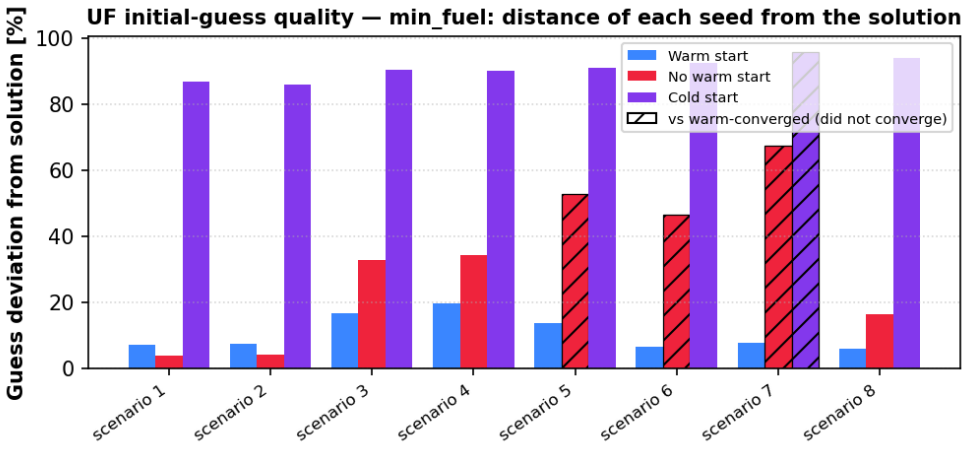


Figure 6.28: Distance of each UF seed from the converged `min_fuel` solution, per scenario. The ordering matches the `min_time` case: the warm start begins close to the optimum, the linear seed starts far once an obstacle appears, and the cold start begins almost entirely apart.

The minimum-fuel picture confirms the minimum-time one in every essential. The warm start again

converges on all eight scenarios, the linear seed clears only the five open missions, and the cold start solves the easy cases but needs several hundred iterations on the hardest geometry. The guess-distance ordering is identical: the warm start begins about a tenth of the way from its answer, the linear seed about a third and much farther around obstacles, and the cold start almost entirely apart. The one difference worth noting is at the cheap end. Where both informed seeds converge, on the open `empty` missions, the warm start is on average faster than the linear seed under minimum fuel, so on this objective it adds a measurable speed gain to the robustness it already provides under minimum time.

Warm Start in the FC Pipeline: The UF Optimum as the FC Seed

The second analysis applies the same principle one stage deeper. The full-control FC formulation is the harder, higher-fidelity problem, and it also requires a seed. The inverse-dynamics layer already produces a complete trajectory, so FC could be seeded from it directly. The analysis therefore determines whether cleaning that guess through the reduced UF solve first is worthwhile. Two ways of starting FC are compared: the full pipeline seeds it with the converged UF optimum, while the direct route seeds it from the inverse-dynamics trajectory and skips UF altogether. As before, the question is settled the same way under both criteria, so the two are presented together.

Minimum-time objective. Table 6.13 reports both routes per scenario, with the FC-solve time and the FC iterations counted on equal terms.

Table 6.13: Computational and iteration overview (FC, min_time): WS→UF→FC (baseline) vs. WS→FC (skip the UF solve). Iteration counts are the FC-solve iterations only on both sides, so the comparison is like-for-like. Runs that did not converge or saturated the computational-time cap (overtime) are shown as – and are excluded from the means.

	Total time [ms]			IPOPT iterations			Dev.
Scenario	WS→UF→FC	WS→FC	Δ%	WS→UF→FC	WS→FC	Δ%	[%]
1	3962	8194	51.6%	32	37	13.5%	3.6
2	4221	10632	60.3%	32	45	28.9%	3.8
3	8900	10427	14.6%	49	42	-16.7%	4.3
4	7642	12901	40.8%	37	47	21.3%	4.2
5	11740	23642	50.3%	43	53	18.9%	3.7
6	13500	17213	21.6%	55	50	-10.0%	3.5
7	10255	17918	42.8%	34	46	26.1%	4.8
8	15244	20526	25.7%	46	50	8.0%	4.7
Mean			**38.5%**			**11.2%**	

Here feasibility is no longer the dividing line. Both routes reach a feasible FC solution on every scenario, because both start from a guess that already respects the geometry. The difference is cost. Seeding FC with the UF optimum makes it converge faster on seven of the eight missions, by roughly forty percent on average and by close to sixty percent on the awkward geometries. The iteration count

moves the same way, since the cleaner seed needs fewer FC steps to settle. Figure 6.30 confirms the two routes converge to essentially the same trajectory: the deviation between the solutions stays a few percent throughout, with the cleaned seed reaching a slightly better trajectory on the constrained cases rather than merely a cheaper one. Figure 6.29 shows the gap at a glance across the suite.

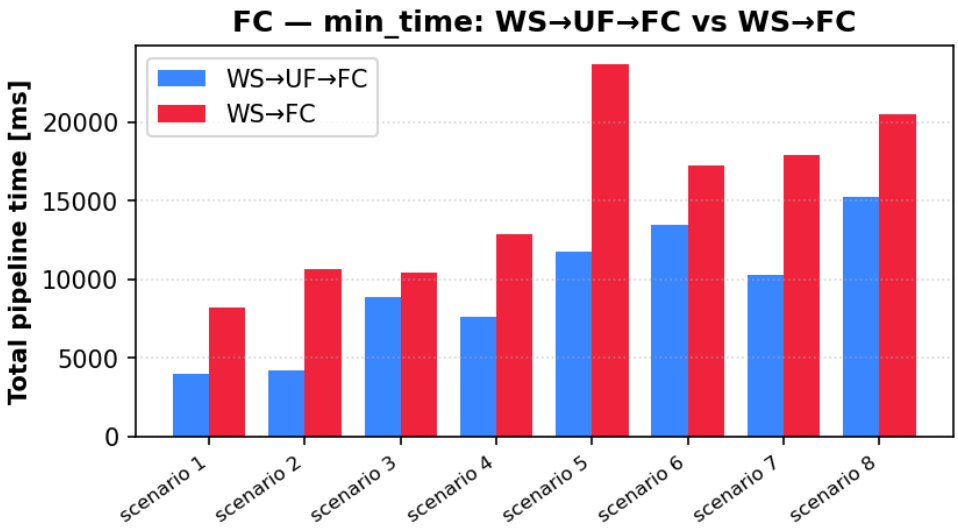


Figure 6.29: FC solve time for the two seeding strategies across all `min_time` scenarios. Seeding FC with the converged UF optimum is consistently faster than seeding it directly from the inverse-dynamics trajectory, and the margin widens on the harder missions.

The reason is the one that governed the UF analysis. Figure 6.30 measures how far each FC seed starts from the converged FC solution. The UF optimum begins about three percent from the FC solution on average; the raw inverse-dynamics guess begins around ten percent away, and much farther on the missions that demand a real climb or turn. That extra distance is exactly the work the FC solver would otherwise have to do itself, and it is the work the intermediate UF solve has already done.

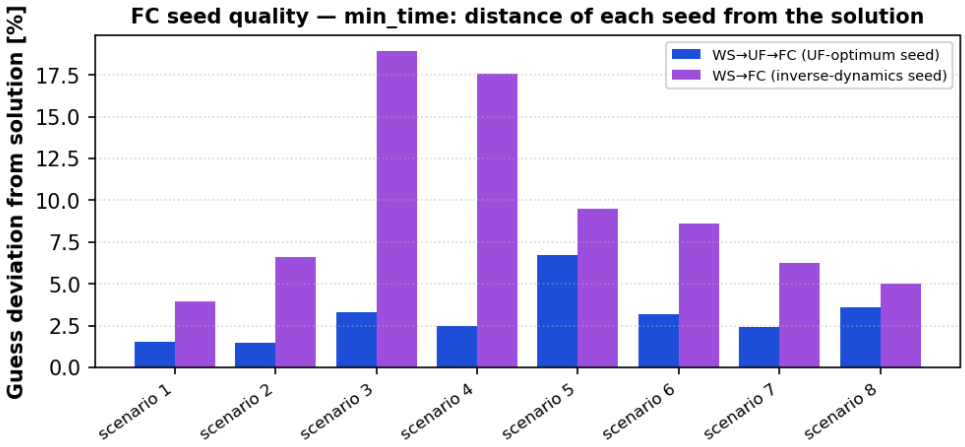


Figure 6.30: Distance of each FC seed from the converged `min_time` solution, per scenario. The UF optimum starts close to the answer everywhere. The raw inverse-dynamics trajectory starts several times farther, most of all on the turning and climbing missions.

Minimum-fuel objective. The minimum-fuel results, in Table 6.14 with the matching guess distances in Figure 6.32, settle the question even more decisively. Both routes stay feasible on every scenario, so again the difference is cost, and here seeding FC with the UF optimum is faster on *all* eight missions, by roughly forty-five percent on average, with fewer FC iterations as well. Figure 6.31 shows the same widening margin seen under minimum time.

Table 6.14: Computational and iteration overview (FC, min_fuel): WS→UF→FC (baseline) vs. WS→FC (skip the UF solve). Iteration counts are the FC-solve iterations only on both sides, so the comparison is like-for-like. Runs that did not converge or saturated the computational-time cap (overtime) are shown as – and are excluded from the means.

	Total time [ms]			IPOPT iterations			Dev.
Scenario	WS→UF→FC	WS→FC	Δ%	WS→UF→FC	WS→FC	Δ%	[%]
1	8475	17280	51.0%	41	48	14.6%	4.0
2	9758	14501	32.7%	44	56	21.4%	4.0
3	9333	15076	38.1%	46	53	13.2%	4.2
4	6839	15927	57.1%	41	61	32.8%	4.1
5	12380	17865	30.7%	50	61	18.0%	6.7
6	15366	20726	25.9%	59	52	-13.5%	3.5
7	11781	18618	36.7%	31	47	34.0%	4.9
8	12695	22687	44.0%	44	56	21.4%	4.1
Mean			**39.5%**			**17.8%**	

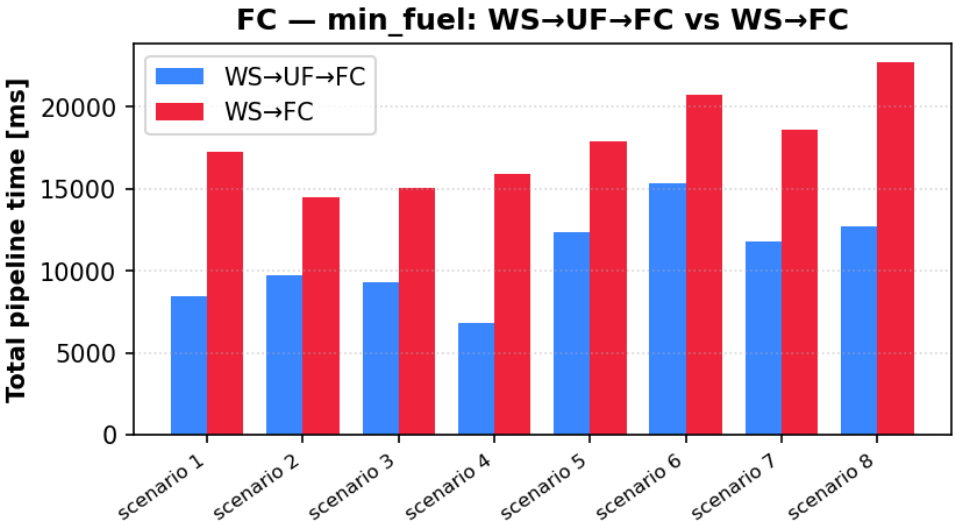


Figure 6.31: FC solve time for the two seeding strategies across all `min_fuel` scenarios. As under minimum time, seeding FC with the converged UF optimum is faster than seeding it directly, on every scenario here.

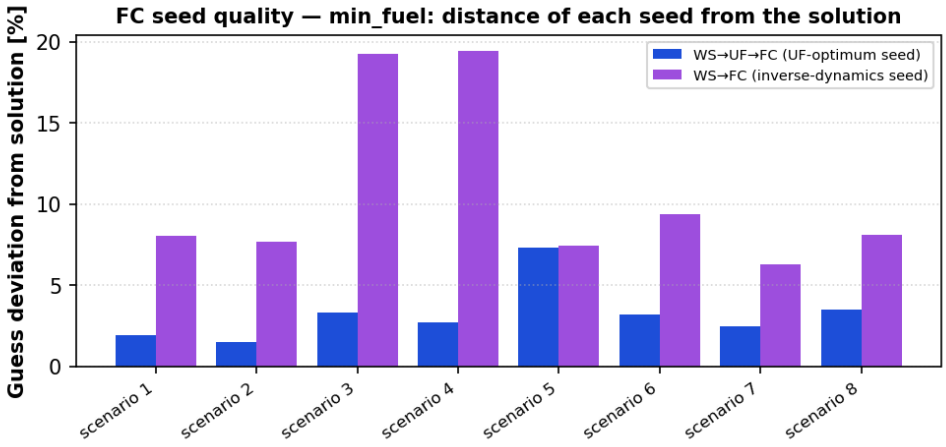


Figure 6.32: Distance of each FC seed from the converged `min_fuel` solution, per scenario. The UF optimum starts close to the answer everywhere, while the raw inverse-dynamics trajectory starts several times farther.

The guess-distance reading is identical to the minimum-time case: the UF optimum begins about three percent from the FC solution, while the raw inverse-dynamics seed begins several times farther. The two converged trajectories agree to within a tenth of a kilogram of end mass, so cleaning the guess through UF buys convergence and cost rather than a different optimum. Across both criteria, then, the same conclusion holds: the UF stage is cheap, far cheaper than the FC iterations it removes, so paying for it is a clear gain.

A warm start is worth exactly as much as the guess it carries, and that statement holds under both tracking criteria without qualification. In the UF stage a geometry-aware guess starts close to the answer, so it converges cheaply and stays feasible where the blind seeds fail—the minimum-time and minimum-fuel tables tell the same story node for node. In the FC stage the same principle decides which seed to use, and the cleanest available guess is the UF optimum itself, which is faster on seven of eight minimum-time missions and on all eight minimum-fuel ones. The value of the whole architecture rests on a single idea that the two objectives confirm in unison: a good initial guess is what makes a local optimizer fast and reliable, and the reduced UF solve is the cheapest way to hand the full-control problem one.

Chapter 7

Budget and Economic Viability

This project is a research-and-development software effort: its deliverable is a reusable trajectory-optimization pipeline rather than a physical prototype. Consequently the budget is dominated by engineering labor, with software licenses, equipment amortization and indirect costs entering as secondary items. The figures below estimate the cost of executing the work between September 2025 and June 2026, a span of roughly ten months that includes the preliminary stage carried out during an internship at CIMNE and the official Bachelor's-thesis period (12 ECTS). The effort estimate is taken directly from the work-package breakdown of Table 1.3.

7.1 Costing Methodology

The budget is built bottom-up and grouped into four categories: *personnel*, *software and licenses*, *equipment* and *indirect costs*. Personnel is costed from the logged engineering hours multiplied by a representative gross hourly rate for the Spanish engineering sector in 2025–2026. Software licenses are prorated to the ten-month project window, and any open-source tool is entered at zero cost. Equipment is amortized linearly over its useful life and charged only for the fraction of that life consumed by the project. A 10% indirect-cost rate covers workspace, electricity, networking and computing power, and a 5% contingency absorbs estimation error. Value-added tax (IVA, 21%) is reported on a separate line so that both the execution cost and the billable figure are visible.

7.2 Personnel Costs

Personnel is the largest line by a wide margin. The author acted as a junior engineer for the full 415 hours of the work-package plan, while academic and industrial supervision (thesis tutor and CIMNE staff) is costed as senior-engineer time. The breakdown is given in Table 7.1.

Table 7.1: Personnel cost, by role, from the logged effort of Table 1.3.

Role	Hours	Rate [EUR/h]	Cost [EUR]
Junior engineer (author)	415	30	12,450
Senior engineer / supervision	25	60	1,500
Personnel subtotal			**13,950**

7.3 Software, Equipment and Indirect Costs

The numerical core relies almost entirely on open-source software — IPOPT and CasADi for the optimization, the C++ toolchain, Python and Git — which carries no license cost. The only paid items are the MATLAB suite used during the early analytical and prototyping stages (academic license, prorated) and standard documentation tools; the manuscript itself was prepared in a free LaTeX environment. Table 7.2 itemizes the license lines.

Table 7.2: Software and license cost, prorated to the ten-month project window.

Item	Cost [EUR]
MATLAB + toolboxes (academic license, prorated)	600
Documentation / office tools	100
Open-source stack (IPOPT, CasADi, Python, C++, Git, LaTeX)	0
Software subtotal	**700**

Equipment is limited to the development workstation and peripherals, amortized linearly over a four-year (48-month) useful life and charged for the ten months of the project; Table 7.3 reports the amortized charge.

Table 7.3: Equipment cost amortized over a four-year useful life, charged for the ten-month project window.

Item	Price [EUR]	Charged fraction	Cost [EUR]
Laptop / workstation	1,600	10/48	333
Monitor and peripherals	300	10/48	63
Equipment subtotal			**396**

7.4 Total Budget

Table 7.4 consolidates the personnel, software and equipment lines of Tables 7.1–7.3 and adds the indirect rate, the contingency margin and IVA.

Table 7.4: Consolidated project budget. Indirect costs are taken as 10% of the direct cost, the contingency as 5%, and IVA at the Spanish standard rate of 21%.

Concept	**Amount [EUR]**
Personnel	13,950
Software and licenses	700
Equipment (amortized)	396
Direct cost	**15,046**
Indirect costs (10%)	1,505
Subtotal	**16,551**
Contingency (5%)	828
Total before IVA	**17,379**
IVA (21%)	3,650
Total project cost	**21,029**

The estimated cost of producing the optimizer is therefore on the order of 21,000 EUR, almost entirely a one-off development expense. It is worth stressing that this is the cost of *building* the tool, not of *using* it: once compiled, a single trajectory solve completes in seconds on a standard laptop (Chapter 6), so the marginal cost per mission plan is negligible. The economic value of the project lies in the reusable software asset, whose cost is amortized across every mission it subsequently plans, and in the operational savings it unlocks — discussed from an environmental standpoint in Chapter 8 — through shorter, lower-fuel trajectories.

Chapter 8

Environmental and Social Impact

The implications of this work are of two distinct kinds. On the environmental side the effect is clearly positive and, importantly, intrinsic to what the optimizer does: a tool whose explicit objective is to minimize fuel or energy cannot help but reduce the environmental footprint of the flights it plans. On the social side the picture is more nuanced, because the work belongs to the wider field of unmanned and autonomous aviation, a field that still faces real public reservations. This chapter summarizes both sides.

8.1 Environmental Impact

The most direct environmental benefit follows from the cost function itself. The minimum-fuel objective formalized in Section 6.3 minimizes the integrated propulsive energy along the trajectory, so every mission planned in that mode is, by construction, the least-consuming trajectory the aircraft can fly within its constraints. Less fuel burnt translates directly into lower CO_2 and pollutant emissions for combustion-powered aircraft, and into lower battery draw — hence smaller batteries or longer endurance — for electric ones. Even the minimum-time mode contributes indirectly, since the geometry-aware planning produces shorter, smoother paths that avoid the wandering, dynamically inconsistent routes a naive planner would generate.

A second, less obvious benefit is acoustic. Because the planner can route around arbitrary exclusion volumes — the no-fly zones of Section 6.2.1 [2] — the same machinery that keeps the aircraft out of restricted airspace can be used to keep it away from noise-sensitive areas such as residential districts, hospitals or protected natural reserves. Noise abatement then becomes a planning constraint rather than an afterthought, and the acoustic footprint of the operation is reduced where it matters most.

Finally, the development process itself is comparatively clean. The entire pipeline is validated in

simulation, so the iterative testing that would otherwise demand repeated physical flights — each one burning fuel and generating emissions — is replaced by numerical solves that consume only a few seconds of laptop-grade computation per trajectory. The direct energy cost of running the optimizer is therefore small and is comfortably outweighed by the operational fuel it saves across the missions it plans.

8.2 Social Impact

The social dimension is where the project meets its principal obstacle. Unmanned aerial vehicles, and autonomous machine-controlled vehicles more generally, are still met with a degree of public scepticism. A significant part of the population is uneasy about aircraft that fly without a human directly at the controls, and that unease is fed by concerns over safety, over privacy, and over the more diffuse worry of ceding decisions to an algorithm. Confidence in machine-controlled vehicles is not yet widespread, and high-profile incidents in adjacent domains tend to reinforce rather than dispel that distrust. Any technology that increases the autonomy of flight inherits this acceptance barrier, and a trajectory optimizer is no exception.

It is precisely here, however, that a constraint-based optimizer can help rather than hurt. The trajectories it produces are not opaque outputs of a black box: they satisfy, by construction, an explicit and auditable set of limits — terrain clearance, no-fly-zone exclusion, the geofence, and the full aerodynamic and structural performance envelope. Behavior that can be specified, bounded and verified in advance is far easier to certify and to explain to the public than behavior that emerges unpredictably, and predictability is the foundation on which trust is usually built. Used responsibly, the same guarantees that make the trajectories safe also make them defensible to a sceptical audience and to a regulator.

Beyond acceptance, the broader social balance of the technology is favorable. Reliable autonomous routing enables applications with clear public value — infrastructure inspection that removes humans from dangerous environments, medical or logistics delivery to poorly connected areas, and search-and-rescue support — and the development of such tools sustains skilled engineering work. None of this removes the legitimate concerns over privacy and safety, which must be addressed through regulation, transparency and careful operational design rather than through the optimizer alone.

8.3 Balance and Outlook

Taken together, the two dimensions point in the same direction. The environmental case is unambiguous: optimization reduces fuel and energy consumption, curbs unnecessary distance, and lets noise be planned away from where people live, all of which is good for the environment. The social case is more demanding: the principal hurdle is not technical but human, namely the limited confidence the public currently places in machine-controlled flight. The most constructive contribution this kind of

work can make to that debate is exactly what an optimizer with explicit, verifiable constraints provides — trajectories that are efficient, predictable and safe by design, and therefore easier to trust as the technology matures.

Chapter 9

Conclusions

9.1 General Conclusions

This thesis has delivered an end-to-end pipeline for three-dimensional minimum-time and minimum-fuel trajectory planning of a fixed-wing aircraft, spanning the climb, cruise and descent phases and supporting both a reduced point-mass Uncoupled Formulation (UF) and a full 6-DoF rigid-body Full-Control Formulation (FC). The objective declared in Section 1.1—a high-performance solver capable of computing optimal flight paths beyond the constant-velocity paradigm, seeded by a hierarchical warm-start strategy and built for real-time operation—has been met and demonstrated across a structured suite of benchmark missions.

The validation campaign confirms that the framework behaves correctly across its full operational envelope. The four reference scenarios converge to physically consistent trajectories that remain within the aircraft flight envelope, and the detailed constraint-satisfaction suite shows that the state, control and collocation-defect bounds are honored throughout. The regularization penalties were shown to act as intended: the ablation study established that removing them buys only a marginal optimality gain—of the order of a few percent in time and energy—at the cost of a far more aggressive maneuver, whereas the regularized optimum yields a markedly smoother and more realistic profile without relocating the constrained solution. The control total-variation metrics, which rise by up to two or three orders of magnitude once the increment penalties are removed, make this distinction unambiguous. The penalties therefore shape smoothness, not the location of the optimum.

The comparison between optimization criteria and between formulations reinforces the same picture of a coherent, well-posed pipeline. The minimum-time and minimum-fuel objectives produce the expected trade-off while sharing a tightly constrained horizontal corridor, and the UF and FC formulations were deliberately designed to pursue the same mission goals without introducing divergences that would

distort the optimization. The two formulations converge to essentially coincident ground tracks and to flight times that agree closely, which confirms that the criterion and the formulation shape the flight profile rather than the route itself. This compatibility is a design property of the pipeline: UF and FC are interchangeable use cases serving a single set of objectives, which is precisely what allows the reduced formulation to act as a faithful precursor to the full one.

That interchangeability is what gives the warm-start architecture its value, and the warm start proved to be not merely beneficial but essential. In the reduced UF stage the geometry-aware Hybrid A* seed, refined through inverse dynamics, was the only initialization that converged on every mission an operational planner would actually face; the simpler linear and cold seeds became infeasible the moment a no-fly zone or an intermediate waypoint blocked the straight line. The same principle, applied one stage deeper, established the homemade cascading pipeline (WS $\rightarrow$ UF $\rightarrow$ FC) as a clear and decisive step in optimal computing time: seeding the full-control problem with the converged UF optimum rather than the raw inverse-dynamics trajectory reduced the FC solve time by roughly forty percent on average, and by close to sixty percent on the most constrained geometries, while converging to the same—and on the hardest cases a slightly better—trajectory. The intermediate UF solve is far cheaper than the FC iterations it removes, so the layered design pays for itself. In short, a good initial guess is what makes a local optimizer fast and reliable, and the purpose-built warm-start pipeline is the cheapest way to provide one.

9.2 Fulfillment of Requirements

This section revisits the technical requirements defined in Section 1.3 and assesses, one by one, the degree to which each has been satisfied by the work reported in this thesis. Table 9.1 summarizes the verdict for every requirement, grouped by the same four families used in the specification, with a short pointer to the supporting evidence. Three outcomes are distinguished: **Met**, requirements demonstrated by the validation campaign or the warm-start benchmark; **Partial**, requirements satisfied by the architecture or shown in part, but not fully exercised within the scope of this work; and **Pending**, requirements deferred to future work, delegated to the companion environmental thesis, or explicitly placed out of scope.

Table 9.1: Fulfillment status of the technical requirements of Section 1.3. *Met*: demonstrated by the results. *Partial*: satisfied by design or shown in part. *Pending*: future work, companion thesis, or out of scope.

#	Requirement	Status	Evidence / Remark
Functional and Dynamic			
F1	Fully coupled 6-DOF formulation	Met	Validated FC formulation; envelope honored
F2	Solver decoupled from the vehicle plant (external datasets)	Partial	Modular by design; validated on the single MSN012 airframe
F3	Enforce state/control constraints, incl. δ and $\dot{\delta}$ limits	Met	Constraint-satisfaction suite and per-run detail (Annex)
F4	Dynamic wind-vector field in the equations of motion	Partial	Modeled in the EOM; deterministic only, full wind is future work
F5	Environmental constraints: DEM, NFZ, geofencing	Partial	NFZ demonstrated (Sc. 5–8); DEM via companion thesis
F6	Multiple, selectable warm-start techniques	Met	Warm/linear/cold seeds benchmarked (Sec. 6.6.6)
F7	Transcription-independent: collocation *and* multiple shooting	Partial	Both formulated; only direct collocation used in results
F8	Multiple integration schemes (Euler, RK4, trapezoidal)	Partial	Trapezoidal used, showed better stability/velocity tradeoff; Euler/RK4 not compared in results
F9	Analytical seed from kinematic primitives (Dubins, straight line)	Met	Dubins heuristic, A* and linear-trim seed implemented
Software and Interface			
S1	JSON-based input architecture	Partial	Present in the pipeline; not results-validated
S2	Export trajectories in WGS84 geodetic coordinates	Partial	NED↔geodetic layer exists; export not shown
S3	Native Linux SiL execution / industrial validation	Partial	Developed through WSL (Windows Subsytem for Lynux); Ongoing QHIL validation
Numerical and Performance			
N1	Convergence within 4.0 s per look-ahead window	Partial	UF meets it; FC exceeds it, targeted by analytical gradients
N2	Convergence tolerance of 10^{-4}	Partial	Benchmark ran at 10^{-2}; production tolerance not exercised
N3	Analytical Jacobians, error $< 10^{-6}$ vs. finite differences	Pending	Results use finite differences; gradients are immediate future work
N4	C++ core with static/pre-allocated memory, no heap in the loop	Pending	C++ migration in scope; static-memory use not evidenced
Safety and Robustness			
R1	Automated range-checking of JSON inputs	Pending	Implementation-level; not demonstrated in the results
R2	Fallback to last valid segment / backup seed on timeout	Pending	Fallback logic not demonstrated
R3	Output trajectory with C^2 continuity	Partial	Smoothness shown via penalty ablation; C^2 not formally proven
R4	Log solver metrics (iterations, residual) per cycle	Met	Iterations and timings logged throughout the benchmark

In aggregate, of the twenty requirements five are fully met, ten are partially satisfied—typically supported by the architecture but not exercised to their full extent within the allocated scope—and five remain pending, deferred either to declared future work (notably the analytical-gradient pipeline and the QHIL validation) or to implementation-level work outside the scope of this report. The two items

most relevant to a critical reading are the real-time threshold (N1), which the reduced UF stage meets but the full FC solve does not yet under the current finite-difference pipeline, and the convergence tolerance (N2), where the benchmark campaign was run at a relaxed 10^{-2} rather than the specified 10^{-4}. Both are addressed directly by the analytical-gradient work discussed in Section 9.4, which is expected to restore the performance margin needed to satisfy them at the production setting.

9.3 Limitations

The results presented in this thesis are subject to a number of limitations, which are stated explicitly here so that the scope of the validated claims is not overstated.

First, the planner is solved in an open-loop, offline setting. It computes an optimal trajectory from a fixed initial state and environment, but it does not yet reject disturbances in real time; closing the loop against the live aircraft state is treated as future work in Section 9.4. Second, the wind field enters the equations of motion as a deterministic vector field, so the framework does not account for turbulence or stochastic gust models. Third, the aerodynamic and propulsive data are drawn exclusively from the MSN012 (Flyox I) airframe: although the solver is modular by design and can in principle be parameterized for any class of fixed-wing aircraft, this generality has not been exercised on a second platform, and porting to a new airframe would require deliberate retuning before execution.

A specific consequence of that retuning concerns the number of discretization nodes, N, which cannot be treated as a fixed hyperparameter. Smaller, more agile aircraft exhibit faster transient dynamics and higher control sensitivities, and resolving them without introducing artificial numerical damping calls for a denser grid, and therefore a larger N. The cost of increasing N is governed by how IPOPT obtains derivative information: in the finite-difference configuration used for the benchmark campaign, each iteration perturbs the decision variables and re-evaluates the full pipeline many times, so the per-solve cost grows unfavorably as the variable space expands. This is the main practical limit on the routes the current pipeline can resolve within the real-time budget, and it is also the limitation most directly targeted by the analytical-gradient work discussed in Section 9.4.

Finally, owing to the project timeline, the framework has not yet been validated in a quasi-hardware-in-the-loop (QHIL) environment. Such a test is needed to cross-check the idealized numerical outputs against the timing and resource constraints of a deployment-ready solver, and it remains an immediate item of future work.

9.4 Future Work and Next Steps

It is important to frame the solver and the results presented in this document as one half of a larger, jointly conceived engineering effort. The pipeline developed here concentrates on the optimal-control core—the UF and FC formulations, the warm-start architecture and the numerical validation—and has

been built side by side with a companion bachelor thesis authored by Marc López, which focuses on the environmental and operational layer of the same Mission Management System: the No-Fly Zone (NFZ) logic, the Digital Elevation Model (DEM) terrain handling, the broader environment constraints and the full control problem they feed into. The two works share a single codebase and a single project objective, and the conclusions of this thesis should be read as the optimization-side contribution to a system whose geospatial and constraint-engine contribution is documented in that parallel work.

Within that shared development, the most immediate next step concerns the incorporation of analytical gradients into the formulation. At the time of writing, the analytical Jacobians for the cost and constraints have already been added to the problem, and preliminary measurements suggest an impressive reduction in computational time—approximately a factor of three faster than the current finite-difference pipeline in the UF formulation. The natural continuation is therefore to analyze rigorously how this gradient information propagates through the complete pipeline and how its implementation interacts with the warm-start and FC stages. The expected benefit is not only raw speed: exact derivatives should allow the number of collocation nodes N to grow without inflating the cost of each solve proportionally, improving the scaling of the problem and thereby enabling longer routes resolved at higher temporal and spatial detail without sacrificing the real-time threshold.

Beyond the gradient work, the project—developed for Singular Aircraft and their UAV platform, as introduced in Chapter 1—is scheduled to continue until December 2026 with a roadmap of further improvements and feature additions. These include the full incorporation of the wind field into the optimization, the exploitation of analytical Dubins use cases as additional warm-start primitives, and the modeling of specific mission maneuvers, among other refinements that extend the operational envelope of the planner.

The clearest longer-term objective is the migration of the framework from an offline solver to a real-time, in-the-loop workflow. The aim is to build a pipeline capable of ingesting the live state of the aircraft as input, executing the planner, and returning the optimized trajectory as output within the operational cycle. The ultimate goal is to couple the behavior of the autopilot directly with the planner running on the on-board computer, closing the loop between guidance and control so that the platform behaves as a fully intelligent agent throughout the mission—continuously re-planning against the current aircraft state and the evolving environment rather than tracking a fixed, pre-computed path. Taken together, these next steps trace a direct line from the validated offline solver delivered here to an autonomous, mission-aware aircraft.

References

[1] NASA Earth Science Data Systems. "Digital elevation / terrain model (dem)," National Aeronautics and Space Administration (NASA), Accessed: Mar. 3, 2026. [Online]. Available: https://www.earthdata.nasa.gov/topics/land-surface/digital-elevation-terrain-model-dem

[2] European Union Aviation Safety Agency. "Geo zones – know where to fly your drone." Official source for aviation safety information for the general public., EASA Light, Accessed: Mar. 3, 2025. [Online]. Available: https://www.easa.europa.eu/en/light/topics/geo-zones-know-where-fly-your-drone

[3] P. Lu, "Inverse dynamics approach to trajectory optimization for an aerospace plane," *Journal of Guidance, Control, and Dynamics*, vol. 16, no. 4, pp. 726–732, 1993. DOI: 10.2514/3.21073 Accessed: Mar. 12, 2026. [Online]. Available: https://arc.aiaa.org/doi/abs/10.2514/3.21073

[4] SESAR Joint Undertaking, "U-space blueprint and european drone outlook study," European Commission, Brussels, Belgium, Technical Report, 2023. Accessed: May 6, 2026. [Online]. Available: https://www.sesarju.eu/

[5] Federal Aviation Administration, "Faa aerospace forecast fiscal years 2024–2044," U.S. Department of Transportation, Washington, D.C., Forecast Report, 2024. Accessed: May 7, 2026. [Online]. Available: https://www.faa.gov/data_research/aviation/aerospace_forecasts/unmanned_aircraft_systems.pdf

[6] A. Nuić, D. Poles, and V. Mouillet, "BADA: An advanced aircraft performance model for present and future ATM systems," *International Journal of Adaptive Control and Signal Processing*, vol. 24, no. 10, pp. 850–866, 2010, Published online in Wiley InterScience. DOI: 10.1002/acs.1176 Accessed: Mar. 12, 2026. [Online]. Available: https://www.eurocontrol.int/sites/default/files/2019-03/overview-bada-apm.pdf

[7] P. E. Hart, N. J. Nilsson, and B. Raphael, "A formal basis for the heuristic determination of minimum cost paths," *IEEE Transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, 1968. DOI: 10.1109/TSSC.1968.300136 Accessed: May 4, 2026. [Online]. Available: https://doi.org/10.1109/TSSC.1968.300136

[8] S. M. LaValle, "Rapidly-exploring random trees: A new tool for path planning," Iowa State University, Technical Report TR 98-11, 1998. Accessed: Mar. 10, 2026. [Online]. Available: https://msl.cs.illinois.edu/~lavalle/papers/Lav98c.pdf

[9] Singular Aircraft. "Flyox i: Technical specifications and mission configurations," Accessed: Mar. 10, 2026. [Online]. Available: https://singularaircraft.com/flyox

[10] Singular Aircraft. "Flyox i: The largest amphibious unmanned aerial platform," Accessed: Mar. 13, 2026. [Online]. Available: https://singularaircraft.com/

[11] F. Santoso, M. A. Garratt, and S. G. Anavatti, "State-of-the-art integrated guidance and control systems in unmanned vehicles: A review," *IEEE Systems Journal*, vol. 15, no. 3, pp. 3312–3323, 2021. DOI: 10.1109/JSYST.2020.3007428 Accessed: May 5, 2026. [Online]. Available: https://doi.org/10.1109/JSYST.2020.3007428

[12] T. A. Johansen, "Introduction to nonlinear model predictive control and moving horizon estimation," Norwegian University of Science and Technology (NTNU), Technical Publication, 2011. Accessed: Mar. 13, 2026. [Online]. Available: https://torarnj.folk.ntnu.no/nonlinear.pdf

[13] Y. Zhang, H. Liang, and J. Li, "Research on UAV path planning based on improved Dijkstra's algorithm," *International Journal for Housing Science and Its Applications*, vol. 46, no. 3, pp. 7473–7482, Aug. 2025. DOI: 10.70517/ijhsa463641 Accessed: Mar. 21, 2026. [Online]. Available: https://doi.org/10.70517/ijhsa463641

[14] ScienceDirect, *Trajectory planning - an overview*, Computer Science Topics, 2024. Accessed: Mar. 21, 2026. [Online]. Available: https://www.sciencedirect.com/topics/computer-science/trajectory-planning

[15] J. Li, C. Yu, Z. Zhang, Z. Sheng, Z. Yan, X. Wu, et al., "Improved A-Star path planning algorithm in obstacle avoidance for the fixed-wing aircraft," *Electronics*, vol. 12, no. 24, p. 5047, 2023. DOI: 10.3390/electronics12245047 Accessed: Mar. 22, 2026. [Online]. Available: https://doi.org/10.3390/electronics12245047

[16] H. Choset, K. M. Lynch, S. Hutchinson, G. Kantor, L. E. Kavraki, and S. Thrun, *Probabilistic roadmap path planning*, Lecture notes based on *Principles of Robot Motion*, MIT Press., Columbia University, Department of Computer Science, 2015. Accessed: Mar. 22, 2026. [Online]. Available: https://www.cs.columbia.edu/~allen/F15/NOTES/Probabilisticpath.pdf

[17] G. Noh, J. Park, D. Han, et al., "Selective goal aiming rapidly exploring random tree path planning for UAVs," *International Journal of Aeronautical and Space Sciences*, vol. 22, no. 6, pp. 1397–1412, 2021. DOI: 10.1007/s42405-021-00406-7 Accessed: Mar. 22, 2026. [Online]. Available: https://doi.org/10.1007/s42405-021-00406-7

[18] S. P. Bradley, A. C. Hax, and J. F. Shapiro, "Applied mathematical programming," in Reading, Mass.: Addison-Wesley, 1977, ch. 13: Non-linear Programming, Available at MIT OpenCourseWare. Accessed: Mar. 22, 2026. [Online]. Available: https://web.mit.edu/15.053/www/AMP-Chapter-13.pdf

[19] A. Wächter and L. T. Biegler, "On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming," *Mathematical Programming*, vol. 106, no. 1, pp. 25–57, 2006. DOI: 10.1007/s10107-004-0559-y Accessed: May 8, 2026. [Online]. Available: https://doi.org/10.1007/s10107-004-0559-y

[20] A. Wächter, C. Laird, and COIN-OR Project, *Ipopt (interior point optimizer) documentation*, https://coin-or.github.io/Ipopt/, 2026. Accessed: Mar. 22, 2026.

[21] J. A. E. Andersson, J. Gillis, G. Horn, J. B. Rawlings, and M. Diehl, "CasADi – A software framework for nonlinear optimization and optimal control," *Mathematical Programming Computation*, vol. 11, no. 1, pp. 1–36, 2019. DOI: 10.1007/s12532-018-0139-4 Accessed: May 7, 2026. [Online]. Available: https://doi.org/10.1007/s12532-018-0139-4

[22] H. Chibueze, *Symbolic regression with expression trees*, Kaggle Code Repository, 2023. Accessed: Mar. 22, 2026. [Online]. Available: https://www.kaggle.com/code/henrychibueze/symbolic-regression-with-expression-trees

[23] H. Kopetz, *Real-Time Systems: Design Principles for Distributed Embedded Applications*, 2nd. New York: Springer, 2011, Reference for deterministic execution and memory constraints., ISBN: 978-1-4419-8237-7. DOI: 10.1007/978-1-4419-8237-7 Accessed: May 8, 2026. [Online]. Available: https://doi.org/10.1007/978-1-4419-8237-7

[24] R. Sambharya, G. Hall, B. Amos, and B. Stellato, "Learning to warm-start fixed-point optimization algorithms," *arXiv preprint*, vol. abs/2309.07835, 2023. arXiv: 2309.07835 [math.OC]. Accessed: May 9, 2026. [Online]. Available: https://arxiv.org/abs/2309.07835

[25] Emergent Mind. "Warm-start initialization," Emergent Mind, Accessed: Mar. 22, 2026. [Online]. Available: https://www.emergentmind.com/topics/warm-start-initialization

[26] I. Lugo-Cardenas, G. Flores, S. Salazar, and R. Lozano, "Dubins path generation for a fixed wing UAV," in *2014 International Conference on Unmanned Aircraft Systems (ICUAS)*, IEEE, 2014, pp. 339–346. DOI: 10.1109/ICUAS.2014.6842272 Accessed: May 5, 2026.

[27] T. Fraichard and A. Scheuer, "From Reeds and Shepp's to continuous-curvature paths," *IEEE Transactions on Robotics*, vol. 20, no. 6, pp. 1025–1035, Dec. 2004. DOI: 10.1109/TRO.2004.833789 Accessed: May 10, 2026. [Online]. Available: https://inria.hal.science/inria-00000009

[28] J. Li, J. Lou, Y. Li, S. Pan, and Y. Xu, "Trajectory tracking of autonomous vehicle using Clothoid curve," *Applied Sciences*, vol. 13, no. 4, p. 2733, 2023, Art. no. 2733. DOI: 10.3390/app13042733 Accessed: May 9, 2026. [Online]. Available: https://www.mdpi.com/2076-3417/13/4/2733

[29] M. Davenport, "The Karush–Kuhn–Tucker (KKT) conditions," *Georgia Tech ECE 3803 Course Notes*, Nov. 2021. Accessed: Mar. 22, 2026. [Online]. Available: https://mdav.ece.gatech.edu/ece-3803-fall2021/notes/17-notes-3803-f21.pdf

[30] A. Albert, L. Imsland, and J. Haugen, "Numerical optimal control mixing collocation with single shooting: A case study," *IFAC-PapersOnLine*, vol. 49, no. 7, pp. 290–295, 2016,

11th IFAC Symposium on Dynamics and Control of Process Systems (DYCOPS-CAB). DOI: 10.1016/j.ifacol.2016.07.307 Accessed: Mar. 22, 2026. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2405896316305146

[31] S. Li, Z. Wang, A. Narayan, R. Kirby, and S. Zhe, *Meta-learning with adjoint methods*, arXiv preprint., 2021. arXiv: 2110.08432 [cs.LG]. Accessed: May 6, 2026. [Online]. Available: https://arxiv.org/abs/2110.08432

[32] A. M. Bradley, "PDE-constrained optimization and the adjoint method," Stanford University, Technical Tutorial, 2010. Accessed: Mar. 22, 2026. [Online]. Available: https://cs.stanford.edu/~ambrad/adjoint_tutorial.pdf

[33] H. C. Elman and G. H. Golub, "Inexact and preconditioned Uzawa algorithms for saddle point problems," *SIAM Journal on Numerical Analysis*, vol. 31, no. 6, pp. 1645–1673, 1994, Dedicated to Seymour Parter. DOI: 10.1137/0731084 Accessed: May 10, 2026. [Online]. Available: https://www.cs.umd.edu/~elman/papers/inexact-uzawa.pdf

[34] J. T. Betts, *Practical Methods for Optimal Control and Estimation Using Nonlinear Programming* (Advances in Design and Control 19), 2nd. Philadelphia, PA: Society for Industrial and Applied Mathematics (SIAM), 2010, ISBN: 978-0-89871-688-7. DOI: 10.1137/1.9780898718577 Accessed: May 16, 2026.

[35] C. R. Hargraves and S. W. Paris, "Direct trajectory optimization using nonlinear programming and collocation," *Journal of Guidance, Control, and Dynamics*, vol. 10, no. 4, pp. 338–342, 1987. DOI: 10.2514/3.20223 Accessed: May 14, 2026.

[36] J. Nocedal and S. J. Wright, *Numerical Optimization* (Springer Series in Operations Research and Financial Engineering), 2nd. New York: Springer, 2006, ISBN: 978-0-387-30303-1. DOI: 10.1007/978-0-387-40065-5 Accessed: May 13, 2026.

[37] L. T. Biegler, *Nonlinear Programming: Concepts, Algorithms, and Applications to Chemical Processes* (MOS-SIAM Series on Optimization). Philadelphia, PA: Society for Industrial and Applied Mathematics (SIAM), 2010, ISBN: 978-0-89871-702-0. DOI: 10.1137/1.9780898719383 Accessed: May 13, 2026.

[38] A. Forsgren, P. E. Gill, and M. H. Wright, "Interior methods for nonlinear optimization," *SIAM Review*, vol. 44, no. 4, pp. 525–597, 2002. DOI: 10.1137/S0036144502414942 Accessed: May 14, 2026.

[39] H. Y. Benson, D. F. Shanno, and R. J. Vanderbei, "A comparative study of large-scale nonlinear optimization algorithms," in *High Performance Algorithms and Software for Nonlinear Optimization*, G. Di Pillo and A. Murli, Eds., Boston, MA: Kluwer Academic Publishers, 2003, pp. 95–127. DOI: 10.1007/978-1-4613-0241-4_5 Accessed: May 16, 2026.

[40] E. A. Yildirim and S. J. Wright, "Warm-start strategies in interior-point methods for linear programming," *SIAM Journal on Optimization*, vol. 12, no. 3, pp. 782–810, 2002. DOI: 10.1137/S1052623400369235 Accessed: May 17, 2026.

[41] D. C. Liu and J. Nocedal, "On the limited memory BFGS method for large scale optimization," *Mathematical Programming*, vol. 45, no. 1–3, pp. 503–528, 1989, Seminal reference for the L-BFGS quasi-Newton update used by interior-point solvers when exact Hessians are unavailable. DOI: 10.1007/BF01589116 Accessed: May 11, 2026. [Online]. Available: https://doi.org/10.1007/BF01589116

[42] A. Griewank and A. Walther, *Evaluating Derivatives: Principles and Techniques of Algorithmic Differentiation*, 2nd. Philadelphia, PA: Society for Industrial and Applied Mathematics (SIAM), 2008, ISBN: 978-0-89871-659-7. DOI: 10.1137/1.9780898717761 Accessed: May 15, 2026.

[43] J. D. Anderson, *Aircraft Performance and Design*. Boston: McGraw-Hill, 1999, Standard reference for the parabolic drag polar, Oswald efficiency factor and point-mass performance equations., ISBN: 978-0-070-01971-3.

[44] B. Etkin and L. D. Reid, *Dynamics of Flight: Stability and Control*, 3rd. New York: John Wiley & Sons, 1996, Canonical reference for stability and control derivatives, body- and wind-axis derivations, and aircraft Newton–Euler equations of motion., ISBN: 978-0-471-03418-6.

[45] A. V. Rao, "A survey of numerical methods for optimal control," *Advances in the Astronautical Sciences*, vol. 135, no. 1, pp. 497–528, 2009. Accessed: May 13, 2026. [Online]. Available: http://www.anilvrao.com/Publications/ConferencePublications/trajectorySurveyAAS.pdf

[46] R. T. Marler and J. S. Arora, "Survey of multi-objective optimization methods for engineering," *Structural and Multidisciplinary Optimization*, vol. 26, no. 6, pp. 369–395, 2004. DOI: 10.1007/s00158-003-0368-6 Accessed: May 17, 2026.

[47] A. N. Tikhonov and V. Y. Arsenin, *Solutions of Ill-Posed Problems*. Washington, DC: V. H. Winston & Sons, 1977, Translated by Fritz John from the 1974 Russian edition., ISBN: 978-0-470-99124-4.

[48] D. Dolgov, S. Thrun, M. Montemerlo, and J. Diebel, "Path planning for autonomous vehicles in unknown semi-structured environments," *The International Journal of Robotics Research*, vol. 29, no. 5, pp. 485–501, 2010, Introduces the Hybrid-state A* algorithm with kinematic constraints. DOI: 10.1177/0278364909359210 Accessed: May 11, 2026. [Online]. Available: https://doi.org/10.1177/0278364909359210

[49] P. Sharma, J. Y. M. Goh, B. Açıkmese, and F. Djeumou. "Transcription-induced failure modes in 6-DOF rocket landing trajectory optimization." arXiv: 2605.08420 [math.OC], Accessed: May 16, 2026. [Online]. Available: https://arxiv.org/abs/2605.08420 preprint.

[50] B. L. Stevens, F. L. Lewis, and E. N. Johnson, *Aircraft Control and Simulation: Dynamics, Controls Design, and Autonomous Systems*, 3rd. Hoboken, NJ: John Wiley & Sons, 2015, ISBN: 978-1-118-87098-3. DOI: 10.1002/9781119174882 Accessed: May 2, 2026. [Online]. Available: https://doi.org/10.1002/9781119174882

[51] M. P. Kelly, "An introduction to trajectory optimization: How to do your own direct collocation," *SIAM Review*, vol. 59, no. 4, pp. 849–904, 2017. DOI: 10.1137/16M1062569 Accessed:

May 16, 2026. [Online]. Available: https://www.matthewpeterkelly.com/research/MatthewKelly_IntroTrajectoryOptimization_SIAM_Review_2017.pdf

[52] C. Goerzen, Z. Kong, and B. Mettler, "A survey of motion planning algorithms from the controls, robotics, and cognitive science communities," *Journal of Intelligent & Robotic Systems*, vol. 57, no. 1–4, pp. 65–100, 2010. DOI: 10.1007/s10846-009-9383-1 Accessed: May 4, 2026.

[53] M. Diehl. "Direct single and direct multiple shooting methods," University of Freiburg, Department of Microsystems Engineering (IMTEK), Accessed: Apr. 10, 2026. [Online]. Available: https://www.imtek.de/professuren/systemtheorie/events/dateien/directshootingmethods

[54] VectorNav Technologies. "Inertial navigation primer: Reference frames," VectorNav Technologies, Accessed: Apr. 12, 2026. [Online]. Available: https://www.vectornav.com/resources/inertial-navigation-primer/math-fundamentals/math-refframes

[55] International Organization for Standardization, *ISO 2533:1975 — standard atmosphere*, International Standard, Defines the International Standard Atmosphere (ISA) used to compute air density as a function of geometric altitude., Geneva, Switzerland, 1975. Accessed: May 3, 2026. [Online]. Available: https://www.iso.org/standard/7472.html

[56] D. E. Hoak and R. D. Finck, "USAF stability and control DATCOM," Air Force Wright Aeronautical Laboratories, Wright-Patterson AFB, Tech. Rep. AFFDL-TR-79-3032, 1978, Revised. Accessed: May 18, 2026. [Online]. Available: https://apps.dtic.mil/sti/citations/ADB072483

[57] P. E. Gill, W. Murray, and M. H. Wright, *Practical Optimization*. London: Academic Press, 1981, ISBN: 978-0-12-283950-4. Accessed: May 15, 2026. [Online]. Available: https://archive.org/details/practicaloptimiz0000gill

[58] A. R. Curtis, M. J. D. Powell, and J. K. Reid, "On the estimation of sparse Jacobian matrices," *Journal of the Institute of Mathematics and Its Applications*, vol. 13, pp. 117–119, 1974. Accessed: May 16, 2026. [Online]. Available: https://academic.oup.com/imamat/article-abstract/13/1/117/719659

[59] S. M. LaValle, *Planning Algorithms*. Cambridge: Cambridge University Press, 2006, ISBN: 978-0-521-86205-9. DOI: 10.1017/CBO9780511546877 Accessed: May 18, 2026. [Online]. Available: https://lavalle.pl/planning/

[60] S. Zhang, Q. Guo, W. Liu, H. Kang, D. Wu, and H. Zhou, "Hybrid A-Star path planning method based on directional guidance and segment-based heuristic weighting," *Applied Sciences*, vol. 15, no. 22, p. 12 298, 2025. DOI: 10.3390/app152212298 Accessed: May 25, 2026. [Online]. Available: https://doi.org/10.3390/app152212298

[61] L. E. Dubins, "On curves of minimal length with a constraint on average curvature, and with prescribed initial and terminal positions and tangents," *American Journal of Mathematics*, vol. 79, no. 3, pp. 497–516, 1957. DOI: 10.2307/2372560 Accessed: May 19, 2026.

[62] National Imagery and Mapping Agency, "Department of defense world geodetic system 1984: Its definition and relationships with local geodetic systems," National Imagery and Mapping Agency (NIMA), Bethesda, MD, USA, Tech. Rep. TR8350.2, 2000, Definition of the WGS-84 reference ellipsoid (semi-major axis a, flattening f). Accessed: May 12, 2026. [Online]. Available: https://earth-info.nga.mil/php/download.php?file=coord-wgs84

[63] B. R. Bowring, "Transformation from spatial to geographical coordinates," *Survey Review*, vol. 23, no. 181, pp. 323–327, 1976, Closed-form initial estimate that drives the iterative latitude solver used in the ECEF→geodetic conversion. DOI: 10.1179/sre.1976.23.181.323 Accessed: May 12, 2026.

Appendix A

Derivation of the Equations of Motion

To analyze the optimal trajectory of the aircraft, we must first establish the relationship between the forces acting on the vehicle and its resultant motion. This process begins with the general Newton-Euler equations in the Body-Fixed frame and concludes with the Wind-Axis formulation required for performance optimization.

A.1 General Body-Axis Dynamics

The motion of an aircraft is governed by the conservation of linear and angular momentum [44], [50]. In a body-fixed reference frame $\mathcal{F}_b$ (with axes x_b, y_b, z_b), the absolute velocity is defined as $\vec{V} = (u, v, w)^T$ and the angular velocity as $\vec{\omega} = (p, q, r)^T$.

Applying Newton's Second Law, the translational dynamics are described by the following scalar equations:

$$\begin{aligned} F_x &= m(\dot{u} - rv + qw) \\ F_y &= m(\dot{v} + ru - pw) \\ F_z &= m(\dot{w} - qu + pv) \end{aligned} \tag{A.1}$$

The external force $\vec{F}$ is the sum of aerodynamic ($\vec{F}_A$), propulsive ($\vec{F}_T$), and gravitational ($\vec{F}_G$) components. Projecting the weight mg from the local horizon frame to the body frame using the pitch (θ) and roll (ϕ) angles, we obtain:

$$\begin{aligned} m(\dot{u} - rv + qw) &= F_{Ax} + F_{Tx} - mg\sin\theta \\ m(\dot{v} + ru - pw) &= F_{Ay} + F_{Ty} + mg\cos\theta\sin\phi \\ m(\dot{w} - qu + pv) &= F_{Az} + F_{Tz} + mg\cos\theta\cos\phi \end{aligned} \tag{A.2}$$

A.2 Transition to the Wind-Axis Frame

For trajectory optimization, we utilize the *Wind-Axis Frame* $\mathcal{F}_w$ [44]. In this frame, the x_w axis is aligned with the velocity vector $\vec{V}$, such that the velocity components become $(V, 0, 0)^T$.

A.2.1 Reference frames and rotation angles

Throughout this annex, the **body frame** $\mathcal{F}_b$ follows the usual aircraft convention: x_b forward (fuselage reference), y_b out the right wing, z_b downward (right-handed). The **wind frame** $\mathcal{F}_w$ has x_w along $\vec{V}$, z_w contained in the plane of symmetry of the aircraft (when $\beta = 0$ it coincides with the body symmetry plane), and y_w completing a right-handed system (positive out the right wing when viewed from behind).

The orientation of $\mathcal{F}_b$ relative to $\mathcal{F}_w$ is parametrized by the *angle of attack* α and the *sideslip angle* β [44]. Equivalently, $\mathcal{F}_w$ is obtained from $\mathcal{F}_b$ by a sequence of two rotations that align x_b with $\vec{V}$: first a rotation about z_b through β (sideslip), then a rotation about the intermediate lateral axis through α (angle of attack). The same direction-cosine matrix maps any bound vector (aerodynamic force, thrust, etc.) from body axes to wind axes.

A more detailed description can be obtained by the reader through the Figure A.1.

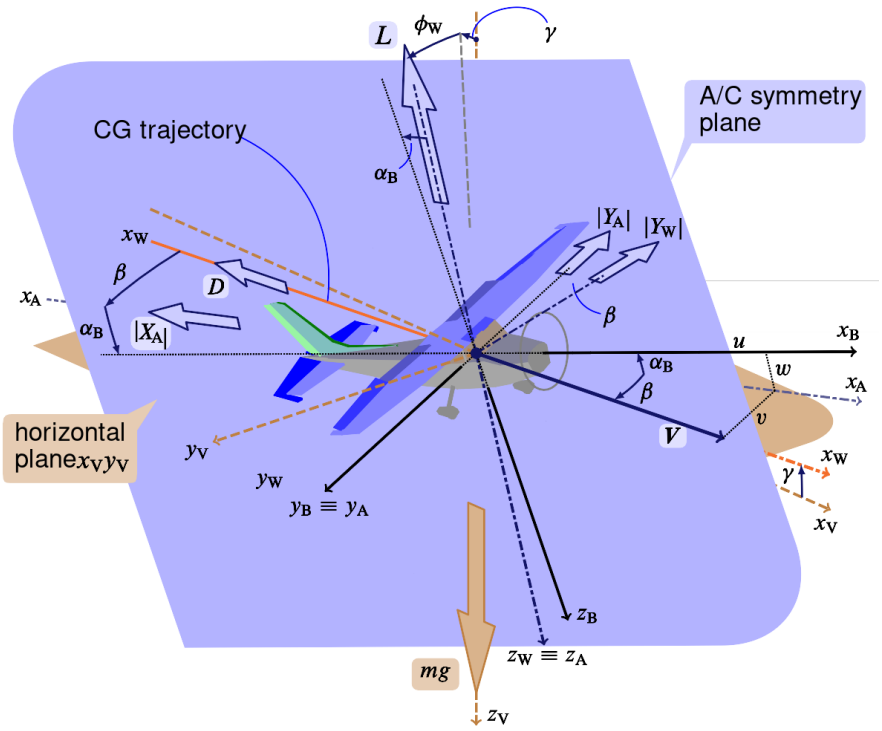


Figure A.1: Body ($\mathcal{F}_B$), aerodynamic ($\mathcal{F}_A$) and wind ($\mathcal{F}_W$) frames of the aircraft, with the angle of attack α_B, the sideslip angle β, the flight-path angle γ, and the aerodynamic forces L, D and the weight mg.

A.2.2 From body-axis forces to drag, sideforce, and lift

Let $\vec{F}_A^{\,b} = (F_{Ax},\, F_{Ay},\, F_{Az})^T$ denote the *aerodynamic* resultant in body axes (force exerted by the air on the aircraft, opposite in sign to the usual "force on the fluid" convention used in some wind-tunnel reports—here F_{Ax} is the axial component in the aircraft-fixed direction). The corresponding components in wind axes $\vec{F}_A^{\,w} = (X_w,\, Y_w,\, Z_w)^T$ are obtained by the orthogonal transformation $\vec{F}_A^{\,w} = \mathbf{R}_{b\to w}\, \vec{F}_A^{\,b}$, with

$$\mathbf{R}_{b\to w}(\alpha,\beta) = \begin{bmatrix} \cos\alpha\cos\beta & \sin\beta & \sin\alpha\cos\beta \\ -\cos\alpha\sin\beta & \cos\beta & -\sin\alpha\sin\beta \\ -\sin\alpha & 0 & \cos\alpha \end{bmatrix}. \tag{A.3}$$

Expanded,

$$X_w = F_{Ax}\cos\alpha\cos\beta + F_{Ay}\sin\beta + F_{Az}\sin\alpha\cos\beta, \tag{A.4}$$

$$Y_w = -F_{Ax}\cos\alpha\sin\beta + F_{Ay}\cos\beta - F_{Az}\sin\alpha\sin\beta, \tag{A.5}$$

$$Z_w = -F_{Ax}\sin\alpha + F_{Az}\cos\alpha. \tag{A.6}$$

The *scalar* aerodynamic coefficients used in performance work refer to the wind-axis decomposition with x_w along the flight path and z_w positive *down*. Under this convention, **Drag** D (positive magnitude, opposing $\vec{V}$) satisfies $X_w = -D$; **Sideforce** Y (positive to the pilot's right) satisfies $Y_w = Y$; and **Lift** L (positive upward, normal to $\vec{V}$ in the mean plane of symmetry) satisfies $Z_w = -L$. Thus

$$D = -X_w, \qquad Y = Y_w, \qquad L = -Z_w, \tag{A.7}$$

with (X_w, Y_w, Z_w) from (A.4)–(A.6). Substituting (A.4)–(A.6) into (A.7) expresses (D, Y, L) directly as linear combinations of (F_{Ax}, F_{Ay}, F_{Az}).

A.2.3 Thrust from body axes to wind axes

The propulsive resultant $\vec{F}_T^{\,b} = (F_{Tx},\, F_{Ty},\, F_{Tz})^T$ is fixed in the airframe (e.g. aligned with x_b for a podded engine with cant angle folded into the model). It transforms as any other bound vector:

$$\vec{F}_T^{\,w} = \mathbf{R}_{b\to w}(\alpha,\beta)\, \vec{F}_T^{\,b}. \tag{A.8}$$

Typical compact notation for the magnitude along the velocity direction uses the *thrust line* relative to body and wind: if the engine is installed with thrust angle corrections $(\epsilon_T,\, \nu_T)$ in body axes, those angles are incorporated into $\vec{F}_T^{\,b}$ before (A.8); the trajectory equations in Section A.3 then use the wind-axis scalars $T\cos\epsilon\cos\nu$, etc., which are shorthand for the projection of $\vec{F}_T^{\,w}$ onto (x_w, y_w, z_w) once α, β, and installation angles are fixed.

A.2.4 Weight and other forces

Gravitational acceleration is most naturally expressed in the local **North–East–Down (NED)** or **local horizontal** frame (vertical along the geoid normal). To combine weight with (A.3), one applies the chain $\mathcal{F}_{\text{NED}} \to \mathcal{F}_b \to \mathcal{F}_w$ using the aircraft attitude Euler angles (ϕ, θ, ψ) for the first leg and (A.3) for the second. Aerodynamic and thrust vectors use (A.3) directly; gravity does not.

A.2.5 Summary

The *same* rotation $\mathbf{R}_{b\to w}(\alpha, \beta)$ maps aerodynamic and propulsive resultants from body to wind. The aerodynamic scalars (D, Y, L) are not the body components; they are the wind-axis splits defined by (A.7). A change of reference is therefore: **(i)** express the physical force in $\mathcal{F}_b$; **(ii)** rotate with $\mathbf{R}_{b\to w}$ to obtain (X_w, Y_w, Z_w); **(iii)** identify (D, Y, L) via (A.7). Thrust follows the identical rotation (A.8).

To transition from the Body-Axis to the Wind-Axis at the level of the reduced trajectory model, we apply the following substitutions based on the assumption that the aircraft's rotation is controlled to maintain the desired flight path:

Table A.1: Substitutions for Wind-Axis Formulation. Wind-axis aerodynamic components follow (A.3)–(A.7).

Variable	Body Axis	Wind Axis Substitution
Velocity	(u, v, w)	$(V, 0, 0)$
Angular Rates	(p, q, r)	(p_w, q_w, r_w)
Aerodynamics	(F_{Ax}, F_{Ay}, F_{Az})	$(X_w, Y_w, Z_w) = (-D,\ Y,\ -L)$
Orientation	(ψ, θ, ϕ)	(χ, γ, μ)

The angular velocity components in the wind frame (p_w, q_w, r_w) are related to the trajectory angles (heading χ, flight path γ, and bank μ) by:

$$\begin{aligned} p_w &= \dot{\mu} - \dot{\chi}\sin\gamma \\ q_w &= \dot{\gamma}\cos\mu + \dot{\chi}\cos\gamma\sin\mu \\ r_w &= -\dot{\gamma}\sin\mu + \dot{\chi}\cos\gamma\cos\mu \end{aligned} \tag{A.9}$$

A.3 Final Trajectory Equations (UF Formulation)

By substituting the wind-axis angular rates into the dynamic force equations and assuming a point-mass model, we derive the system used for the Uncoupled (UF) formulation.

A.3.1 Dynamic Equations

The acceleration along the velocity vector and the curvature of the flight path are given by:

$$\dot{V} = \frac{T\cos\epsilon\cos\nu - D - mg\sin\gamma}{m} \tag{A.10}$$

$$\dot{\gamma} = \frac{L\cos\mu + T\sin(\epsilon+\alpha)\cos\mu - mg\cos\gamma}{mV} \tag{A.11}$$

$$\dot{\chi} = \frac{L\sin\mu + T\sin(\epsilon+\alpha)\sin\mu}{mV\cos\gamma} \tag{A.12}$$

In standard performance studies, the thrust angle of attack ϵ and sideslip ν are considered small, so $\cos\epsilon \approx 1$ and $\cos\nu \approx 1$, which simplifies these expressions considerably.

A.3.2 Kinematic Equations

The change in geodetic position in the North-East-Down (NED) frame is:

$$\begin{aligned} \dot{x}_e &= V\cos\gamma\cos\chi \\ \dot{y}_e &= V\cos\gamma\sin\chi \\ \dot{z}_e &= -V\sin\gamma \quad \text{(where altitude } h = -z_e\text{)} \end{aligned} \tag{A.13}$$

A.3.3 Mass and Energy Relation

Finally, we account for the consumption of energy (fuel or electricity). For a fuel-based system with mass flow rate φ:

$$\dot{m} = -\varphi(h, V, T) \tag{A.14}$$

A.4 Problem Statement and Degrees of Freedom

The resulting system consists of 7 state variables $(x_e, y_e, h, V, \chi, \gamma, m)$ and 6 potential control variables $(\alpha, \beta, \mu, \epsilon, \nu, \pi)$. In a standard uncoupled control problem, we fix the engine geometry (ϵ, ν) and focus on the maneuverability controls (α, μ, π), resulting in a system with 3 mathematical degrees of freedom ($N = 8$ variables $-$ 5 equations in a vertical plane, or $N = 13 - 7 = 6$ in full 3D flight).

A.4.1 Comparison of the Two Formulations

The Uncoupled Formulation (UF, denoted UF in Chapter 5.2) can be interpreted as a reduced version of the Full Control (FC) model of Section 5.3. In UF/UF, the rigid-body rotational dynamics are not integrated explicitly; instead, the model is rewritten in the wind-axis frame and trajectory-level variables are propagated directly.

By contrast, FC keeps the complete Newton–Euler structure for a 6-DoF rigid body in body axes: three translational equations plus three rotational equations, together with kinematic attitude propagation and actuator-level controls. Hence, UF/UF prioritises reduced complexity and faster optimisation, while FC prioritises physical fidelity and direct control-surface consistency.

Appendix B

Coordinate Frame Transformations

B.1 Motivation

Trajectory optimisers for fixed-wing aircraft almost always integrate the equations of motion in a *local* Cartesian frame—here the North–East–Down (NED) frame fixed to a reference point on the Earth—because the dynamics are written in metres per second, load factors, and bank angles that are naturally expressed in locally flat coordinates [50], [54]. Mission data from operators and avionics, by contrast, are typically specified in geodetic latitude φ, longitude λ, and altitude h above the WGS-84 ellipsoid [62]. Bridging these representations is therefore a prerequisite for ingesting waypoints, initial conditions, and terminal constraints, and for exporting optimised trajectories back to global coordinates.

The same split appears in software architecture: outer layers (mission databases, map displays, ATC interfaces) reason in degrees and metres above ellipsoid, while the nonlinear programming kernel and collocation defects are assembled in NED metres relative to a user-chosen origin. Keeping conversions explicit and bidirectional avoids silent duplication of Earth-radius constants and documents the vertical datum assumed in the cost and constraint functions.

The local tangent-plane model trades exact global geometry for computational simplicity: straight lines in NED are slightly curved on the ellipsoid, and verticals from two widely separated points are not strictly parallel. Over a sufficiently small operational patch—typical for tactical UAS legs—NED errors remain small compared with sensor noise and wind-model error, while retaining millimetre-level algebraic consistency if the ellipsoid and rotation formulas below are implemented in double precision. When missions span hundreds of kilometres, the curvature correction of Section 5 becomes relevant; for global validity one would eventually replace the single tangent plane with multiple patches or work directly in ECEF, at higher implementation cost.

B.2 Reference ellipsoid: WGS-84

The World Geodetic System 1984 (WGS-84) defines an oblate ellipsoid of revolution with equatorial semi-major axis [62]

$$a = 6\,378\,137.0 \text{ m}, \tag{B.1}$$

flattening

$$f = \frac{1}{298.257\,223\,563}, \tag{B.2}$$

and (first) eccentricity squared

$$e^2 = 2f - f^2, \tag{B.3}$$

which follows from the geometric definition $e^2 = (a^2 - b^2)/a^2$ with polar semi-minor axis $b = a(1-f)$. The ellipsoid is a surface of revolution: every meridian is an ellipse with equatorial radius a and polar radius b, while parallels are circles.

The *prime vertical radius of curvature* at geodetic latitude φ is

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}. \tag{B.4}$$

Geometrically, $N(\varphi)$ is the distance from the surface point along the *ellipsoid normal* to its intersection with the polar (rotation) axis; it appears whenever a point is displaced from the ellipsoid along that normal. It is distinct from the meridian radius of curvature $M(\varphi)$; only $N(\varphi)$ enters the standard ECEF conversion used here.

B.3 Geodetic to ECEF (*toECEF*)

Let φ denote geodetic latitude, λ longitude (positive east), and h ellipsoidal height (measured along the normal above the ellipsoid). The Earth-Centred Earth-Fixed (ECEF) Cartesian coordinates $(X, Y, Z)^T$ (origin at Earth's centre, Z along the North pole, X through $\lambda = 0$) are

$$X = \big(N(\varphi) + h\big) \cos\varphi \cos\lambda, \tag{B.5}$$

$$Y = \big(N(\varphi) + h\big) \cos\varphi \sin\lambda, \tag{B.6}$$

$$Z = \big(N(\varphi)(1 - e^2) + h\big) \sin\varphi. \tag{B.7}$$

The factor $(1 - e^2)$ in the Z term shortens the normal intercept on the polar axis for an oblate ellipsoid: without it, the construction would correspond to a sphere. The X and Y components share the same radial distance $N + h$ in the equatorial plane and differ only through $\cos\lambda$ and $\sin\lambda$, reflecting longitude as a rotation about the Z axis.

B.4 Geodetic to NED (*geodeticToNed*)

The conversion from geodetic coordinates to local NED is performed by routing both points through ECEF as a common intermediate representation. A reference point $(\varphi_{\text{ref}}, \lambda_{\text{ref}}, h_{\text{ref}})$ and the target point are each converted to ECEF using the formulas of the previous section, yielding position vectors $\mathbf{r}_{\text{ref}}$ and $\mathbf{r}_{\text{tgt}}$. Their displacement

$$\Delta\mathbf{r} = \mathbf{r}_{\text{tgt}} - \mathbf{r}_{\text{ref}} = \begin{bmatrix} \Delta X & \Delta Y & \Delta Z \end{bmatrix}^T \tag{B.8}$$

is expressed in geocentric Cartesian coordinates and must be rotated into the local frame tangent to the ellipsoid at the reference point. Setting $\varphi \equiv \varphi_{\text{ref}}$ and $\lambda \equiv \lambda_{\text{ref}}$, the ECEF-to-NED direction-cosine matrix is

$$\mathbf{R}_{\text{ECEF}\to\text{NED}}(\varphi, \lambda) = \begin{bmatrix} -\sin\varphi\cos\lambda & -\sin\varphi\sin\lambda & \cos\varphi \\ -\sin\lambda & \cos\lambda & 0 \\ -\cos\varphi\cos\lambda & -\cos\varphi\sin\lambda & -\sin\varphi \end{bmatrix}, \tag{B.9}$$

so that $[n,\ e,\ d]^T = \mathbf{R}_{\text{ECEF}\to\text{NED}}\,\Delta\mathbf{r}$, i.e.

$$n = -\sin\varphi\cos\lambda\,\Delta X - \sin\varphi\sin\lambda\,\Delta Y + \cos\varphi\,\Delta Z, \tag{B.10}$$

$$e = -\sin\lambda\,\Delta X + \cos\lambda\,\Delta Y, \tag{B.11}$$

$$d = -\cos\varphi\cos\lambda\,\Delta X - \cos\varphi\sin\lambda\,\Delta Y - \sin\varphi\,\Delta Z. \tag{B.12}$$

The rows are the unit vectors North, East, and Down expressed in ECEF; the Down axis points along the local vertical. In this document the ellipsoidal normal is used; separation between normal and plumb line (deflection of the vertical) is ignored, which is adequate for trajectory planning unless sub-metre geoid fidelity is required.

The matrix $\mathbf{R}_{\text{ECEF}\to\text{NED}}$ is orthogonal: its rows are orthonormal in ECEF, and $\det\mathbf{R} = +1$ for the axis ordering chosen (North–East–Down in that sequence).

In the implementation used with the trajectory model, the *Down* coordinate equals negative target altitude expressed in the NED sense: if h is physical altitude (positive upward), then $d = -h_{\text{tgt}}$ when the reference is on the ellipsoid at the same vertical datum, because NED z is positive downward while altitude is positive upward.

B.5 Altitude curvature correction

Over long horizontal distances, treating the Earth as a flat tangent plane introduces a vertical error: two points at the same ellipsoidal height but separated along the surface follow a slightly curved normal. A

practical correction uses the horizontal ground distance in NED,

$$d_{\text{horiz}} = \sqrt{n^2 + e^2}, \tag{B.13}$$

an approximate subtended angle (small-angle approximation)

$$\theta \approx \frac{d_{\text{horiz}}}{N(\varphi_{\text{ref}})}, \tag{B.14}$$

and a correction added to the Down component,

$$h_{\text{corr}} = N(\varphi_{\text{ref}})\big(1 - \cos\theta\big). \tag{B.15}$$

Without this term, the flat-Earth vertical error grows roughly with the sagitta of the Earth slice: for $d_{\text{horiz}} \gtrsim 50$ km the uncorrected Down component can drift on the order of 10 m or more relative to the ellipsoid, whereas shorter legs ($\lesssim$ 10–20 km) usually remain sub-metre in the vertical for many applications. The correction is inexpensive relative to a full geodesic solver and stabilises consistency when the same code must serve both short tactical profiles and occasional long transits.

B.6 NED to geodetic (*nedToGeodetic*)

The inverse mapping recovers geodetic coordinates from local NED offsets by reversing the two-stage ECEF route. Let N, E, and D_{corr} denote North, East, and Down with the curvature correction included in the Down component where applicable. Since $\mathbf{R}_{\text{ECEF}\to\text{NED}}$ is proper orthogonal, the ECEF displacement is recovered by transposing the rotation matrix:

$$\Delta\mathbf{r} = \mathbf{R}_{\text{ECEF}\to\text{NED}}^{T} \begin{bmatrix} N \\ E \\ D_{\text{corr}} \end{bmatrix}, \tag{B.16}$$

which expands as

$$\Delta X = -\sin\varphi\cos\lambda\, N - \sin\lambda\, E - \cos\varphi\cos\lambda\, D_{\text{corr}}, \tag{B.17}$$

$$\Delta Y = -\sin\varphi\sin\lambda\, N + \cos\lambda\, E - \cos\varphi\sin\lambda\, D_{\text{corr}}, \tag{B.18}$$

$$\Delta Z = \cos\varphi\, N - \sin\varphi\, D_{\text{corr}}. \tag{B.19}$$

The absolute ECEF position then follows as $\mathbf{r}_{\text{tgt}} = \mathbf{r}_{\text{ref}} + \Delta\mathbf{r}$, and the geodetic coordinates are recovered from this Cartesian vector by inverting the ellipsoidal map. Writing $\mathbf{r}_{\text{tgt}} = (X, Y, Z)^T$ and $p = \sqrt{X^2 + Y^2}$, longitude is

$$\lambda = \operatorname{atan2}(Y, X). \tag{B.20}$$

Latitude and height follow from an iterative Bowring/Vermeille-type scheme [63]. Initialise

$$\varphi_0 = \operatorname{atan2}\!\left(\frac{Z}{p},\; 1 - e^2\right). \tag{B.21}$$

For $i = 0, 1, \ldots$ until $|\varphi_{i+1} - \varphi_i| < 10^{-12}$ rad (typically fewer than five iterations), compute

$$N_i = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi_i}}, \tag{B.22}$$

$$h = \frac{p}{\cos \varphi_i} - N_i, \tag{B.23}$$

$$\varphi_{i+1} = \operatorname{atan2}\!\left(\frac{Z}{p},\; 1 - \frac{e^2 N_i}{N_i + h}\right). \tag{B.24}$$

Then $\varphi = \varphi_{i+1}$ and h is the converged ellipsoidal height. With the 10^{-12} rad tolerance, residual errors are at the millimetre level in horizontal position for typical Earth radii when all arithmetic is performed in IEEE double precision.

B.7 Algorithm schema

Figure B.1 summarises the forward and inverse data flow used in software.

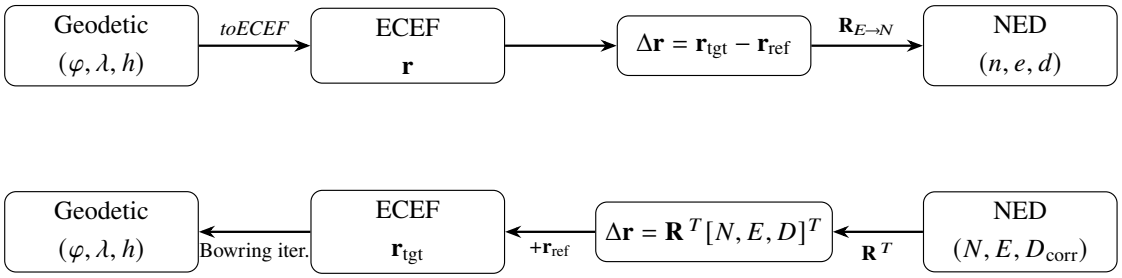


Figure B.1: Forward path (top): geodetic coordinates for reference and target are converted to ECEF, differenced, and projected with $\mathbf{R}_{\text{ECEF}\to\text{NED}}$ evaluated at the reference. Inverse path (bottom): NED offsets are mapped back through $\mathbf{R}^{\,T}$, referenced to ECEF, and converted to geodetic latitude, longitude, and height.

B.8 Summary table

Table B.1: Reference conversion paths between geodetic coordinates, ECEF, and local NED.

Conversion	Input	Intermediate	Output	Key formula
Geodetic → ECEF	φ, λ, h	$N(\varphi)$	X, Y, Z	$X = (N+h)\cos\varphi\cos\lambda$, etc.
Geodetic → NED	$(\varphi,\lambda,h)_{\text{ref}}$, $(\varphi,\lambda,h)_{\text{tgt}}$	$\mathbf{r}_{\text{ref}}$, $\mathbf{r}_{\text{tgt}}$; $\Delta\mathbf{r}$	n, e, d	$[n,e,d]^T = \mathbf{R}_{\text{ECEF}\to\text{NED}}(\varphi_{\text{ref}},\lambda_{\text{ref}})\,\Delta\mathbf{r}$
NED → Geodetic	N, E, D_{corr}, reference $(\varphi_{\text{ref}},\lambda_{\text{ref}},h_{\text{ref}})$	$\Delta\mathbf{r}$, then $\mathbf{r}_{\text{tgt}}$	φ, λ, h	Bowring iteration on (X,Y,Z); $\lambda = \operatorname{atan2}(Y,X)$

Appendix C

Aircraft Specifications — MSN012 (Flyox I)

This annex documents the technical specifications, aerodynamic coefficients and performance constraints of the MSN012 aircraft, a full-scale implementation of the Flyox I platform intended for autonomous heavy-payload operations. Unless stated otherwise, MSN012 is the reference aircraft used in every example, test and simulation reported in this thesis; the parameters listed below define the plant model and the envelope constraints exercised by the optimisation and validation results.



Figure C.1: Flyox I platform used as the MSN012 reference aircraft throughout the thesis.

C.1 Airframe and performance envelope

The MSN012 features a high-aspect-ratio wing and a substantial payload capacity. The mass properties below correspond to the Maximum Take-Off Weight (MTOW) configuration; the centre of gravity and the power-train location are expressed in the body-fixed frame relative to the Mean Aerodynamic Chord (MAC). Table C.1 collects the geometric and inertial data, and Table C.2 summarises the flight envelope, including velocity limits, attitude bounds and structural load factors used as hard constraints during optimisation.

Table C.1: Geometric and mass properties of MSN012.

Parameter	Symbol	Value	Unit
Maximum Take-Off Mass	m	4000.0	kg
Basic Empty Weight	BEW	2450.0	kg
Max Payload Mass	$PM_{\max}$	1425.0	kg
Max Fuel Mass	$FM_{\max}$	400.0	kg
Wing Reference Area	S_w	28.0	m^2
Wing Span	b	14.0	m
Mean Aerodynamic Chord	c	2.0	m
Aspect Ratio	AR	7.0	—
Oswald Efficiency	e	0.839	—
Moment of Inertia (x)	I_{xx}	18340.0	kg·m^2
Moment of Inertia (y)	I_{yy}	20980.0	kg·m^2
Moment of Inertia (z)	I_{zz}	29390.0	kg·m^2
Product of Inertia	I_{xz}	0.0	kg·m^2

Table C.2: Performance limits and operational constraints.

Constraint	Symbol	Value	Unit
Never Exceed Speed	V_{NE}	72.016	m/s
Cruise Speed	V_{cruise}	50.411	m/s
Stall Speed	V_{stall}	38.066	m/s
Minimum Altitude	$h_{\min}$	150.0	m (AGL)
Max Pitch Angle	$\theta_{\max}$	15.0	deg
Min Pitch Angle	$\theta_{\min}$	-10.0	deg
Bank Angle Limit	$\phi_{\max}$	±35.0	deg
Load Factor (Flaps Up)	n_{up}	[-1.4, 3.5]	g
Load Factor (Flaps Down)	n_{down}	[0.0, 2.5]	g

C.2 Aerodynamic and propulsion data

The aerodynamic model is described by a set of non-dimensional stability and control derivatives. The lift coefficient C_L depends on the angle of attack α and on the elevator deflection δ_e; the drag polar follows the quadratic form $C_D = C_{D0} + k_2 C_L^2$. Table C.3 reports the longitudinal, lateral-directional and control derivatives used in the rigid-body formulation. The propulsion model in Table C.4 couples a

fixed-pitch propeller with a piston/turboprop unit, characterised by its maximum shaft power, propeller efficiency η_p and specific fuel consumption.

Table C.3: Aerodynamic stability and control derivatives.

Coeff.	Value	Coeff.	Value	Coeff.	Value
$C_{L,0}$	0.410	$C_{l,\beta}$	-0.112	$C_{n,\beta}$	0.300
$C_{L,\alpha}$	3.340	$C_{l,p}$	-0.550	$C_{n,p}$	-0.072
C_{L,δ_e}	0.750	$C_{l,r}$	0.230	$C_{n,r}$	-0.600
$C_{D,0}$	0.048	C_{l,δ_a}	0.340	C_{n,δ_a}	-0.312
$C_{k,2}$	0.054	C_{l,δ_r}	0.004	C_{n,δ_r}	-0.200
$C_{m,0}$	0.061	$C_{Y,\beta}$	-0.570	$C_{Y,p}$	-0.226
$C_{m,\alpha}$	-1.030	C_{m,δ_e}	-2.640	C_{Y,δ_r}	0.260

Table C.4: Propulsion and power-plant specifications.

Parameter	Symbol	Value	Unit
Maximum Power	$P_{\max}$	521.99	kW
Propeller Efficiency	η_p	0.8	—
Specific Fuel Consumption	SFC	0.210	kg/(kW·h)
Thrust Position (z)	PT_z	-1.1	m

Appendix D

Regularisation penalties: full per-scenario validation

This annex extends Section 6.5, where the impact of cost regularisation on the optimum was illustrated on scenario SC-03 under the minimum-time criterion. The body chapter retains only that representative case for brevity; the present annex closes the loop by reporting *every* benchmark scenario (SC-00 to SC-03) for both the Uncoupled and Full-Control formulations, under the two cost criteria $\alpha = 0$ (min-time) and $\alpha = 1$ (min-fuel). The derivation of the regularisation container $J_\star$ is not repeated here, as it is covered in Section 6.3.2 of the body chapter; this annex is purely a numerical reference.

D.1 Weight tuning protocol

The per-aircraft regularisation weights are tuned by the heuristic procedure outlined in Section 6.3.2, which is described here in full. First, every benchmark scenario is solved with all smoothness weights set to zero ($w_{\Delta(\cdot)} = 0$) and the total variation $\mathrm{TV}(u_i)$ of each control axis is recorded as the unregularised reference. Next, for every axis u_i the smoothness weight $w_{\Delta u_i}$ is grid-searched on the geometric ladder $\{10^{-4}, 10^{-3}, \ldots, 10^{1}\}$, holding the remaining weights at zero; the converged $\mathrm{TV}(u_i)$ and the relative weight of the regularisation term in the cost, P_c/J_{total}, are stored at the initial point x_0. The smallest $w_{\Delta u_i}$ for which $\mathrm{TV}(u_i)$ drops below an order of magnitude of its unregularised reference is then retained. Finally, the procedure is repeated with all axes active simultaneously and the global bound $P_c + P_s \lesssim 15\,\%$ is verified.

The tuned weights used in this work are the values committed to the aircraft configuration files distributed with the validation suite; the corresponding weight-grid tables can be appended once exported from the sweep script.

D.2 Reading guide

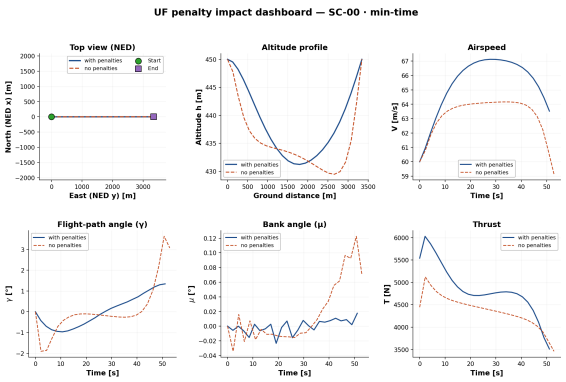
Each of the four scenario sections that follow has the same internal layout:

- A 2×2 figure with the multi-panel dashboards for the Uncoupled (UF) and Full-Control (FC) formulations under both cost criteria. Within every dashboard the solid blue curves correspond to the regularised production setting ($\beta_c = \beta_s = 1$) and the dashed red curves to the unregularised ablation ($\beta_c = \beta_s = 0$). The dashboards are sourced from the canonical `cross_mode` artefact tree shared with Chapter 6.

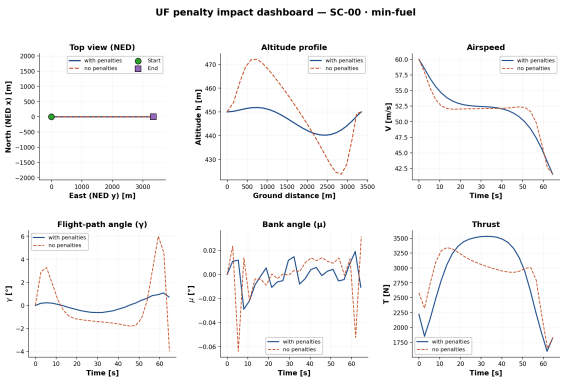
- Two diagnostic tables — one per formulation — that condense the relevant numerical signature of the ablation: the upper block reports the control total variation $\mathrm{TV}(u) = \sum_k |\Delta u_k|$, which drops sharply when the smoothness weights are activated; the lower block reports the per-state normalised trapezoidal defect, with the row $\|\Delta\|_\infty$ listing the global ℓ_∞ norm already reported in the aggregated summary of Section D.7. Heading angles are unwrapped before differencing so that 2π jumps do not inflate the defect.

The qualitative picture is the same across all scenarios: the bank angle and the FC actuator deflections concentrate the chatter that the regularisation suppresses, while the global trajectory in altitude, airspeed and flight-path angle is essentially invariant. The tables quantify this with two stable patterns: the TV of the chatter-prone control axes (notably C_L in UF and δ_a, δ_r in FC) collapses by one to four orders of magnitude when the penalties are turned on, and the $\|\Delta\|_\infty$ defect norm remains within the prescribed interior-point tolerance in both regimes, confirming that the regularisation does not relax the collocation accuracy.

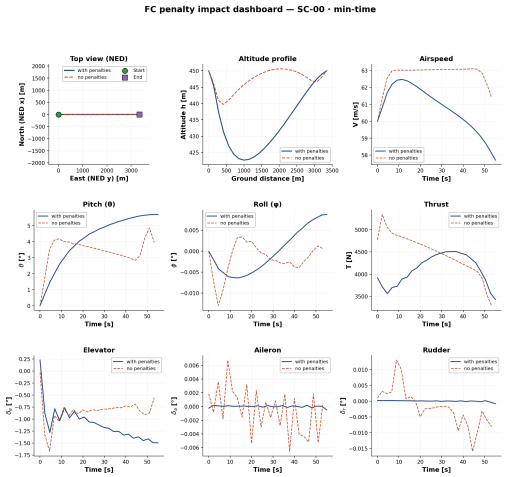
D.3 Scenario SC-00



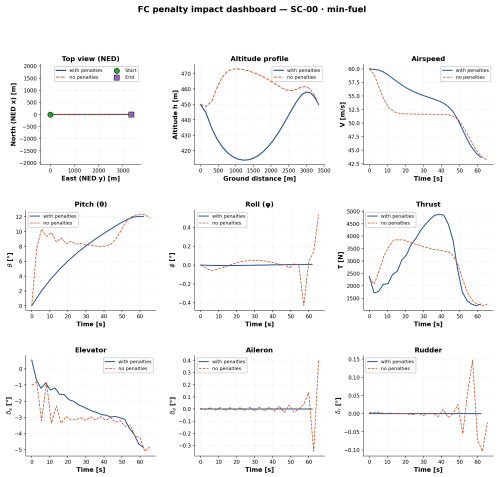
(a) UF, min-time.



(b) UF, min-fuel.



(c) FC, min-time.



(d) FC, min-fuel.

Figure D.1: Penalty-ablation dashboards for SC-00 (straight-and-level baseline). Solid blue: regularised; dashed red: unregularised.

Tables D.1 and D.2 report the UF and FC diagnostics.

Table D.1: Penalty ablation diagnostics for the UF formulation on SC-00. The upper block reports the control total variation $\sum_i |u_{i+1} - u_i|$ (higher values indicate more chatter); the lower block reports the per-state normalised trapezoidal discretisation defect. Each criterion (min-time, min-fuel) is shown as a pair of columns comparing the regularised run ($\beta_c = \beta_s = 1$) against the unregularised ablation ($\beta_c = \beta_s = 0$). The row $\|\Delta\|_\infty$ matches the global ℓ_∞ norm reported in the aggregated summary.

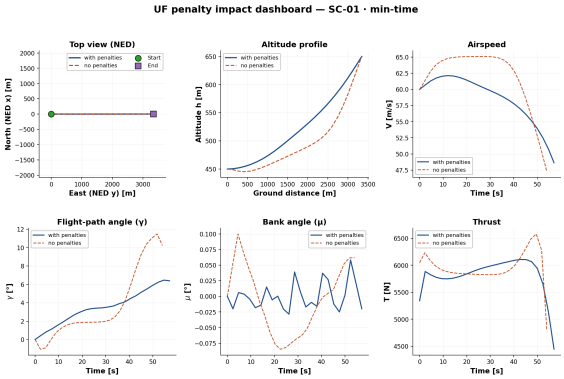
	Min-time		**Min-fuel**	
	Reg.	No reg.	Reg.	No reg.
Control total variation				
TV(T)	1688.531	2241.612	7671.860	14678.546
TV(C_L)	0.195	1.188	1.388	0.720
TV(p)	7.722e-04	0.238	1.022e-04	0.008
Per-state discretisation defect				
x	2.2315e-06	1.7717e-05	6.0017e-08	7.1285e-06
y	3.1029e-09	6.7316e-08	1.3098e-11	1.5314e-08
z	2.0714e-05	5.7600e-05	3.4972e-08	2.6371e-05
V	5.6325e-09	1.3780e-05	8.7435e-11	1.8759e-07
μ	3.2773e-08	9.0445e-09	1.6811e-10	1.2062e-08
γ	5.7681e-08	2.1224e-06	1.4836e-09	2.7581e-07
ξ	7.1412e-09	2.5040e-06	4.7643e-11	1.9489e-08
m	1.1535e-12	1.1751e-11	1.4711e-13	2.2043e-11
C	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00
$\|\Delta\|_\infty$	2.0714e-05	5.7600e-05	6.0017e-08	2.6371e-05

Table D.2: Penalty ablation diagnostics for the FC formulation on SC-00. The upper block reports the control total variation $\sum_i |u_{i+1} - u_i|$ (higher values indicate more chatter); the lower block reports the per-state normalised trapezoidal discretisation defect. Each criterion (min-time, min-fuel) is shown as a pair of columns comparing the regularised run ($\beta_c = \beta_s = 1$) against the unregularised ablation ($\beta_c = \beta_s = 0$). The row $\|\Delta\|_\infty$ matches the global ℓ_∞ norm reported in the aggregated summary.

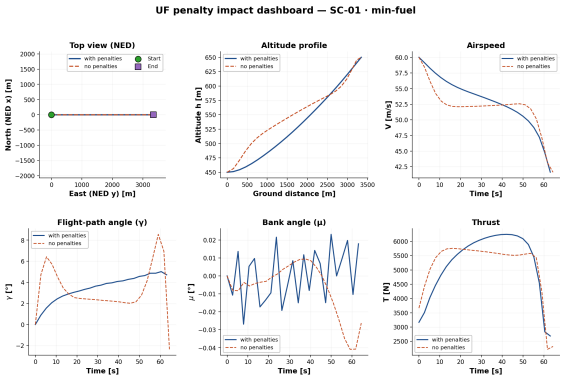
	Min-time		**Min-fuel**	
	Reg.	No reg.	Reg.	No reg.
Control total variation				
TV(T)	1470.674	2262.071	9546.196	14713.078
TV(δ_e)	0.051	0.194	0.084	0.586
TV(δ_a)	3.600e-05	0.065	7.703e-05	4.514
TV(δ_r)	2.210e-05	0.094	2.600e-05	1.776
Per-state discretisation defect				
x	1.5310e-09	7.3327e-11	7.1704e-09	1.4775e-07
y	1.0849e-11	2.4858e-13	4.8559e-12	5.6220e-11
z	2.7709e-08	2.5044e-10	1.3053e-08	1.7860e-07
u	1.5733e-10	2.3692e-11	4.5211e-10	5.2453e-09
v	3.1767e-10	3.3511e-12	2.0258e-10	4.9761e-09
w	1.4093e-10	1.8404e-11	3.8554e-10	7.8847e-09
ϕ	5.5000e-11	3.8448e-13	5.4923e-11	4.9645e-10
θ	2.6239e-11	2.1304e-12	8.1109e-11	1.8978e-09
ψ	1.0056e-11	5.1231e-13	3.0803e-11	5.2215e-10
p	1.7813e-08	2.1715e-10	1.6235e-08	2.8646e-07
q	6.9861e-09	1.5815e-10	3.5244e-09	2.3719e-07
r	5.9117e-09	2.0196e-11	3.0865e-09	1.8799e-07
m	2.8715e-14	3.9917e-16	1.2598e-13	1.0745e-13
C	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00
$\|\Delta\|_\infty$	2.7709e-08	2.5044e-10	1.6235e-08	2.8646e-07

D.4 Scenario SC-01

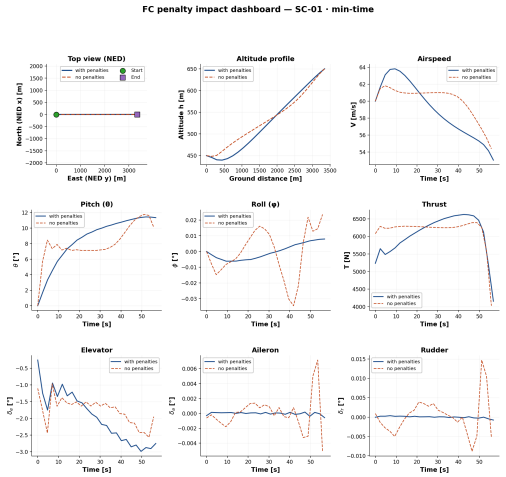
The dashboards below complete the SC-01 record for cross-scenario comparison; the representative case discussed in Section 6.5 is SC-03 (climb-and-turn, minimum-time).



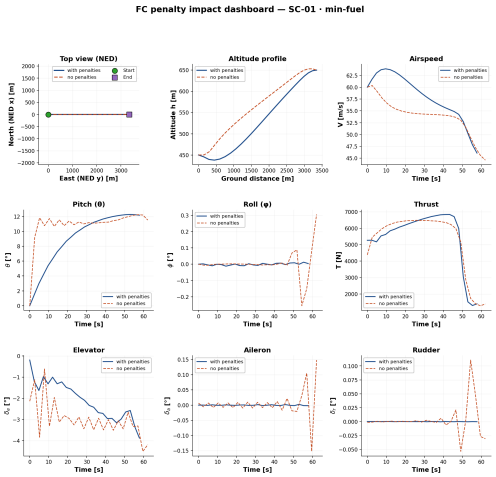
(a) UF, min-time.



(b) UF, min-fuel.



(c) FC, min-time.



(d) FC, min-fuel.

Figure D.2: Penalty-ablation dashboards for SC-01 (straight climb). Solid blue: regularised; dashed red: unregularised.

Tables D.3 and D.4 report the UF and FC diagnostics.

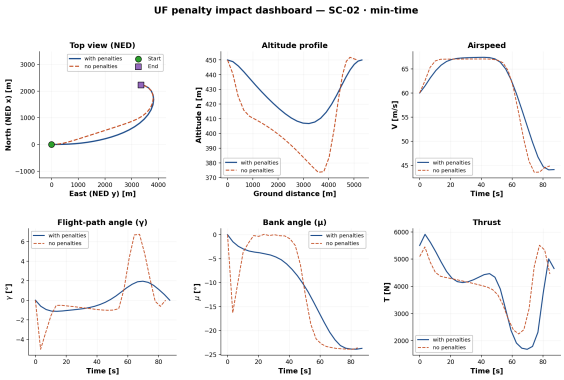
Table D.3: Penalty ablation diagnostics for the UF formulation on SC-01. The upper block reports the control total variation $\sum_i |u_{i+1} - u_i|$ (higher values indicate more chatter); the lower block reports the per-state normalised trapezoidal discretisation defect. Each criterion (min-time, min-fuel) is shown as a pair of columns comparing the regularised run ($\beta_c = \beta_s = 1$) against the unregularised ablation ($\beta_c = \beta_s = 0$). The row $\|\Delta\|_\infty$ matches the global ℓ_∞ norm reported in the aggregated summary.

	Min-time		**Min-fuel**	
	Reg.	No reg.	Reg.	No reg.
Control total variation				
TV(T)	1493.499	3829.613	12184.777	12903.727
TV(C_L)	0.348	1.035	1.868	0.695
TV(p)	2.403e-04	0.131	4.007e-04	0.018
Per-state discretisation defect				
x	4.6024e-07	7.8518e-04	1.8334e-06	1.8043e-04
y	1.9924e-10	3.4463e-07	3.0590e-10	3.2565e-07
z	3.9229e-09	3.7783e-06	1.3892e-08	2.8191e-06
V	1.1785e-09	3.0906e-06	2.1177e-09	1.9991e-06
μ	1.0266e-08	3.2163e-07	6.1993e-09	1.3888e-06
γ	1.5227e-08	7.0628e-06	8.4646e-08	8.4874e-06
ξ	2.4373e-09	7.3139e-07	5.9488e-09	4.0679e-07
m	7.2229e-13	6.7977e-11	3.4599e-12	2.3415e-10
C	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00
$\|\Delta\|_\infty$	4.6024e-07	7.8518e-04	1.8334e-06	1.8043e-04

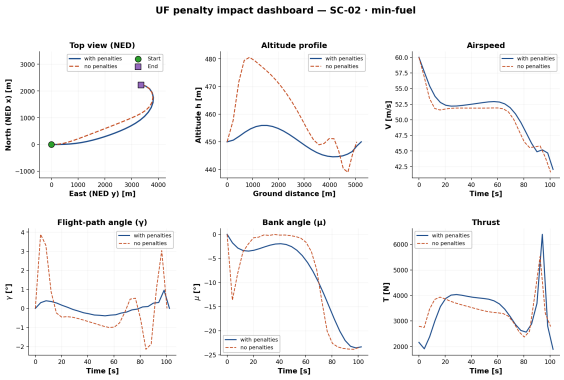
Table D.4: Penalty ablation diagnostics for the FC formulation on SC-01. The upper block reports the control total variation $\sum_i |u_{i+1} - u_i|$ (higher values indicate more chatter); the lower block reports the per-state normalised trapezoidal discretisation defect. Each criterion (min-time, min-fuel) is shown as a pair of columns comparing the regularised run ($\beta_c = \beta_s = 1$) against the unregularised ablation ($\beta_c = \beta_s = 0$). The row $\|\Delta\|_\infty$ matches the global ℓ_∞ norm reported in the aggregated summary.

	Min-time		**Min-fuel**	
	Reg.	No reg.	Reg.	No reg.
Control total variation				
TV(T)	1773.109	2920.184	8545.358	13931.952
TV(δ_e)	0.088	0.323	0.099	0.500
TV(δ_a)	1.673e-05	0.604	4.947e-05	3.743
TV(δ_r)	1.721e-05	0.553	1.717e-05	1.185
Per-state discretisation defect				
x	1.7329e-11	2.5491e-09	1.6652e-09	9.8794e-08
y	1.4140e-12	3.8628e-13	3.3955e-12	2.4978e-11
z	1.5322e-11	3.1254e-11	1.1129e-10	2.3703e-10
u	1.3011e-10	4.0964e-11	2.3063e-10	2.4090e-09
v	1.8480e-12	5.0073e-11	1.5691e-10	1.2260e-09
w	1.3498e-10	4.4777e-10	2.4898e-10	1.1171e-09
ϕ	2.9115e-13	5.2902e-12	2.2112e-11	1.7681e-10
θ	5.0376e-12	1.3092e-11	2.1899e-11	3.3407e-10
ψ	8.7885e-14	9.6150e-12	1.0940e-11	6.5821e-11
p	4.6318e-11	8.9223e-10	7.2654e-09	8.0655e-08
q	2.0492e-09	2.7218e-09	9.9642e-09	4.9759e-08
r	2.4762e-11	1.7415e-10	3.2970e-09	2.1273e-08
m	2.5097e-14	1.0503e-15	1.2804e-13	8.8209e-15
C	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00
$\|\Delta\|_\infty$	2.0492e-09	2.7218e-09	9.9642e-09	9.8794e-08

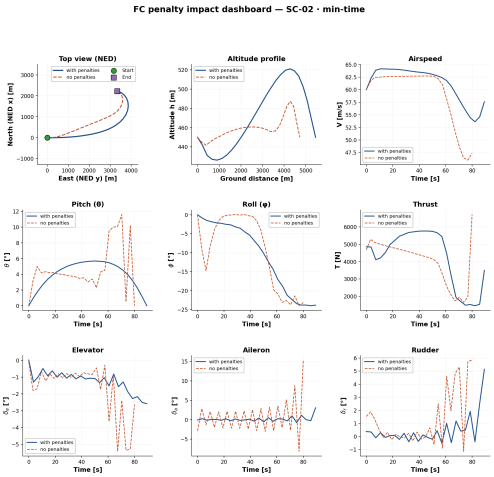
D.5 Scenario SC-02



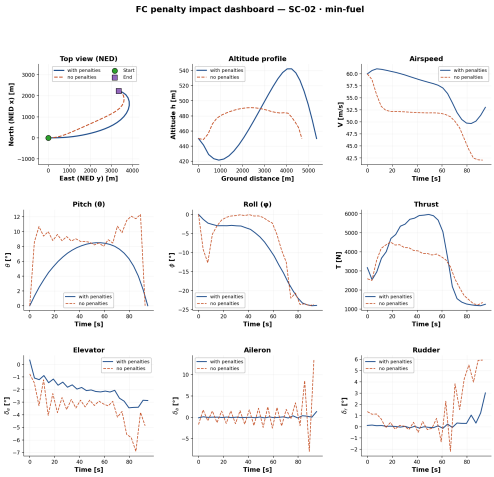
(a) UF, min-time.



(b) UF, min-fuel.



(c) FC, min-time.



(d) FC, min-fuel.

Figure D.3: Penalty-ablation dashboards for SC-02 (heading reversal at level flight). Solid blue: regularised; dashed red: unregularised.

Tables D.5 and D.6 report the UF and FC diagnostics.

Table D.5: Penalty ablation diagnostics for the UF formulation on SC-02. The upper block reports the control total variation $\sum_i |u_{i+1} - u_i|$ (higher values indicate more chatter); the lower block reports the per-state normalised trapezoidal discretisation defect. Each criterion (min-time, min-fuel) is shown as a pair of columns comparing the regularised run ($\beta_c = \beta_s = 1$) against the unregularised ablation ($\beta_c = \beta_s = 0$). The row $\|\Delta\|_\infty$ matches the global ℓ_∞ norm reported in the aggregated summary.

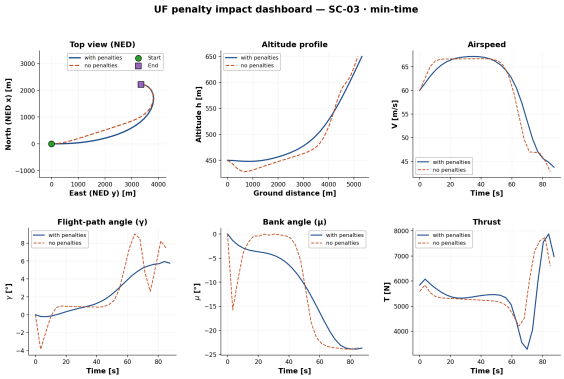
	Min-time		**Min-fuel**	
	Reg.	No reg.	Reg.	No reg.
Control total variation				
TV(T)	2584.470	3033.028	9157.164	13521.556
TV(C_L)	0.233	1.953	1.785	0.750
TV(p)	0.010	1.615	0.012	0.930
Per-state discretisation defect				
x	1.5599e-11	9.4206e-11	1.1591e-10	1.6781e-10
y	1.0308e-11	1.0103e-10	1.0432e-10	2.4054e-10
z	1.1138e-08	1.2548e-08	5.2979e-08	2.7047e-07
V	1.7834e-11	2.7369e-09	8.9926e-11	1.3013e-09
μ	4.1296e-11	2.4835e-10	3.8341e-10	5.0584e-10
γ	3.2328e-10	1.7877e-08	5.1181e-09	6.2334e-09
ξ	6.5021e-11	2.4561e-09	1.3042e-10	1.0712e-09
m	1.3650e-13	8.7624e-15	2.7566e-13	3.0184e-13
C	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00
$\|\Delta\|_\infty$	1.1138e-08	1.7877e-08	5.2979e-08	2.7047e-07

Table D.6: Penalty ablation diagnostics for the FC formulation on SC-02. The upper block reports the control total variation $\sum_i |u_{i+1} - u_i|$ (higher values indicate more chatter); the lower block reports the per-state normalised trapezoidal discretisation defect. Each criterion (min-time, min-fuel) is shown as a pair of columns comparing the regularised run ($\beta_c = \beta_s = 1$) against the unregularised ablation ($\beta_c = \beta_s = 0$). The row $\|\Delta\|_\infty$ matches the global ℓ_∞ norm reported in the aggregated summary.

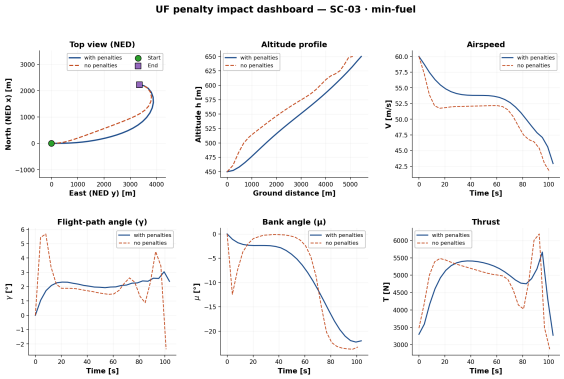
	Min-time		**Min-fuel**	
	Reg.	No reg.	Reg.	No reg.
Control total variation				
TV(T)	2511.389	1994.909	11623.603	25382.013
TV(δ_e)	0.063	0.368	0.095	1.011
TV(δ_a)	0.008	5.845	0.009	6.837
TV(δ_r)	0.014	0.902	0.016	1.485
Per-state discretisation defect				
x	1.4555e-10	6.7834e-12	1.7406e-11	1.5531e-10
y	2.9591e-10	1.3393e-11	9.3113e-12	7.2313e-11
z	1.9236e-06	4.5504e-08	6.2721e-08	1.2778e-07
u	3.9902e-09	2.4362e-09	1.4031e-09	2.6982e-09
v	5.7473e-09	5.6061e-09	5.9040e-10	2.6763e-09
w	2.2587e-08	3.2508e-09	9.9080e-10	2.9582e-09
ϕ	9.7037e-10	1.5729e-10	1.8594e-10	3.7563e-10
θ	3.9311e-10	4.3140e-10	7.7758e-11	5.8802e-10
ψ	9.8095e-10	2.0832e-10	6.8380e-11	6.3051e-10
p	2.9887e-07	1.2974e-07	4.4867e-08	2.3196e-07
q	2.8235e-07	2.5863e-08	2.2040e-08	5.5470e-08
r	1.0529e-07	2.3215e-08	1.2298e-08	4.8476e-08
m	3.2283e-12	1.3161e-14	1.8947e-13	5.0552e-14
C	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00
$\|\Delta\|_\infty$	1.9236e-06	1.2974e-07	6.2721e-08	2.3196e-07

D.6 Scenario SC-03

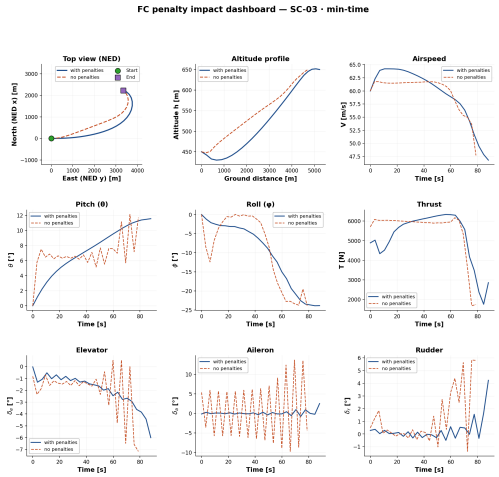
This is the representative case of Section 6.5 (climb-and-turn, minimum-time); the per-formulation diagnostic tables below expand Table 6.7 of the body chapter.



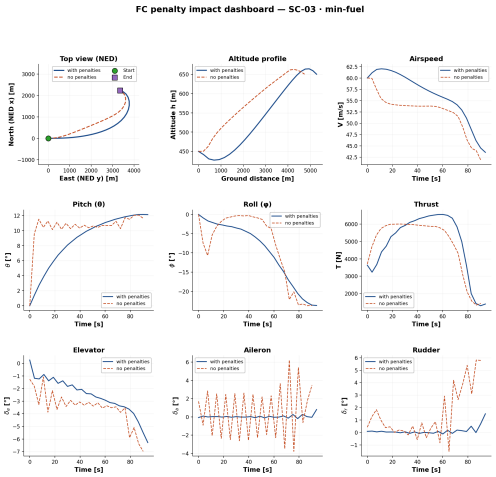
(a) UF, min-time.



(b) UF, min-fuel.



(c) FC, min-time.



(d) FC, min-fuel.

Figure D.4: Penalty-ablation dashboards for SC-03 (climb plus heading reversal). Solid blue: regularised; dashed red: unregularised.

Tables D.7 and D.8 report the UF and FC diagnostics.

Table D.7: Penalty ablation diagnostics for the UF formulation on SC-03. The upper block reports the control total variation $\sum_i |u_{i+1} - u_i|$ (higher values indicate more chatter); the lower block reports the per-state normalised trapezoidal discretisation defect. Each criterion (min-time, min-fuel) is shown as a pair of columns comparing the regularised run ($\beta_c = \beta_s = 1$) against the unregularised ablation ($\beta_c = \beta_s = 0$). The row $\|\Delta\|_\infty$ matches the global ℓ_∞ norm reported in the aggregated summary.

	Min-time		**Min-fuel**	
	Reg.	No reg.	Reg.	No reg.
Control total variation				
TV(T)	1670.522	4097.340	10376.231	12819.727
TV(C_L)	0.385	1.690	2.365	0.716
TV(p)	0.012	1.034	0.009	1.103
Per-state discretisation defect				
x	1.8717e-11	7.9695e-10	3.6987e-12	2.7007e-08
y	3.9002e-11	8.1264e-10	1.2316e-12	2.3339e-08
z	1.8355e-10	8.6494e-10	1.4916e-11	6.1518e-08
V	2.0719e-10	1.1751e-07	8.5236e-12	1.4301e-07
μ	3.7724e-10	7.0267e-10	5.3382e-11	2.0891e-07
γ	7.2138e-10	1.3559e-08	3.8274e-10	5.8207e-07
ξ	1.7686e-10	1.4167e-08	7.8083e-12	1.0678e-07
m	1.4009e-13	2.2845e-14	7.2667e-15	1.1129e-11
C	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00
$\|\Delta\|_\infty$	7.2138e-10	1.1751e-07	3.8274e-10	5.8207e-07

Table D.8: Penalty ablation diagnostics for the FC formulation on SC-03. The upper block reports the control total variation $\sum_i |u_{i+1} - u_i|$ (higher values indicate more chatter); the lower block reports the per-state normalised trapezoidal discretisation defect. Each criterion (min-time, min-fuel) is shown as a pair of columns comparing the regularised run ($\beta_c = \beta_s = 1$) against the unregularised ablation ($\beta_c = \beta_s = 0$). The row $\|\Delta\|_\infty$ matches the global ℓ_∞ norm reported in the aggregated summary.

	Min-time		**Min-fuel**	
	Reg.	No reg.	Reg.	No reg.
Control total variation				
TV(T)	2373.003	2498.186	8094.212	15103.328
TV(δ_e)	0.088	0.610	0.107	0.945
TV(δ_a)	0.005	5.648	0.006	5.893
TV(δ_r)	0.010	0.609	0.011	1.123
Per-state discretisation defect				
x	7.4561e-10	1.8237e-09	6.5068e-10	4.3262e-09
y	8.8819e-10	1.0385e-09	7.0290e-10	2.7076e-09
z	2.6840e-09	1.3212e-08	3.4443e-09	1.4471e-07
u	6.3354e-09	2.7757e-08	4.8339e-09	8.6087e-08
v	5.7737e-09	1.5792e-08	1.9819e-08	1.0107e-07
w	6.1586e-09	2.6557e-08	8.6539e-09	2.4576e-07
ϕ	9.2146e-10	3.0418e-09	2.7125e-09	2.3338e-08
θ	1.1244e-09	6.5832e-09	1.8100e-09	1.0037e-07
ψ	1.8495e-09	1.5309e-08	2.0009e-09	4.4420e-08
p	2.6875e-07	1.2307e-06	9.4777e-07	7.1120e-06
q	9.7628e-08	2.3180e-07	5.2153e-07	1.4689e-06
r	7.4456e-08	1.7100e-07	2.7218e-07	1.1888e-06
m	2.2057e-12	6.3605e-14	2.3313e-12	8.9881e-13
C	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00
$\|\Delta\|_\infty$	2.6875e-07	1.2307e-06	9.4777e-07	7.1120e-06

D.7 Cross-scenario tables

For reference, the same data are also collected in the original per-metric layout (rows = quantity, scenarios spanning column blocks) that was used to publish the validation suite. This view makes trends across scenarios easier to spot at a glance and is the source the per-scenario tables above are sliced from. Tables D.9–D.16 collect the cross-scenario aggregates.

UF: control total variation

Table D.9: Total variation $\sum_i |u_{i+1} - u_i|$ of the UF control signals under the min-time criterion. Higher values indicate more chatter. Within each scenario block, the first column gives the regularised run ($\beta_c = \beta_s = 1$) and the second the unregularised ablation ($\beta_c = \beta_s = 0$).

	SC-00		**SC-01**		**SC-02**		**SC-03**	
	Reg.	No reg.	Reg.	No reg.	Reg.	No reg.	Reg.	No reg.
TV(T)	1688.531	2241.612	1493.499	3829.613	2584.470	3033.028	1670.522	4097.340
TV(C_L)	0.195	1.188	0.348	1.035	0.233	1.953	0.385	1.690
TV(p)	7.722e-04	0.238	2.403e-04	0.131	0.010	1.615	0.012	1.034

Table D.10: Total variation $\sum_i |u_{i+1} - u_i|$ of the UF control signals under the min-fuel criterion. Higher values indicate more chatter. Within each scenario block, the first column gives the regularised run ($\beta_c = \beta_s = 1$) and the second the unregularised ablation ($\beta_c = \beta_s = 0$).

	SC-00		**SC-01**		**SC-02**		**SC-03**	
	Reg.	No reg.	Reg.	No reg.	Reg.	No reg.	Reg.	No reg.
TV(T)	7671.860	14678.546	12184.777	12903.727	9157.164	13521.556	10376.231	12819.727
TV(C_L)	1.388	0.720	1.868	0.695	1.785	0.750	2.365	0.716
TV(p)	1.022e-04	0.008	4.007e-04	0.018	0.012	0.930	0.009	1.103

FC: control total variation

Table D.11: Total variation $\sum_i |u_{i+1} - u_i|$ of the FC control signals under the min-time criterion. Higher values indicate more chatter. Within each scenario block, the first column gives the regularised run ($\beta_c = \beta_s = 1$) and the second the unregularised ablation ($\beta_c = \beta_s = 0$).

	SC-00		**SC-01**		**SC-02**		**SC-03**	
	Reg.	No reg.	Reg.	No reg.	Reg.	No reg.	Reg.	No reg.
TV(T)	1470.674	2262.071	1773.109	2920.184	2511.389	1994.909	2373.003	2498.186
TV(δ_e)	0.051	0.194	0.088	0.323	0.063	0.368	0.088	0.610
TV(δ_a)	3.600e-05	0.065	1.673e-05	0.604	0.008	5.845	0.005	5.648
TV(δ_r)	2.210e-05	0.094	1.721e-05	0.553	0.014	0.902	0.010	0.609

Table D.12: Total variation $\sum_i |u_{i+1} - u_i|$ of the FC control signals under the min-fuel criterion. Higher values indicate more chatter. Within each scenario block, the first column gives the regularised run ($\beta_c = \beta_s = 1$) and the second the unregularised ablation ($\beta_c = \beta_s = 0$).

	SC-00		**SC-01**		**SC-02**		**SC-03**	
	Reg.	No reg.	Reg.	No reg.	Reg.	No reg.	Reg.	No reg.
TV(T)	9546.196	14713.078	8545.358	13931.952	11623.603	25382.013	8094.212	15103.328
TV(δ_e)	0.084	0.586	0.099	0.500	0.095	1.011	0.107	0.945
TV(δ_a)	7.703e-05	4.514	4.947e-05	3.743	0.009	6.837	0.006	5.893
TV(δ_r)	2.600e-05	1.776	1.717e-05	1.185	0.016	1.485	0.011	1.123

UF: per-state defect breakdown

Table D.13: Per-state trapezoidal discretisation defect for the UF formulation under the min-time criterion. Angular states are unwrapped (2π jumps are not counted as defects). All values are normalised; the row $\|\Delta\|_\infty$ lists the corresponding global ℓ_∞ norm reported in the aggregated summary. Within each scenario block, the first column gives the regularised run ($\beta_c = \beta_s = 1$) and the second the unregularised ablation ($\beta_c = \beta_s = 0$).

	SC-00		**SC-01**		**SC-02**		**SC-03**	
	Reg.	No reg.	Reg.	No reg.	Reg.	No reg.	Reg.	No reg.
x	2.2315e-06	1.7717e-05	4.6024e-07	7.8518e-04	1.5599e-11	9.4206e-11	1.8717e-11	7.9695e-10
y	3.1029e-09	6.7316e-08	1.9924e-10	3.4463e-07	1.0308e-11	1.0103e-10	3.9002e-11	8.1264e-10
z	2.0714e-05	5.7600e-05	3.9229e-09	3.7783e-06	1.1138e-08	1.2548e-08	1.8355e-10	8.6494e-10
V	5.6325e-09	1.3780e-05	1.1785e-09	3.0906e-06	1.7834e-11	2.7369e-09	2.0719e-10	1.1751e-07
μ	3.2773e-08	9.0445e-09	1.0266e-08	3.2163e-07	4.1296e-11	2.4835e-10	3.7724e-10	7.0267e-10
γ	5.7681e-08	2.1224e-06	1.5227e-08	7.0628e-06	3.2328e-10	1.7877e-08	7.2138e-10	1.3559e-08
ξ	7.1412e-09	2.5040e-06	2.4373e-09	7.3139e-07	6.5021e-11	2.4561e-09	1.7686e-10	1.4167e-08
m	1.1535e-12	1.1751e-11	7.2229e-13	6.7977e-11	1.3650e-13	8.7624e-15	1.4009e-13	2.2845e-14
C	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00
$\|\Delta\|_\infty$	2.0714e-05	5.7600e-05	4.6024e-07	7.8518e-04	1.1138e-08	1.7877e-08	7.2138e-10	1.1751e-07

Table D.14: Per-state trapezoidal discretisation defect for the UF formulation under the min-fuel criterion. Angular states are unwrapped (2π jumps are not counted as defects). All values are normalised; the row $\|\Delta\|_\infty$ lists the corresponding global ℓ_∞ norm reported in the aggregated summary. Within each scenario block, the first column gives the regularised run ($\beta_c = \beta_s = 1$) and the second the unregularised ablation ($\beta_c = \beta_s = 0$).

	SC-00		**SC-01**		**SC-02**		**SC-03**	
	Reg.	No reg.	Reg.	No reg.	Reg.	No reg.	Reg.	No reg.
x	6.0017e-08	7.1285e-06	1.8334e-06	1.8043e-04	1.1591e-10	1.6781e-10	3.6987e-12	2.7007e-08
y	1.3098e-11	1.5314e-08	3.0590e-10	3.2565e-07	1.0432e-10	2.4054e-10	1.2316e-12	2.3339e-08
z	3.4972e-08	2.6371e-05	1.3892e-08	2.8191e-06	5.2979e-08	2.7047e-07	1.4916e-11	6.1518e-08
V	8.7435e-11	1.8759e-07	2.1177e-09	1.9991e-06	8.9926e-11	1.3013e-09	8.5236e-12	1.4301e-07
μ	1.6811e-10	1.2062e-08	6.1993e-09	1.3888e-06	3.8341e-10	5.0584e-10	5.3382e-11	2.0891e-07
γ	1.4836e-09	2.7581e-07	8.4646e-08	8.4874e-06	5.1181e-09	6.2334e-09	3.8274e-10	5.8207e-07
ξ	4.7643e-11	1.9489e-08	5.9488e-09	4.0679e-07	1.3042e-10	1.0712e-09	7.8083e-12	1.0678e-07
m	1.4711e-13	2.2043e-11	3.4599e-12	2.3415e-10	2.7566e-13	3.0184e-13	7.2667e-15	1.1129e-11
C	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00
$\|\Delta\|_\infty$	6.0017e-08	2.6371e-05	1.8334e-06	1.8043e-04	5.2979e-08	2.7047e-07	3.8274e-10	5.8207e-07

FC: per-state defect breakdown

Table D.15: Per-state trapezoidal discretisation defect for the FC formulation under the min-time criterion. Angular states are unwrapped (2π jumps are not counted as defects). All values are normalised; the row $\|\Delta\|_\infty$ lists the corresponding global ℓ_∞ norm reported in the aggregated summary. Within each scenario block, the first column gives the regularised run ($\beta_c = \beta_s = 1$) and the second the unregularised ablation ($\beta_c = \beta_s = 0$).

	SC-00		**SC-01**		**SC-02**		**SC-03**	
	Reg.	No reg.	Reg.	No reg.	Reg.	No reg.	Reg.	No reg.
x	1.5310e-09	7.3327e-11	1.7329e-11	2.5491e-09	1.4555e-10	6.7834e-12	7.4561e-10	1.8237e-09
y	1.0849e-11	2.4858e-13	1.4140e-12	3.8628e-13	2.9591e-10	1.3393e-11	8.8819e-10	1.0385e-09
z	2.7709e-08	2.5044e-10	1.5322e-11	3.1254e-11	1.9236e-06	4.5504e-08	2.6840e-09	1.3212e-08
u	1.5733e-10	2.3692e-11	1.3011e-10	4.0964e-11	3.9902e-09	2.4362e-09	6.3354e-09	2.7757e-08
v	3.1767e-10	3.3511e-12	1.8480e-12	5.0073e-11	5.7473e-09	5.6061e-09	5.7737e-09	1.5792e-08
w	1.4093e-10	1.8404e-11	1.3498e-10	4.4777e-10	2.2587e-08	3.2508e-09	6.1586e-09	2.6557e-08
ϕ	5.5000e-11	3.8448e-13	2.9115e-13	5.2902e-12	9.7037e-10	1.5729e-10	9.2146e-10	3.0418e-09
θ	2.6239e-11	2.1304e-12	5.0376e-12	1.3092e-11	3.9311e-10	4.3140e-10	1.1244e-09	6.5832e-09
ψ	1.0056e-11	5.1231e-13	8.7885e-14	9.6150e-12	9.8095e-10	2.0832e-10	1.8495e-09	1.5309e-08
p	1.7813e-08	2.1715e-10	4.6318e-11	8.9223e-10	2.9887e-07	1.2974e-07	2.6875e-07	1.2307e-06
q	6.9861e-09	1.5815e-10	2.0492e-09	2.7218e-09	2.8235e-07	2.5863e-08	9.7628e-08	2.3180e-07
r	5.9117e-09	2.0196e-11	2.4762e-11	1.7415e-10	1.0529e-07	2.3215e-08	7.4456e-08	1.7100e-07
m	2.8715e-14	3.9917e-16	2.5097e-14	1.0503e-15	3.2283e-12	1.3161e-14	2.2057e-12	6.3605e-14
C	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00
$\|\Delta\|_\infty$	2.7709e-08	2.5044e-10	2.0492e-09	2.7218e-09	1.9236e-06	1.2974e-07	2.6875e-07	1.2307e-06

Table D.16: Per-state trapezoidal discretisation defect for the FC formulation under the min-fuel criterion. Angular states are unwrapped (2π jumps are not counted as defects). All values are normalised; the row $\|\Delta\|_\infty$ lists the corresponding global ℓ_∞ norm reported in the aggregated summary. Within each scenario block, the first column gives the regularised run ($\beta_c = \beta_s = 1$) and the second the unregularised ablation ($\beta_c = \beta_s = 0$).

	SC-00		**SC-01**		**SC-02**		**SC-03**	
	Reg.	No reg.	Reg.	No reg.	Reg.	No reg.	Reg.	No reg.
x	7.1704e-09	1.4775e-07	1.6652e-09	9.8794e-08	1.7406e-11	1.5531e-10	6.5068e-10	4.3262e-09
y	4.8559e-12	5.6220e-11	3.3955e-12	2.4978e-11	9.3113e-12	7.2313e-11	7.0290e-10	2.7076e-09
z	1.3053e-08	1.7860e-07	1.1129e-10	2.3703e-10	6.2721e-08	1.2778e-07	3.4443e-09	1.4471e-07
u	4.5211e-10	5.2453e-09	2.3063e-10	2.4090e-09	1.4031e-09	2.6982e-09	4.8339e-09	8.6087e-08
v	2.0258e-10	4.9761e-09	1.5691e-10	1.2260e-09	5.9040e-10	2.6763e-09	1.9819e-08	1.0107e-07
w	3.8554e-10	7.8847e-09	2.4898e-10	1.1171e-09	9.9080e-10	2.9582e-09	8.6539e-09	2.4576e-07
ϕ	5.4923e-11	4.9645e-10	2.2112e-11	1.7681e-10	1.8594e-10	3.7563e-10	2.7125e-09	2.3338e-08
θ	8.1109e-11	1.8978e-09	2.1899e-11	3.3407e-10	7.7758e-11	5.8802e-10	1.8100e-09	1.0037e-07
ψ	3.0803e-11	5.2215e-10	1.0940e-11	6.5821e-11	6.8380e-11	6.3051e-10	2.0009e-09	4.4420e-08
p	1.6235e-08	2.8646e-07	7.2654e-09	8.0655e-08	4.4867e-08	2.3196e-07	9.4777e-07	7.1120e-06
q	3.5244e-09	2.3719e-07	9.9642e-09	4.9759e-08	2.2040e-08	5.5470e-08	5.2153e-07	1.4689e-06
r	3.0865e-09	1.8799e-07	3.2970e-09	2.1273e-08	1.2298e-08	4.8476e-08	2.7218e-07	1.1888e-06
m	1.2598e-13	1.0745e-13	1.2804e-13	8.8209e-15	1.8947e-13	5.0552e-14	2.3313e-12	8.9881e-13
C	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00
$\|\Delta\|_\infty$	1.6235e-08	2.8646e-07	9.9642e-09	9.8794e-08	6.2721e-08	2.3196e-07	9.4777e-07	7.1120e-06

Appendix E

Per-run constraint detail

This annex consolidates the per-run constraint-margin tables emitted by the validation pipeline for every converged (SC, formulation, criterion) triple under the `with_penalties` mode. The headline scoreboards in Sections 6.4.7 and 6.4.7 aggregate these data; the tables below provide the per-row, per-node detail the reader needs to audit any specific run. The optimizer's global box bounds and hard-constraint limits for UF and FC are collected first, in Section E.1.

Each block reports three stacked tables, mirroring the optimizer's constraint partition:

Group A Decision-variable box bounds. One row per (variable, side). The *Worst margin* column is $\max_n (x_n - \bar{x})/\Delta$ for upper bounds and $\max_n (\underline{x} - x_n)/\Delta$ for lower bounds, with $\Delta = \bar{x} - \underline{x}$.

Group B Hard (path) constraints written as $g_i(x, u) \leq 0$, reported with the same sign convention as Group A.

Group C Trapezoidal collocation residuals — equality residuals satisfied to solver tolerance, hence always `OK`. The reported magnitudes are the infinity norms of the per-state residual after non-dimensionalisation.

A positive *Worst margin* entry is flagged `ACTIVE` and a non-positive entry `OK`. There is no tolerance band: `ACTIVE` corresponds to a boundary touched or violated; `OK` quantifies the remaining room.

E.1 Decision-variable and hard-constraint bounds

The validation pipeline emits, for each formulation, a compact reference listing the box bounds applied to every entry of the decision vector and the algebraic limits attached to each hard (path) constraint family. Values are in the scaled non-dimensional space the interior-point solver uses; open sides are rendered as $\pm\infty$ where the optimizer leaves a bound inactive. These tables are cited from the UF and

FC constraint-verification subsections in Chapter 6 but are reproduced here so the margin tables in the following sections can be read against a fixed boundary specification.

Table E.1: UF decision-variable boundaries applied by the optimizer (non-dimensional; widest per-node window).

Decision variable	Lower bound	Upper bound
Final Time (tf)	0.0151	1.5064e+17
Thrust (T)	0.0000e+00	1.0000
Lift Coeff (CL)	-0.5000	1.5791
Roll Rate (p)	-2.0000	2.0000
Pos X (North)	-2.9883e+15	2.9883e+15
Pos Y (East)	-2.9883e+15	2.9883e+15
Pos Z (Down)	-4.4824	-0.0448
Velocity (V)	0.7551	1.4286
Bank (mu)	-0.6109	0.6109
Gamma	-0.1745	0.2618
Heading (xi)	-7.5000	8.5000
Mass (m)	0.1000	1.2000
Fuel/Cost (C)	0.0000e+00	1.1000

Table E.2: UF hard (path) constraint boundaries applied by the optimizer (non-dimensional).

Hard constraint	Lower bound	Upper bound
V_max (V/V_NE)	0.0000e+00	1.0000
V_stall	1.0000	$+\infty$
Power_max	$-\infty$	1.0000
Alt_min (h_min)	1.0000	$+\infty$

Uncoupled formulation (UF).

Table E.3: FC decision-variable boundaries applied by the optimizer (non-dimensional; widest per-node window).

Decision variable	Lower bound	Upper bound
Final Time (tf)	0.0211	2.1090e+17
Thrust (T)	0.0000e+00	1.0000
Elevator (de)	-1.0000	1.0000
Aileron (da)	-1.0000	1.0000
Rudder (dr)	-1.0000	1.0000
Pos X (North)	-2.9883e+15	2.9883e+15
Pos Y (East)	-2.9883e+15	2.9883e+15
Pos Z (Down)	-4.4824	-0.0448
Mass (m)	-2.5000e+15	2.5000e+15
Vel U (body)	0.0000e+00	1.0204
Vel V (body)	-1.4169e+17	1.4169e+17
Vel W (body)	-1.4169e+17	1.4169e+17
Roll (phi)	-1.0000	1.0000
Pitch (theta)	-0.6667	1.0000
Rate P (roll)	-0.0198	0.0198
Rate Q (pitch)	-0.0057	0.0057
Rate R (yaw)	-0.0397	0.0397
Fuel/Cost (C)	0.0000e+00	1.1000

Table E.4: FC hard (path) constraint boundaries applied by the optimizer (non-dimensional).

Hard constraint	Lower bound	Upper bound
V_max (V/V_NE)	0.0000e+00	1.0000
V_stall	1.0000	$+\infty$
CL_max	-0.5000	1.0000
Power_max	$-\infty$	1.0000
Alt_min (h_min)	1.0000	$+\infty$

Full-control formulation (FC).

E.2 Worst-case margin comparison across criteria and formulations

The per-run tables of the following sections list every (variable, side) margin in full. To support the comparative reading of the body, this section first condenses the *binding* constraints—the few that actually approach their limits—into three cross-cut tables. Throughout, $V/V_{\text{stall}} - 1$ is the fractional airspeed margin above the stall (larger is safer), $P/P_{\max}$ is the peak power as a fraction of the available ceiling ($\rightarrow 1$ means power-limited), the lift-coefficient margin is the room left below $C_{L,\max}$, and the

control-surface and roll-rate figures are peak magnitudes as a fraction of their actuator limits ($\rightarrow 1$ means saturated). Every run is feasible, so all entries sit on the satisfied side of their bound.

Table E.5 is the criterion cut: with the formulation fixed, it places the minimum-time and minimum-fuel envelope margins side by side. The two binding quantities move in opposite directions—minimum-time presses the power ceiling while keeping a wide stall margin, minimum-fuel collapses the stall margin while backing the power well off—which is the numerical signature of the time–energy trade-off.

Table E.5: Criterion comparison (formulation held constant): worst-case flight-envelope margins per scenario, minimum-time versus minimum-fuel. Minimum-time rides the power ceiling at a wide stall margin; minimum-fuel flies near the stall at low power.

		$V/V_{\text{stall}} - 1$		$P/P_{\max}$	
Form.	SC	min-time	min-fuel	min-time	min-fuel
UF	SC-00	0.577	0.090	0.990	0.517
UF	SC-01	0.389	0.090	0.993	0.893
UF	SC-02	0.145	0.106	0.984	0.636
UF	SC-03	0.157	0.115	0.989	0.831
FC	SC-00	0.577	0.283	0.995	0.830
FC	SC-01	0.462	0.278	0.999	0.997
FC	SC-02	0.577	0.471	0.998	0.995
FC	SC-03	0.464	0.216	0.998	0.996

The rigid-body model carries attitude-control variables the point-mass model does not. Table E.6 reports them for the FC under both criteria. The lift-coefficient margin tightens under minimum-fuel (more lift is needed at low speed), and the lateral actuators—aileron, rudder and roll rate—are essentially idle on the straight missions and only load up on the turning ones, yet they remain far from saturation under either criterion.

Table E.6: Criterion comparison, FC attitude-control margins. Lift-coefficient margin below $C_{L,\max}$ and peak control-surface/roll-rate magnitudes as a fraction of their limits. The lateral channels activate only in the turning scenarios and stay well clear of saturation.

SC	Crit.	$1 - C_L/C_{L,\max}$	$\lvert\delta_e\rvert$	$\lvert\delta_a\rvert$	$\lvert\delta_r\rvert$	$\lvert p\rvert$
SC-00	min-time	0.616	0.112	7.2×10^{-6}	2.4×10^{-5}	1.6×10^{-4}
SC-00	min-fuel	0.408	0.330	1.6×10^{-5}	1.4×10^{-5}	1.4×10^{-4}
SC-01	min-time	0.547	0.317	6.5×10^{-6}	1.8×10^{-5}	1.3×10^{-4}
SC-01	min-fuel	0.434	0.302	9.0×10^{-6}	1.4×10^{-5}	1.3×10^{-4}
SC-02	min-time	0.587	0.116	3.8×10^{-3}	3.4×10^{-2}	4.3×10^{-2}
SC-02	min-fuel	0.506	0.204	5.6×10^{-3}	4.0×10^{-2}	4.1×10^{-2}
SC-03	min-time	0.505	0.253	2.7×10^{-3}	2.4×10^{-2}	7.5×10^{-2}
SC-03	min-fuel	0.284	0.523	2.1×10^{-3}	1.6×10^{-2}	9.9×10^{-2}

Table E.7 is the formulation cut: with the criterion fixed, it pairs the UF and FC envelope margins. The two formulations sit at comparable distances from the same physical limits; the FC generally keeps a little more stall margin and rides the power ceiling marginally higher, but neither buys feasibility at the

expense of the other's bound.

Table E.7: Formulation comparison (criterion held constant): worst-case flight-envelope margins per scenario, UF versus FC. The two formulations respect the same envelope with comparable slack.

		$V/V_{\text{stall}} - 1$		$P/P_{\max}$	
Crit.	SC	UF	FC	UF	FC
min-time	SC-00	0.577	0.577	0.990	0.995
min-time	SC-01	0.389	0.462	0.993	0.999
min-time	SC-02	0.145	0.577	0.984	0.998
min-time	SC-03	0.157	0.464	0.989	0.998
min-fuel	SC-00	0.090	0.283	0.517	0.830
min-fuel	SC-01	0.090	0.278	0.893	0.997
min-fuel	SC-02	0.106	0.471	0.636	0.995
min-fuel	SC-03	0.115	0.216	0.831	0.996

The collocation defects underpinning every one of these runs stay negligible (worst-case infinity norm below 10^{-5} throughout, and typically several orders smaller), and tend to shrink further under the minimum-fuel criterion, consistent with its smoother control histories.

Finally, Figures E.1 and E.2 render the same verdict visually, gathering every worst-case margin of each formulation across all four scenarios and both criteria on a single chart. The bars are renormalised to $[-1, 0]$ purely for visualisation, under the convention explained in Section 6.4.7.

The full per-row detail follows.

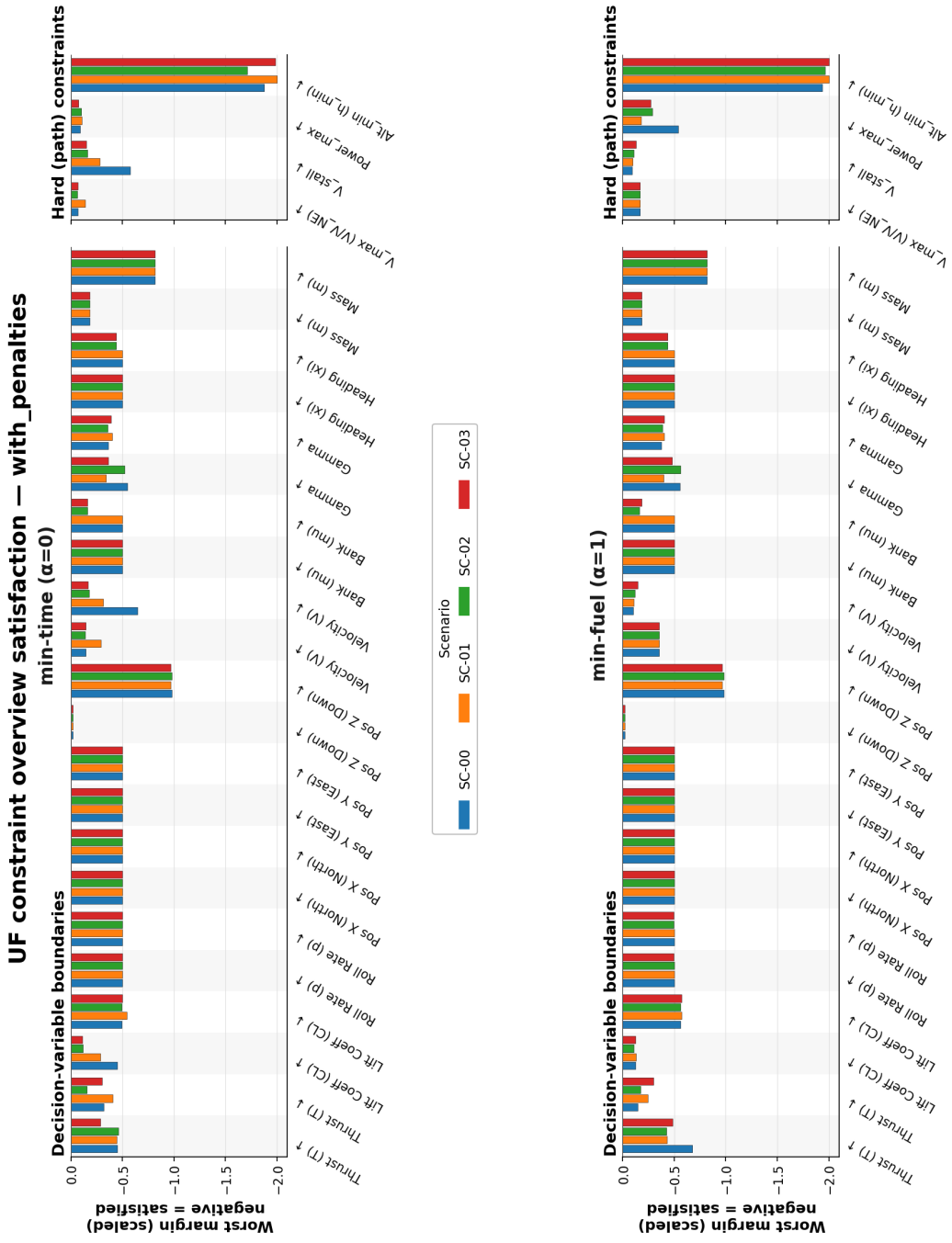


Figure E.1: Normalised worst-case constraint margins for the UF across the four reference scenarios under the minimum-time and minimum-fuel objectives. Bars are renormalised to $[-1, 0]$ for visualisation purposes.

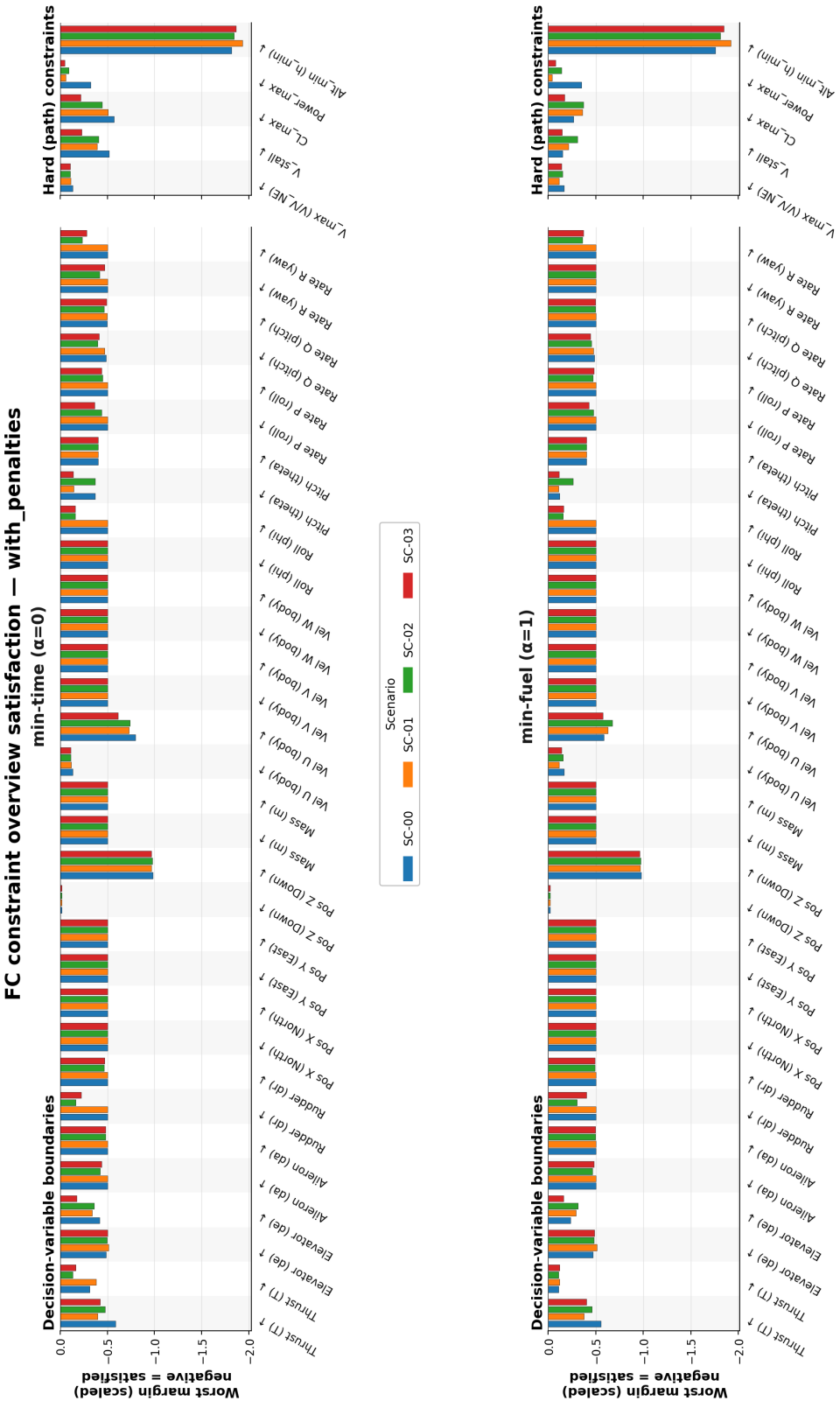


Figure E.2: Normalised worst-case constraint margins for the FC across the four reference scenarios under the minimum-time and minimum-fuel objectives. Bars are renormalised to $[-1, 0]$ for visualisation purposes.

E.3 SC-00 — straight-line level flight

E.3.1 Uncoupled formulation (UF)

Table E.8: Group A — decision-variable boundaries for UF on SC-1 (min-time); positive worst margin = **ACTIVE**.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
Final Time (tf)	5.1092e-18	5.1092e-18	Upper	-1.0000	–	OK
Final Time (tf)	5.1092e-18	5.1092e-18	Lower	-5.0092e-18	–	OK
Thrust (T)	0.3201	0.5487	Upper	-0.4513	11	OK
Thrust (T)	0.3201	0.5487	Lower	-0.3201	249	OK
Lift Coeff (CL)	0.2549	0.3112	Upper	-0.4483	0	OK
Lift Coeff (CL)	0.2549	0.3112	Lower	-0.4954	156	OK
Roll Rate (p)	-8.5635e-05	1.1934e-04	Upper	-0.4999	104	OK
Roll Rate (p)	-8.5635e-05	1.1934e-04	Lower	-0.4999	114	OK
Pos X (North)	-1.8813e-27	3.8607e-20	Upper	-0.5000	0	OK
Pos X (North)	-1.8813e-27	3.8607e-20	Lower	-0.5000	0	OK
Pos Y (East)	-4.1117e-28	1.6732e-16	Upper	-0.5000	187	OK
Pos Y (East)	-4.1117e-28	1.6732e-16	Lower	-0.5000	0	OK
Pos Z (Down)	-0.0303	-0.0290	Upper	-0.0189	135	OK
Pos Z (Down)	-0.0303	-0.0290	Lower	-0.9798	0	OK
Velocity (V)	1.7673	1.9771	Upper	-0.1441	135	OK
Velocity (V)	1.7673	1.9771	Lower	-0.6461	0	OK
Bank (mu)	-3.2160e-04	2.5102e-04	Upper	-0.4997	249	OK
Bank (mu)	-3.2160e-04	2.5102e-04	Lower	-0.4997	93	OK
Gamma	-0.0387	0.0537	Upper	-0.5463	249	OK
Gamma	-0.0387	0.0537	Lower	-0.3613	52	OK
Heading (xi)	0.0312	0.0313	Upper	-0.5000	0	OK
Heading (xi)	0.0312	0.0313	Lower	-0.5000	135	OK
Mass (m)	0.9088	0.9091	Upper	-0.1818	0	OK
Mass (m)	0.9088	0.9091	Lower	-0.8179	249	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Upper	-1.0000	0	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Lower	0.0000e+00	0	OK

Table E.9: Group B — hard (path) constraints for UF on SC-1 (min-time); sign convention as Table E.8.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
V_max (V/V_NE)	0.8331	0.9321	Upper	-0.0679	135	OK
V_max (V/V_NE)	0.8331	0.9321	Lower	-0.8331	0	OK
V_stall	1.5765	1.7638	Lower	-0.5765	0	OK
Power_max	0.5548	0.9128	Upper	-0.0872	11	OK
Alt_min (h_min)	2.8748	3.0000	Lower	-1.8748	135	OK

Table E.10: Group C — dynamic constraints (trapezoidal collocation residuals, ℓ_∞ per state) for UF on SC-1 (min-time); equalities satisfied to tolerance (status OK).

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
x	9.475661e-05	9.475661e-05	Equality	9.475661e-05	7	OK
y	9.544814e-08	9.544814e-08	Equality	9.544814e-08	3	OK
z	2.051685e-04	2.051685e-04	Equality	2.051685e-04	4	OK
v	2.322866e-07	2.322866e-07	Equality	2.322866e-07	1	OK
mu	1.990578e-06	1.990578e-06	Equality	1.990578e-06	7	OK
gamma	2.828747e-06	2.828747e-06	Equality	2.828747e-06	6	OK
xi	2.415447e-07	2.415447e-07	Equality	2.415447e-07	1	OK
m	7.195983e-11	7.195983e-11	Equality	7.195983e-11	4	OK
C	0.000000e+00	0.000000e+00	Equality	0.000000e+00	0	OK

Table E.11: Group A — decision-variable boundaries for UF on SC-1 (min-fuel); positive worst margin = **ACTIVE**.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
Final Time (tf)	6.4486e-18	6.4486e-18	Upper	-1.0000	–	OK
Final Time (tf)	6.4486e-18	6.4486e-18	Lower	-6.3486e-18	–	OK
Thrust (T)	0.1466	0.3218	Upper	-0.6782	125	OK
Thrust (T)	0.1466	0.3218	Lower	-0.1466	239	OK
Lift Coeff (CL)	0.3224	0.6328	Upper	-0.1267	249	OK
Lift Coeff (CL)	0.3224	0.6328	Lower	-0.5629	0	OK
Roll Rate (p)	-9.0267e-05	7.8894e-05	Upper	-0.4999	42	OK
Roll Rate (p)	-9.0267e-05	7.8894e-05	Lower	-0.4999	249	OK
Pos X (North)	-1.5920e-21	3.8607e-20	Upper	-0.5000	0	OK
Pos X (North)	-1.5920e-21	3.8607e-20	Lower	-0.5000	0	OK
Pos Y (East)	-2.4830e-28	1.6732e-16	Upper	-0.5000	181	OK
Pos Y (East)	-2.4830e-28	1.6732e-16	Lower	-0.5000	0	OK
Pos Z (Down)	-0.0304	-0.0296	Upper	-0.0195	176	OK
Pos Z (Down)	-0.0304	-0.0296	Lower	-0.9797	52	OK
Velocity (V)	1.2250	1.7673	Upper	-0.3539	0	OK
Velocity (V)	1.2250	1.7673	Lower	-0.1037	249	OK
Bank (mu)	-4.0683e-04	2.6257e-04	Upper	-0.4997	238	OK
Bank (mu)	-4.0683e-04	2.6257e-04	Lower	-0.4996	31	OK
Gamma	-0.0247	0.0429	Upper	-0.5571	238	OK
Gamma	-0.0247	0.0429	Lower	-0.3753	114	OK
Heading (xi)	0.0312	0.0313	Upper	-0.5000	21	OK
Heading (xi)	0.0312	0.0313	Lower	-0.5000	104	OK
Mass (m)	0.9089	0.9091	Upper	-0.1818	0	OK
Mass (m)	0.9089	0.9091	Lower	-0.8180	249	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Upper	-1.0000	0	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Lower	0.0000e+00	0	OK

Table E.12: Group B — hard (path) constraints for UF on SC-1 (min-fuel); sign convention as Table E.11.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
V_max (V/V_NE)	0.5775	0.8331	Upper	-0.1669	0	OK
V_max (V/V_NE)	0.5775	0.8331	Lower	-0.5775	249	OK
V_stall	1.0928	1.5765	Lower	-0.0928	249	OK
Power_max	0.1743	0.4598	Upper	-0.5402	124	OK
Alt_min (h_min)	2.9346	3.0123	Lower	-1.9346	176	OK

Table E.13: Group C — dynamic constraints (trapezoidal collocation residuals, ℓ_∞ per state) for UF on SC-1 (min-fuel); equalities satisfied to tolerance (status OK).

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
x	1.046839e-05	1.046839e-05	Equality	1.046839e-05	19	OK
y	4.708446e-09	4.708446e-09	Equality	4.708446e-09	2	OK
z	1.027124e-05	1.027124e-05	Equality	1.027124e-05	18	OK
v	3.006324e-08	3.006324e-08	Equality	3.006324e-08	3	OK
mu	6.822248e-08	6.822248e-08	Equality	6.822248e-08	2	OK
gamma	1.560006e-07	1.560006e-07	Equality	1.560006e-07	10	OK
xi	3.168561e-08	3.168561e-08	Equality	3.168561e-08	9	OK
m	8.638508e-12	8.638508e-12	Equality	8.638508e-12	5	OK
C	0.000000e+00	0.000000e+00	Equality	0.000000e+00	0	OK

E.3.2 Full-control formulation (FC)

Table E.14: Group A — decision-variable boundaries for FC on SC-1 (min-time); positive worst margin = **ACTIVE**.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
Final Time (tf)	5.5038e-18	5.5038e-18	Upper	-1.0000	–	OK
Final Time (tf)	5.5038e-18	5.5038e-18	Lower	-5.4038e-18	–	OK
Thrust (T)	0.3128	0.4111	Upper	-0.5889	166	OK
Thrust (T)	0.3128	0.4111	Lower	-0.3128	249	OK
Elevator (de)	-0.0813	0.0125	Upper	-0.4875	0	OK
Elevator (de)	-0.0813	0.0125	Lower	-0.4187	249	OK
Aileron (da)	-1.2486e-05	4.0218e-06	Upper	-0.5000	207	OK
Aileron (da)	-1.2486e-05	4.0218e-06	Lower	-0.5000	249	OK
Rudder (dr)	-5.2624e-05	1.5223e-05	Upper	-0.5000	0	OK
Rudder (dr)	-5.2624e-05	1.5223e-05	Lower	-0.4999	249	OK
Pos X (North)	0.0000e+00	3.8607e-20	Upper	-0.5000	0	OK
Pos X (North)	0.0000e+00	3.8607e-20	Lower	-0.5000	0	OK
Pos Y (East)	0.0000e+00	1.6732e-16	Upper	-0.5000	185	OK
Pos Y (East)	0.0000e+00	1.6732e-16	Lower	-0.5000	0	OK
Pos Z (Down)	-0.0303	-0.0285	Upper	-0.0184	73	OK
Pos Z (Down)	-0.0303	-0.0285	Lower	-0.9798	0	OK
Mass (m)	1.9995e-16	2.0000e-16	Upper	-0.5000	0	OK
Mass (m)	1.9995e-16	2.0000e-16	Lower	-0.5000	0	OK
Vel U (body)	0.7977	0.8657	Upper	-0.1343	52	OK
Vel U (body)	0.7977	0.8657	Lower	-0.7977	249	OK
Vel V (body)	-8.1601e-23	2.4292e-23	Upper	-0.5000	0	OK
Vel V (body)	-8.1601e-23	2.4292e-23	Lower	-0.5000	0	OK
Vel W (body)	0.0000e+00	2.7100e-19	Upper	-0.5000	0	OK
Vel W (body)	0.0000e+00	2.7100e-19	Lower	-0.5000	0	OK
Roll (phi)	-9.0969e-05	1.2528e-04	Upper	-0.4999	249	OK
Roll (phi)	-9.0969e-05	1.2528e-04	Lower	-0.4999	62	OK
Pitch (theta)	0.0000e+00	0.2279	Upper	-0.3721	249	OK
Pitch (theta)	0.0000e+00	0.2279	Lower	-0.4000	0	OK
Rate P (roll)	-7.5246e-05	2.6491e-05	Upper	-0.5000	176	OK
Rate P (roll)	-7.5246e-05	2.6491e-05	Lower	-0.4999	10	OK
Rate Q (pitch)	-0.0013	0.0138	Upper	-0.4862	10	OK
Rate Q (pitch)	-0.0013	0.0138	Lower	-0.4987	83	OK
Rate R (yaw)	-3.1033e-05	6.0757e-05	Upper	-0.4999	239	OK
Rate R (yaw)	-3.1033e-05	6.0757e-05	Lower	-0.5000	42	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Upper	-1.0000	0	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Lower	0.0000e+00	0	OK

Table E.15: Group B — hard (path) constraints for FC on SC-1 (min-time); sign convention as Table E.14.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
V_max (V/V_NE)	0.8013	0.8676	Upper	-0.1324	52	OK
V_max (V/V_NE)	0.8013	0.8676	Lower	-0.8013	249	OK
V_stall	1.5164	1.6417	Lower	-0.5164	249	OK
CL_max	0.2504	0.4272	Upper	-0.5728	249	OK
CL_max	0.2504	0.4272	Lower	-0.7504	0	OK
Power_max	0.4902	0.6787	Upper	-0.3213	145	OK
Alt_min (h_min)	2.8178	3.0000	Lower	-1.8178	73	OK

Table E.16: Group C — dynamic constraints (trapezoidal collocation residuals, ℓ_∞ per state) for FC on SC-1 (min-time); equalities satisfied to tolerance (status OK).

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
x	2.562734e-08	2.562734e-08	Equality	2.562734e-08	8	OK
y	1.823875e-09	1.823875e-09	Equality	1.823875e-09	19	OK
z	2.431555e-05	2.431555e-05	Equality	2.431555e-05	18	OK
m	4.803176e-12	4.803176e-12	Equality	4.803176e-12	18	OK
u	8.564121e-08	8.564121e-08	Equality	8.564121e-08	23	OK
v	2.658919e-10	2.658919e-10	Equality	2.658919e-10	8	OK
w	1.344480e-07	1.344480e-07	Equality	1.344480e-07	19	OK
phi	4.520124e-11	4.520124e-11	Equality	4.520124e-11	7	OK
theta	6.115732e-09	6.115732e-09	Equality	6.115732e-09	19	OK
psi	7.492921e-11	7.492921e-11	Equality	7.492921e-11	8	OK
p	1.186246e-08	1.186246e-08	Equality	1.186246e-08	9	OK
q	5.180718e-06	5.180718e-06	Equality	5.180718e-06	22	OK
r	1.641249e-09	1.641249e-09	Equality	1.641249e-09	7	OK
C	0.000000e+00	0.000000e+00	Equality	0.000000e+00	0	OK

Table E.17: Group A — decision-variable boundaries for FC on SC-1 (min-fuel); positive worst margin = **ACTIVE**.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
Final Time (tf)	6.1833e-18	6.1833e-18	Upper	-1.0000	–	OK
Final Time (tf)	6.1833e-18	6.1833e-18	Lower	-6.0833e-18	–	OK
Thrust (T)	0.1097	0.4442	Upper	-0.5558	156	OK
Thrust (T)	0.1097	0.4442	Lower	-0.1097	239	OK
Elevator (de)	-0.2639	0.0294	Upper	-0.4706	0	OK
Elevator (de)	-0.2639	0.0294	Lower	-0.2361	249	OK
Aileron (da)	-3.8798e-06	3.8840e-06	Upper	-0.5000	145	OK
Aileron (da)	-3.8798e-06	3.8840e-06	Lower	-0.5000	135	OK
Rudder (dr)	-1.8764e-05	6.6508e-06	Upper	-0.5000	31	OK
Rudder (dr)	-1.8764e-05	6.6508e-06	Lower	-0.5000	249	OK
Pos X (North)	0.0000e+00	3.8607e-20	Upper	-0.5000	0	OK
Pos X (North)	0.0000e+00	3.8607e-20	Lower	-0.5000	0	OK
Pos Y (East)	0.0000e+00	1.6732e-16	Upper	-0.5000	179	OK
Pos Y (East)	0.0000e+00	1.6732e-16	Lower	-0.5000	0	OK
Pos Z (Down)	-0.0308	-0.0279	Upper	-0.0178	83	OK
Pos Z (Down)	-0.0308	-0.0279	Lower	-0.9793	218	OK
Mass (m)	1.9996e-16	2.0000e-16	Upper	-0.5000	0	OK
Mass (m)	1.9996e-16	2.0000e-16	Lower	-0.5000	0	OK
Vel U (body)	0.5864	0.8331	Upper	-0.1669	0	OK
Vel U (body)	0.5864	0.8331	Lower	-0.5864	249	OK
Vel V (body)	-2.7495e-23	2.0764e-23	Upper	-0.5000	0	OK
Vel V (body)	-2.7495e-23	2.0764e-23	Lower	-0.5000	0	OK
Vel W (body)	0.0000e+00	5.5916e-19	Upper	-0.5000	0	OK
Vel W (body)	0.0000e+00	5.5916e-19	Lower	-0.5000	0	OK
Roll (phi)	-8.2309e-05	9.7637e-05	Upper	-0.4999	249	OK
Roll (phi)	-8.2309e-05	9.7637e-05	Lower	-0.4999	52	OK
Pitch (theta)	0.0000e+00	0.4812	Upper	-0.1188	239	OK
Pitch (theta)	0.0000e+00	0.4812	Lower	-0.4000	0	OK
Rate P (roll)	-5.8575e-05	6.1771e-05	Upper	-0.4999	249	OK
Rate P (roll)	-5.8575e-05	6.1771e-05	Lower	-0.4999	10	OK
Rate Q (pitch)	-9.0695e-04	0.0157	Upper	-0.4843	10	OK
Rate Q (pitch)	-9.0695e-04	0.0157	Lower	-0.4991	249	OK
Rate R (yaw)	-2.2810e-05	5.2702e-05	Upper	-0.4999	239	OK
Rate R (yaw)	-2.2810e-05	5.2702e-05	Lower	-0.5000	42	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Upper	-1.0000	0	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Lower	0.0000e+00	0	OK

Table E.18: Group B — hard (path) constraints for FC on SC-1 (min-fuel); sign convention as Table E.17.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
V_max (V/V_NE)	0.6066	0.8331	Upper	-0.1669	0	OK
V_max (V/V_NE)	0.6066	0.8331	Lower	-0.6066	249	OK
V_stall	1.1480	1.5765	Lower	-0.1480	249	OK
CL_max	0.2529	0.7343	Upper	-0.2657	249	OK
CL_max	0.2529	0.7343	Lower	-0.7529	0	OK
Power_max	0.1289	0.6486	Upper	-0.3514	155	OK
Alt_min (h_min)	2.7594	3.0539	Lower	-1.7594	83	OK

Table E.19: Group C — dynamic constraints (trapezoidal collocation residuals, ℓ_∞ per state) for FC on SC-1 (min-fuel); equalities satisfied to tolerance (status OK).

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
x	1.339310e-07	1.339310e-07	Equality	1.339310e-07	7	OK
y	2.109114e-10	2.109114e-10	Equality	2.109114e-10	12	OK
z	7.306350e-07	7.306350e-07	Equality	7.306350e-07	10	OK
m	6.050599e-14	6.050599e-14	Equality	6.050599e-14	23	OK
u	2.398461e-08	2.398461e-08	Equality	2.398461e-08	15	OK
v	4.794758e-09	4.794758e-09	Equality	4.794758e-09	11	OK
w	9.354341e-09	9.354341e-09	Equality	9.354341e-09	10	OK
phi	1.178415e-09	1.178415e-09	Equality	1.178415e-09	17	OK
theta	1.380999e-09	1.380999e-09	Equality	1.380999e-09	17	OK
psi	1.419685e-09	1.419685e-09	Equality	1.419685e-09	12	OK
p	1.009636e-07	1.009636e-07	Equality	1.009636e-07	13	OK
q	1.514719e-07	1.514719e-07	Equality	1.514719e-07	11	OK
r	3.939022e-08	3.939022e-08	Equality	3.939022e-08	14	OK
C	0.000000e+00	0.000000e+00	Equality	0.000000e+00	0	OK

E.4 SC-01 — straight-line steady climb

E.4.1 Uncoupled formulation (UF)

Table E.20: Group A — decision-variable boundaries for UF on SC-2 (min-time); positive worst margin = **ACTIVE**.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
Final Time (tf)	5.7119e-18	5.7119e-18	Upper	-1.0000	–	OK
Final Time (tf)	5.7119e-18	5.7119e-18	Lower	-5.6119e-18	–	OK
Thrust (T)	0.4051	0.5567	Upper	-0.4433	187	OK
Thrust (T)	0.4051	0.5567	Lower	-0.4051	249	OK
Lift Coeff (CL)	0.3024	0.4751	Upper	-0.2844	249	OK
Lift Coeff (CL)	0.3024	0.4751	Lower	-0.5429	52	OK
Roll Rate (p)	-2.0984e-04	1.6444e-04	Upper	-0.4998	218	OK
Roll Rate (p)	-2.0984e-04	1.6444e-04	Lower	-0.4998	135	OK
Pos X (North)	2.4903e-27	3.9956e-20	Upper	-0.5000	0	OK
Pos X (North)	2.4903e-27	3.9956e-20	Lower	-0.5000	0	OK
Pos Y (East)	1.0135e-28	1.6732e-16	Upper	-0.5000	182	OK
Pos Y (East)	1.0135e-28	1.6732e-16	Lower	-0.5000	0	OK
Pos Z (Down)	-0.0438	-0.0303	Upper	-0.0202	0	OK
Pos Z (Down)	-0.0438	-0.0303	Lower	-0.9663	249	OK
Velocity (V)	1.4322	1.8300	Upper	-0.2912	52	OK
Velocity (V)	1.4322	1.8300	Lower	-0.3110	249	OK
Bank (mu)	-4.0625e-04	8.1465e-04	Upper	-0.4992	228	OK
Bank (mu)	-4.0625e-04	8.1465e-04	Lower	-0.4996	114	OK
Gamma	3.8126e-12	0.2585	Upper	-0.3415	239	OK
Gamma	3.8126e-12	0.2585	Lower	-0.4000	0	OK
Heading (xi)	0.0312	0.0313	Upper	-0.5000	239	OK
Heading (xi)	0.0312	0.0313	Lower	-0.5000	166	OK
Mass (m)	0.9088	0.9091	Upper	-0.1818	0	OK
Mass (m)	0.9088	0.9091	Lower	-0.8179	249	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Upper	-1.0000	0	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Lower	0.0000e+00	0	OK

Table E.21: Group B — hard (path) constraints for UF on SC-2 (min-time); sign convention as Table E.20.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
V_max (V/V_NE)	0.6752	0.8627	Upper	-0.1373	52	OK
V_max (V/V_NE)	0.6752	0.8627	Lower	-0.6752	249	OK
V_stall	1.2778	1.6325	Lower	-0.2778	249	OK
Power_max	0.5466	0.8968	Upper	-0.1032	114	OK
Alt_min (h_min)	3.0000	4.3333	Lower	-2.0000	0	OK

Table E.22: Group C — dynamic constraints (trapezoidal collocation residuals, ℓ_∞ per state) for UF on SC-2 (min-time); equalities satisfied to tolerance (status OK).

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
x	1.668392e-04	1.668392e-04	Equality	1.668392e-04	20	OK
y	1.551074e-07	1.551074e-07	Equality	1.551074e-07	4	OK
z	1.379641e-06	1.379641e-06	Equality	1.379641e-06	5	OK
v	6.245133e-07	6.245133e-07	Equality	6.245133e-07	3	OK
mu	1.630313e-05	1.630313e-05	Equality	1.630313e-05	16	OK
gamma	1.599455e-05	1.599455e-05	Equality	1.599455e-05	16	OK
xi	1.340129e-06	1.340129e-06	Equality	1.340129e-06	15	OK
m	7.163961e-11	7.163961e-11	Equality	7.163961e-11	7	OK
C	0.000000e+00	0.000000e+00	Equality	0.000000e+00	0	OK

Table E.23: Group A — decision-variable boundaries for UF on SC-2 (min-fuel); positive worst margin = **ACTIVE**.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
Final Time (tf)	6.3120e-18	6.3120e-18	Upper	-1.0000	–	OK
Final Time (tf)	6.3120e-18	6.3120e-18	Lower	-6.2120e-18	–	OK
Thrust (T)	0.2450	0.5693	Upper	-0.4307	166	OK
Thrust (T)	0.2450	0.5693	Lower	-0.2450	249	OK
Lift Coeff (CL)	0.3321	0.6319	Upper	-0.1276	249	OK
Lift Coeff (CL)	0.3321	0.6319	Lower	-0.5725	0	OK
Roll Rate (p)	-1.7180e-04	1.9550e-04	Upper	-0.4998	114	OK
Roll Rate (p)	-1.7180e-04	1.9550e-04	Lower	-0.4998	0	OK
Pos X (North)	7.8657e-28	3.8892e-20	Upper	-0.5000	0	OK
Pos X (North)	7.8657e-28	3.8892e-20	Lower	-0.5000	0	OK
Pos Y (East)	-1.4710e-28	1.6732e-16	Upper	-0.5000	181	OK
Pos Y (East)	-1.4710e-28	1.6732e-16	Lower	-0.5000	0	OK
Pos Z (Down)	-0.0438	-0.0303	Upper	-0.0202	0	OK
Pos Z (Down)	-0.0438	-0.0303	Lower	-0.9663	249	OK
Velocity (V)	1.2262	1.7673	Upper	-0.3539	0	OK
Velocity (V)	1.2262	1.7673	Lower	-0.1050	249	OK
Bank (mu)	-3.7681e-04	3.2376e-04	Upper	-0.4997	197	OK
Bank (mu)	-3.7681e-04	3.2376e-04	Lower	-0.4996	31	OK
Gamma	5.2864e-13	0.2002	Upper	-0.3998	238	OK
Gamma	5.2864e-13	0.2002	Lower	-0.4000	0	OK
Heading (xi)	0.0312	0.0313	Upper	-0.5000	249	OK
Heading (xi)	0.0312	0.0313	Lower	-0.5000	145	OK
Mass (m)	0.9088	0.9091	Upper	-0.1818	0	OK
Mass (m)	0.9088	0.9091	Lower	-0.8179	249	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Upper	-1.0000	0	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Lower	0.0000e+00	0	OK

Table E.24: Group B — hard (path) constraints for UF on SC-2 (min-fuel); sign convention as Table E.23.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
V_max (V/V_NE)	0.5781	0.8331	Upper	-0.1669	0	OK
V_max (V/V_NE)	0.5781	0.8331	Lower	-0.5781	249	OK
V_stall	1.0940	1.5765	Lower	-0.0940	249	OK
Power_max	0.2830	0.8230	Upper	-0.1770	145	OK
Alt_min (h_min)	3.0000	4.3333	Lower	-2.0000	0	OK

Table E.25: Group C — dynamic constraints (trapezoidal collocation residuals, ℓ_∞ per state) for UF on SC-2 (min-fuel); equalities satisfied to tolerance (status OK).

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
x	1.135104e-05	1.135104e-05	Equality	1.135104e-05	5	OK
y	5.738664e-08	5.738664e-08	Equality	5.738664e-08	6	OK
z	1.764381e-07	1.764381e-07	Equality	1.764381e-07	3	OK
v	2.421133e-07	2.421133e-07	Equality	2.421133e-07	2	OK
mu	1.234361e-06	1.234361e-06	Equality	1.234361e-06	22	OK
gamma	3.435064e-06	3.435064e-06	Equality	3.435064e-06	6	OK
xi	7.187639e-08	7.187639e-08	Equality	7.187639e-08	8	OK
m	5.359679e-11	5.359679e-11	Equality	5.359679e-11	0	OK
C	0.000000e+00	0.000000e+00	Equality	0.000000e+00	0	OK

E.4.2 Full-control formulation (FC)

Table E.26: Group A — decision-variable boundaries for FC on SC-2 (min-time); positive worst margin = **ACTIVE**.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
Final Time (tf)	5.6732e-18	5.6732e-18	Upper	-1.0000	–	OK
Final Time (tf)	5.6732e-18	5.6732e-18	Lower	-5.5732e-18	–	OK
Thrust (T)	0.3787	0.6038	Upper	-0.3962	187	OK
Thrust (T)	0.3787	0.6038	Lower	-0.3787	249	OK
Elevator (de)	-0.1621	-0.0141	Upper	-0.5141	0	OK
Elevator (de)	-0.1621	-0.0141	Lower	-0.3379	218	OK
Aileron (da)	-1.4424e-05	4.0929e-06	Upper	-0.5000	207	OK
Aileron (da)	-1.4424e-05	4.0929e-06	Lower	-0.5000	249	OK
Rudder (dr)	-5.0129e-05	2.3236e-05	Upper	-0.5000	31	OK
Rudder (dr)	-5.0129e-05	2.3236e-05	Lower	-0.4999	249	OK
Pos X (North)	0.0000e+00	3.8608e-20	Upper	-0.5000	0	OK
Pos X (North)	0.0000e+00	3.8608e-20	Lower	-0.5000	0	OK
Pos Y (East)	0.0000e+00	1.6732e-16	Upper	-0.5000	183	OK
Pos Y (East)	0.0000e+00	1.6732e-16	Lower	-0.5000	0	OK
Pos Z (Down)	-0.0438	-0.0296	Upper	-0.0195	31	OK
Pos Z (Down)	-0.0438	-0.0296	Lower	-0.9663	249	OK
Mass (m)	1.9993e-16	2.0000e-16	Upper	-0.5000	0	OK
Mass (m)	1.9993e-16	2.0000e-16	Lower	-0.5000	0	OK
Vel U (body)	0.7299	0.8842	Upper	-0.1158	41	OK
Vel U (body)	0.7299	0.8842	Lower	-0.7299	249	OK
Vel V (body)	-7.7942e-23	2.9064e-23	Upper	-0.5000	0	OK
Vel V (body)	-7.7942e-23	2.9064e-23	Lower	-0.5000	0	OK
Vel W (body)	0.0000e+00	3.5962e-19	Upper	-0.5000	0	OK
Vel W (body)	0.0000e+00	3.5962e-19	Lower	-0.5000	0	OK
Roll (phi)	-8.7843e-05	1.1305e-04	Upper	-0.4999	249	OK
Roll (phi)	-8.7843e-05	1.1305e-04	Lower	-0.4999	42	OK
Pitch (theta)	0.0000e+00	0.4576	Upper	-0.1424	228	OK
Pitch (theta)	0.0000e+00	0.4576	Lower	-0.4000	0	OK
Rate P (roll)	-7.1368e-05	3.0090e-05	Upper	-0.5000	124	OK
Rate P (roll)	-7.1368e-05	3.0090e-05	Lower	-0.4999	10	OK
Rate Q (pitch)	-0.0020	0.0304	Upper	-0.4696	10	OK
Rate Q (pitch)	-0.0020	0.0304	Lower	-0.4980	62	OK
Rate R (yaw)	-3.8839e-05	6.7837e-05	Upper	-0.4999	239	OK
Rate R (yaw)	-3.8839e-05	6.7837e-05	Lower	-0.5000	42	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Upper	-1.0000	0	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Lower	0.0000e+00	0	OK

Table E.27: Group B — hard (path) constraints for FC on SC-2 (min-time); sign convention as Table E.26.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
V_max (V/V_NE)	0.7367	0.8861	Upper	-0.1139	41	OK
V_max (V/V_NE)	0.7367	0.8861	Lower	-0.7367	249	OK
V_stall	1.3943	1.6767	Lower	-0.3943	249	OK
CL_max	0.2465	0.4925	Upper	-0.5075	249	OK
CL_max	0.2465	0.4925	Lower	-0.7465	0	OK
Power_max	0.5524	0.9403	Upper	-0.0597	145	OK
Alt_min (h_min)	2.9312	4.3333	Lower	-1.9312	31	OK

Table E.28: Group C — dynamic constraints (trapezoidal collocation residuals, ℓ_∞ per state) for FC on SC-2 (min-time); equalities satisfied to tolerance (status OK).

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
x	6.290847e-08	6.290847e-08	Equality	6.290847e-08	10	OK
y	8.610386e-10	8.610386e-10	Equality	8.610386e-10	9	OK
z	8.519524e-09	8.519524e-09	Equality	8.519524e-09	5	OK
m	2.412358e-13	2.412358e-13	Equality	2.412358e-13	14	OK
u	2.491743e-08	2.491743e-08	Equality	2.491743e-08	23	OK
v	7.461242e-10	7.461242e-10	Equality	7.461242e-10	19	OK
w	2.560687e-08	2.560687e-08	Equality	2.560687e-08	23	OK
phi	3.164201e-10	3.164201e-10	Equality	3.164201e-10	22	OK
theta	1.282625e-09	1.282625e-09	Equality	1.282625e-09	22	OK
psi	5.601454e-10	5.601454e-10	Equality	5.601454e-10	23	OK
p	4.241920e-08	4.241920e-08	Equality	4.241920e-08	19	OK
q	3.768174e-07	3.768174e-07	Equality	3.768174e-07	22	OK
r	1.347897e-08	1.347897e-08	Equality	1.347897e-08	23	OK
C	0.000000e+00	0.000000e+00	Equality	0.000000e+00	0	OK

Table E.29: Group A — decision-variable boundaries for FC on SC-2 (min-fuel); positive worst margin = **ACTIVE**.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
Final Time (tf)	5.7739e-18	5.7739e-18	Upper	-1.0000	–	OK
Final Time (tf)	5.7739e-18	5.7739e-18	Lower	-5.6739e-18	–	OK
Thrust (T)	0.1182	0.6236	Upper	-0.3764	186	OK
Thrust (T)	0.1182	0.6236	Lower	-0.1182	239	OK
Elevator (de)	-0.2104	-0.0105	Upper	-0.5105	0	OK
Elevator (de)	-0.2104	-0.0105	Lower	-0.2896	249	OK
Aileron (da)	-4.3796e-05	6.3192e-05	Upper	-0.4999	228	OK
Aileron (da)	-4.3796e-05	6.3192e-05	Lower	-0.5000	249	OK
Rudder (dr)	-2.2398e-05	1.7956e-05	Upper	-0.5000	124	OK
Rudder (dr)	-2.2398e-05	1.7956e-05	Lower	-0.5000	166	OK
Pos X (North)	-5.2750e-23	3.8608e-20	Upper	-0.5000	0	OK
Pos X (North)	-5.2750e-23	3.8608e-20	Lower	-0.5000	0	OK
Pos Y (East)	0.0000e+00	1.6732e-16	Upper	-0.5000	180	OK
Pos Y (East)	0.0000e+00	1.6732e-16	Lower	-0.5000	0	OK
Pos Z (Down)	-0.0438	-0.0295	Upper	-0.0194	31	OK
Pos Z (Down)	-0.0438	-0.0295	Lower	-0.9663	249	OK
Mass (m)	1.9993e-16	2.0000e-16	Upper	-0.5000	0	OK
Mass (m)	1.9993e-16	2.0000e-16	Lower	-0.5000	0	OK
Vel U (body)	0.6251	0.8850	Upper	-0.1150	42	OK
Vel U (body)	0.6251	0.8850	Lower	-0.6251	249	OK
Vel V (body)	-2.0841e-22	2.0935e-22	Upper	-0.5000	0	OK
Vel V (body)	-2.0841e-22	2.0935e-22	Lower	-0.5000	0	OK
Vel W (body)	0.0000e+00	4.9107e-19	Upper	-0.5000	0	OK
Vel W (body)	0.0000e+00	4.9107e-19	Lower	-0.5000	0	OK
Roll (phi)	-1.6960e-04	1.7213e-04	Upper	-0.4998	239	OK
Roll (phi)	-1.6960e-04	1.7213e-04	Lower	-0.4998	62	OK
Pitch (theta)	0.0000e+00	0.4907	Upper	-0.1093	228	OK
Pitch (theta)	0.0000e+00	0.4907	Lower	-0.4000	0	OK
Rate P (roll)	-6.9588e-04	4.4626e-04	Upper	-0.4996	207	OK
Rate P (roll)	-6.9588e-04	4.4626e-04	Lower	-0.4993	249	OK
Rate Q (pitch)	-6.4964e-04	0.0261	Upper	-0.4739	10	OK
Rate Q (pitch)	-6.4964e-04	0.0261	Lower	-0.4994	239	OK
Rate R (yaw)	-1.3758e-04	1.8162e-04	Upper	-0.4998	145	OK
Rate R (yaw)	-1.3758e-04	1.8162e-04	Lower	-0.4999	187	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Upper	-1.0000	0	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Lower	0.0000e+00	0	OK

Table E.30: Group B — hard (path) constraints for FC on SC-2 (min-fuel); sign convention as Table E.29.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
V_max (V/V_NE)	0.6398	0.8869	Upper	-0.1131	42	OK
V_max (V/V_NE)	0.6398	0.8869	Lower	-0.6398	249	OK
V_stall	1.2109	1.6782	Lower	-0.2109	249	OK
CL_max	0.2471	0.6403	Upper	-0.3597	249	OK
CL_max	0.2471	0.6403	Lower	-0.7471	0	OK
Power_max	0.1539	0.9586	Upper	-0.0414	145	OK
Alt_min (h_min)	2.9194	4.3333	Lower	-1.9194	31	OK

Table E.31: Group C — dynamic constraints (trapezoidal collocation residuals, ℓ_∞ per state) for FC on SC-2 (min-fuel); equalities satisfied to tolerance (status OK).

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
x	6.132762e-06	6.132762e-06	Equality	6.132762e-06	21	OK
y	9.056234e-10	9.056234e-10	Equality	9.056234e-10	4	OK
z	1.823610e-08	1.823610e-08	Equality	1.823610e-08	3	OK
m	2.060187e-11	2.060187e-11	Equality	2.060187e-11	20	OK
u	2.592855e-08	2.592855e-08	Equality	2.592855e-08	23	OK
v	6.341187e-08	6.341187e-08	Equality	6.341187e-08	22	OK
w	2.695552e-08	2.695552e-08	Equality	2.695552e-08	22	OK
phi	3.615795e-09	3.615795e-09	Equality	3.615795e-09	16	OK
theta	8.164756e-09	8.164756e-09	Equality	8.164756e-09	13	OK
psi	1.017582e-08	1.017582e-08	Equality	1.017582e-08	13	OK
p	2.356229e-06	2.356229e-06	Equality	2.356229e-06	21	OK
q	4.029171e-06	4.029171e-06	Equality	4.029171e-06	21	OK
r	5.085654e-07	5.085654e-07	Equality	5.085654e-07	23	OK
C	0.000000e+00	0.000000e+00	Equality	0.000000e+00	0	OK

E.5 SC-02 — 180° horizontal turn

E.5.1 Uncoupled formulation (UF)

Table E.32: Group A — decision-variable boundaries for UF on SC-3 (min-time); positive worst margin = **ACTIVE**.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
Final Time (tf)	8.7163e-18	8.7163e-18	Upper	-1.0000	–	OK
Final Time (tf)	8.7163e-18	8.7163e-18	Lower	-8.6163e-18	–	OK
Thrust (T)	0.1538	0.5375	Upper	-0.4625	10	OK
Thrust (T)	0.1538	0.5375	Lower	-0.1538	207	OK
Lift Coeff (CL)	0.2531	0.6414	Upper	-0.1181	249	OK
Lift Coeff (CL)	0.2531	0.6414	Lower	-0.4936	93	OK
Roll Rate (p)	-0.0028	2.5322e-04	Upper	-0.4997	259	OK
Roll Rate (p)	-0.0028	2.5322e-04	Lower	-0.4972	166	OK
Pos X (North)	-1.3572e-27	1.1111e-16	Upper	-0.5000	170	OK
Pos X (North)	-1.3572e-27	1.1111e-16	Lower	-0.5000	0	OK
Pos Y (East)	-3.3280e-28	1.9102e-16	Upper	-0.5000	141	OK
Pos Y (East)	-3.3280e-28	1.9102e-16	Lower	-0.5000	0	OK
Pos Z (Down)	-0.0303	-0.0274	Upper	-0.0173	145	OK
Pos Z (Down)	-0.0303	-0.0274	Lower	-0.9798	0	OK
Velocity (V)	1.2972	1.9853	Upper	-0.1359	124	OK
Velocity (V)	1.2972	1.9853	Lower	-0.1760	249	OK
Bank (mu)	-0.3411	9.7157e-13	Upper	-0.5000	0	OK
Bank (mu)	-0.3411	9.7157e-13	Lower	-0.1589	248	OK
Gamma	-0.0455	0.0778	Upper	-0.5222	207	OK
Gamma	-0.0455	0.0778	Lower	-0.3545	41	OK
Heading (xi)	-0.0312	0.0313	Upper	-0.5000	0	OK
Heading (xi)	-0.0312	0.0313	Lower	-0.4375	259	OK
Mass (m)	0.9087	0.9091	Upper	-0.1818	0	OK
Mass (m)	0.9087	0.9091	Lower	-0.8178	259	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Upper	-1.0000	0	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Lower	0.0000e+00	0	OK

Table E.33: Group B — hard (path) constraints for UF on SC-3 (min-time); sign convention as Table E.32.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
V_max (V/V_NE)	0.6116	0.9359	Upper	-0.0641	124	OK
V_max (V/V_NE)	0.6116	0.9359	Lower	-0.6116	249	OK
V_stall	1.1574	1.7712	Lower	-0.1574	249	OK
Power_max	0.2217	0.9021	Upper	-0.0979	11	OK
Alt_min (h_min)	2.7117	3.0000	Lower	-1.7117	145	OK

Table E.34: Group C — dynamic constraints (trapezoidal collocation residuals, ℓ_∞ per state) for UF on SC-3 (min-time); equalities satisfied to tolerance (status OK).

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
x	3.079935e-07	3.079935e-07	Equality	3.079935e-07	17	OK
y	2.456129e-07	2.456129e-07	Equality	2.456129e-07	17	OK
z	2.601519e-04	2.601519e-04	Equality	2.601519e-04	18	OK
v	2.998766e-07	2.998766e-07	Equality	2.998766e-07	21	OK
mu	1.030998e-06	1.030998e-06	Equality	1.030998e-06	2	OK
gamma	4.744631e-06	4.744631e-06	Equality	4.744631e-06	19	OK
xi	7.065580e-07	7.065580e-07	Equality	7.065580e-07	19	OK
m	1.311352e-10	1.311352e-10	Equality	1.311352e-10	17	OK
C	0.000000e+00	0.000000e+00	Equality	0.000000e+00	0	OK

Table E.35: Group A — decision-variable boundaries for UF on SC-3 (min-fuel); positive worst margin = **ACTIVE**.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
Final Time (tf)	1.0233e-17	1.0233e-17	Upper	-1.0000	–	OK
Final Time (tf)	1.0233e-17	1.0233e-17	Lower	-1.0133e-17	–	OK
Thrust (T)	0.1723	0.5760	Upper	-0.4240	238	OK
Thrust (T)	0.1723	0.5760	Lower	-0.1723	259	OK
Lift Coeff (CL)	0.3219	0.6546	Upper	-0.1050	259	OK
Lift Coeff (CL)	0.3219	0.6546	Lower	-0.5624	0	OK
Roll Rate (p)	-0.0036	0.0012	Upper	-0.4988	259	OK
Roll Rate (p)	-0.0036	0.0012	Lower	-0.4964	207	OK
Pos X (North)	3.6415e-28	1.1111e-16	Upper	-0.5000	173	OK
Pos X (North)	3.6415e-28	1.1111e-16	Lower	-0.5000	0	OK
Pos Y (East)	-6.2823e-29	1.9052e-16	Upper	-0.5000	149	OK
Pos Y (East)	-6.2823e-29	1.9052e-16	Lower	-0.5000	0	OK
Pos Z (Down)	-0.0307	-0.0299	Upper	-0.0198	197	OK
Pos Z (Down)	-0.0307	-0.0299	Lower	-0.9794	72	OK
Velocity (V)	1.2386	1.7673	Upper	-0.3539	0	OK
Velocity (V)	1.2386	1.7673	Lower	-0.1174	259	OK
Bank (mu)	-0.3364	-5.6122e-13	Upper	-0.5000	0	OK
Bank (mu)	-0.3364	-5.6122e-13	Lower	-0.1636	249	OK
Gamma	-0.0156	0.0366	Upper	-0.5634	249	OK
Gamma	-0.0156	0.0366	Lower	-0.3844	135	OK
Heading (xi)	-0.0312	0.0312	Upper	-0.5000	0	OK
Heading (xi)	-0.0312	0.0312	Lower	-0.4375	259	OK
Mass (m)	0.9088	0.9091	Upper	-0.1818	0	OK
Mass (m)	0.9088	0.9091	Lower	-0.8179	259	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Upper	-1.0000	0	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Lower	0.0000e+00	0	OK

Table E.36: Group B — hard (path) constraints for UF on SC-3 (min-fuel); sign convention as Table E.35.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
V_max (V/V_NE)	0.5839	0.8331	Upper	-0.1669	0	OK
V_max (V/V_NE)	0.5839	0.8331	Lower	-0.5839	259	OK
V_stall	1.1051	1.5765	Lower	-0.1051	259	OK
Power_max	0.1977	0.7097	Upper	-0.2903	238	OK
Alt_min (h_min)	2.9639	3.0395	Lower	-1.9639	197	OK

Table E.37: Group C — dynamic constraints (trapezoidal collocation residuals, ℓ_∞ per state) for UF on SC-3 (min-fuel); equalities satisfied to tolerance (status OK).

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
x	9.535400e-09	9.535400e-09	Equality	9.535400e-09	16	OK
y	3.623076e-08	3.623076e-08	Equality	3.623076e-08	9	OK
z	1.504510e-05	1.504510e-05	Equality	1.504510e-05	19	OK
v	1.982705e-08	1.982705e-08	Equality	1.982705e-08	1	OK
mu	5.343517e-08	5.343517e-08	Equality	5.343517e-08	17	OK
gamma	6.788205e-07	6.788205e-07	Equality	6.788205e-07	17	OK
xi	5.450713e-08	5.450713e-08	Equality	5.450713e-08	17	OK
m	8.913281e-12	8.913281e-12	Equality	8.913281e-12	15	OK
C	0.000000e+00	0.000000e+00	Equality	0.000000e+00	0	OK

E.5.2 Full-control formulation (FC)

Table E.38: Group A — decision-variable boundaries for FC on SC-3 (min-time); positive worst margin = **ACTIVE**.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
Final Time (tf)	8.9253e-18	8.9253e-18	Upper	-1.0000	–	OK
Final Time (tf)	8.9253e-18	8.9253e-18	Lower	-8.8253e-18	–	OK
Thrust (T)	0.1327	0.5258	Upper	-0.4742	125	OK
Thrust (T)	0.1327	0.5258	Lower	-0.1327	238	OK
Elevator (de)	-0.1403	0.0011	Upper	-0.4989	0	OK
Elevator (de)	-0.1403	0.0011	Lower	-0.3597	259	OK
Aileron (da)	-0.0164	0.0769	Upper	-0.4231	259	OK
Aileron (da)	-0.0164	0.0769	Lower	-0.4836	218	OK
Rudder (dr)	-0.0323	0.3335	Upper	-0.1665	259	OK
Rudder (dr)	-0.0323	0.3335	Lower	-0.4677	166	OK
Pos X (North)	0.0000e+00	1.1111e-16	Upper	-0.5000	182	OK
Pos X (North)	0.0000e+00	1.1111e-16	Lower	-0.5000	0	OK
Pos Y (East)	0.0000e+00	1.9323e-16	Upper	-0.5000	141	OK
Pos Y (East)	0.0000e+00	1.9323e-16	Lower	-0.5000	0	OK
Pos Z (Down)	-0.0351	-0.0287	Upper	-0.0186	41	OK
Pos Z (Down)	-0.0351	-0.0287	Lower	-0.9750	197	OK
Mass (m)	1.9991e-16	2.0000e-16	Upper	-0.5000	0	OK
Mass (m)	1.9991e-16	2.0000e-16	Lower	-0.5000	0	OK
Vel U (body)	0.7382	0.8893	Upper	-0.1107	32	OK
Vel U (body)	0.7382	0.8893	Lower	-0.7382	238	OK
Vel V (body)	-9.9092e-20	4.3548e-19	Upper	-0.5000	0	OK
Vel V (body)	-9.9092e-20	4.3548e-19	Lower	-0.5000	0	OK
Vel W (body)	0.0000e+00	4.4459e-19	Upper	-0.5000	0	OK
Vel W (body)	0.0000e+00	4.4459e-19	Lower	-0.5000	0	OK
Roll (phi)	-0.3427	0.0000e+00	Upper	-0.5000	0	OK
Roll (phi)	-0.3427	0.0000e+00	Lower	-0.1573	249	OK
Pitch (theta)	0.0000e+00	0.2277	Upper	-0.3723	145	OK
Pitch (theta)	0.0000e+00	0.2277	Lower	-0.4000	0	OK
Rate P (roll)	-0.0481	0.0585	Upper	-0.4415	228	OK
Rate P (roll)	-0.0481	0.0585	Lower	-0.4519	166	OK
Rate Q (pitch)	-0.0370	0.1035	Upper	-0.3965	249	OK
Rate Q (pitch)	-0.0370	0.1035	Lower	-0.4630	238	OK
Rate R (yaw)	-0.2653	0.0841	Upper	-0.4159	238	OK
Rate R (yaw)	-0.2653	0.0841	Lower	-0.2347	249	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Upper	-1.0000	0	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Lower	0.0000e+00	0	OK

Table E.39: Group B — hard (path) constraints for FC on SC-3 (min-time); sign convention as Table E.38.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
V_max (V/V_NE)	0.7450	0.8908	Upper	-0.1092	31	OK
V_max (V/V_NE)	0.7450	0.8908	Lower	-0.7450	238	OK
V_stall	1.4099	1.6857	Lower	-0.4099	238	OK
CL_max	0.2488	0.5546	Upper	-0.4454	228	OK
CL_max	0.2488	0.5546	Lower	-0.7488	0	OK
Power_max	0.1931	0.9099	Upper	-0.0901	124	OK
Alt_min (h_min)	2.8422	3.4735	Lower	-1.8422	41	OK

Table E.40: Group C — dynamic constraints (trapezoidal collocation residuals, ℓ_∞ per state) for FC on SC-3 (min-time); equalities satisfied to tolerance (status OK).

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
x	2.764840e-10	2.764840e-10	Equality	2.764840e-10	23	OK
y	3.583260e-11	3.583260e-11	Equality	3.583260e-11	24	OK
z	6.352489e-08	6.352489e-08	Equality	6.352489e-08	17	OK
m	9.188190e-14	9.188190e-14	Equality	9.188190e-14	16	OK
u	1.983026e-09	1.983026e-09	Equality	1.983026e-09	22	OK
v	6.033928e-09	6.033928e-09	Equality	6.033928e-09	23	OK
w	1.051770e-09	1.051770e-09	Equality	1.051770e-09	23	OK
phi	1.339411e-10	1.339411e-10	Equality	1.339411e-10	17	OK
theta	1.210088e-10	1.210088e-10	Equality	1.210088e-10	18	OK
psi	2.351605e-10	2.351605e-10	Equality	2.351605e-10	23	OK
p	6.227332e-08	6.227332e-08	Equality	6.227332e-08	24	OK
q	3.109183e-08	3.109183e-08	Equality	3.109183e-08	24	OK
r	2.121316e-08	2.121316e-08	Equality	2.121316e-08	22	OK
C	0.000000e+00	0.000000e+00	Equality	0.000000e+00	0	OK

Table E.41: Group A — decision-variable boundaries for FC on SC-3 (min-fuel); positive worst margin = **ACTIVE**.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
Final Time (tf)	9.4801e-18	9.4801e-18	Upper	-1.0000	–	OK
Final Time (tf)	9.4801e-18	9.4801e-18	Lower	-9.3801e-18	–	OK
Thrust (T)	0.1075	0.5429	Upper	-0.4571	135	OK
Thrust (T)	0.1075	0.5429	Lower	-0.1075	248	OK
Elevator (de)	-0.1873	0.0187	Upper	-0.4813	0	OK
Elevator (de)	-0.1873	0.0187	Lower	-0.3127	218	OK
Aileron (da)	-0.0035	0.0335	Upper	-0.4665	259	OK
Aileron (da)	-0.0035	0.0335	Lower	-0.4965	197	OK
Rudder (dr)	-0.0086	0.1956	Upper	-0.3044	259	OK
Rudder (dr)	-0.0086	0.1956	Lower	-0.4914	166	OK
Pos X (North)	0.0000e+00	1.1111e-16	Upper	-0.5000	176	OK
Pos X (North)	0.0000e+00	1.1111e-16	Lower	-0.5000	0	OK
Pos Y (East)	0.0000e+00	1.9256e-16	Upper	-0.5000	142	OK
Pos Y (East)	0.0000e+00	1.9256e-16	Lower	-0.5000	0	OK
Pos Z (Down)	-0.0365	-0.0284	Upper	-0.0183	42	OK
Pos Z (Down)	-0.0365	-0.0284	Lower	-0.9736	196	OK
Mass (m)	1.9992e-16	2.0000e-16	Upper	-0.5000	0	OK
Mass (m)	1.9992e-16	2.0000e-16	Lower	-0.5000	0	OK
Vel U (body)	0.6756	0.8453	Upper	-0.1547	21	OK
Vel U (body)	0.6756	0.8453	Lower	-0.6756	228	OK
Vel V (body)	-1.8701e-20	1.8736e-19	Upper	-0.5000	0	OK
Vel V (body)	-1.8701e-20	1.8736e-19	Lower	-0.5000	0	OK
Vel W (body)	0.0000e+00	5.0071e-19	Upper	-0.5000	0	OK
Vel W (body)	0.0000e+00	5.0071e-19	Lower	-0.5000	0	OK
Roll (phi)	-0.3425	0.0000e+00	Upper	-0.5000	0	OK
Roll (phi)	-0.3425	0.0000e+00	Lower	-0.1575	249	OK
Pitch (theta)	0.0000e+00	0.3398	Upper	-0.2602	155	OK
Pitch (theta)	0.0000e+00	0.3398	Lower	-0.4000	0	OK
Rate P (roll)	-0.0283	0.0265	Upper	-0.4735	228	OK
Rate P (roll)	-0.0283	0.0265	Lower	-0.4717	10	OK
Rate Q (pitch)	-0.0040	0.0466	Upper	-0.4534	228	OK
Rate Q (pitch)	-0.0040	0.0466	Lower	-0.4960	145	OK
Rate R (yaw)	-0.1405	0.0000e+00	Upper	-0.5000	0	OK
Rate R (yaw)	-0.1405	0.0000e+00	Lower	-0.3595	228	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Upper	-1.0000	0	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Lower	0.0000e+00	0	OK

Table E.42: Group B — hard (path) constraints for FC on SC-3 (min-fuel); sign convention as Table E.41.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
V_max (V/V_NE)	0.6899	0.8477	Upper	-0.1523	21	OK
V_max (V/V_NE)	0.6899	0.8477	Lower	-0.6899	228	OK
V_stall	1.3056	1.6041	Lower	-0.3056	228	OK
CL_max	0.2513	0.6288	Upper	-0.3712	228	OK
CL_max	0.2513	0.6288	Lower	-0.7513	0	OK
Power_max	0.1467	0.8622	Upper	-0.1378	134	OK
Alt_min (h_min)	2.8102	3.6154	Lower	-1.8102	42	OK

Table E.43: Group C — dynamic constraints (trapezoidal collocation residuals, ℓ_∞ per state) for FC on SC-3 (min-fuel); equalities satisfied to tolerance (status OK).

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
x	9.355514e-11	9.355514e-11	Equality	9.355514e-11	18	OK
y	9.498992e-11	9.498992e-11	Equality	9.498992e-11	6	OK
z	6.084440e-08	6.084440e-08	Equality	6.084440e-08	19	OK
m	1.551797e-13	1.551797e-13	Equality	1.551797e-13	17	OK
u	1.041553e-09	1.041553e-09	Equality	1.041553e-09	24	OK
v	8.720355e-10	8.720355e-10	Equality	8.720355e-10	19	OK
w	1.484100e-09	1.484100e-09	Equality	1.484100e-09	19	OK
phi	8.479360e-11	8.479360e-11	Equality	8.479360e-11	10	OK
theta	9.145922e-11	9.145922e-11	Equality	9.145922e-11	21	OK
psi	8.984415e-11	8.984415e-11	Equality	8.984415e-11	10	OK
p	7.056765e-08	7.056765e-08	Equality	7.056765e-08	18	OK
q	1.053111e-07	1.053111e-07	Equality	1.053111e-07	19	OK
r	2.519083e-08	2.519083e-08	Equality	2.519083e-08	19	OK
C	0.000000e+00	0.000000e+00	Equality	0.000000e+00	0	OK

E.6 SC-03 — combined climb-and-turn

E.6.1 Uncoupled formulation (UF)

Table E.44: Group A — decision-variable boundaries for UF on SC-4 (min-time); positive worst margin = **ACTIVE**.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
Final Time (tf)	8.7511e-18	8.7511e-18	Upper	-1.0000	–	OK
Final Time (tf)	8.7511e-18	8.7511e-18	Lower	-8.6511e-18	–	OK
Thrust (T)	0.3000	0.7159	Upper	-0.2841	248	OK
Thrust (T)	0.3000	0.7159	Lower	-0.3000	207	OK
Lift Coeff (CL)	0.2565	0.6465	Upper	-0.1130	249	OK
Lift Coeff (CL)	0.2565	0.6465	Lower	-0.4970	93	OK
Roll Rate (p)	-0.0028	4.1194e-04	Upper	-0.4996	259	OK
Roll Rate (p)	-0.0028	4.1194e-04	Lower	-0.4972	176	OK
Pos X (North)	-2.9786e-31	1.1111e-16	Upper	-0.5000	170	OK
Pos X (North)	-2.9786e-31	1.1111e-16	Lower	-0.5000	0	OK
Pos Y (East)	-1.1378e-30	1.9110e-16	Upper	-0.5000	141	OK
Pos Y (East)	-1.1378e-30	1.9110e-16	Lower	-0.5000	0	OK
Pos Z (Down)	-0.0438	-0.0302	Upper	-0.0201	41	OK
Pos Z (Down)	-0.0438	-0.0302	Lower	-0.9663	259	OK
Velocity (V)	1.2892	1.9786	Upper	-0.1426	104	OK
Velocity (V)	1.2892	1.9786	Lower	-0.1680	259	OK
Bank (mu)	-0.3409	-8.5019e-16	Upper	-0.5000	0	OK
Bank (mu)	-0.3409	-8.5019e-16	Lower	-0.1591	248	OK
Gamma	-0.0086	0.2371	Upper	-0.3629	249	OK
Gamma	-0.0086	0.2371	Lower	-0.3914	20	OK
Heading (xi)	-0.0312	0.0313	Upper	-0.5000	0	OK
Heading (xi)	-0.0312	0.0313	Lower	-0.4375	259	OK
Mass (m)	0.9086	0.9091	Upper	-0.1818	0	OK
Mass (m)	0.9086	0.9091	Lower	-0.8177	259	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Upper	-1.0000	0	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Lower	0.0000e+00	0	OK

Table E.45: Group B — hard (path) constraints for UF on SC-4 (min-time); sign convention as Table E.44.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
V_max (V/V_NE)	0.6078	0.9328	Upper	-0.0672	104	OK
V_max (V/V_NE)	0.6078	0.9328	Lower	-0.6078	259	OK
V_stall	1.1503	1.7652	Lower	-0.1503	259	OK
Power_max	0.4433	0.9295	Upper	-0.0705	11	OK
Alt_min (h_min)	2.9861	4.3333	Lower	-1.9861	41	OK

Table E.46: Group C — dynamic constraints (trapezoidal collocation residuals, ℓ_∞ per state) for UF on SC-4 (min-time); equalities satisfied to tolerance (status OK).

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
x	1.230607e-10	1.230607e-10	Equality	1.230607e-10	7	OK
y	1.334895e-10	1.334895e-10	Equality	1.334895e-10	4	OK
z	4.963455e-10	4.963455e-10	Equality	4.963455e-10	5	OK
v	2.314813e-10	2.314813e-10	Equality	2.314813e-10	22	OK
mu	2.674408e-10	2.674408e-10	Equality	2.674408e-10	15	OK
gamma	4.459980e-09	4.459980e-09	Equality	4.459980e-09	21	OK
xi	4.859055e-10	4.859055e-10	Equality	4.859055e-10	1	OK
m	7.570384e-14	7.570384e-14	Equality	7.570384e-14	7	OK
C	0.000000e+00	0.000000e+00	Equality	0.000000e+00	0	OK

Table E.47: Group A — decision-variable boundaries for UF on SC-4 (min-fuel); positive worst margin = **ACTIVE**.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
Final Time (tf)	1.0347e-17	1.0347e-17	Upper	-1.0000	–	OK
Final Time (tf)	1.0347e-17	1.0347e-17	Lower	-1.0247e-17	–	OK
Thrust (T)	0.2983	0.5151	Upper	-0.4849	238	OK
Thrust (T)	0.2983	0.5151	Lower	-0.2983	259	OK
Lift Coeff (CL)	0.3295	0.6373	Upper	-0.1222	259	OK
Lift Coeff (CL)	0.3295	0.6373	Lower	-0.5700	0	OK
Roll Rate (p)	-0.0037	0.0021	Upper	-0.4979	259	OK
Roll Rate (p)	-0.0037	0.0021	Lower	-0.4963	186	OK
Pos X (North)	-1.3798e-29	1.1111e-16	Upper	-0.5000	178	OK
Pos X (North)	-1.3798e-29	1.1111e-16	Lower	-0.5000	0	OK
Pos Y (East)	2.9514e-29	1.9403e-16	Upper	-0.5000	141	OK
Pos Y (East)	2.9514e-29	1.9403e-16	Lower	-0.5000	0	OK
Pos Z (Down)	-0.0438	-0.0303	Upper	-0.0202	0	OK
Pos Z (Down)	-0.0438	-0.0303	Lower	-0.9663	259	OK
Velocity (V)	1.2647	1.7673	Upper	-0.3539	0	OK
Velocity (V)	1.2647	1.7673	Lower	-0.1435	259	OK
Bank (mu)	-0.3177	4.7369e-13	Upper	-0.5000	0	OK
Bank (mu)	-0.3177	4.7369e-13	Lower	-0.1823	249	OK
Gamma	-9.2217e-13	0.1206	Upper	-0.4794	249	OK
Gamma	-9.2217e-13	0.1206	Lower	-0.4000	0	OK
Heading (xi)	-0.0312	0.0313	Upper	-0.5000	0	OK
Heading (xi)	-0.0312	0.0313	Lower	-0.4375	259	OK
Mass (m)	0.9086	0.9091	Upper	-0.1818	0	OK
Mass (m)	0.9086	0.9091	Lower	-0.8177	259	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Upper	-1.0000	0	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Lower	0.0000e+00	0	OK

Table E.48: Group B — hard (path) constraints for UF on SC-4 (min-fuel); sign convention as Table E.47.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
V_max (V/V_NE)	0.5962	0.8331	Upper	-0.1669	0	OK
V_max (V/V_NE)	0.5962	0.8331	Lower	-0.5962	259	OK
V_stall	1.1285	1.5765	Lower	-0.1285	259	OK
Power_max	0.3554	0.7285	Upper	-0.2715	93	OK
Alt_min (h_min)	3.0000	4.3333	Lower	-2.0000	0	OK

Table E.49: Group C — dynamic constraints (trapezoidal collocation residuals, ℓ_∞ per state) for UF on SC-4 (min-fuel); equalities satisfied to tolerance (status OK).

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
x	2.370448e-08	2.370448e-08	Equality	2.370448e-08	7	OK
y	4.492820e-08	4.492820e-08	Equality	4.492820e-08	5	OK
z	1.427648e-07	1.427648e-07	Equality	1.427648e-07	9	OK
v	5.125591e-08	5.125591e-08	Equality	5.125591e-08	24	OK
mu	2.114119e-07	2.114119e-07	Equality	2.114119e-07	19	OK
gamma	5.538264e-07	5.538264e-07	Equality	5.538264e-07	3	OK
xi	4.826705e-08	4.826705e-08	Equality	4.826705e-08	4	OK
m	1.088206e-11	1.088206e-11	Equality	1.088206e-11	22	OK
C	0.000000e+00	0.000000e+00	Equality	0.000000e+00	0	OK

E.6.2 Full-control formulation (FC)

Table E.50: Group A — decision-variable boundaries for FC on SC-4 (min-time); positive worst margin = **ACTIVE**.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
Final Time (tf)	8.8087e-18	8.8087e-18	Upper	-1.0000	–	OK
Final Time (tf)	8.8087e-18	8.8087e-18	Lower	-8.7087e-18	–	OK
Thrust (T)	0.1636	0.5777	Upper	-0.4223	166	OK
Thrust (T)	0.1636	0.5777	Lower	-0.1636	249	OK
Elevator (de)	-0.3262	-0.0013	Upper	-0.5013	0	OK
Elevator (de)	-0.3262	-0.0013	Lower	-0.1738	259	OK
Aileron (da)	-0.0176	0.0627	Upper	-0.4373	259	OK
Aileron (da)	-0.0176	0.0627	Lower	-0.4824	218	OK
Rudder (dr)	-0.0308	0.2745	Upper	-0.2255	259	OK
Rudder (dr)	-0.0308	0.2745	Lower	-0.4692	166	OK
Pos X (North)	0.0000e+00	1.1111e-16	Upper	-0.5000	179	OK
Pos X (North)	0.0000e+00	1.1111e-16	Lower	-0.5000	0	OK
Pos Y (East)	0.0000e+00	1.8782e-16	Upper	-0.5000	145	OK
Pos Y (East)	0.0000e+00	1.8782e-16	Lower	-0.5000	0	OK
Pos Z (Down)	-0.0439	-0.0289	Upper	-0.0188	31	OK
Pos Z (Down)	-0.0439	-0.0289	Lower	-0.9662	249	OK
Mass (m)	1.9990e-16	2.0000e-16	Upper	-0.5000	0	OK
Mass (m)	1.9990e-16	2.0000e-16	Lower	-0.5000	0	OK
Vel U (body)	0.6144	0.8903	Upper	-0.1097	41	OK
Vel U (body)	0.6144	0.8903	Lower	-0.6144	259	OK
Vel V (body)	-7.6031e-20	3.5615e-19	Upper	-0.5000	0	OK
Vel V (body)	-7.6031e-20	3.5615e-19	Lower	-0.5000	0	OK
Vel W (body)	0.0000e+00	6.7378e-19	Upper	-0.5000	0	OK
Vel W (body)	0.0000e+00	6.7378e-19	Lower	-0.5000	0	OK
Roll (phi)	-0.3415	0.0000e+00	Upper	-0.5000	0	OK
Roll (phi)	-0.3415	0.0000e+00	Lower	-0.1585	249	OK
Pitch (theta)	0.0000e+00	0.4623	Upper	-0.1377	259	OK
Pitch (theta)	0.0000e+00	0.4623	Lower	-0.4000	0	OK
Rate P (roll)	-0.0591	0.1366	Upper	-0.3634	228	OK
Rate P (roll)	-0.0591	0.1366	Lower	-0.4409	217	OK
Rate Q (pitch)	-0.0074	0.0881	Upper	-0.4119	228	OK
Rate Q (pitch)	-0.0074	0.0881	Lower	-0.4926	166	OK
Rate R (yaw)	-0.2201	0.0304	Upper	-0.4696	238	OK
Rate R (yaw)	-0.2201	0.0304	Lower	-0.2799	228	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Upper	-1.0000	0	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Lower	0.0000e+00	0	OK

Table E.51: Group B — hard (path) constraints for FC on SC-4 (min-time); sign convention as Table E.50.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
V_max (V/V_NE)	0.6498	0.8918	Upper	-0.1082	32	OK
V_max (V/V_NE)	0.6498	0.8918	Lower	-0.6498	259	OK
V_stall	1.2299	1.6875	Lower	-0.2299	259	OK
CL_max	0.2484	0.7843	Upper	-0.2157	259	OK
CL_max	0.2484	0.7843	Lower	-0.7484	0	OK
Power_max	0.2118	0.9502	Upper	-0.0498	134	OK
Alt_min (h_min)	2.8649	4.3445	Lower	-1.8649	31	OK

Table E.52: Group C — dynamic constraints (trapezoidal collocation residuals, ℓ_∞ per state) for FC on SC-4 (min-time); equalities satisfied to tolerance (status OK).

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
x	2.401565e-08	2.401565e-08	Equality	2.401565e-08	23	OK
y	4.696898e-09	4.696898e-09	Equality	4.696898e-09	23	OK
z	1.274794e-07	1.274794e-07	Equality	1.274794e-07	24	OK
m	4.544866e-11	4.544866e-11	Equality	4.544866e-11	23	OK
u	5.217365e-08	5.217365e-08	Equality	5.217365e-08	24	OK
v	2.081717e-07	2.081717e-07	Equality	2.081717e-07	24	OK
w	4.437957e-08	4.437957e-08	Equality	4.437957e-08	23	OK
phi	1.210097e-08	1.210097e-08	Equality	1.210097e-08	24	OK
theta	1.484277e-08	1.484277e-08	Equality	1.484277e-08	10	OK
psi	1.903139e-08	1.903139e-08	Equality	1.903139e-08	12	OK
p	3.782804e-06	3.782804e-06	Equality	3.782804e-06	23	OK
q	5.573942e-06	5.573942e-06	Equality	5.573942e-06	23	OK
r	1.032232e-06	1.032232e-06	Equality	1.032232e-06	21	OK
C	0.000000e+00	0.000000e+00	Equality	0.000000e+00	0	OK

Table E.53: Group A — decision-variable boundaries for FC on SC-4 (min-fuel); positive worst margin = **ACTIVE**.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
Final Time (tf)	9.3966e-18	9.3966e-18	Upper	-1.0000	–	OK
Final Time (tf)	9.3966e-18	9.3966e-18	Lower	-9.2966e-18	–	OK
Thrust (T)	0.1180	0.5980	Upper	-0.4020	165	OK
Thrust (T)	0.1180	0.5980	Lower	-0.1180	249	OK
Elevator (de)	-0.3414	0.0138	Upper	-0.4862	0	OK
Elevator (de)	-0.3414	0.0138	Lower	-0.1586	259	OK
Aileron (da)	-0.0044	0.0201	Upper	-0.4799	259	OK
Aileron (da)	-0.0044	0.0201	Lower	-0.4956	218	OK
Rudder (dr)	-0.0075	0.0971	Upper	-0.4029	259	OK
Rudder (dr)	-0.0075	0.0971	Lower	-0.4925	166	OK
Pos X (North)	0.0000e+00	1.1111e-16	Upper	-0.5000	178	OK
Pos X (North)	0.0000e+00	1.1111e-16	Lower	-0.5000	0	OK
Pos Y (East)	0.0000e+00	1.8875e-16	Upper	-0.5000	143	OK
Pos Y (East)	0.0000e+00	1.8875e-16	Lower	-0.5000	0	OK
Pos Z (Down)	-0.0447	-0.0288	Upper	-0.0187	31	OK
Pos Z (Down)	-0.0447	-0.0288	Lower	-0.9654	238	OK
Mass (m)	1.9990e-16	2.0000e-16	Upper	-0.5000	0	OK
Mass (m)	1.9990e-16	2.0000e-16	Lower	-0.5000	0	OK
Vel U (body)	0.5740	0.8590	Upper	-0.1410	31	OK
Vel U (body)	0.5740	0.8590	Lower	-0.5740	259	OK
Vel V (body)	-1.7502e-20	1.0561e-19	Upper	-0.5000	0	OK
Vel V (body)	-1.7502e-20	1.0561e-19	Lower	-0.5000	0	OK
Vel W (body)	0.0000e+00	6.8213e-19	Upper	-0.5000	0	OK
Vel W (body)	0.0000e+00	6.8213e-19	Lower	-0.5000	0	OK
Roll (phi)	-0.3383	0.0000e+00	Upper	-0.5000	0	OK
Roll (phi)	-0.3383	0.0000e+00	Lower	-0.1617	259	OK
Pitch (theta)	0.0000e+00	0.4861	Upper	-0.1139	249	OK
Pitch (theta)	0.0000e+00	0.4861	Lower	-0.4000	0	OK
Rate P (roll)	-0.0202	0.0694	Upper	-0.4306	249	OK
Rate P (roll)	-0.0202	0.0694	Lower	-0.4798	10	OK
Rate Q (pitch)	-0.0016	0.0562	Upper	-0.4438	259	OK
Rate Q (pitch)	-0.0016	0.0562	Lower	-0.4984	62	OK
Rate R (yaw)	-0.1320	0.0000e+00	Upper	-0.5000	0	OK
Rate R (yaw)	-0.1320	0.0000e+00	Lower	-0.3680	249	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Upper	-1.0000	0	OK
Fuel/Cost (C)	0.0000e+00	0.0000e+00	Lower	0.0000e+00	0	OK

Table E.54: Group B — hard (path) constraints for FC on SC-4 (min-fuel); sign convention as Table E.53.

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
V_max (V/V_NE)	0.6051	0.8612	Upper	-0.1388	31	OK
V_max (V/V_NE)	0.6051	0.8612	Lower	-0.6051	259	OK
V_stall	1.1453	1.6297	Lower	-0.1453	259	OK
CL_max	0.2506	0.8281	Upper	-0.1719	259	OK
CL_max	0.2506	0.8281	Lower	-0.7506	0	OK
Power_max	0.1401	0.9221	Upper	-0.0779	135	OK
Alt_min (h_min)	2.8497	4.4263	Lower	-1.8497	31	OK

Table E.55: Group C — dynamic constraints (trapezoidal collocation residuals, ℓ_∞ per state) for FC on SC-4 (min-fuel); equalities satisfied to tolerance (status OK).

Variable / Constraint	Min value	Max value	Boundary	Worst margin	Node	Status
x	3.064511e-09	3.064511e-09	Equality	3.064511e-09	23	OK
y	2.249463e-09	2.249463e-09	Equality	2.249463e-09	22	OK
z	1.407923e-08	1.407923e-08	Equality	1.407923e-08	20	OK
m	1.160464e-11	1.160464e-11	Equality	1.160464e-11	21	OK
u	2.550602e-08	2.550602e-08	Equality	2.550602e-08	23	OK
v	5.480238e-08	5.480238e-08	Equality	5.480238e-08	21	OK
w	6.095829e-08	6.095829e-08	Equality	6.095829e-08	20	OK
phi	5.733406e-09	5.733406e-09	Equality	5.733406e-09	17	OK
theta	1.521915e-08	1.521915e-08	Equality	1.521915e-08	18	OK
psi	2.868493e-09	2.868493e-09	Equality	2.868493e-09	10	OK
p	1.696214e-06	1.696214e-06	Equality	1.696214e-06	21	OK
q	1.824688e-06	1.824688e-06	Equality	1.824688e-06	23	OK
r	3.614640e-07	3.614640e-07	Equality	3.614640e-07	23	OK
C	0.000000e+00	0.000000e+00	Equality	0.000000e+00	0	OK

Appendix F

Min-time vs min-fuel comparison: per-scenario tables

This annex collects the detailed numerical comparison between the minimum-time ($\alpha = 0$) and minimum-fuel ($\alpha = 1$) solutions for every benchmark scenario and for both formulations. The global, condensed view used in the body chapter is reported in Figures 6.5 and 6.6 of Section 6.4.5; the present annex backs those figures with the underlying per-scenario tables in their original per-(scenario, formulation) layout.

The tables share a common structure: rows list the standard performance, kinematic, propulsive and computational metrics; columns report the converged value under the min-time and min-fuel criteria, followed by the relative deviation $\Delta = 100\,(\text{min-fuel} - \text{min-time})/\text{min-time}$.

F.1 Scenario SC-00 — straight-and-level baseline

Table F.1: Min-time vs min-fuel comparison for UF on SC-1; Δ is min-fuel vs min-time [%].

Metric	**Min-time** ($\alpha = 0$)	**Min-fuel** ($\alpha = 1$)	**Δ [%]**
Flight time t_f [s]	51.09	64.49	26.2
Energy consumed [J]	1.62e+07	9.83e+06	-39.4
Path length [m]	3346.7	3346.5	-0.0
Mean airspeed $\bar{V}$ [m/s]	65.35	51.85	-20.7
Mean thrust $\bar{T}$ [N]	4839.58	2888.97	-40.3
Max. thrust $T_{\max}$ [N]	6029.40	3530.43	-41.4
Max. shaft power $P_{\max}$ [W]	3.7e+05	1.8e+05	-49.6
Mean bank $\overline{\lvert\phi\rvert}$ [°]	0.01	0.01	22.8
Mean flight-path $\bar{\gamma}$ [°]	0.02	0.03	49.8
Pipeline solve time [s]	0.64	0.78	22.0

Table F.2: Min-time vs min-fuel comparison for FC on SC-1; Δ is min-fuel vs min-time [%].

Metric	**Min-time** ($\alpha = 0$)	**Min-fuel** ($\alpha = 1$)	**Δ [%]**
Flight time t_f [s]	55.04	61.83	12.3
Energy consumed [J]	1.38e+07	1.03e+07	-25.6
Path length [m]	3347.1	3347.9	0.0
Mean airspeed $\bar{V}$ [m/s]	60.74	54.05	-11.0
Mean thrust $\bar{T}$ [N]	4100.34	2987.44	-27.1
Max. thrust $T_{\max}$ [N]	4510.25	4874.04	8.1
Max. shaft power $P_{\max}$ [W]	2.7e+05	2.6e+05	-3.9
Mean bank $\overline{\lvert\phi\rvert}$ [°]	0.00	0.00	-20.1
Mean pitch $\bar{\theta}$ [°]	4.17	7.54	80.9
Pipeline solve time [s]	3.25	6.03	85.3

F.2 Scenario SC-01 — straight climb

Table F.3: Min-time vs min-fuel comparison for UF on SC-2; Δ is min-fuel vs min-time [%].

Metric	**Min-time** ($\alpha = 0$)	**Min-fuel** ($\alpha = 1$)	**Δ [%]**
Flight time t_f [s]	57.12	63.12	10.5
Energy consumed [J]	1.96e+07	1.78e+07	-9.1
Path length [m]	3354.0	3353.2	-0.0
Mean airspeed $\bar{V}$ [m/s]	58.54	53.03	-9.4
Mean thrust $\bar{T}$ [N]	5802.60	5215.78	-10.1
Max. thrust $T_{\max}$ [N]	6106.74	6245.39	2.3
Max. shaft power $P_{\max}$ [W]	3.6e+05	3.3e+05	-8.6
Mean bank $\overline{\lvert\phi\rvert}$ [°]	0.02	0.01	-26.7
Mean flight-path $\bar{\gamma}$ [°]	3.50	3.45	-1.3
Pipeline solve time [s]	0.54	0.74	37.0

Table F.4: Min-time vs min-fuel comparison for FC on SC-2; Δ is min-fuel vs min-time [%].

Metric	**Min-time** ($\alpha = 0$)	**Min-fuel** ($\alpha = 1$)	**Δ [%]**
Flight time t_f [s]	56.73	57.74	1.8
Energy consumed [J]	2.04e+07	1.90e+07	-7.2
Path length [m]	3354.6	3355.3	0.0
Mean airspeed $\bar{V}$ [m/s]	59.03	57.91	-1.9
Mean thrust $\bar{T}$ [N]	6040.31	5483.34	-9.2
Max. thrust $T_{\max}$ [N]	6623.77	6840.90	3.3
Max. shaft power $P_{\max}$ [W]	3.8e+05	3.8e+05	1.9
Mean bank $\overline{\lvert\phi\rvert}$ [°]	0.00	0.01	19.9
Mean pitch $\bar{\theta}$ [°]	8.55	9.04	5.7
Pipeline solve time [s]	3.80	6.75	77.8

F.3 Scenario SC-02 — 180° heading reversal at level flight

Table F.5: Min-time vs min-fuel comparison for UF on SC-3; Δ is min-fuel vs min-time [%].

Metric	**Min-time** ($\alpha = 0$)	**Min-fuel** ($\alpha = 1$)	**Δ [%]**
Flight time t_f [s]	87.16	102.33	17.4
Energy consumed [J]	2.11e+07	1.81e+07	-14.4
Path length [m]	5311.6	5229.8	-1.5
Mean airspeed $\bar{V}$ [m/s]	60.77	51.28	-15.6
Mean thrust $\bar{T}$ [N]	3946.93	3396.23	-14.0
Max. thrust $T_{\max}$ [N]	5910.27	6391.94	8.1
Max. shaft power $P_{\max}$ [W]	3.6e+05	2.9e+05	-20.5
Mean bank $\overline{\lvert\phi\rvert}$ [°]	11.11	8.44	-24.0
Mean flight-path $\bar{\gamma}$ [°]	0.09	0.01	-92.0
Pipeline solve time [s]	1.39	1.74	25.3

Table F.6: Min-time vs min-fuel comparison for FC on SC-3; Δ is min-fuel vs min-time [%].

Metric	**Min-time** ($\alpha = 0$)	**Min-fuel** ($\alpha = 1$)	**Δ [%]**
Flight time t_f [s]	89.25	94.80	6.2
Energy consumed [J]	2.36e+07	2.10e+07	-10.9
Path length [m]	5455.8	5370.9	-1.6
Mean airspeed $\bar{V}$ [m/s]	61.21	56.82	-7.2
Mean thrust $\bar{T}$ [N]	4235.02	3750.07	-11.5
Max. thrust $T_{\max}$ [N]	5767.84	5957.75	3.3
Max. shaft power $P_{\max}$ [W]	3.7e+05	3.5e+05	-5.1
Mean bank $\overline{\lvert\phi\rvert}$ [°]	10.80	9.72	-10.0
Mean pitch $\bar{\theta}$ [°]	3.92	5.82	48.4
Pipeline solve time [s]	6.33	7.82	23.6

F.4 Scenario SC-03 — climb plus heading reversal

Table F.7: Min-time vs min-fuel comparison for UF on SC-4; Δ is min-fuel vs min-time [%].

Metric	**Min-time** ($\alpha = 0$)	**Min-fuel** ($\alpha = 1$)	**Δ [%]**
Flight time t_f [s]	87.51	103.47	18.2
Energy consumed [J]	2.89e+07	2.70e+07	-6.4
Path length [m]	5323.0	5448.2	2.4
Mean airspeed $\bar{V}$ [m/s]	60.65	52.78	-13.0
Mean thrust $\bar{T}$ [N]	5479.78	4879.81	-10.9
Max. thrust $T_{\max}$ [N]	7873.63	5663.51	-28.1
Max. shaft power $P_{\max}$ [W]	3.7e+05	2.9e+05	-22.1
Mean bank $\overline{\lvert\phi\rvert}$ [°]	11.01	8.66	-21.4
Mean flight-path $\bar{\gamma}$ [°]	2.39	2.08	-13.0
Pipeline solve time [s]	2.90	1.27	-56.2

Table F.8: Min-time vs min-fuel comparison for FC on SC-4; Δ is min-fuel vs min-time [%].

Metric	**Min-time** ($\alpha = 0$)	**Min-fuel** ($\alpha = 1$)	**Δ [%]**
Flight time t_f [s]	88.09	93.97	6.7
Energy consumed [J]	2.79e+07	2.68e+07	-4.0
Path length [m]	5238.2	5286.0	0.9
Mean airspeed $\bar{V}$ [m/s]	59.39	56.24	-5.3
Mean thrust $\bar{T}$ [N]	5186.71	4872.37	-6.1
Max. thrust $T_{\max}$ [N]	6337.71	6560.86	3.5
Max. shaft power $P_{\max}$ [W]	3.8e+05	3.7e+05	-2.9
Mean bank $\overline{\lvert\phi\rvert}$ [°]	10.40	9.65	-7.2
Mean pitch $\bar{\theta}$ [°]	7.33	8.76	19.5
Pipeline solve time [s]	8.92	6.01	-32.6